Meeting of the Board
30 June – 3 July 2025
Port Moresby, Papua New Guinea
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GCF/B.42/02/Add.12

9 June 2025

Consideration of funding proposals – Addendum XII

Funding proposal package for FP270

Summary

This addendum contains the following seven parts:

a) A funding proposal titled "Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR) in Cambodia";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Environmental and social report(s) disclosure;
d) Secretariat's assessment;
e) Independent Technical Advisory Panel's assessment;
f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
g) Gender documentation.

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Funding Proposal

Project/Programme title:	Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR) in Cambodia
Country:	Cambodia
Accredited Entity:	International Fund for Agriculture Development
Date of first submission:	
Date of current submission	
Version number	V.1- 13052025

GREEN CLIMATE FUND

GREEN CLIMATE FUND

Contents

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

LIST OF ABBREVIATIONS

ADB	Asian Development Bank
AE	Accredited Agency
AFLOU	Agriculture Forestry and Other Land Use
AIIB	Asian Infrastructure Investment Bank
AMA	Accreditation Master Agreement
AWD	Alternate Wetting and Drying
AIMS	Accelerating Inclusive Markets for Smallholders
ASPIRE	Agricultural Services Programme for Innovation, Resilience and Extension
ASPIRE AT	Agriculture Services Programme for an Inclusive Rural Economy and Agricultural Trade
CAEP	Climate Action Enhancement Package
CARD	Climate Adaptation in Rural Development
CAISAR	Climate Adaptive Irrigation and Sustainable Agriculture for Resilience
CAVAC	Cambodia Agriculture Value Chain Project
CCAP	Cambodia Climate Change Action Plan
CCCSP	Cambodia Climate Change Strategic Plan
CBME	Community Based Monitoring and Evaluation
CR	Climate Resilient
CSA	Climate Smart Agriculture
CSDGs	Cambodia Sustainable Development Goals
DED	Detailed Engineering Design
DFAT	Department of Foreign Affairs and Trade
DMSP	District Multi-Stakeholder Platform
EA/IA	Executing Agency/Implementing Agency
EE	Executing Entity
EFA	Economic and Financial Analysis
ESCP	Environmental and Social Commitment Plan
ESF	Environmental and Social Framework
ESMF	Environmental and Social Management Framework
ESMP	Environmental and Social Management Plan
ESO	Environmental and Social Officers
ESS	Environmental and Social Standards
FAO	Food and Agriculture Organization
FFBS	Farmers' Field and Business School
FOLU	Forestry and Other Land Use
FIRM	Feasibility Study and Project Preparation
FPIC	Free Prior and Informed Consent
FWS	Farmer's Water School approach
FWUC	Farmer Water User Committee
GASIP	Gender Assessment and Social Inclusion Plan
GDP	Gross Domestic Product
IDA	International Development Association
IFAD	International Fund for Agricultural Development

ILO	International Labour Organization
IP	Indigenous People
IPP	Indigenous Peoples' Plan
IPPF	Indigenous Peoples Planning Framework
IPPU	Industrial Processes and Product Use
IWRM	Integrated Water Resource Management
LDC	Least Developed Countries
LLL	Laser Land Levelling
MAFF	Ministry of Agriculture, Forestry and Fisheries
mDSR	Mechanized Dry Seeded Rice
M&E	Monitoring and Evaluation
MoE	Ministry of Environment
MoWRAM	Ministry of Water Resources and Meteorology
MoWAs	Ministry of Women's Affairs
MSP	Multi-Stakeholder Platform
MTs	Master Trainers
NAMAs	Nationally Appropriate Mitigation Actions
NAPs	National Adaptation Plan
NCDDS	National Committee for Sub-National Democratic Development Secretariat
NDA	National Designated Authority
NDCs	Nationally Determined Contributions
NGO	Non-Government Organization
NSPGC	National Strategic Plan on Green Growth
PEARL	Public-Social-Private Partnership for Ecologically Sound Agriculture and Resilient Livelihood project
PDAFF	Provincial Department of Agriculture, Forestry and Fisheries
PDOC	Provincial Department of Commerce
PDWRAM	Provincial Department of Water Resources and Meteorology
PIU	Project Implementation Unit
PMU	Project Management Unit
PPC	Project Preparation Consultants
PPPPs or 4Ps	Public Private Producer Partnerships
RCP	Representative Concentration Pathways
RGC	Royal Government of Cambodia
RP	Resettlement Plan
RPF	Resettlement Planning Framework
SA	Social Assessment
SAAMBAT	Sustainable Assets for Agriculture Markets, Business and Trade Project
SCADA	Supervisory Control and Data Acquisition
SEP	Stakeholder Engagement Plan
SMSs	Subject Matter Specialists
TOTs	Training of Trainers
UNDRR	United Nations Office for Disaster Risk Reduction
WB	World Bank

D

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Project	**A.2. Public or private sector**	Public
A.3. Request for Proposals (RFP)	Not applicable		

A.4. Result area(s)

*Check the applicable GCF result area(s) that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of **GCF and Co-financers' contribution** devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.*

	GCF contribution	Co-financers' contribution[1]
Mitigation total	**33.2%**	**31.5%**
☐ Energy generation and access	0%	0%
☐ Low-emission transport	0%	0%
☐ Buildings, cities, industries and appliances	0 %	0 %
☒ Forestry and land use	33.2%	31.5%
Adaptation total	**66.8 %**	**68%**
☒ Most vulnerable people and communities	17.4%	21.3%
☒ Health and well-being, and food and water security	13.7 %	13.9 %
☐ Infrastructure and built environment	0 %	0 %
☒ Ecosystems and ecosystem services	35.7 %	33.3 %

A.5. Expected mitigation outcome *(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)*	Reduction of **-1,006,507.43 tCO2-eq over 20 years** [2]	**A.6. Expected adaptation outcome** *(Core indicator 2: direct and indirect beneficiaries reached)*	*Indicate total number of direct and indirect beneficiaries.* *1,714,375 people*	
			Direct beneficiaries 562,757 people (225,103 female)	*Indicate number of indirect beneficiaries* 1151618people (575,809 female)
			Indicate % of direct beneficiaries vis-à-vis total population 3.16% of the Cambodian population.	*Indicate % of indirect beneficiaries vis-à-vis total population* 6.47% of the population
A.7. Total financing (GCF + co-finance)	240 USD million	**A.9. Project size**	Medium (Upto USD 250 million)	
A.8. Total GCF funding requested	80 USD *million*			

[1] Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.
[2] EX-ACT Analysis. Annex 22.

A.10. Financial instrument(s) requested for the GCF funding	*Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.* ☒ Grant 40 mn USD ☒ Loan 40 mn USD ☐ Guarantee Enter number ☐ Equity Enter number ☐ Results-based payment Enter number		
A.11. Implementation period	*7 years*	**A.12. Total lifespan**	*30 years.*[3]
A.13. Expected date of AE internal approval	Click or tap to enter a date.	**A.14. ESS category**	A
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	**A.18. Is this FP included in the country programme?**	Yes ☒ No ☐
A.19. Complementarity and coherence	*Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.* Yes ☒ No ☐		

[3] The project lifespan is for 30 years aligned with the lifespan of the irrigation infrastructure. However, the mitigation potential of the project in terms of reduction in CO_{2e} is based on and will be reported for a 20-year period (consistent with IPCC guidance/approach). Further, in the EFA, to ensure alignment with evaluation period, the GHG estimates are extended to 30 years using a linear assumption for the 30-year evaluation scenario.

A.20. Executing Entity information	The Royal Government of Cambodia, acting through the Ministry of Economy and Finance, the Ministry of Water Resources and Meteorology, and the National Committee for Sub-National Democratic Development Secretariat will be the Executing Entity. In 1996, the Ministry of Economy and Finance (MEF) was established by law. Its organization and functioning is defined by a sub-decree of the Royal Government of Cambodia.[4] The minister responsible for the MEF is selected by the prime minister and nominated by the King. Following a vote of confidence by the National Assembly, the minister is appointed along with the Council of Ministers by the royal decree (Preah Reach Kret). The role of the MEF is to implement and contribute to the government's economic and financial policy. Included in the main responsibilities of the ministry are the establishment of the country's uniform financial system, preparation and implementation of the national budget, distribution and redistribution of the total national revenues, inspection of the public's finances, and monitoring of the government's economic and financial policies.[5] The Ministry of Water Resources and Meteorology (MOWRAM) will be the main Executing Entity (EE) for CAISAR. Established in 1999, MOWRAM is a key government body in Cambodia, led by a Minister. Its responsibilities include developing strategic directions for the water sector, analyzing technical climate and hydrology data, identifying innovative techniques for water use and treatment, and collaborating in the management of the Mekong Basin concerning water resources and meteorology. The Ministry comprises ten departments and one technical center, with Provincial Departments of Water Resources and Meteorology (PDWRAMs) in each of Cambodia's 24 provinces for on-the-ground implementation. The National Committee for Sub-National Democratic Development Secretariat (NCDDS), which was established to implement decentralization reform policies, local governance, and capacity development, will also be an Executing Entity for the project. NCDDS is responsible for developing regulations and providing finance to local governments and will play a crucial role in the implementation of Component 1 and output 3.2 of Component 3, in close collaboration with provincial departments and service providers.

A.21. Executive summary

Climate change problem

i. Cambodia's agriculture sector is crucial for the economy but also highly susceptible to the effects of climate change. The country is ranked 149th out of 187 countries in the 2022 ND-GAIN Index, was the 12th most disaster-prone country among 172 countries between 1999–2018, and remains one of the few Least Developed Countries (LDCs) in Asia. The World Risk Index of 15.8% indicates a very high risk of disaster as a result of extreme natural events, with an exposure index of 27% and a vulnerability index of 59%. Approximately 80% of the country is within the Mekong River and Tonle Sap basins, increasing exposure to floods, storms, and droughts. The Tonle Sap Lake Basin, often referred to as the rice basket of Cambodia, undergoes cycles of severe droughts and floods. This has resulted in lower

[4] Sub-decree No.488 on Organization and Functioning of the Ministry of Economy and Finance dated on October 16, 2013.

[5] Ibid, supra-note 2, Sub-decree on Organization and Functioning of the Ministry of Economy and Finance, article 5.

agricultural productivity and reduced cultivation for smallholder farmers. Similar to other monsoon-dependent countries, climate change is altering Cambodia's monsoon calendar, increasing total drought days and shifting the start of the rainy season. While annual rainfall is expected to increase, the number of days with rain is decreasing and daily rain intensity is increasing, resulting in extreme rainfall events and flooding risks. Furthermore, identified hazards are assessed as high across the whole country for floods, cyclones and extreme heat. By mid-century, climate change will result in more frequent heavy precipitation days in the rainy season, and an increase in the number of consecutive dry days. The median temperature is projected to increase by 1.83°C by 2040-59 under RCP8.5 (World Bank CCKP).[6]

ii. The country has a high reliance on the agriculture sector, which accounted for nearly 22% of its GDP in 2022. The rural population represents 75.8% of the national population in Cambodia. Since 2020, the negative impacts of climate change and the pandemic have led to a 10% loss in Gross Domestic Product (GDP).[7] In addition to negative climate impacts, stagnating global agricultural commodity prices, rising labour costs and the limited scope for cropland expansion in the country pose challenges for the agriculture sector. Relatively low yields, coupled with frequent natural disasters, contribute to temporary food shortages for vulnerable communities. Seven of Cambodia's 25 provinces (including Phnom Penh) are classified as acutely food insecure, and an additional seven are moderately insecure. Livelihoods rely heavily on rain-fed agriculture and non-poor households are vulnerable to falling back into poverty in the event of extreme or frequent climate shocks.[8] Women and children are among the most vulnerable groups to climate change.

iii. The lack of irrigation water is a crucial barrier for farmers, alongside projected increase in dry days and rising temperatures aggravating climate change-induced water shortages, which prevent the cultivation of more than a single crop each year. Studies suggest that shifting to irrigated cultivation could result in annual overall production increases of up to 40%. Furthermore, adoption of intermittent drainage could reduce the greenhouse gas emissions from rice cultivation. The lack of timely information on droughts and floods (e.g. seasonal forecasts, the changing length of each event) is a major problem. Farmers lack the knowledge and tools to adapt their farming practices and production techniques to climate change as they are largely unaware of climate resilient practices, stress-tolerant seed varieties and improved planting materials, efficient input management, or the benefits of applying these practices. Cambodia is unlikely to achieve 5% of its annual agricultural growth target by 2030 without more investment in climate adaptation, sustainable irrigation, flood control and drainage schemes. Persistent poverty, limited access to finance, and insufficient institutional capacities have left the rural agrarian population vulnerable to the negative impacts of climate change, resulting economic slowdown, and environmental degradation.

Proposed interventions

iv. The Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR) project originated from the RGC's National Water Resources Management and Sustainable Irrigation Road Map and Investment Program (2019 -2023). The project is designed to assist the Ministry of Water Resources and Meteorology (MOWRAM) with its strategic plans to modernize the irrigation sector. The overall goal of the project is to make the agriculture sector in Cambodia climate resilient and sustainable. The specific objective of the project is to modernize the irrigation sector by installing resilient irrigation systems and enable farmers to adapt to climate risks and mitigate crop emissions through use of energy- and water-efficient technologies and practices, timely weather information and improved market integration. CAISAR's Theory of Change is premised on the experience that addressing the complex impacts of climate change on rain-fed and irrigated agriculture requires action at three levels: farm level, irrigation scheme level and at the national level by creating a strong institutional base and an enabling environment.

[6] World Bank. Climate Change Knowledge portal, Cambodia. https://climateknowledgeportal.worldbank.org/country/cambodia
[7] World Bank Group, 2015. Cambodia Economic Update, October 2015: Adapting to Stay Competitive. World Bank, Phnom Penh. http://hdl.handle.net/10986/22934
[8] https://docs.wfp.org/api/documents/WFP-0000147767/download/?_ga=2.188002408.1116277075.1681899242-636470978.1669479646

v. This objective of system modernization and improvement of farmer resilience will be achieved by implementing three components which intend to address climate change vulnerabilities, reduce GHG emissions, increase agriculture productivity, and develop institutional capacities: Component 1: Farm-level climate adaptation focusing on better crop-water management practices, capacity enhancement on select value chains, including rice, vegetable, poultry and aquaculture, providing opportunities for greater market integration through value chain development and rural road rehabilitation, and enhancing access to agro-meteorological information and services. Component 2: Upgrading and climate-proofing of water infrastructure which will focus on modernization of irrigation schemes and ponds, flood-proofing and drainage improvements, and the establishment and training of Farmers Water User Communities (FWUC) and Component 3: Institutional strengthening of government institutions, principally MOWRAM, tand the National Committee for Sub-National Democratic Development Secretariat (NCDDS) as well as the FWUCs. While the focus for MOWRAM will be on upgrading technical capacity in various aspects of climate resilient irrigation design and management, river basin management and water accounting,

Climate results/benefits

vi. The Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR) project will be focusing on climate change adaptation with cross-cutting benefits through the reduction of GHG emissions linked primarily to more efficient use of energy and water resources in rice cultivation – the latter is underpinned by extension and advisory to farmers on intermittent flooding (e.g., AWD), early warning and climate-informed advisory services, and high-efficiency smart irrigation systems. It is expected that the CAISAR project will benefit 562,757 rural people directly by making smallholders more resilient to climate change and 1.15 million people would benefit indirectly through the multiplier impacts that are generated from the investments in enhanced smallholder adaptation, market integration and improved irrigation systems and strengthening of early warning and climate information systems. Some of the benefits that will emerge include (i) reduction in GHG emissions due to changes in the cropping pattern, adoption of intermittent flooding in rice cultivation, switching to solar pumps and reduction in use of diesel pumps; (ii) increase in incomes due to improved yields and reduction in crop losses, (iii) improved health and well-being due to increased incomes and production of food crops, (iv) increased community collaboration and more active participation of women in decision-making regarding their livelihoods and (v) the active participation of the private sector in the water and agriculture economy. The project is also expected to generate GHG mitigation benefits as a result of reducing emissions from agriculture by **-1,006,507.43** tCO2-e over 20 years.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context

Country and Agricultural Context:

1. Cambodia is highly vulnerable to climate change and disaster risks. It is ranked 149 out of 187 countries in the 2022 ND-GAIN Index, was the 12th most disaster-prone country among 172 countries for 1999–2018 and remains one of the few Least Developed Countries (LDCs) in Asia. Annual economic losses resulting from natural disasters in the country were estimated at 0.7% of GDP in 2011[9]. The World Risk Index of 15.8% indicates a very high risk of disaster as a result of extreme natural events, with an exposure index of 27% and a vulnerability index of 59%[10]. Approximately 80% of the country is within the Mekong River and Tonle Sap basins, increasing exposure to floods, storms, and droughts. Identified hazards are assessed as high across the whole country for river floods and coastal floods, cyclones and extreme heat. In terms of the proportion of the population affected, Cambodia is one the world's most flood exposed countries in the world.[11] It is estimated that around 4 million people, or around 25% of the population, are affected when an extreme river flood occurs.[12] A WB study estimates the increase of population exposed to floods by 2050 under RCP8.5 at 19%.[13] The United Nations Office for Disaster Risk Reduction (UNDRR) estimates that Cambodia experiences over USD 250 million in average annual losses due to damage caused by floods (just over 1% of GDP).[14] Projected climate change trends indicate more severe floods and droughts, which is expected to affect Cambodia's GDP by nearly 10% by 2050.[15]

2. Agriculture is the primary source of livelihood for the communities living in most provinces of Cambodia and it accounts for 22.5% of the country's Gross Domestic Product[16] and employs over 3 million people. By mid-century, climate change will result in more frequent heavy precipitation days, from 2.4 days to 4.9 days in the rainy season, and an increase in the number of consecutive dry days in the same season from 0.8 days to 4.4 days.[17] The average monthly temperature is projected to increase by 1.83°C by 2040-59 under RCP8.5.[18] Cambodia's limited financial resources, infrastructure, and technical expertise hinder its ability to adapt to climate change. Farmers lack access to climate-resilient seeds, irrigation technologies, and weather information. In Cambodia, relatively low yields, coupled with frequent natural disasters, contribute to temporary food shortages for vulnerable communities. Based on food availability and access, a mapping of food security has been conducted for the country. Seven of Cambodia's 25 provinces (including Phnom Penh) are classified as acutely food insecure, and an additional seven are moderately insecure. Livelihoods rely heavily on rain-fed agriculture and non-poor households are vulnerable to falling back into poverty in the event of extreme or frequent climate shocks[19][20]

3. Farming remains mostly subsistence-based and rain-fed, with low productivity. Only 7 – 8% of the total potential agricultural land area is under full irrigation in Cambodia. Further, only 15% of the cultivated rice area is irrigated (compared to 28% in Thailand and 33% in Viet Nam), and 85% of the rice cropping area is vulnerable to changing rainfall patterns.[21] While rice accounts for 50% of agricultural GDP, yields are lower than in neighbouring countries. The lack of water for irrigation is a crucial barrier for farmers to deal with climate change-induced water shortages, and to cultivate more than a single crop per year - limiting their resilience and adaptive capacity. Some farmers do engage in some autonomous adaptation actions but

[9] World Bank and GFDRR (2019). Disaster Risk Finance Country Diagnostic Note: Cambodia. https://www.gfdrr.org/en/publication/disaster-risk-finance-country-diagnostic-note-cambodia
[10] World Risk Report (2022). https://weltrisikobericht.de/wp-content/uploads/2022/09/WorldRiskReport-2022_Online.pdf
[11] Climate Risk Profile: Cambodia (2021): The World Bank Group and Asian Development Bank
[12] Willner, S., Levermann, A., Zhao, F., Frieler, K. (2018). Adaptation required to preserve future high-end river flood risk at present levels. Science Advances: 4:1.
[13] Winsemius, Hessel C.; Jongman, Brenden; Veldkamp, Ted I.E.; Hallegatte, Stephane; Bangalore, Mook; Ward, Philip J. (2015). Disaster risk, climate change, and poverty: assessing the global exposure of poor people to floods and droughts (English). Policy Research working paper; no. WPS 7480. Washington, D.C.: World Bank Group
[14] Climate Risk Profile: Cambodia (2021): The World Bank Group and Asian Development Bank
[15] Climate Risk Country Profile. Cambodia. ADB. 2021
[16] UNDESA (2018) The Least Developed Country Category: 2018 Country Snapshots. at: https://www.un.org/development/desa/dpad/wp-content/uploads/sites/45/Snapshots2018.pdf.
[17] World Bank CCKP.
[18] World Bank CCKP.

[20] https://docs.wfp.org/api/documents/WFP-0000147767/download/?_ga=2.188002408.1116277075.1681899242-636470978.1669479646
[21] Kono, A. and Chey, T. 2019. Rapid assessment of upland cashew, mango, organic rice, and vegetable production. Actions for Climate-resilient and Sustainable Agricultural Production in the Northern Tonle Sap Basin. FAO.

these often come at the cost of yields and economic returns. For example, because wet season rice depends on irrigation water, farmers cultivate medium maturity (120-150 days) or late maturity (more than 150 days) rice. Medium maturity rice allows farmers—typically those situated at higher elevated locations—to avoid potential damage from water shortages in late November, since rice is harvested in early November. Late maturing varieties, which have a better resistance to flood, are chosen by farmers in areas that normally experience longer / bigger floods and is predominantly used for household consumption (with the expectation that yield will be lower than dry season rice).

4. Largely, however, farmers lack the knowledge and tools to adapt their farming practices to climate change; they are largely unaware of climate resilient practices (e.g., alternate wetting and drying, direct rice seeding and land leveling), stress-tolerant seed varieties and improved planting materials, efficient input management, etc. Cambodia's agricultural census in 2019, a representative survey of farmers across 25 provinces, revealed that only 26% of farmers had received agricultural training[22]. Other significant barriers for small-scale farmers include access to credit on attractive and appropriate terms, and market intelligence. Cambodia is unlikely to achieve 5% of its annual agricultural growth target by 2030, without more investment in climate adaptation, sustainable irrigation, flood control and drainage schemes. Persistent poverty, limited access to finance, and insufficient institutional capacities have left the rural agrarian population vulnerable to the negative impacts of climate change, economic slowdown, and environmental degradation.

5. In Cambodia, GHG emissions are dominated by the Forestry and Other Land Use (FOLU) sector (60.90%) and agriculture (16.9%), followed by energy (12.10%), Industrial Processes and Product Use (IPPU) (7.90%) and waste (2.20%) sectors. GHG emissions without the FOLU sector are dominated by the agriculture sector (43.5%), followed by energy (31.0%), waste (5.5%), and IPPU (20.3%). Under a business-as-usual (BAU) scenario, Cambodia forecasts an increase in emissions from 125.2 MtCO2eq in 2016 to 155.0 MtCO2eq in 2030.[23] Agriculture is a significant source of methane (from rice paddies) and nitrous oxide (from fertilizer use). Limited adoption of climate-smart agricultural practices to reduce emissions, and in the absence of the project about 1.5% of farmers are assumed to autonomously shift to improved flooded rice management based on regional studies[24]. Farmers also have poor fertilizer management: the amount of fertilizer applied is not adjusted to soil fertility characteristics and excessive use of fertilizers—contributing to higher GHG emission intensity of rice—is driven by the need for better growth in a shorter growing period and to ensure return on other agri-input investments. The use of organic fertilizers (livestock manure) occurs to a limited extent for wet season rice, but not at all in dry season rice. Balancing food security with emission reduction remains a challenge. Deforestation and land use change, particularly for agriculture, logging, and urban expansion, are significant sources of GHG emissions. These are due to loss of carbon sinks, such as forests and wetlands and reduce the country's capacity to absorb CO_2. Weak enforcement of land-use regulations and illegal logging exacerbate the problem. There is low participation in mitigation initiatives, such as energy conservation and sustainable land management. Limited funding for mitigation projects, particularly at the local level.

Climate change affects women, men, and vulnerable people differently, and their anticipatory actions and responses to its impacts also differ, including in relation to safeguarding food security and livelihoods and coping with hazards and risks. Women and children are among the most vulnerable groups to climate change. In the project area, women have significantly less access than men to financial instruments, land, natural resources, extension information and adaptive agricultural technologies, education (low literacy levels), and other development services for successful adaptation to climate change; this despite women having higher participation in agriculture than men. Cambodia's Resilience Index found women less resilient in times of disaster than men due to women's unstable and insecure sources of income, limited access to shelters or safe places, inadequate climate information and early warning systems, poor housing, weak social safety nets and absence

[22] Khun, C. and Lim, S. 2023. Productivity and market participation: Cambodian rice farmers. Journal of Asian Economics, 88: 101646. https://doi.org/10.1016/j.asieco.2023.101646

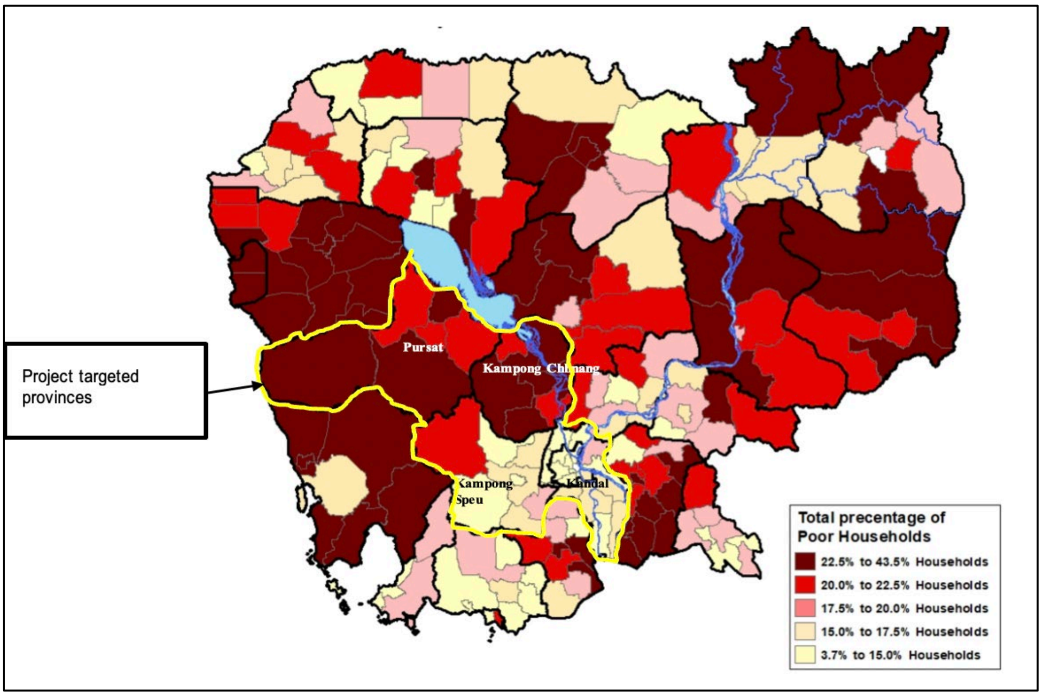
[23] RGC, 2020. Cambodia's Updated NDC. Retrieve from: https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Cambodia%20First/20201231_NDC_Update_Cambodia.pdf

[24] Ramírez Villegas, J., Rosentock, T., Steward, P., Thornton, P.K., Loboguerrero Rodriguez, A.M. and Jarvis, A., 2021. Projected climate adaptation benefits of One CGIAR. https://cgspace.cgiar.org/bitstream/handle/10568/115780/projected-adaptation-benefits-15Apr2021v2.pdf

of emergency networks. Furthermore, women's responsibilities in managing household resources positions them well to contribute to livelihood strategies adapted to changing environmental realities. However, lack of information, access to planning processes, and domestic and care obligations limit their economic empowerment and ability to contribute.

6. The CAISAR target provinces are Kampong Speu, Kampong Chhnang, Kandal, and Pursat. These provinces[25] have the highest population density, high total percentage of poor households, and a high percentage of women-headed households (Figure 1). The livelihoods of the poorest and most vulnerable people in the region depend mainly on agricultural production and water availability through precipitation, river flow, and irrigation schemes.

Figure 1: Map of Project Area



7. The project area was already under stress prior to the COVID-19 pandemic with small-scale farmers and larger food producers struggling to produce food and access markets. Food production faces additional pressure from climate change with serious droughts[26] during the dry months and heavy precipitation during the rainfall season. Rural communities highlight major water problems due to aging infrastructure and reduced water availability in rivers and ponds. The proposed underlying investment project will support the government's efforts to sustain food security and rural livelihoods through the restoration and enhancement of climate-resilient irrigation systems.

8. Since 2020, the negative impacts of climate change and the pandemic have led to a 10% loss in GDP.[27] In addition to negative climate impacts, stagnating global agricultural commodity prices, rising labor costs and the limited scope for cropland expansion in the country pose challenges for the agriculture sector[28] Production and distribution costs of agricultural products are high compared to the neighbouring countries, largely due to high energy and transport costs. Private sector investment in water saving or energy efficient technologies in irrigation, and their engagement in extension services remains low due to an unfavorable

[25] The Asia Foundation (2018). cambodia-atlas-on-gender-and-environment.pdf.pdf (opendevelopmentmekong.net)
[26] Kampong Spoeu, one of the four target provinces, reported the highest number of droughts (one of the eight main natural hazards in Cambodia) between 1996 and 2013. (https://www.unccd.int/sites/default/files/country_profile_documents/1%2520FINAL_NDP_Cambodia%255B1157%255D.pdf).
[27] World Bank, 2015. Cambodia Economic Update, October 2015: Adapting to Stay Competitive. https://openknowledge.worldbank.org/entities/publication/78d3d276-3a5f-5bf2-915d-2d0381a13930
[28] World Bank, 2015.

agri-business environment. Large agri-businesses face high costs and shortages of working capital. Many agri-business firms operate below capacity and potential new investors are reluctant to invest due to low profitability and a high risk of failure

Climate Change Impacts and Vulnerability of the Agriculture Sector

9. The Tonle Sap Lake Basin, often referred to as the rice basket of Cambodia, undergoes cycles of severe droughts and floods. This has resulted in lower agricultural productivity and reduced cultivation for smallholder farmers. Similar to other monsoon-dependent countries, climate change is altering Cambodia's monsoon calendar, increasing total drought days and shifting the start of the rainy season. While annual rainfall is expected to increase, the number of days with rain is decreasing and daily rain intensity is increasing, resulting in extreme rainfall events and flooding risks.[29] The World Resource Institute (WRI) rates the upper and lower Tonle Sap watershed as extremely high for water risk.[30] Inter-annual variability in rainfall significantly influences river discharges and between one and four million hectares of Cambodia's floodplain, of which about 25% falls in the project areas, may be submerged during the wet season, reaching critical thresholds for irrigation and ecosystem stability.

10. On average, annual damages and loss of wet season rice caused by floods and droughts is approximately 120,501 hectares, of which 70 percent of the damage and loss is due to floods and 30 percent is the result of droughts. Flooding in the project areas generally occurs between July and October, driven by rainfall as well as high water levels in the Mekong River and Tonle Sap Lake. Observed impacts of climate change trends in recent years include longer dry seasons and more intense "El Niño" related droughts, delayed onset of the monsoon season, increased rain intensity and frequency – leading to floods and unexpected dry periods during the rainy season. Annual mean temperature has increased at a rate of 0.23 °C per decade since 1950[31,] with a higher increase during the dry season i.e., the rate of temperature change is notable in the dry season[32] According to the International Rice Research Institute (IRRI), rice grain yield will decline by 10% for each 1°C increase in growing-season minimum (night) temperature in the dry season. The increase in mean annual temperature and decrease in dry season rainfall have manifested in droughts and water shortages significantly rainfed agriculture due to a late start and early ending of the wet season.[33] Floods also damage roads, collapse ponds and cause canal breakages – impacting the irrigation infrastructure directly, and this then results in inadequate water supply for cultivation in subsequent years. Of the estimated US$ 356.23 million loss and damage from 2013 floods, water and irrigation infrastructure accounted for US$ 52 million (14.5%) and national / rural roads for another US$ 79.61 million (22.3%)[34].

11. Drought has significant social and economic impacts in Cambodia, in particular in the project areas. NCDM estimated that at least 50 percent of districts were affected by the drought and in 2017 due to El Nino in 2015-2016, 18 provinces out of 25 provinces in Cambodia were severely affected by drought and 2.5 million people were affected (CEDMHA. 2017). The drought has had a significant impact on communities with crop damage and loss of livelihoods negatively affecting poor subsistent farmers and small land holders (Sharon. 2016). Agriculture would be one main sector which has strong relationship with drought. In 2009, 13 out of 24 provinces were affected by severe droughts. There were 57,965 hectares of rice crops affected and 2,621 hectares were destroyed. In 2010, 12 provinces out of 24 were affected by severe drought, as well as 14,103 hectares of transplanted rice. In 2011, drought affected 3804 hectares of rice fields while in 2012, drought affected 14,190 hectare of rice fields and destroyed 3151 hectares (CEDMHA. 2017). Drought events have become more frequent and are expected to intensify in future.

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[29] Cambodia Country Profile. World Bank and ADB. 2021.

[30] WRI 2023. Aquaduct https://www.wri.org/applications/aqueduct/country- rankings/?country=KHM&indicator=rfr

[31] Thoeun, H. C. (2015) Observed and projected changes in temperature and rainfall in Cambodia. Weather and Climate Extremes. 7:61. [old reference]

[32] WRI 2023. Aquaduct https://www.wri.org/applications/aqueduct/country- rankings/?country=KHM&indicator=rfr

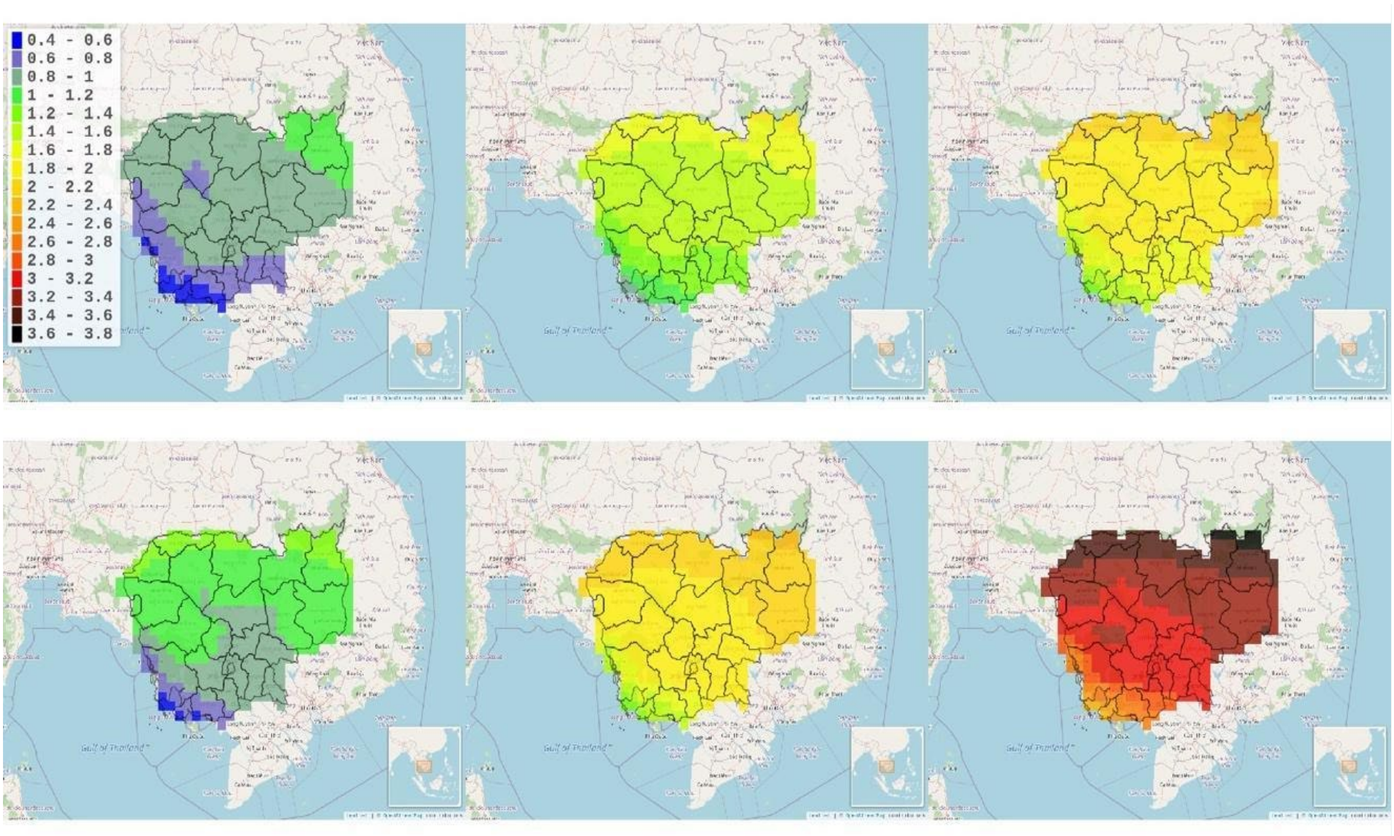
[33] MAFF (2017) MAFF Annual Report 2016-17

[34] https://www.undp.org/sites/g/files/zskgke326/files/migration/kh/Cambodia-post-flood-recovery-need-assessment-report.pdf

12. Agricultural sector accounts for 94-98% of the water use in Cambodia, with 1-2% for industry and 4% for the domestic use sectors.[35,36] Water use across sectors is 2-7% of available water that flows through the country annually, and water scarcity results primarily from limited access to irrigation and a lack of water storage capacity, particularly in the dry season (ADB 2021).[37] Projections in 2003 suggested that industrial and domestic uses will increase to 15% and 11% respectively by 2025 leading to increased competition for water resources due to climate impacts and economic changes upstream of the river basins (industrial use, deforestation, etc.) that are transboundary or national. Therefore, enhancing water use efficiency in agriculture and agro-industrial sectors is critical – including through tailoring advisories linked to agro-meteorological information and climate forecasts.

13. Future projections of climate change in the project area under the RCP4.5 scenario (Figure 2) indicate a temperature increase of 1.4-1.6°C by mid-century (2040-2070) and up to 2.2°C by the end of the 21st century (2070-2100) compared to baseline (1980-2005); the increase is much higher under RCP8.5 scenario (3.6°C, 2070-2100). Nights exposed to high temperatures are projected to become more frequent during the 21st century (+114 nights by mid-century and +151 nights by the 2080s).

Figure 2. Temperature maximum deviation from historical baseline period (1980-2005)



Under RCP 4.5 (top row) for the period 2011-2040 (left), 2040-2070 (middle) and 2070-2100 (right) and RCP8.5 (bottom row) for the same time periods. The deviation highlights the projected changes into the future in degrees Celsius based on projections from bias corrected regional CORDEX data (FAO analysis, CAVA tool).

14. Across Southeast Asia, a modest increase in precipitation is projected – from 1-8% by 2100 (IPCC AR5), but there are some uncertainties in the direction of changes in annual precipitation and river flows for the

[35] Cambodia National Mekong Committee (CNMC) Water Security, Food Security and Livelihoods in Cambodia and the Lower Mekong Basin. https://ppcrcambodia.files.wordpress.com/2012/11/watt-1106-w-food-security.pdf.
[36] https://www.worldometers.info/water/cambodia-water/
[37] ADB Cambodia Agriculture, Natural Resources, and Rural Development, Sector Assessment, Strategy and Road Map. https://www.adb.org/sites/default/files/publication/718806/cambodia-agriculture-rural-development-road-map.pdf

Mekong River Basin.[38] In general, it can be said that for the CAISAR observation points, models[39] show a reduction in rainfall during the dry season but an increase in rainfall during the wet season and an increase in frequency and intensity of heavy rainfall events.

15. Since the number of days with rain is decreasing and daily rain intensity is increasing, it results in extreme rainfall events and flooding risks.[40] For the Lower Mekong Basin, both an increase in drought days by 2100[41] and an increase in drought frequency is predicted,[42] and flooding is expected to increase in frequency, severity, and duration.[43] Studies find more severe flood inundation magnitude in the Lower Mekong Basin, in the future (2075-2099) compared to 1979-2003, under four RCP scenarios and sea surface temperature patterns.[44]

16. The projected climate change is leading to the change in all water balance components which are expected to increase compared to the baseline period. Water yield is expected to increase higher than other water balance components dependent on the water basins. Most significant change in water balance is expected to occur under RCP8.5 by 2026-2050 compared to baseline period (1995-2021). Based on the projection results, the monthly flow change is projected to increase during the rainy and dry seasons for 4RCM-RCP4.5 and 4RCM-RCP8.5 scenarios. The highest monthly flow is predicted to increase in October by up to 22%-55% for the 4RCM-RCP4.5, and 4RCM-RCP8.5 compared to the baseline period (1995-2021). The lowest monthly flow will increase in February by 34%-44% for 4RCM-RCP4.5 and 4RCM-RCP8.5. The project area is likely to experience changes in flood regimes due to the projected increases in temperature and changes in precipitation patterns.

17. Population, crop and livestock systems will also experience heat extremes more frequently in comparison to the reference period (42 days by mid-century and reaching 62 days by the 2080s). The changes in precipitation and temperature patterns induced by climate change in the project area could affect the climatic suitability of cereals, vegetables and starchy crops produced. On average, for most crops analyzed for this project, the temperature suitability could progressively decrease with some crops falling outside their optimal temperature levels. Overall, this change in patterns could contribute to a reduction in crop yield for both irrigated and non-irrigated crops. For example, in the Kampong Chhnang region, rice yield could decrease by anywhere from 9% to 12%, with and without irrigation, respectively, by 2050 (a period of 20 years centered around 2050) compared to 2020. These projections are for the RCP8.5 scenario and without CO_2 fertilization. As smallholder farmers' incomes are largely dependent on their yield, it is anticipated that farmers' income could become more volatile with a decreasing trend over time. It is projected that vegetables will be increasingly exposed to heat-stress conditions during the pre-monsoon and early wet-season months (March-June). Over these months, heat-stress conditions (>35ºC) will be exceeded more than 50 days during the growing season, lasting approximately 90 days. Women have higher participation in agriculture than men (women are 48% of the labour force, of which 30% women are employed in agriculture) but receive lower benefits and have lower access to land, extension services, financial services, markets, and climate-resilient technologies. One of the challenges in agricultural extension is that it dominated by men who cater to male farmers, and do not tailor services or advisories to women farmers (e.g., they may be more involved in horticulture, fish or poultry requiring different timing and content of weather forecasts and advisories as well as technologies)[45]. Rural youth face difficulties in accessing land, which also limits access to credit. Thus, a

[38] Piman, T., Cochrane, T.A., Arias, M.E., Dat, N.D. and Vonnarart, O., 2014. Managing Hydropower under Climate Change in the Mekong Tributaries. Chapter 11 in S. Shrestha et al. (Ed.), Managing Water Resources under Climate Uncertainty: Examples from Asia, Europe, Latin America, and Australia, In Press, Springer. https://doi.org/10.1007/978-3-319-10467-6_11
[39] Global Climate Models and Regional Climate Models comparing data for 1995-2001 with 2026-2050 projections. WAPCOS and Phnom Penh International Consultants, 2023. CAISAR Feasibility Report D8, Version 1.2.
[40] World Bank and ADB, 2021.
[41] Hirabayashi, Y., Kanae, S., Emori, S., Oki, T., and Kimoto, M., 2008. Global projections of changing risks of floods and droughts in a changing climate. Hydrological Sciences, 53 (4): 754-772. https://doi.org/10.1623/hysj.53.4.754
[42] MRC, 2010. State of the Basin Report 2010. Mekong River Commission: Vientiane.
[43] Thoeun, 2015.
[44] Try, S., Tanaka, S., Tanaka, K., Sayama, T., Lee, G., and Oeurng, C., 2020. Assessing the effects of climate change on flood inundation in the lower Mekong Basin using high-resolution AGCM outputs. Progress in Earth and Planetary Science, 2020 (7): 34. https://doi.org/10.1186/s40645-020-00353-z
[45] MAFF, 2022b. Policy and Strategic Framework for Gender Mainstreaming in Agriculture, 2022-2026. https://elibrary.maff.gov.kh/book/64f9302c036dc

high proportion migrate to urban areas in search of work. Poverty incidence among ethnic minorities is not significantly higher than among the majority, but they suffer disadvantages in access to health, education and other services.

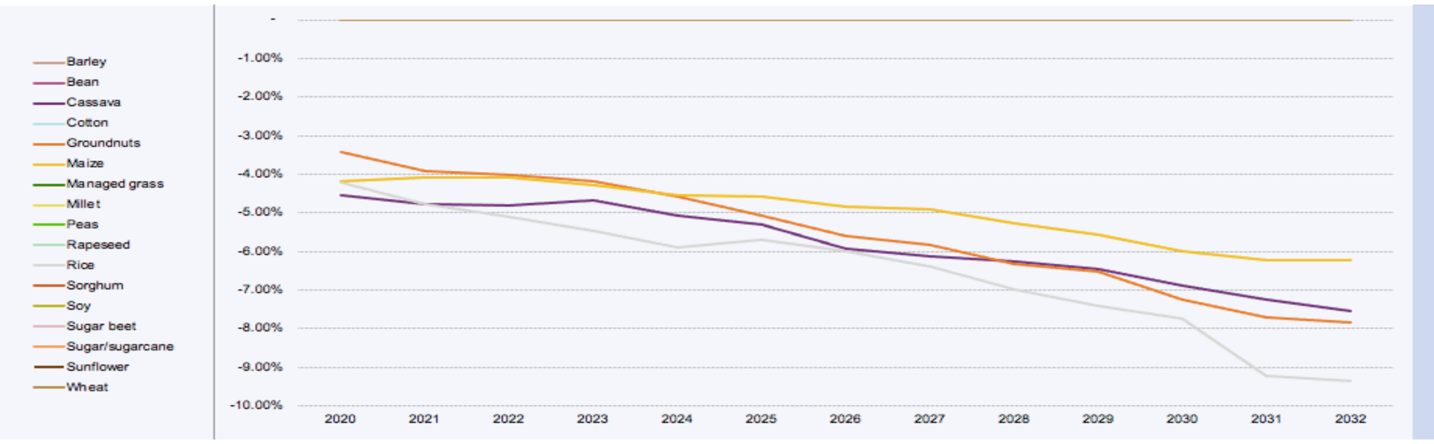
18. The impact of climate on various crops in Cambodia is outlined in Table 1 together with the impact of the climate change and the impact on specific crops.

Table 1: The Impact of Climate Exposure to Selected Crops

Crop	Climate Exposure	Impacts	Source
Rice	High temperature, rainfall variations	low yields, loss of crop	IFAD CARD
Maize	High temperature, drought, rainfall variations	Reduced yields, water stress, loss of crop	IFAD CARD
Cassava	Drought, high temperature	Reduced yields, water stress	IFAD CARD
Soybeans	High temperature, drought, rainfall variations	Reduced yields, water stress	IFAD CARD
Sugar cane	High temperature, rainfall variations	Droughts, decreased yields, water stress	IFAD CARD
Vegetables	Temperature extremes, water availability	Reduced yields, increased pest pressure, loss of crop	Alvar-Beltan et al (2022)[46]
Fruits	Temperature extremes, water availability	Reduced yields, increased pest pressure, loss of crop, fruit fall.	

19. Data from IFAD's Climate Adaptation in Rural Development – Assessment Tool (CARD) which explores the effects of climate change on the yield of major crops. CARD shows that the yields of all crops are expected to decrease without irrigation (Figure 3).

Figure 3: Projected Crop Yields without Irrigation in Cambodia



Key Barriers to Climate resilience

20. Climate analysis has highlighted four main risks for farmers: (i) a decline in the rainy season days, despite increasing precipitation the project area is likely to experience more extreme dry periods; (ii) rapidly increasing temperatures causing deteriorating suitability conditions for many crops, and further drying in the wet season (through evapotranspiration); (iii) increasing risk of heavy precipitation potentially leading to flooding, erosion and loss in crop yields, livestock productivity and damage to infrastructure; and (iv) unpredictability of rainy season onset and the timing of rainfall within season. The project design is taking

[46] Alvar-Beltan, J. et al 2022. Climate change impacts on irrigated crops in Cambodia https://doi.org/10.1016/j.agrformet.2022.109105

these risks into consideration through climate change adaptation measures that focus on balancing the watershed's ecosystem services and the climate-proofing of existing combined irrigation and flood control infrastructures. Where possible, the project will employ smart irrigation systems that collect and distribute key aggregated data to maximize irrigation and flood control, key weather data for water resource managers and farmers, and outreach to last-mile smallholder farmer groups on adaptive water and crop management methods. The project promotes viable business models for the private sector to offer affordable smart water solutions and adapted farming technologies respecting agro-ecological context and climate-resilient principles. The project design will be informed by hydrologic and hydraulic modelling to assess water sources, water availability, existing and new canal networks, and optimize the irrigation and drainage/flood control system in the project area.

21. A major barrier to sustainable and climate-resilient development of irrigation and water infrastructure in the project area is insufficient investment in planning and long-term maintenance of infrastructure in a systematic way. Many of the schemes were never completed up to tertiary irrigation networks which affects their efficiency. In addition, the technical capacities of relevant national institutions and Farmer Water Use Communities (FWUCs) are rarely assessed in detail at the design stage, nor addressed during the construction phase. In most irrigation schemes, there is lack of a systematic cost-recovery system leading to little incentives for the FWUCs to manage the infrastructure. The design process of the proposed project included a capacity and needs assessment of MOWRAM, PDWRAMs and relevant FWUCs/FOs in the region to ensure targeted interventions and suitable training programs are included as part of output 3.1 under component 3 in the project to build the skills needed to undertake O&M responsibilities and make the arrangements more sustainable and inclusive.[47]

22. Another important barrier in the project area is the lack of extension services at the provincial and commune level. Agricultural extension services lack adequate staff, capacity and network to access remote communities. In addition, current extension personnel possess an inadequate understanding of climate change and climate -resilient practices and technologies that are suitable for local contexts. Farmers have limited knowledge of technologies and techniques that are more sustainable and climate resilient. The technologies offered or available are not always appropriate and farmers are generally resistant to adopting them when the technology is not climatically fit or it is labour intensive, costly or not suitable to smallholder farmers. Agricultural expansion has been possible through unsustainable practices, including mono-cropping, excessive application of chemical fertilizers and pesticides, and poor soil nutrient management. Farmers, agricultural cooperatives and SMEs have limited access to finance, which can facilitate the adoption of appropriate technologies and techniques. Access to finance remains a major barrier as small holders, farmers and agribusinesses often cannot meet the high collateral requirements of financial institutions and higher interest rates and short tenors reflect both the risks inherent in agriculture activities as well as a less nuanced approach to managing those risks. Further lack of farmer credit histories of low financial literacy contributes to poor financial access.[48] The project will address the specific challenges women, youth and people from other vulnerable groups face in accessing extension services and external credit lines through facilitating links with existing facilities.

23. The local private sector also fails to offer technologies that farmers need (e.g., stress tolerant seeds, bio-inputs) and engage in value addition (e.g., rice milling prior to exports) adequately and sufficiently. Production and distribution costs of agricultural products are high compared to the neighboring countries, largely due to high energy and transport costs. Private sector investment in water saving micro technologies in irrigation and extension services remains low due to an unfavorable agri-business environment, such as access to appropriate SME finance, poor communication and cooperation between government and private sector, and the absence of a clear regulatory framework for the establishment, operation and pricing for the private management of water. Large agri-businesses face high costs and shortages of working capital. Many agri-business firms operate below capacity and potential new investors are reluctant to invest due to low profitability and high risk of failure.

[47] ADB (2021) Cambodia Agriculture, Natural Resources, and Rural Development Sector Assessment, Strategy, and Road Map (adb.org)

[48] ADB (2021) Cambodia Agriculture, Natural Resources, and Rural Development Sector Assessment, Strategy, and Road Map (adb.org)

Alignment with ongoing projects and programs

24. The project is aligned with IFAD's Strategic Framework (2025-2030) and relevant Sustainable Development Goals (SDGs) 6, 12, and 13 and will support the United Nations Sustainable Development Cooperation Framework. The Project will directly link to the strategic objectives of IFAD's Country Strategy in Cambodia (COSOP 2022-27) by supporting the development and adoption of climate-resilient and environmentally sustainable agriculture techniques, promote agriculture diversification and climate resilience of poor rural people's economic activities. The project is closely aligned with the RGC's priorities on low-carbon and climate resilient development as outlined in the RGC's overarching Pentagonal **Strategy Realizing the Cambodia Vision 20250** and with the thematic portfolio on Climate Change and Natural Resources Management of the Mekong Region Cooperation Programme (MRCP) 2022-2025.

25. The project design utilizes lessons and experiences from IFAD's on-going projects in the country. These lessons illustrate that public extension services are generally underfunded and understaffed, and mainly focus on production-related advisory services. While there is need to strengthen extension services and expand outreach, there is also need to provide greater focus on value addition and increased access to markets. Smallholders can compete in the market provided they are supported to better understand market demand in terms of quality and volume of production required, access to resources that can help them add value and better link to markets through information technology. IFAD's experience in Cambodia, shows the great potential for small-scale farmers to contribute to economic growth provided they are assisted in dealing with their multiple constraints (Agricultural Services Programme for Innovation, Resilience and Extension (ASPIRE)). This requires, among other things, investments in infrastructure, building value chains, irrigation facilities, creating markets and ensuring access to financial services. It is also possible to empower rural women by establishing clear targets for their inclusion. The lessons included the importance of adopting a demand driven approach learned from ASPIRE on agriculture extension, the need for support to market infrastructure, digital technology, and youth programmes (Sustainable Assets for Agriculture Markets, Business and Trade Project

26. (SAAMBAT)). Lessons from Accelerating Inclusive Markets for Smallholders project (AIMS) regarding the most effective strategies for market access such as the cluster approach, use of MMSPs and sales contracts for improved market access will be deployed. Projects can also enable farmers to become more resilient and recover faster after a disaster with appropriate resources that can help them in the recovery (ASPIRE).

27. The design also builds on similar ongoing and planned projects and programmes in the country financed by other development partners. The project builds on the available lessons from the ongoing Climate-Friendly Agribusiness Value Chain Sector Project (GCF FP 076) on application of CSA technologies and market access. It will further employ key lessons and feedback from the ongoing ASEAN Catalytic Green Finance Facility (ACGF): green Recovery Program (GCF FP 156). The project will work together with the PEARL (GCF FP199) in areas of value addition and market linkages for improved business environment for farmers. The elements of early warning systems and climate information systems will also be closely linked to the activities in PEARL – specifically, it will leverage PEARL's outputs on tailored agro- meteorological advisory services to support decision making by agricultural producers and value chain actors.

28. Both IFAD and PEARL's feasibility study assessed that despite Cambodia's robust network of hydrometeorological stations to provide general forecasting services and early warning on multiple hazards, significant infrastructural and capacity gaps remain in the provision and dissemination of agrometeorological forecasting and advisory services. CAISAR will specifically focus on advisory related to water availability and water use efficiency. CAISAR utilizes key lessons from the GEF project Promoting Climate Resilient Livelihoods in Rice-Based Communities in the Tonle Sap Region and will link with future projects on rice landscape and the proposed new GEF project "Building climate resilience of communities in Cambodia's protected landscapes: bio-diversity friendly crop and livestock systems for adaptation" that aims to

strengthen climate resilience of local communities, ecosystems and livelihoods in Cambodia's protected landscapes

29. Projects that closely align with this project include the ADB-assisted Irrigated Agriculture Improvement Project in Battambang, Kampong Cham, Kampong Thom, and Takeo provinces which aims to modernize and improve the climate and disaster resilience of irrigation systems; the proposed WB-supported Cambodia Water Security Improvement Project (CWSIP) in 3S and 5P basins in North-East part of the country and the ongoing JICA assisted West Tonle Sap Irrigation and Drainage Rehabilitation and Improvement Project. The CAISAR project design is also closely aligned with the design of the Integrated Water Resource Management Project (IWRMP) funded by ADB with co-funding from the AIIB to ensure strong synergy between the two projects. The_IWRMP will be implemented in Moung Ruessei, Pursat, Stueng Sangkae, Svay Doun Kaev river basin. The overall goal of IWRMP is enhanced resilience of Cambodia's water resources. The project will strengthen planning, coordination, and climate change adaptation capacities of water resources management while also reducing flood and drought risk in the target areas. It will collaborate with the ADB's Community Resiliency Partnership Program (CRPP) making use of innovative approaches developed out of knowledge and action research and in community capacity development initiatives.

30. The International Rice Research Institute (IRRI) has been long engaged in Cambodia in promoting locally specific climate resilient rice cultivation technologies and a wide range of activities under the Excellence in Agronomy (EiA) Initiative have been implemented since its launch in 2020. IRRI's lessons and recommendation in areas of Directly Seeded Rice (especially in areas of mechanization, mDSR[49]), AWD, Land Laser Levelling (LLL) and rice straw management have been incorporated in the project activities. The project will continue its collaboration with IRRI during implementation phase for enhanced synergy and to utilize its knowledge and products in implementing climate resilient technologies and practices.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

31. The Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR) project will be focusing on climate change adaptation with cross-cutting benefits through the reduction of GHG emissions linked to the use of smart irrigation systems, efficient irrigation management in rice (intermittent flooding enabled by, direct seeding and land leveling), use of renewable energy for water pumping, and cover techniques (retained residue) to reduce evapotranspiration – underpinned by timely and relevant early warning and climate information services. Cambodia's rainfed agriculture faces increasing challenges from adverse impacts of climate change, especially the changes in rainfall patterns, duration and timing of the rainy season, and climate-induced water disasters such as floods and droughts. Climate Resilient (CR) and low emission practices and investments in agriculture and water management are therefore crucial to protect and enhance Cambodia's agricultural production and productivity which will in turn contribute to poverty reduction and increased food security.

32. The overall goal of the project is to make the agriculture sector in Cambodia climate resilient and sustainable. The specific objective of the project is to modernize the irrigation sector by installing smart irrigation systems and enable farmers to adapt to climate risks and mitigate crop emissions through use of energy- and water-efficient technologies and practices, timely weather information and improved market integration. CAISAR's Theory of Change is premised on the experience that addressing the complex impacts of climate change on rain-fed and irrigated agriculture requires action at three levels: farm level, irrigation scheme level and at the national level by creating a strong institutional base and an enabling environment.

33. The theory of change of the project has premised its investments on an understanding of the pathways that can help to reduce GHG emissions and help smallholder farmers adapt to climate change by applying climate resilient agricultural technologies. The Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR) project will adopt a three-pronged strategy to bring about a paradigm shift in the

[49] Creating a path for scaling mechanized direct-seeded rice in Cambodia - Rice Today (irri.org) https://ricetoday.irri.org/creating-a-path-for-scaling-mechanized-direct-seeded-rice-in-cambodia/

farming systems and the water management governance in Cambodia. The integrated actions combining the various elements of the strategy can have a transformative impact on reducing vulnerability of water and agriculture systems to climate change impacts while also reducing GHG emissions and enhancing the livelihood of rural populations who primarily depend on agriculture.

34. The Theory of Change of the project is based on the assumption **IF** the irrigation system is upgraded and modernized, farmers are made aware of climate risks and extended support for mitigation and adaptation **THEN** 1.7 million people will improve their health, livelihoods and food security and 1 million tonne of CO2-eq emission will be reduced, **BECAUSE,** of enhanced resilience of the agriculture sector to climate change resulting from improved agricultural productivity practices, more resilient irrigation infrastructure, water and soil management, better market linkage, and institutional support.

35. The theory of change postulates the following with respect to each of the three key pathways that CAISAR has chosen for investment.

Pathway 1: Farm Level:

36. At the farm level, the lack of knowledge about adaptation technologies and practices and limited access to improved adaptation technologies and clean energy sources leads to low productivity, inability to cope with climate risks and high rates of GHG emissions. Reduction of GHG emissions from rice production can be achieved by shifting away from fully flooded irrigation using traditional varieties to shorter duration modern varieties coupled with mechanized direct seeding and alternate wetting and drying; improving fertilizer management and enhancing the use of organic inputs, diversifying to other crops, promoting residue retention, and deploying solar irrigation pumps (to pump water from canals to fields) instead of the diesel pumps currently used. In addition, the project will encourage innovation by providing start-up support to innovative entrepreneurs who will invest in the provinces based on Value Chain Action Plans (VCAPs) that will be developed in the first year of implementation. Through a composite set of activities, the project will increase farmer capacity in climate smart agriculture, assist in developing climate adaptive value chains through public private partnerships and increased access to finance, enable increased access to and use of climate-informed advisory for climate responsive water-use and crop planning, strengthen widespread awareness of and utilization of early warnings (especially, floods and droughts), and access to markets and services vis-à-vis increased resilience of farm road infrastructure to climate change. The expected outcome from these outputs will be reduction of GHG emissions of -1,006,507tCO2eq over 20 years (detailed in Annex 22), reduction in crop losses and improved production and enhancement of dietary diversity and improved health. The investment in agricultural practices and the irrigation approach will irrigate approximately 32,056 hectares (ha) in the rainy season and about 22,051 ha in the dry season – increasing cropping intensity from 137% at baseline to 192% with project.

37. Increased adoption of new and improved climate resilient agriculture technologies and practices can be achieved through working closely with smallholders. Often the barriers that farmers face in adopting the technology are not properly understood and farmers are resistant to adoption because of incompatibility with their farm conditions, affordability or high labour requirements. The technological packages that will be provided under CAISAR will keep the barriers to adoption of new technologies in mind and provide the technologies and practices that are appropriate and enhance their adoption. CAISAR will link the farmers with research institutions and value chain actors in the private sector who provide technologies locally, better integrating them into the value chains and increasing access to finance to enable them to continue to use these inputs and strengthening the value chain of key agriculture commodities. CAISAR will organize its investment activities to assist smallholders to assess their vulnerability and in increasing their awareness about adaptation to changing rainfall patterns and increasing drought conditions, providing them access to water and roads as well as early warning and climate informed advisories that support their on-farm decision-making, and provide start-up support to innovative entrepreneurs that offer technologies or engage in value chain activities that enhance smallholder adaptation. The reduction of losses through better farm-level adaptation and drought / flood proofing (irrigation, drainage) will help farmers reduce their investment risks, increase production – thereby increasing their incomes, their cash flows and food security, and enable

them to sustain farming as a key source of livelihood. These adaptation practices will help in diversifying their agricultural practices as well as variety of food production. This will support the households diversify their consumption of food and better nutrition for the families.

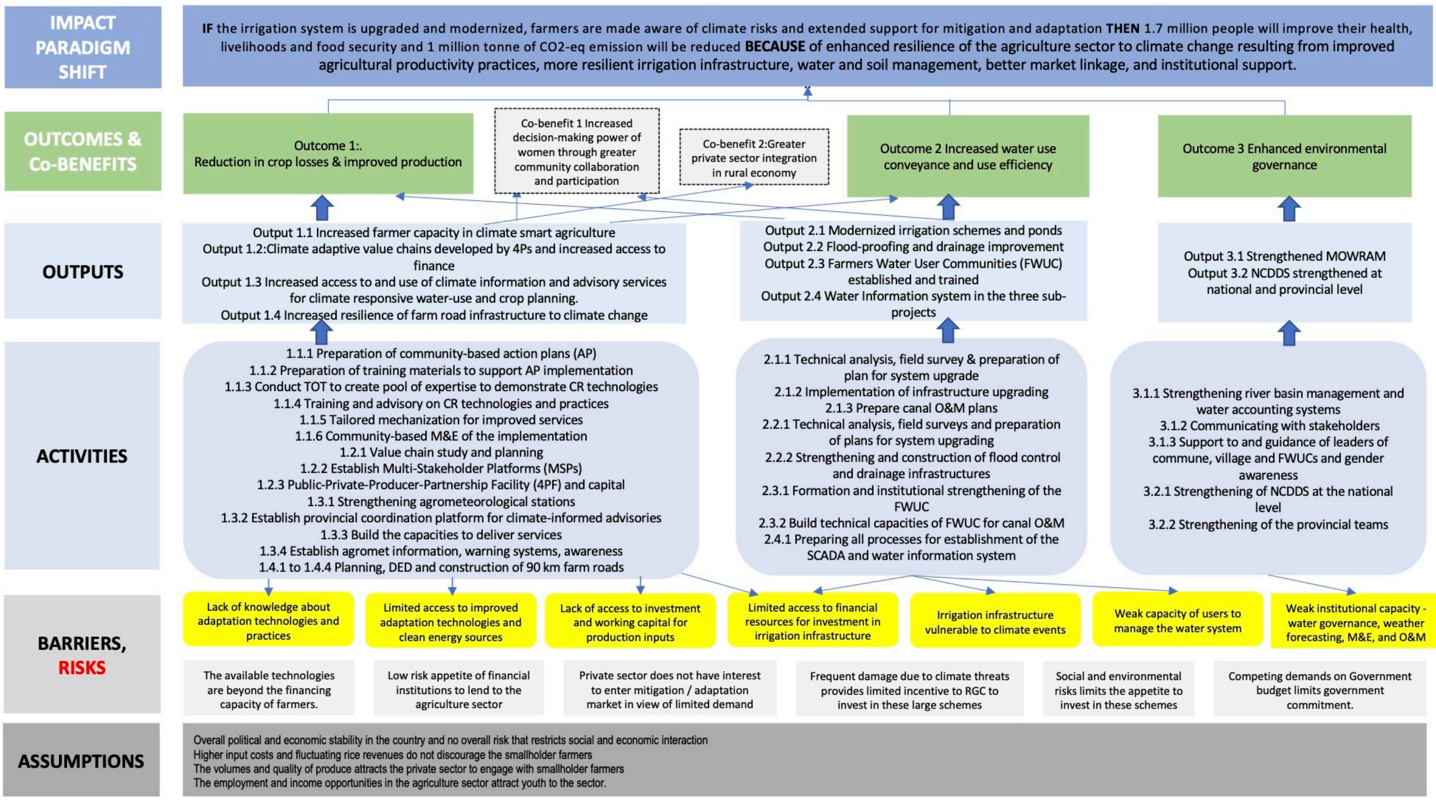
Pathway 2: Irrigation Scheme Level:

38. Much of the irrigation infrastructure in Cambodia is outdated and in need of rehabilitation and upgrading, and much of the older infrastructure is not climate proofed and is vulnerable to damages from floods that also result in silting of canals and poor drainage – particularly, during extreme precipitation and flooding events; this adds to recurrent operation and maintenance expenditures that strain public budgets. Also, most of the irrigation schemes in Cambodia were never completed and do not always have the appropriate tertiary irrigation distribution networks. This leads to unreliability of water supply, inefficient water use and high conveyance losses which aggravates the impact of climate risk. The Theory of Change of CAISAR builds on IFAD experience that investments in modernizing irrigation systems and flood proofing drainage systems will increase the efficiency and reliability of the systems, protect the natural capital stocks, reduce the need for frequent rehabilitation and reconstruction owing to frequent breakdown, and lower the cost of operation and maintenance. In addition, the project will strengthen the capacity of FWUCs for water / irrigation system management and governance as this will make the systems more sustainable through cost recovery and enhance the technical and financial sustainability of the system. The more efficient delivery of irrigation water will increase production and reduce losses, cropping system diversity, and enhance food security. Currently, women are excluded from FWUCs and the project will target their inclusion in FWUC management structure and meaningful participation in decision making. Women's participation in decision making process for food production helps in improved nutrition for the households. The project will also replace diesel pumps with solar pumps that will contribute, albeit limitedly, to reduced GHG emissions and generate other co-benefits. The solar pumps are unlikely to lead to any type of maladaptation as the solar pumps will not be used for pumping ground water but simply to help convey the water from the main canals to farmer's fields as well as to drain the flood water.

Pathway 3: Institutional and Enabling Environment Level

39. At the national level, the institutional and policy framework in Cambodia can be strengthened through strategic investments in developing staff capacity for improved irrigation infrastructure designs and planning (river basin management), use of Information Communication Technology (ICT), deploying water accounting and audit systems, strengthening the policy and regulatory frameworks for climate action – including the preparation of a loss and damage strategy, enhancing M&E capacity and encouraging greater collaboration between NDA and other stakeholders. The theory of change also postulates that developing a post-project strategy for sustainability and engaging the private sector in its design and implementation can help develop innovative and sustainable models for management and operation and maintenance of the irrigation system in the country. It is expected that improved institutional strengthening of the sector institutions will make them more aware of these opportunities and open the way for innovation in the long-term management of the sector.

Figure 4. Theory of Change



B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1: Reduction in crop losses & improved production	☐	☐	☐	☒	☒	☒	☐	☒
Outcome 2: Increased water use conveyance and use efficiency	☐	☐	☐	☒	☒	☒	☐	☒
Outcome 3: Enhanced environmental governance	☐	☐	☐	☒	☒	☒	☐	☒

If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

	Co-benefit

Co-benefit number	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Increased decision-making power of women through greater community collaboration and participation	☐	☒	☐	☒	☐	☐
Co-benefit 2 Greater private sector integration in rural economy	☐	☐	☒	☐	☐	☐

B.3. Project/programme description

40. The project's results will be achieved through the implementation of three components. The first component will focus on building farm level resilience and investments at the local level in the command area of the selected irrigation schemes and their immediate surroundings, the second component will focus on building resilience at the irrigation flood control/prevention system level and will involve modernizing and climate proofing of infrastructure to strengthen the sustainability of water resource systems in both the wet and dry season. The first component is designed to bring about a transformation in these farming systems through awareness, capacity building and re-orientation of farmers production practices and assist in both improved adaptation and climate change mitigation. The second component will focus on physical infrastructure investments. The third component will deal with institutional strengthening of the National Designated Authority (NDA), the Ministry of Water Resources and Meteorology (MOWRAM) and the National Committee on Sub-National Democratic Development Secretariat (NCDDS) and will directly contribute to both implementation and sustainability of the results generated from components 1 and 2.

41. The project area is expected to comprise 32,056 Hectares. The schemes or sub-projects under the project will include the Ou Ta Paong (OTP) and Lum Harch and 4 smaller sub-projects in the Krang Ponley valley. The cultivation/ utilization of these areas varies by season and planting cycle. While the entire project area is expected to plant wet season rice crop, the area in which a second irrigated dry season rice crop can be established is 22,051 Ha. In the remaining 10,000 Ha in the dry season, vegetable production, mainly homestead gardening, and poultry production as well as aquaculture can be established. In several of the 4 Krang Ponley sub-projects commercial vegetables are planned due to the close vicinity to Phnom Penh city and other conditions. Components 1 and 2 are expected to work in close synergy to maximize the benefits from the investments.

Table 2: Details of 6 Irrigation sub-projects

Irrigation scheme	Component 1 *Same as Comp 2.1+2.2		Component 2						Total	
			Component 2.1		Component 2.2 *Additional area of C2.1		Component 2.3 *Additional area of C2.1+C2.2 ** Not included in the total area			
	Area(ha)	Ratio (%)	Area(ha)	Ratio (%)	Area(ha)	Ratio (%)	Area(ha)	Ratio (%)	Area(ha)	Ratio (%)
Ou Ta Paong	17,079	53%	14,874	67%	2,205	22%	80,000	100%	17,079	53%
Lum Harch	6,350	20%	3,900	18%	2,450	24%	-	-	6,350	20%
Kropeu Trom	1,140	4%	690	3%	450	5%	-	-	1,140	4%
Yotasast	2,693	8%	593	2%	2,100	21%	-	-	2,693	8%
Stoeung Krang Bat	1,394	4%	994	5%	400	4%	-	-	1,394	4%
Brambei Mom	3,400	11%	1,000	5%	2,400	24%	-	-	3,400	11%
Total	**32,056**	**100%**	**22,051**	**100%**	**10,005**	**100%**	**80,000**	**100%**	**32,056**	**100%**

C2.2* River training: length of river training work * 400 m (width of impact) converted in ha.
C2.2* Polder: length of polder (m) * 1000 m (width of impact) – converted in ha.
C2.3* Masterplan area potential impact (80.000 ha) – OTP scheme area (not included in total area)

Component 1. *Farm-level climate adaptation and resilience.* (NCDD-S)

42. The objective of this component is to build climate resilience (CR) of smallholder farmers and enhance sustainable production through evidence-based planning and context-relevant climate resilient practices at the farm level. This component is designed to address the lack of knowledge and skills to deploy technologies and practices at farm level by farmers and the lack of appropriate extension services to propagate them. It will introduce farmers with various climate resilient technologies and practices for both rice and non-rice commodities such as vegetable, poultry and aquaculture. The component will assist in value addition, facilitate market linkages, introduce agro-meteorological information and services for climate informed agriculture planning and improve both physical and digital connectivity to facilitate improved production and marketing. Component 1 is expected to benefit over 110,000 farming households. In addition, some of the technologies and practices, for example the Alternate Wetting and Drying technique (AWD), direct seeded rice (DSR), Land Laser Levelling (LLL), and rice straw management will directly reduce methane emissions contributing to GHG reduction. The component will be implemented by NCDDS through four outputs described in the following paragraphs. The activities under this component will also capitalize on the presence of existing platforms, studies and plans in order to avoid potential overlapping, duplication or spending unnecessarily resources in planning activities. The agencies responsible for implementation this component will undertake the tasks of identifying existing resources at the outset.

Outcome 1: Reduction in crop losses & improved production

<u>Output 1.1: Increased farmer capacity in climate resilient agriculture</u> *(NCDD-S)*

43. This output will focus on developing farmer's capacity in deploying climate resilient technologies and practices to transform the agricultural production system to adapt to the changing climate context. Separate dialogues will be held with women to ensure that their concerns and needs are addressed in designing the training curricula, the schedule and delivery methodology. The main objective is to strengthen adaptive management strategies to mitigate climate variability risks. Both men and women farmer's will be trained to first develop Action Plans (AP) to re-orient farmer behaviour and assist them in transforming the agriculture production system in a manner that is better adapted to factoring in the agro-ecological context and expected climate change impacts. Farmer capacities will be strengthened in CR technologies mainly focusing on rice cultivation. However, for those parts of the project areas which have insufficient irrigation water supply in the dry season for a second rice-crop, alternatives will be supported such as vegetable production, poultry and aquaculture especially for farmers with small land holdings. Crops of interest to women will be included in the selection. Farmers are generally resistant to adopting new technologies due to their inability to afford the technologies or technological incompatibility i.e. when the technology is not climatically fit or it is labour intensive, costly or not suitable for smallholder farmers. Those designing the courses and the technological packages under CAISAR will be mindful of these barriers. The plans will be refined over time based on the local context and crop choices and emerging adaptation technologies. The project will use the Farmer's Field and Business School approach (FFBS) and design context specific training materials to enhance the capacities of the farming communities. The FFBS will run for both rice and non-rice crops and will use a cascade approach to disseminate the training messages to a wider audience using key farmers including both women and men leaders. It is expected that 40,000 farmers will be re-oriented of which at least 40% will be women. The improvement of the irrigation systems will be critical to fully capitalise on the training provided to farmers.

Activity 1.1.1 Preparation of community-based action plans (AP) to transform agriculture with CR practices and technologies (NCDD-S)

44. The project will utilize a Climate Smart (CS) Mapping approach[50] to develop awareness materials on climate change and its impacts on men and women separately. IRRI will conduct CS Mapping in the first year as a part of their TA contract, combining participatory data with GIS and crop modeling to create risk maps identifying areas vulnerable to climate stresses like drought and flood. These maps will also outline

[50] An approach that helps guide actions to transform agri-food systems towards green and climate-resilient practices.

recommended adaptation options and be gender disaggregated where required to ensure that the needs of women are reflected. Additionally, the project will facilitate the creation of Value Chain Action Plans (VCAP) at the provincial level, which will inform annual commune-level Action Plans (AP). There is currently no agricultural value chain planning at the provincial level. The VCAP will help the target provinces to conduct VC planning and the communes in augmenting their development plans with considerations on agronomic and market opportunities. The APs will be developed through participatory sessions involving both men and women farmers, Producer Organizations, water users' groups, service providers, and NGOs, ensuring representation from men, women, and minority groups. Communities will identify and prioritize potential adaptation options and prepare a plan to implement them. Such options may include crop diversification such as vegetable or fruit production, poultry and aquaculture production, rotation or changing varieties (for e.g. flood or drought resistant varieties in rice) or improved water management (for e.g. AWD) and new practices such as Laser Land Levelling (LLL), mechanized direct seeded rice (mDSR), soil health management options, and Integrated Pest Management (IPM) in both rice and non-rice crops, rice straw management, etc.

45. The annual AP will specify preferred rice varieties, vegetable choices, and adaptation measures like poultry production and aquaculture, along with capacity support needed for the selected value chains. It will also outline implementation schedules, targets, and monitoring arrangements, with revisions made annually based on new data and community feedback. The APs will ensure the inclusion of women and women-headed-households in their development. A community-based monitoring and evaluation system will track AP outcomes, integrating findings from climate information systems and groups of both men and women farmers. Furthermore, representatives from each commune involved in AP preparation will participate in district-level multi-stakeholder platforms (DMSPs) to enhance value-chain coordination, ensuring effective communication between the commune and district levels. Special efforts will be made to ensure the inclusion of at least 30% women and 20% youth in these forums. MSMEs will also be a key part of these forums given their vital role in agriculture development in Cambodia. CAISAR will include elements of the APs into commune plans and build local capacity to enable them to develop them on their own. Representatives from other communes will be invited to learn from the experience for further dissemination. Provincial level staff will also be trained in the development of these VCAP to enable them to develop them on their own. Medium, Small MSMEs[51] will be engaged in the exercise to enable the private sector to work with local farmer groups and cooperatives and develop them independently. Representatives from other provinces and communes will be invited to learn from the experience and ensure that the MSMEs and private sector service providers are able to expand the idea to other areas due to the value they add in helping them increase their revenue and increase farmer incomes. Lead farmers are expected to be an important resource to demonstrate the impact on agriculture productivity.

Activity 1.1.2 Preparation of training materials to support implementation of the AP (NCDD-S)

46. The project will create and update training curricula and materials to support farmer adoption of climate-resilient (CR) technologies and practices outlined in action plans. Emphasizing user-friendly, visual learning materials like charts, leaflets, and short notes, the initiative aims to accommodate men and women farmers with lower literacy rates. Training materials will be regularly revised based on field experiences, new practices, varieties, and climate information and feedback from the participants particularly women for the crops they are involved in producing. IRRI and a service provider will develop these materials for rice and non-rice commodities, respectively, which will be suitable for Training of Trainers (ToTs) and Farmer Field and Business School (FFBS) training. The materials will encompass both technical and facilitation aspects of crop, livestock, and agribusiness management, featuring hands-on activities related to production and business planning. The specific tasks that women perform in each crop will be considered in designing the training programmes.

47. Key training topics will include best practices for irrigated and rainfed lowland rice, vegetable production techniques (e.g., drip irrigation), fish farming, and improving household chicken production. Where required

[51] According to the government's definition, MSMEs are recognized as agribusinesses, which may include businesses operated by organized farmer and youth groups, encompassing aggregation, value-added, and service enterprises.

separate training sessions will be conducted for men and women. Existing materials from GDA, CARDI, and IRRI will be adapted to the specific contexts of target provinces and needs by gender. TOT will also draw on activity 1.3.3, 1.3.4, and 1.3.5 for CIEWs. Collaborative input will be sought from various stakeholders, including service providers and men and women farmers, through consultative meetings to ensure the training materials are context-specific and relevant. The training materials and course content will be provided to both NCDDS and the MAFF for further dissemination and use. The staff from both these institutions will be invited to attend these courses to build their capacity and introduce them to other areas.

Activity 1.1.3 Conduct training of trainers to create a pool of expertise to demonstrate and propagate the CR technologies and practices (NCDD-S)

48. Once the training materials are prepared, ToTs will be organized to create a pool of expertise to propagate the CR technologies and practices, and outreach to farmers and disseminate information about CIEWS (including, awareness of mobile app(s))About 200 champions from interested NGOs, PDAF, universities or any external institutions and private entities will be trained, who will be able to work as Master Trainers (MTs). At least 40% of the TOTs will be women. The training will be organized by the SMSs who will continue to provide support to the MTs in testing and adapting to new technologies and practices, and create awareness and understanding of how to use CIEWS in decision making.

49. The project will organize two ToT sites in each province or a total of 8 in the project area. The 8 ToT sites will be retained until the fifth year of the project to strengthen the knowledge base of the MTs and to test, modify and demonstrate new technologies and practices as required based on the changing context. The pool of expertise developed will help address the extension gaps in CR practices and technologies and will be able to disseminate and expand the evolving practices even after project completion.

Activity 1.1.4 Training and advisory on CR technologies and practices (NCDD-S)

50. The core team trained will start training farmers using the training materials developed. Altogether, 40,000 farmers will be trained through 1600 Farmer's Field Schools. Given the need to cover such huge numbers of farmers in a short time, a cascade approach will be followed. The details of the cascade system are described in Annex 2. It will begin with a 160 FFS in the first year implemented by the Master Trainers (MTs), and will be expanded utilizing newly trained key farmers, which include women and youth under the supervision of the MTs. Inclusion of women as Master Trainers and separate sessions for women will be considered wherever this is preferred by the participants. The program will continue until the 7th year of the project. The project will enter into agreement with training entities who have been trained earlier (MT trainings above) to implement the FFS programs. Arrangements will be made to keep a roster of the MTs with both MAFF and NCDDs for further dissemination of the techniques and technologies disseminated through the FFS in other areas for programmes that can help finance the training.

Table 3: Outreach of FFS

Year	Target groups	FFS target	Beneficiary target	Objective	Remarks
1	Master trainers only (NGOs, /PDAFF, research institutes, young ag. graduates etc.)	8	200	Develop master trainers (MTs) to conduct training to farmers	Trained by subject matter specialists (SMS) at IRRI and a service provider in collaboration with commune facilitators. These MTs will be responsible for training to farmers from year 2.
2	Farmers	160	4000	Build farmers 'capacity on CR technologies	Trained by MTs, supported/ supervised by SMS
3	Farmers	320	8000	Build farmers 'capacity on CR technologies	Trained by key farmers and MTs, overall supervision by SMS

4	Farmers	320	8000	Build farmers 'capacity on CR technologies	Trained by key farmers and MTs, overall supervision by SMS
5	Farmers	320	8000	Build farmers 'capacity on CR technologies	Trained by key farmers and MTs, overall supervision by SMS
6	Farmers	320	8000	Build farmers 'capacity on CR technologies	Trained by key farmers and MTs, overall supervision by SMS
7	Farmers	160	4000	Build farmers 'capacity on CR technologies	Trained by key farmers and MTs, overall supervision by SMS
		1600	40,000		
About 40% of FFS will be focused on women groups only (vegetable production, mushrooms, chicken production, aquaculture etc.), FFS will have at least 40% women participation and at least 15 % youth participation (age group till 30 years)					

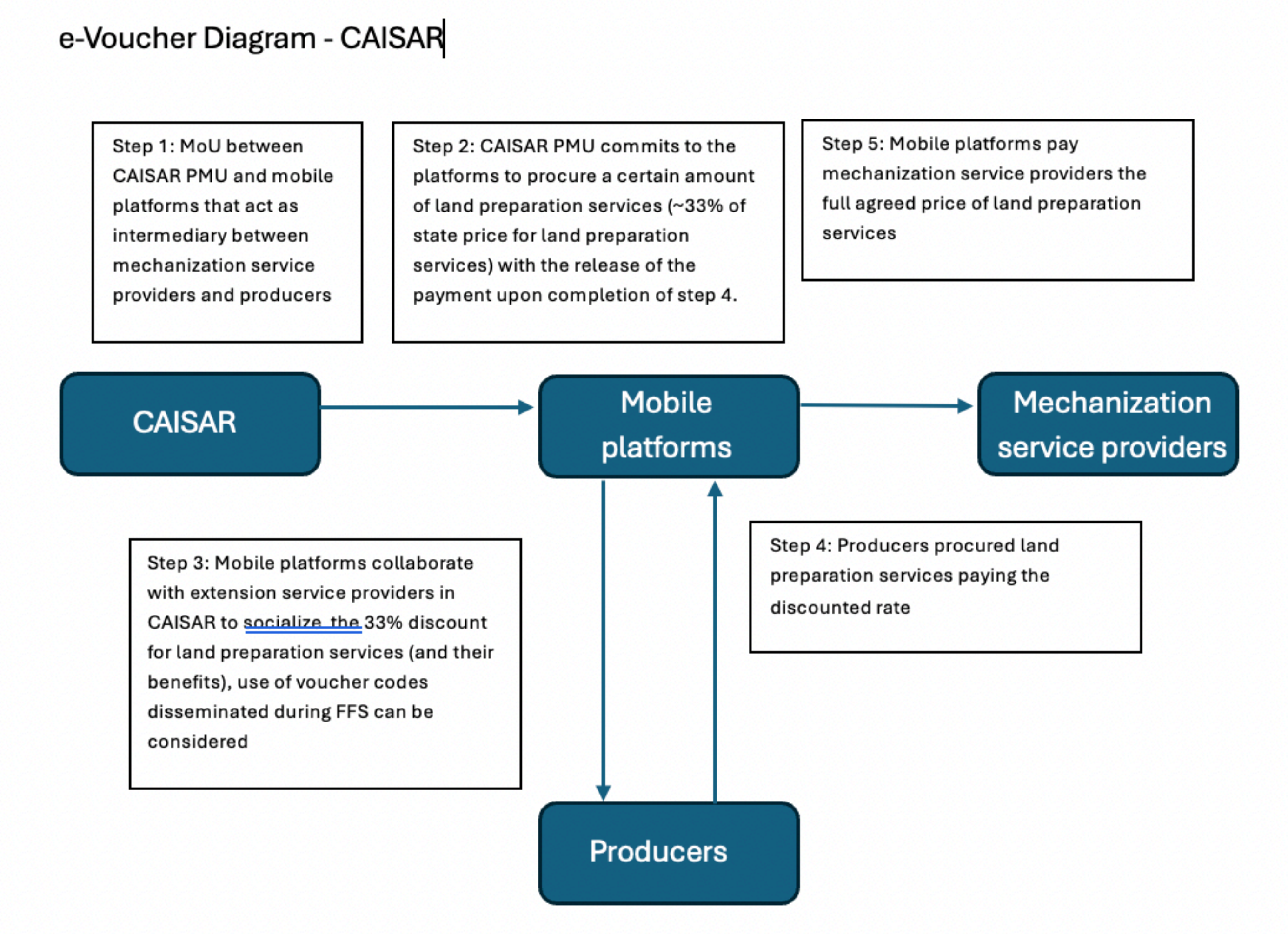
51. The FFBS activities will be used to link the farmers with markets and build their capacities in value chain development covered under Output 1.2. An important element of the training will be the integration of some of these alternative crops in the rice farming system and the focus on post-harvest management and marketing. To further fortify the capacity development post-FFBS, practical video library and online advisory services on rice and non-rice commodities will be developed through mobile application, allowing farmers to access information and troubleshoot via their mobile phones.

Activity 1.1.5 Strengthening and fostering tailored mechanization service providers for improved mechanization service delivery (NCDD-S)

52. This activity aims to enhance the availability of high-quality mechanization services to farmers, especially to support them to shift towards the mechanized direct seeded rice (mDSR) IRRI.
53. The mechanization service providers will be linked with the machinery dealers and mechanics for smooth service delivery to farmers. As explained in Annex 2, there is already a strong desire among farmers and willingness to pay to shift mDSR from existing manual broadcasting method.

54. For LLL, there is a growing interest among farmers, but the technique is still seen as expensive by farmers. In addition, farmers have seen benefit of LLL mostly in dry season rice or in dry direct seeded rice practices. The project will therefore procure 20 LLL units (five for each province) for demonstration purpose and will integrate it with the service providers so that interested farmers can more readily access LLL. This approach has already worked well to establish successful private industry laser levelling providers in Cambodia. IRRI will be engaged for high level knowledge products and capacity support to commune agricultural officers. IRRI will organize training for commune agricultural officers on facilitating the growth of mechanization service providers at the commune level, provide continuous backstop and develop knowledge products and training materials on the application of mDSR.

55. The project will strengthen the commune agricultural officers' current extension model to promote mechanization especially in rice cultivation. MAFF's commune level extension workers will be trained on promotion of mDSR and other technologies and will co-ordinate the services of service providers through capacity support and in assisting them in developing viable business models. A mobile platform connecting service providers and farmers will be developed in mobile app and farmers will be incentivized to try the mDSR services through eVoucher in the app. By demonstrating the value of these technologies, the project will increase demand which will encourage Private sector to provide these services on an on-going basis. The demonstration of increased yields from the use of these technologies will increase demand beyond the project area and scale their use through provision by private sector.



Activity 1.1.6 Community-based monitoring and evaluation (CBME) of the implementation (NCDD-S)

56. Local farming communities play a crucial role in implementing climate-resilient measures, and their involvement in monitoring outcomes is vital for the sustained adoption of these practices. The Community Based Monitoring and Evaluation (CBME) approach enhances community ownership from planning to evaluation. Performance indicators will be developed in collaboration with men and women farmers and stakeholders to track the effectiveness of interventions. These indicators may include metrics on training participation, engagement of female farmers in various production activities, and the introduction of new crop varieties. The indicators will be validated for integration into the project's monitoring and evaluation framework and national Climate Action Monitoring systems.

57. Results and lessons from the CBME will be linked to a Web-based monitoring system. This will facilitate information flow both ways, allowing feedback from national levels to inform community-based systems. The feedback from agriculture cooperative and farmer groups will also be reported. By the sixth year, discussions will be initiated at the sub-national level to explore the potential for integrating community action planning and CBME processes into routine operations beyond the project's duration. The M&E system will report also on the project performance in the inclusion of women and youth in the training programmes. Commune level officers and MAFF and NCDDs staff will be included in understanding the monitoring system so that they can then use it for their own purposes. The participation of the MAFF and NCDDs staff in the commune monitoring system may lead to its upscaling to other areas.

Output 1.2 **Climate adaptive value chains developed by 4Ps and increased access to finance** *(NCDD-S)*

58. This output aims to address financial barriers to invest and improve sales of smallholder farmers, farmer groups, agricultural cooperatives and related MSMEs by helping them leverage capital for adaptation and value chain investments. The scaling up of climate smart practices, services, and technologies demonstrated under output 1.1 will be supported under this output. This output consists of three key and synergistic activities: (i) Value chain (VC) study that aims to identify the gaps and improve VC governance; (ii) Establish and facilitate multi-stake holder platforms to improve VC coordination and stimulate investment from the financial sector to invest in climate change adaptation through VC development; (iii) Public – Private – Producer – Partnership Facility (4PF) that crowd in, de risk, and co-finance investment with MSMEs and farmers in support of climate-sensitive commodity development and rural employment generation. The development of these value chains activities will be undertaken through a gender lens to capture the engagement of women in the different agriculture commodities and their specific role in each value chain and how they can benefit from the activities of the project.

Activity 1.2.1. Value chain study and planning (NCDD-S)

59. The project will focus on enhancing the value chains for rice, vegetables, native chicken, and fish in Cambodia, with the possibility of adding more commodities later. Within the first six months, the Agri-Enterprise Development (AED) team, supported by international experts, will train provincial and district partners to develop concise value chain action plans (VCAP). These plans will involve mapping commodity actors, identifying value-added processes, and pinpointing challenges and opportunities. The VCAPs aim to highlight interventions that provide higher income potential, are inclusive of vulnerable groups, and are climate resilient.

60. The VCAP development process will consist of five steps: conducting background research, mapping commodity sectors, analyzing structural and dynamic factors, performing financial and economic analysis, and hosting a workshop to validate findings. Each target province will produce at least four approved VCAPs, which will guide commodity development integration into provincial programs, provide technical support, mentor value chain actors, and facilitate business operations. The VCAPs will also inform community-based adaptation planning, ensuring ongoing updates and relevance to agribusiness entrepreneurs in the region.

Activity 1.2.2 – Establish Multi-Stakeholder Platforms (MSPs) (NCDD-S)

61. The project will establish Multi-Stakeholder Platforms (MSPs) to facilitate collaboration among value chain actors, focusing on enhancing governance and improving relationships within selected agricultural value chains. Two types of MSPs will be created: District Commodity-Based Multi-Stakeholder Platforms (DMSPs) and Provincial Agro-Financing Workshops. DMSPs shall strengthen coordination among stakeholders such as farmers, traders, and financial institutions, and Provincial Agro-Financing Workshops to connect local financiers and producers. The DMSPs aim to organize at least 1,000 events throughout the project to improve market linkages, enhance transparency, and foster partnerships, while the Agro-Financing Workshops will occur bi-annually at the provincial level to inform participants about financing options and opportunities.

62. The DMSPs will provide a structured approach to identify challenges and opportunities within the value chains, facilitating discussions on potential interventions and joint actions among stakeholders. These will be undertaken from a gender perspective. The outcomes may include improved market information, stronger partnerships, coordinated actions, and conflict resolution among value chain actors. Follow-up actions will be tailored to stakeholder needs, and the success of these platforms will rely on effective facilitation and engagement. How women can best be engaged in the value chains will be an area of focus. The Provincial Agro-Financing Workshops will be coordinated with local and regional financial institutions and relevant stakeholders to share innovative financing options, ensuring the participation of MSMES, and relevant Financial Institutions. It is expected that some sustainable relationships will be built after the organization of the large number of events throughout the project lifecycle which will foster partnerships and can be continued on their own. In case some of these partnerships are successful they are expected to be scaled up beyond the project area due to mutual benefit. While the bi-annual workshops at the provincial level may not continue after the end of the project it is expected that some of the introductions made during

the process will lead to better access to financial institutions and some of the access to financing options are expected to continue beyond the project period.

Activity 1.2.3. Public Private Producer Partnership Facility (4PF) and leveraging of capital (NCDD-S)

63. Cambodia's agricultural sector has experienced substantial growth over the past two decades, leading to significant poverty reduction and improved food security. However, concerns persist regarding the quality, sustainability, and environmental impact of this growth. In response, the Royal Government of Cambodia (RGC) is evolving its agricultural and rural development strategy to emphasize sustainable development through the "Triple Bottom Line" approach, focusing on economic, social, and environmental goals. The RGC aims to foster public-private partnerships to enhance investments, improve the quality of services, encourage innovation, and manage resources more effectively.

64. The Public Private Partnership Facility (4PF) is designed to strengthen micro, small, and medium enterprises (MSMEs) in the agricultural sector, particularly focusing on supporting young rural entrepreneurs and women entrepreneurs. Through grants and technical assistance, the 4PF encourages investments that benefit both businesses and local men and women farmers. Eligible enterprises can receive financial support based on their investment size, with specific criteria for applicant selection, including economic viability, equity participation and potential impact on target groups. The types of investments which are eligible for financing will include both production-oriented activities and value-added activities that improve the capacity for processing, access to markets, improve product quality, reliability and produce in sufficient scale to attract the private sector. The initiative aims to support at least 120 MSMEs, with a goal of achieving an 80% increase in net turnover among recipients and a return on investment exceeding 10%. At least 40% of these enterprises will be those which are led by women or have a majority of women shareholders or employees. To ensure inclusion of youth, a target of 20% will be established for inclusion of youth. Separate windows will be established to ensure the inclusion of women and youth in the grant application call. The 4PF will be managed, likely by a third party contractor, to ensure a private sector approach, including due diligence and standards of service. It will operate in close collaboration with other project components to ensure that supported investments positively affect local farmers and agricultural practices. Details of the grant and/or TA mechanism are outlined in Annex 2. The grant facility is envisaged as an incentive for demonstration purposes, to derisk first responders and those with significant climate risks. The project will provide technical assistance for a specific period and grant support of a specific amount after which it will only provide mentoring and supervisory support during the project period. The role of the financial institutions will be to subsequently upscale the investments through a system that provides finance on a more sustainable basis through loans and equity of the investors.

65. Building on IFAD's ongoing projects, such as AIMS and SAAMBAT, the initiative will encourage the financial sector to identify and invest in climate change adaptation and rural value chain development opportunities. While MFIs report having sufficient capital for medium and long-term investments, they lack risk management tools to engage in climate adaptation efforts. To bridge this gap, CAISAR aims to attract financing from various financial institutions by enhancing their understanding of rural activities and investments including capitalizing on increased business opportunities which the project seeks to stimulate at the farm and business level by organizing semi-annual Provincial Agro-Finance Workshops to improve information flow between value chain actors and financial institutions and address asymmetries of information. Further, commercial finance will also be integrated into the 4p investments where possible, creating viable pipelines for finance and both reinforcing existing, and stimulating new, value chain finance. This proactive approach is expected to unlock substantial financial resources for agricultural investments, significantly impacting rural economies and poverty alleviation in the target provinces.

Output 1.3. Increased access to climate information and advisory services for climate responsive water-use and crop planning *(NCDD-S)*

66. In Cambodia, the capacity to utilize weather and climate data for agricultural applications is limited, despite significant investments in data generation. The PEARL project by the GCF identifies critical gaps in agro-meteorological services, particularly in value-added services and their uptake by farmers. While some NGOs have provided climate-informed advisories, these efforts tend to be project-specific and lack

sustainability. Existing platforms, such as the PRISM initiative by the WFP, focuses primarily on social protection against climate events rather than comprehensive farm-level advisory services. There is a pressing need to enhance the capabilities of the Department of Meteorology and the Department of Hydrology and River Works to improve data collection, modeling, and the translation of weather forecasts into actionable agricultural advisories.

67. To address these challenges, Component 2.4 of the project will focus on strengthening national institutions to improve water and weather information systems, enabling more effective forecasts and early warnings for floods and droughts. This output aims to develop and disseminate relevant agrometeorological information tailored to local agricultural needs, facilitating better climate risk management for farmers. It will draw on approaches such as PICSA (Participatory Integrated Climate Services for Agriculture) that allows for co-designing advisories with intermediate and end-users, ensuring timely, relevant and actionable information. The information needs of women and the most appropriate channels of communication for them will be examined to ensure they access the information they need given their key role in crop and livestock management. A diversity of dissemination channels and mechanisms has been proposed to take unique circumstances into account. In general, information will be accessible through mobile apps but also disseminated through traditional communication channels where appropriate and feasible (extension system, local radio, village commune leaders). Within the mobile apps, the project will explore alternatives to text-based content – using visual and verbal options. In addition, reinforcement on availability of information and its utility will occur through FFS (in which 40% of target participants are women) and FWUCs, with FFS sessions dedicated to discussing forecasts and advisories at critical junctures in the cropping season or when specific risks are triggered. A coordination platform will be established at the provincial level to integrate national advisories into localized services, benefiting approximately 90,000 people (40% women) across four provinces. This approach will ensure a collaborative and adaptive learning process, promoting effective agricultural practices and enhancing resilience to climate change.

Activity 1.3.1 Strengthening agro-meteorological stations in the project area (NCDD-S)

68. This activity involves the procurement and rehabilitation of six automatic hydromet stations and four agrometeorological stations, enhancing data collection capabilities in areas not covered by the Ou Ta Paong SCADA system. Additionally, four automatic weather stations will be established. The network will generate reliable data to support decision-making across national, provincial, district, and local levels, while improving connectivity and data-sharing mechanisms among various departments. Equipped with sensors to measure temperature, water levels, rainfall, turbidity and soil pH-moisture sensor, soil temperature sensor, evaporation pan, pyranometer, etc as required. The infrastructure will be managed by (MoWRAM) through its General Department of Meteorology (GDM) and the data collected will be transmitted to a server as part of the SCADA system, facilitating better water management and providing farmers with quality agrometeorological information. To ensure data sharing and operability at national and provincial levels, CAISAR will leverage and align with similar activities of existing projects (e.g., PEARL).

Activity 1.3.2 Establish Provincial-Level Coordination Platform for climate-informed agricultural advisories (NCDD-S)

69. A process to integrate meteorological with agricultural data and produce national level agrometeorological advisory is being developed outside this project. This component will facilitate the translation of national level agrometeorological advisories into district level advisories. Provincial coordination platforms will be established to prepare Agricultural Meteorological Advisory Services (AMAS) at the district level, tailored to specific target crops. This platform will include representatives from provincial departments of water resources, meteorology, agriculture, research institutes, key farmers, NGOs, and relevant private sector members. It will integrate weather, climate, and soil data to produce routine, crop-specific advisory services, verify past forecasts, enhance dissemination efforts, and gather feedback from farmers to improve these services. The provincial coordination platform will receive technical support from organizations like IRRI that are involved in the project's implementation.

Activity 1.3.3 Build the capacities to deliver services (NCDD-S)

70. Capacity development will focus on enhancing the technical skills necessary for climate data collection and management, as well as translating this data into agricultural advisories through modeling and statistical analyses. The initiative will involve focus group discussions to facilitate co-creation among various government entities and farming communities, primarily targeting staff and professionals in climate and agriculture data management, public services, research, and NGOs. The program aims to train approximately 250 individuals through structured training courses, including short-term summer courses on GIS and data validation, as well as advanced training in areas like crop modeling. Both formal training and on-the-job learning approaches will be employed to build these capacities effectively. Inclusion of women will be an area of focus for capacity building ensuring that at least 40% of the individuals are women.

Activity 1.3.4. Establish the agrometeorology information and warning systems and the mechanisms including awareness raising (NCDD-S)

71. The dissemination of advisory services will be structured as a daily, bi-weekly and/or weekly bulletin (interval will be confirmed during implementation), utilizing multiple channels to reach farmers and stakeholders effectively. Some bulletins will be seasonal in nature to allow FWUCs to plan cropping area and project irrigation requirements. Given the existing gaps in systematic communication within Cambodia, the project will leverage various methods to ensure that essential information is widely shared. This includes posting advisories on the websites of the Ministry of Water Resources and Meteorology (MOWRAM), as well as local government platforms. Additionally, advisories will be broadcast through local radio and television programs, tailored into accessible audio and video formats for greater community reach. Special effort will be made to identify the channels of communication most accessible to women and use them as well. To further enhance accessibility, SMS messages will be sent to men and women farmers, commune leaders, and Farmer Water User Communities (FWUCs), who will help disseminate the information to the wider farming population. The advisories will also be integrated into ongoing Farmer Field Schools (FFS) to educate and inform participants directly. Furthermore, mobile applications (e.g., Chamkar app that is used in other IFAD-funded projects in Cambodia) will be employed to provide real-time access to advisory services, ensuring that farmers can easily obtain relevant information and make informed decisions about their agricultural practices.

72. To enhance farmers' awareness of agrometeorological advisories, an effective delivery mechanism will be established through participatory and cross-disciplinary approaches. This involves organizing awareness programs and seminars that bring together men and women farmers, research institutions, and other stakeholders as equal partners. These initiatives aim to educate farmers about the weather forecast process, its challenges, and how to utilize climate information in their decision-making. Existing platforms, such as FFBS and Agricultural Planning (AP) formulation processes, will be leveraged for outreach, incorporating field days and visits to explain the significance of climate-informed practices in simple, relatable terms. Additionally, targeted training sessions on climate-informed agricultural decision-making will be conducted, involving 320 key farmers and stakeholders. These trained individuals will serve as resource persons, promoting the importance of climate awareness during the FFBS and AP processes. The key farmers will also include 40% of women farmers. Feedback and lessons learned from these interactions will be continuously integrated into the advisory services, ensuring that the information provided remains relevant and effective for the farming community.

Output 1.4: Increased resilience of farm road infrastructure to climate change *(NCDD-S)*

73. Approximately 90 kilometers of rural roads in targeted project areas will be upgraded to enhance climate resilience, providing a safer and more cost-effective road network with year-round access to markets and social services. These improvements will differ from "inspection roads" constructed atop embankments of primary and secondary irrigation canals, which also serve local communities and aid in the operation and maintenance of upgraded irrigation systems under Component 2. When feasible, flood defense embankments will be designed to serve dual purposes of flood control and road access. This initiative will benefit about 10,673 farmers across 35 villages directly and another 1.15 million people indirectly by improving connectivity.

74. Road designs will take into account current and future climate change risks, including increased flooding and drought incidents. Drought conditions significantly affect unpaved roads, leading to higher dust levels, reduced visibility, and poorer air quality. Addressing flooding and soil moisture concerns will be prioritized to protect road investments. Resilience will be enhanced through climate-resilient designs, embankment raising, surface strengthening, drainage improvements, and erosion protection using vegetation and bio-engineering methods.

75. A list of potential farm roads has been identified, with feasibility studies completed (See Annex 2). Improving local road networks is a key-supporting factor for socio-economic development and will be integrated with Component 2, particularly along irrigation canals allowing farmers to better monitor crop progress and water levels. Independent farm-to-market roads fall under Component 1, managed by NCDDS. Activities for these roads will include:

Activity 1.4.1 Identification, selection and initial planning (NCDD-S)

76. The initial planning will involve consultative meetings with the communities and stakeholders to gather insights on local contexts, opportunities and challenges. This engagement will inform the design process, utilizing lessons from the Community Participation Framework (ADB 2018) prepared under the Rural Roads Improvements Project III. Key data on traffic volumes, climate change impacts, and existing conditions will be collected, along with future maintenance needs.

Activity 1.4.2 Technical survey and design considerations, preparation of cost estimation (NCDD-S)

77. In this activity, the design of rural roads will prioritize climate resilience, addressing the anticipated increases in flooding and drought due to climate change. The hydrology of the project areas indicates that both the frequency and intensity of flooding will rise during wet seasons, while droughts will become more prevalent in dry seasons. To mitigate these impacts, design approaches will incorporate advanced techniques, such as raising embankments, enhancing drainage, and utilizing bioengineering solutions for erosion control. The integration of climate resilience features will ensure roads remain functional year-round, with specific design elements including sufficient drainage capacity, flexible pavement materials, and vegetation-based erosion protection.

78. Three types of roads will be considered for improvement: polder embankment roads, canal embankment roads, and standalone rural roads (Type 3 and Type 4). For polder embankments, the design will maintain a minimum top width of 4.5 meters with a slope of 1:2.5, while ensuring adequate side protection using local vegetation. Canal embankments will follow similar criteria, reinforcing their dual function as transport routes. Type 3 and Type 4 roads will incorporate features such as increased embankment heights, improved cross drainage, and the selection of permeable materials to enhance resilience against flooding. Overall, the design will be informed by hydrological data and best practices to ensure sustainable, climate-proof road infrastructure.

Activity 1.4.3 Improve 90 Kilometers of farm roads (NCDD-S)

79. The construction of the road works will be executed through competitively procured local contractors and supervised by both technical experts and community members for proper quality assurance. The local communities will be engaged in monitoring and reporting on the construction works and will be trained for the purpose by the consultants.

Activity 1.4.4 Handing over of the completed works (NCDD-S)

80. The completed roads will be handed over to the Ministry of Rural Development (MRD). The MRD will commit to the operation and maintenance of the roads in line with the sustainable road maintenance program. This will be agreed prior to construction in an MOU between the project and MRD. The costs of the annual O&M will be estimated and indicated in the MOU.

Component 2: Upgrading and climate-proofing water infrastructure for increased resilience. (MOWRAM)

81. Component 2 is linked with Component 1 such that it facilitates the implementation of CR on farm crop and water management practices through improved field level water supply delivery and drainage. It will focus on rehabilitating and modernizing of irrigation and flood protection/drainage infrastructure in the 3 sub-projects[52], including irrigation and drainage canals, flood control embankments, and ponds, to provide high-efficiency climate-resilient irrigated agriculture systems for adapting to both increasing flood and drought conditions.
82. The expected outcome of this component is that within the 6 irrigation schemes the water management condition throughout the year is improved and can be controlled, meaning that there will be no major flooding in the wet season and no severe drought in the dry season. As concluded during the re-evaluation of the Feasibility Study and Project Preparation (FIRM) study, the Overall Feasibility Study (Annex 2) proposes to modernize and rehabilitate the irrigation and drainage infrastructure serving 32,056 has. The anticipated cropping calendar is based on a rain-fed/partially irrigated rice crop in the wet season in the entire project area, while in the dry season in around 70% of the total project area a second dry season irrigated rice crop can be grown, with in the remaining 30% of the project area (mainly the Lum Hach irrigation scheme) the other 3 value chains will be providing sustainable livelihoods as availability of irrigation water in this area is limited. The climate adaptation activities will ensure that the projected increase in extreme wet days and more frequent upstream flooding do not lead to damage and destruction of irrigation infrastructure and also transform this risk into a green opportunity by storing more water in irrigation ponds, improved drainage facilities to drain the excess flood and increased access to water through solar pumping. Additionally, more efficient irrigation technologies will ensure that farmers can irrigate larger parts of the schemes for longer periods throughout the year.
83. This component will be implemented through four integrated outputs. Output 2.1 is dedicated to the enhancement, establishment, and modernization of hydraulic infrastructure to elevate irrigation practices within the framework of the three sub-projects. In addition, component 2.2 is geared towards enhancing infrastructure that protects the project areas against flooding by means of encompassing rivers, drainage systems, and embankments. Consequently, the project aims to deliver irrigation systems that are highly efficient and resilient to climate change, catering to both increased flood risks and drought conditions. These two components aim to ensure and boost agricultural production by improving irrigation supply and safeguarding crops from water-related disasters.
84. The sustainability of the irrigation scheme will be ensured by strengthening the O&M capacities of the FWUC (Output 2.3) and PDoWRAM with the creation of a SCADA system for Ou Ta Paong scheme (output 2.4). The proposed areas that combine adequate potential with greatest need for assistance have been identified in consultation with MOWRAM and PDWRAMs. For ensuring that local information is taken into account, the FIRM study undertook field surveys and stakeholder consultations. Output 2.3 aims to strengthen the FWUC for improved irrigation management while the last output provides farmers relevant water information and SCADA based canal operation. Based on available water and weather information, farmers will be supported with the agro meteorological services as described in output 1.3. These four outputs are described in more detail in the following paragraphs.

Outcome 2: Increased water use conveyance and use efficiency

Output 2.1: Modernized irrigation schemes and ponds (MOWRAM)

85. Rehabilitating, upgrading and, where water and land resources are available and socially and environmentally possible, constructing new irrigation facilities and storage ponds within the 6 irrigation schemes, which will all have positive climate adaptation and mitigation impacts. Thus, this output has two main benefits: (i) increasing water availability and restoring existing and introducing new storage capacity for irrigation in the form of ponds while decreasing the destructiveness of floods on downstream locations; and (ii) the possibility of implementing crop diversification and new activities to increase farming incomes such as vegetables, poultry production and/or aquaculture. It will also allow for creation of employment opportunities, especially for young people. This measure would be part of integrated and

[52] The 3-sub projects include Ou Ta Paong, Lum Harch and Krang Poneley, with 6 sub schemes (Ou Ta Paong, Lum Harch, Kropeu Trom, Yotasast, Sooeung Krang Bat and Brambel Mom)

resilient water resources management and climate proofing to secure the sustainability of the overall irrigation system against negative climate impacts of droughts and dry spells as well as floods during the monsoon season. Output 2.1 and 2.2 will be implemented in three sub-projects within a total of six separate irrigation schemes. The flood protection works will directly benefit an area of 10,005 ha, including a proposed semi-polder area of around 2,550 ha (Output 2.2).

Activity 2.1.1: Technical analysis, field surveys and preparation of plans for system upgrading *(MOWRAM)*

86. After the detailed topo-surveys have been completed in all 6 irrigation schemes a detailed DED for all irrigation and drainage/flood control systems will be prepared taking into consideration the detailed social- and environmental impact assessment to minimize the amount of private land to be used and plot the new infrastructure as much as possible on public land. For the larger schemes, tertiary distribution systems are foreseen. This DED will primarily be based on gravity-driven flow irrigation but in case the natural slopes are insufficient to allow for gravity flow, solar-driven pumps will be considered at the beginning of the secondary or tertiary canals to allow for sufficient head to have good distribution of the available water over the entire scheme.

87. In this final DED the different layout options as identified in the FIRM phase will also be reinvestigated for technical, environmental and social/safeguard considerations. The implementation will focus on maximizing ecosystem-based solutions, reducing energy consumption, and consequently reducing GHG emissions. In addition, current diesel pumps lifting water to tertiary and secondary canals will be replaced where possible by collective solar pumping systems when the pre-conditions are acceptable with low pressure water delivery at the plot level to increase the water efficiency, avoid land losses and improve water use efficiency. Moreover, these technologies will allow to save time and labour, especially for women, and offer opportunities to engage rural youth.

Activity 2.1.2: Implementation of infrastructure upgrading *(MOWRAM)*

88. The implementation will follow a participatory approach involving farmers' at all stages from the survey, design, construction, implementation and operations. It will also cover all cross-cutting issues such as gender, environmental and social issues, resettlement, etc. The land rights aspects in construction-related activities will be guided by the Resettlement Planning Framework (RPF). The project will minimize land acquisition by following existing rights of way, where land acquisition is required, it will implement fair and transparent processes under the RPF. The ESCMF includes detailed procedures for managing land acquisition, including socio-economic surveys, eligibility and entitlement criteria, consultation, grievance redress, and monitoring, ensuring access to recourse for affected persons. This subcomponent will be financed by AIIB and Government.

Activity 2.1.3: Preparation of canal O&M plans including application of ICT and SCADA for operation *(MOWRAM)*

89. Upon completion of construction works, schemes will be handed over to the Irrigated Agriculture Department (IAD) of MOWRAM, and O&M responsibilities will be shared between the concerned Provincial Department of Water Resources and Meteorology (PDWRAM) and FWUC. MOWRAM's Irrigated Agriculture Department will be assigned overall management responsibility and required to introduce a system of on-going M&E and reporting of scheme performance. FWUCs will receive financial support until they become financially independent. PDWRAMs will provide technical backstopping to FWUCs. MOWRAM will arrange government budget for funding routine and periodic maintenance of the infrastructure retained under government control and for providing financial support over the first five years after FWUC formation. MOWRAM will authorize the transfer of management responsibility to a FWUC after training and assessment of the FWUC being financially independent (Prakas 306/2000-PID-RGOC Annex 21). It is expected that the impact of the project on increase water use efficiency through modernization of the schemes will encourage greater allocation of resources for expanding the modernization of irrigation schemes in other areas.

Output 2.2: Flood-proofing and drainage improvement (MOWRAM)

90. It focuses on improving disaster prevention and protection of farmlands and assets by establishment of early warning systems and improvement of the existing drainage and flood protection systems. Existing drainage and river training networks will be improved based on the results of the hydrological modelling and consultation with stakeholders living in the project areas.

Activity 2.2.1 Technical analysis, field surveys and preparation of plans for system upgrading *(MOWRAM)*

91. This output focuses on equipping the 6 schemes with facilities to withstand the currently annually recurring flooding of all parts, but mostly the downstream parts of the 6 schemes and also disaster prevention and protection of farmlands and assets by the establishment of flood protection embankments and drainage canals helped by early warning systems to improve capacities of the existing drainage networks and flood embankments. This flooding is partially due to the water inflow from the downstream end of the scheme from the Tonle Sap Lake and partly from local runoff of the higher areas upstream of the scheme for which no proper drainage infrastructure was made. Existing drainage and river training networks (especially in the Ou Ta Paong and Lum Harch schemes) will be improved based on the results of the hydrological modelling and consultation with stakeholders living in the project areas. Also, at the downstream end of these subprojects in the new semi-polder area solar-driven water pumps will be installed wherever necessary to pump excess drainage water to the outer area.

Activity 2.2.2: Strengthening and construction of flood control and drainage infrastructures *(MOWRAM)*

92. The applicability of nature-based solutions in flood management (e.g., revival of old river channels like Ou Ta Paong River, embankment improvement with vegetation strips, etc.) will be reviewed and subsequently used to inform the investment planning under this component. Farmers will also be supported to restore, and construct farm level drainages as required while project will install sub surface drainages in needy areas. In addition, it will also support strengthening of river training and their embankments to make them resilient against intensifying flood threat under the changing climate context. These improved embankments will be linked to the flood risk assessments improved in Component 1 and the water-related data will feed into the national database to improve flood early warnings for target areas (delivered through output 1.3). New flood protection embankments will be constructed in Ou Ta Paong and Stoeung Krang Bat irrigation schemes to avoid high season water levels in the neighboring Tonle Sap River into the command area.

93. The levels of the embankments have been based on the topo surveys recently carried out and the 2 water models focusing on the impact of the climate change on the flood levels in the Tonle Sap Lake and the rivers crossing the various project areas. For more information reference is made to the D.4 report of the FIRM study. One additional very considerable benefit of these embankments will be improved accessibility for the communities in these lower parts of the scheme as now they will be able to use the newly built embankments for local transport while in the past they had to use small boats. This will facilitate local communities to access markets and the flow of agriculture inputs and outputs. Annex 2 provides more details of the drainage infrastructure at the sub-basin level.

Output 2.3: Farmers Water User Communities (FWUC) established and trained (MOWRAM)

94. Establishment and training of FWUC: FWUCs are critical for ensuring the sustainable and effective operation and maintenance of irrigation schemes, in particular at the distributary network system (secondary tertiary and quaternary levels. With the rehabilitation of the irrigation systems and the creation of a flood protection embankment system of the three subprojects, the project will support the establishment and capacity building of the FWUC from the participatory approach of FWUC creation in the design, development and sustainable use of the rehabilitated irrigation schemes to be rehabilitated by the Project as these organisations are crucial for O&M management.

Activity 2.3.1: Formation and institutional strengthening of the FWUC *(MOWRAM)*

95. FWUCs are critical for ensuring the sustainable and effective operation and maintenance of irrigation schemes, in particular at the secondary and tertiary levels. With the rehabilitation of the irrigation system

and the creation of a flood protection/drainage embankment and canal system in the lower parts of the 3 sub-projects, the project will also focus on establishing new FWUCs and strengthening the existing FWUCs as these organizations are crucial for water management. This will include investments in improving the capacities in O&M activities, water use efficiency, improving the arrangement of water rights, water allocation, use of green technology, service fee collection and making them more inclusive. Furthermore, the project will train women and men farmers on using sustainable local water resources, climate adaptation strategies, and mitigation tools to reduce GHG emissions and increase agriculture productivity without negatively impacting the environment. The activities will be designed to strengthen the capacity of FWUCs for sustainable operation of the climate-resilient and climate-proofed irrigation systems. The inclusion of women in the FWUCs will be encouraged in both leadership positions and as members.

Activity 2.3.2: Build technical capacities of FWUCs for canal O&M *(MOWRAM)*

96. Build technical capacities of FWCUs for canal structure O&M and 2.3.3. prepare long term financing plan for O&M of the systems including the WUAS. The participation of women in these forums will be encouraged in discussion with women farmers. This will require an identification of the capacity needs, membership, and functionality of the FWUCs in the area. The concept of water pricing is embodied in the *Water Law*. However, mechanisms for water pricing have yet to be implemented. Thus, CAISAR will assist them in developing a business model for the operation and maintenance of each irrigation scheme on a self-sustaining basis in partnership with the farming communities and the private sector. This will entail the development of an O&M plan for the infrastructure investments, assessment of the willingness and ability of the users to pay for the services and a regulatory framework that gives the WUAs the authority to collect the fee and assess the basis on which water fee will be paid as Irrigation fees that can be based on the size of the land being irrigated or the volume of water used. Fees will be charged only when the scheme becomes operational.

Output 2.4: Water information system established and operational in three sub-projects[53] (MOWRAM)

97. The SCADA model is crucial to tackle the significant challenges in the agricultural sector on the inefficiency of irrigation practices, characterised by both indiscriminate and irregular water distribution, leading to substantial water wastage. This approach offers real-time information on water levels and flow rates, enabling more accurate and timely responses to fluctuations in water demand. The SCADA presents a contemporary solution for the modernization of existing irrigation systems, prevent flooding and it does so with a fundamental, cost-effective approach. Implementing the Water Information System allows for command-and-control functionalities while collecting valuable data on irrigation. This, in turn, will rationalise and streamline the irrigation process, significantly mitigating the wastage of water.

Activity 2.4.1 Preparing all processes for establishment of the SCADA and water information system *(MOWRAM)*

98. This output aims to establish a water information and warning system such that farmers and other relevant stakeholders are fully aware of climate condition, water availability, silt condition so as to effectively plan their irrigation scheduling and operation. The water and weather information will further help develop crop specific advisory services as described in component 1. However, the forthcoming IWRM, CWISP programs in the MoWRAM will be developing area specific flood and drought warning system, water information and will also engage in capacity support. The project will directly use these available systems to help farmers in the project areas better plan their agriculture and protect their livelihood assets. Working together with the above projects, a communication system using mobile phones will be established so that farmers are timely informed on evolving threats and challenges.

[53] Three sub-projects consisting of six schemes namely the Ou Ta Paong (1 scheme), Lum Hach (1 scheme), and Krang Ponley (4 schemes) to be established under the project.

99. At irrigation system level, it will install Supervisory Control and Data Acquisition systems (SCADA) in the Pursat hydraulic system network that includes the Ou Ta Pong irrigation scheme. The SCADA system will allow the water manager to monitor the real-time information on water levels and flow rates and the possibility to respond to fluctuations more accurately and timely. It provides command and control and collect data on irrigation to rationalize and streamline the process of irrigation service delivery. The proposed SCADA system will involve remote terminal units (RTU) to collect local data (water levels, flow rates, gate positions) and command local equipment (gates); a command center, to supervise/manage all the RTU; a communication system, to link the RTUs to the command center. Further details are explained in Annex 2.

100. The component 2 will be implemented by the Project Management Unit (PMU) at MOWRAM and its Project Implementation Units (PIUs) at the provincial level, including training and capacity development on development partners' policies and standards, monitoring, evaluation, reporting and verification, including climate rating and climate finance. Thus, PMU and PIUs will actively disseminate knowledge and green solutions in climate adaptation and mitigation among farmers, policy- and decision-makers, and rural communities.

Component 3. Strengthened institutional and regulatory capacity for low-emission climate-resilient development pathways.

101. This component aims to strengthen Government institutions, mainly the MOWRAM, MoE the NCDDS and the FWUCs. The focus for MOWRAM will be on upgrading technical capacity in various aspects of CR irrigation design and management. The key focus area under NCDDS will be on climate resilient crop production.

Outcome 3: Enhanced environmental governance

Output 3.1 Strengthened MOWRAM *(MOWRAM)*

102. This output supports, training and capacity development for the project management unit (PMU) and will enhance the institutional capacity of the Ministry of Water Resources and Meteorology (MOWRAM) and Provincial Departments of Water Resources and Meteorology (PDWRAM). This includes preparation of design manuals, capacity building and training in river basin management and water accounting, the operation of Supervisory Control and Data Acquisition (SCADA) systems, in the design and implementation of climate-resilient green technology, low-carbon pathways, climate-proof irrigation systems and gender mainstreaming and social inclusion. The development of standard designs and construction manuals will enhance the replicability and long-term sustainability of the project. In addition, it will build capacities to employ digital platforms and the best-available climate and water data to monitor and boost the operational performance of the irrigation systems and future maintenance plans.

Activity 3.1.1 Strengthening river basin management and water accounting systems *(MOWRAM)*

103. CAISAR's investment in irrigation development envisages improved water availability to and throughout each of the schemes. This will require careful management of water resources storage, allocation, and delivery, and careful distribution of delivered water to the farmer/water users. Therefore, CAISAR will support capacity strengthening for water accounting in the river basins where CAISAR will rehabilitate and climate proof the irrigation infrastructure. CAISAR Component 2.4 will support MOWRAM with establishing a water information system, and this will serve and needs to be designed in such a manner that it provides the water accounting systems that will be strengthened with the data these systems need. The CAISAR activity will support MOWRAM and the four PDWRAMs in reviewing their current water accounting system in the project-related river basins and to plan/implement improvements where desirable and feasible. Activity 3.1.2. Communicating with stakeholders

Activity 3.1.2 Communication with stakeholders *(MOWRAM)*

104. MOWRAM's PDOWRAMs participate actively in the various Provincial-level meetings where water-related aspects are relevant in the discussions. Key items are the flood management plan for the wet season and the water allocation plan for the dry season. For MOWRAM/PDOWRAM to prepare and implement these plans properly, it will make use of the water information system and water accounting system mentioned above. This will be achieved efficiently and effectively by publishing the water situation on a website that can easily be consulted by the stakeholders. MOWRAM already has such an information website, namely for Flood and Drought Forecasting and is in the process of establishing a water data and information centre, which will result in more data coming available. These are centrally managed, with data being provided by PDWRAMs and other MOWRAM units, closer to 'the field'.

Activity 3.1.3. Support to and guidance of leaders of commune, village and FWUCs and gender awareness *(MOWRAM)*

105. The development of standard designs and construction manuals will enhance the replicability and long-term sustainability of the investment. Building capacities to employ digital platforms and the available climate and water data will also benefit other irrigation systems. More details are provided in Annex 2 section 6-part B page no194-195.

Output 3.2 NCDDS strengthened at national and provincial level

06. NCDDS plays a crucial role in promoting democratic development through decentralization and deconcentration of reforms throughout the country. To effectively carry out this mission, the secretariat of NCDDS has been actively engaged in capacity development initiatives aimed at enhancing the skills and knowledge of sub-national government officials

Activity 3.2.1 Strengthening of NCDDS at the national level (NCDDS)

107. The capacity development efforts of NCDDS will include:

- Strengthening NCDDS' project management capacity: This includes training on project cycle management, financial management, monitoring and evaluation, human resources management, climate change and gender perspectives.
- Policy formulation and implementation: Enhancing the capacity to develop and implement effective policies and programs at the sub-national level, especially the Sub decree 182 on the functions transferred to the district level. Disseminate and knowledge sharing the role and responsibility of district and municipality to support farmers.
- Resource mobilization: Strengthening skills in resource mobilization, and planning at the sub-national level. GCF is another window where institutions can access to Climate Fund, so capacity building of NCDDS staff on project feasibility study and preparing full project proposal to GCF can help NCDDS to mobilize more resources.
- Institutional strengthening and coordination: Promoting efficient institutional structures, coordination mechanisms, and inter-agency/ministry collaboration at the national level and sub-national level.

Activity 3.2.2 Strengthening of the provincial teams (NCDDS)

108. This will involve the following sub-activities:

- Conduct a detailed capacity development needs assessment - The full evaluation of the current capacity of PDAFF, PDOC, PDWA to support extension service will be done by a consultant recruited for this purpose. Tools to be developed and administered to provincial staff and a report capturing the findings will be drafted. The purpose of a capacity development needs assessment is to identify the gaps between the current capacity of the Provincial Staff and the capacity required to achieve its objectives.

- Training in technical concepts and cross-cutting subjects - Training and capacity building for Provincial staff in areas such as policy analysis, project management, market systems development, use of ICT tools and incorporating gender perspectives. Support to the adoption of ICT tools to improve the project efficiency, effectiveness, and transparency. To effectively carry out their duties, Provincial staff need to be well-versed in a wide range of technical concepts and cross-cutting subjects. In addition to the gaps, topics like - Sustainable agriculture, climate change adaptation and Information and communication technology (ICT) and gender analysis to be included in the training. The project will provide structured support to ensure that the participants are able to conduct some of the activities independently. The trainings will be structured to provide skills adequate for the functions that NCDDS staff are expected to complete. The increase in management capacity will be useful for deploying in other areas and the resource mobilization capacity will ensure the generation of resources for scaling up. The strengthening of the provincial teams will enable them to scale up and expand to other areas.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Accredited Entity (AE)

109. The International Fund for Agriculture Development (IFAD) has a long history of implementing rural development projects in Cambodia, which became a member in 1992. IFAD has supported twelve investment projects for a total value of USD 940 million in collaboration with a host of co-financing partners and its own financing of USD 309 million benefitting 1,565 million households. IFAD will serve as the Accredited Entity (AE) for this project and will be responsible for the overall management of the project, including: (i) administrative, financial and technical oversight and supervision throughout project implementation; (ii) ensuring funds are effectively managed to deliver results and achieve objectives; (iii) ensuring the quality of project monitoring, as well as the timeliness and quality of reporting to GCF; and (iv) undertake project mid-term and completion evaluation. IFAD will ensure that each of these areas of responsibility are undertaken in accordance with the detailed provisions outlined in the Accreditation Master Agreement (AMA) between IFAD and GCF. IFAD's role as AE will be assigned to its regional office in Bangkok, and its country office in Cambodia. AIIB is co-financing the project and will provide funds directly to MOWRAM. In its capacity as co-financier, AIIB will undertake joint supervision of the project with IFAD and also participate in and contribute to the evaluation of the project.

Executing Entity

110. The AE will enter into a financing agreement (i.e. a subsidiary agreement) with the Royal Government of Cambodia, acting through the MEF. The Executing entity is the Royal Government of Cambodia, acting through (i) MEF; (ii) MOWRAM; and (iii) NCDDS. (EE(s)). The MEF is the ministry authorized to enter into financing agreements on behalf of the RGC. The financing agreement will set out the roles and responsibilities of MEF, MOWRAM and NCDDS. MOWRAM will establish a Project Management Unit for the coordination of all components and consolidation of the project planning and reporting. In addition, the National Committee for Sub-National Democratic Development Secretariat (NCDDS) will be responsible for the implementation of component 1 and output 3.1 of component 3 which is designed to improve farm-level climate adaptation, resilience, water use efficiency and institutional strengthening. The NCDDS is a Direct Access Accredited Entity (DAE) to the GCF and will have a Project Implementation Unit (PIU) for implementing the component 1 and output 3.2 of component 3. The budget allocation between MOWRAM and NCDD-S will be determined based on prepared cost tables. Designated accounts will be established for both MOWRAM and NCDDS, and funds will be transferred directly by IFAD to MOWRAM and NCDDS.

Project Oversight and Steering Arrangements

111. A Project Steering Committee (PSC) will be established for inter-ministerial coordination. The role of the PSC, chaired by the minister of MoWRAM, is primarily to provide strategic oversight of the project's overall progress toward its intended goals and outcomes, to make decisions regarding the project's overall directions, and to address high-level issues and risks that are beyond the immediate control of the project. The PSC makes decisions based on consensus. It will be chaired by the Minister of MOWRAM and comprise of senior officials from MOWRAM, the Ministry of Economy and Finance (MEF), the Ministry of Interior (MOI) which is responsible for the NCDDS, Ministry of Agriculture, Forestry and Fisheries (MAFF), Ministry of Environment (MOE), and the Provincial Governor's Offices of Pursat,

Kampong Chhnang, Kandal, and Kampong Speu provinces. An Inter-Ministerial Resettlement Committee (IRC), chaired by MEF, will manage resettlement and land acquisition issues where they arise. The National Project Director of the Project Management Unit (PMU) will serve as a secretariat to the PSC.

Composition of the Project Steering Committee (PSC):

Chair: Minister, MOWRAM
Secretary: Secretary of State, MOWRAM National Project Director,

Members:

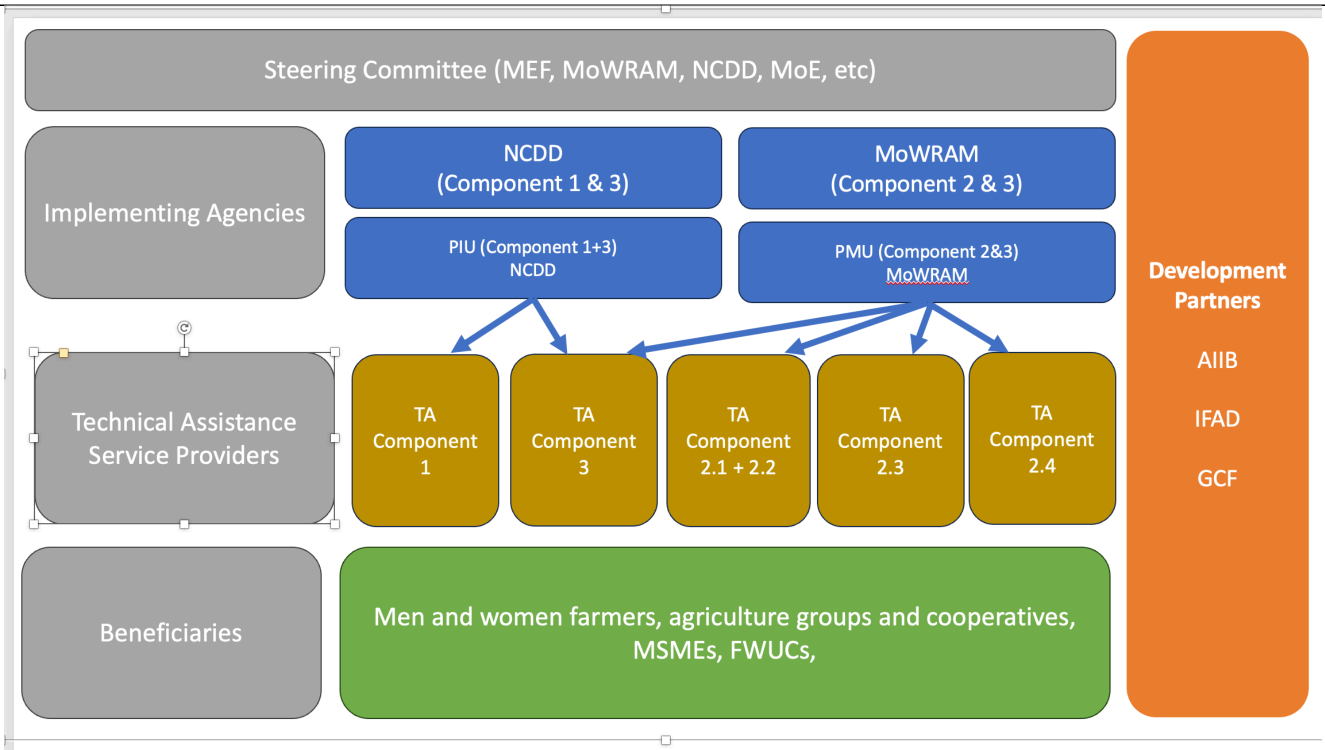
H.E. Secretary of State, MOWRAM
H.E. Secretary of State, MOI (NCDDS)
H.E. Secretary of State, MEF
H.E. Secretary of State, MOE
H.E. Under Secretary of State, MEF
H.E. Under Secretary State, MAFF
H.E. Director General, MEF
H.E. Four Provincial Governors
Director of Department of Multilateral Cooperation, MEF

Management Arrangements

112. MOWRAM will establish a Project Management Unit (PMU) for the implementation of the CAISAR project. The PMU will be led by the Secretary of State (MOWRAM) as the National Project Director (NPD) and managed by an appointed National Project Manager (NPM). The NPM will be responsible for overall project management and coordination. Within the PMU, project-recruited staff and staff seconded by the government will collectively comprise a project delivery team that will ensure sound and effective project implementation. The PMU will also include an Operations Officers, Gender Specialist, Finance Officer, Human Resources and Administration Officer, Procurement Officer and Monitoring and Evaluation Officer. Consulting firms and individual consultants will be recruited to support the project implementation and undertake detailed design services and construction supervision. Start-up orientation sessions will be organized to orient staff with regards to the procedures of GCF, IFAD and AIIB. Capacity building sessions in understanding the procurement, financial management and monitoring and evaluation sessions will be held as needed.

113. The Project Management Unit will consist of personnel from MOWRAM, the Department of Hydrology and River Works (DHRW) of MOWRAM, and Provincial Departments of Water Resources and Meteorology (PDWRAMs) of the four provinces in which the project will be implemented. The PDWRAMs will be responsible for coordinating all field activities with Farmer Water User Committees (FWUCs). The Department of Farmer Water User Committees (DFWUC) has personnel at the central level but no office or personnel at the Province and District level. For the effective operation of the committees, the Department of FWUC will appoint additional personnel at the provincial and district levels.

Figure 5: CAISAR Organization Chart



114. The PMU will be responsible for the overall management of the project; it will coordinate directly with EEs and project stakeholders, be responsible for providing support to the execution of day-to-day activities and be responsible for the technical quality of the project outputs, activities, financial management, procurement, monitoring and evaluation. The National Committee for Sub-National Democratic Development Secretariat (NCDDS) will be responsible for the implementation of Component 1 and output 3.2 under component 3 and will establish a Project Implementation Unit and contract Service providers at the provincial level as required. The PMU will be directly responsible for the implementation of Component 2 (irrigation infrastructure) and output 3.1 under Component 3. There will be monthly coordination between MOWRAM and NCDDS to ensure the synergies between the various components.

115. The roles and responsibilities of each organization have been clearly outlined, with MOWRAM responsible for overall supervision and guidance, and the DFWUC responsible for preparing the annual work plan and budgets, initiating and coordinating communication, coordinating with the NCDDS PIU on agriculture support activities, and implementing social and environmental safeguards, among other tasks. The PDWRAM will assist the PMU in disseminating information and coordinating with the Ministry of Economy and Finance-General Department of Resettlement (MEF-GDR) to implement the resettlement plan, among other tasks. NCDDS PIU will work in close coordination with the provincial and district agencies for agriculture to implement the activities under Component 1. NCDDS PIU will coordinate with the MAFF/ASPIRE program team, MoC/AIMS team and MRD/SAAMBAT team to ensure strong synergy with the extension and farmer organizations that have been organized under these projects as well as GCF-PEARL for agrometeorological information and advisory services. The role and relevance of the NCDDS in CAISAR underscores its institutional capacity in successfully implementing farm-level components which are one of its mandates under the Decentralization and De-concentration (D&D) Management Policy which the NCDDS is tasked with discharging. NCDD-S will at the outset establish a mechanism for the identification and approval for selecting sub-grant recipients and determining the eligible activities to be undertaken by the sub-grantees. NCDD-s was used by IFAD under its Agriculture Services Programme for Innovation, Resilience and Extension (ASPIRE) in Cambodia. NCCDS has built considerable experience over the last 24 years in implementing a range of local level projects with a range of partners such as UNDP, SIDA and DFID, DANIDA, JICA, GEF, WFP, ADB, WB and IFAD

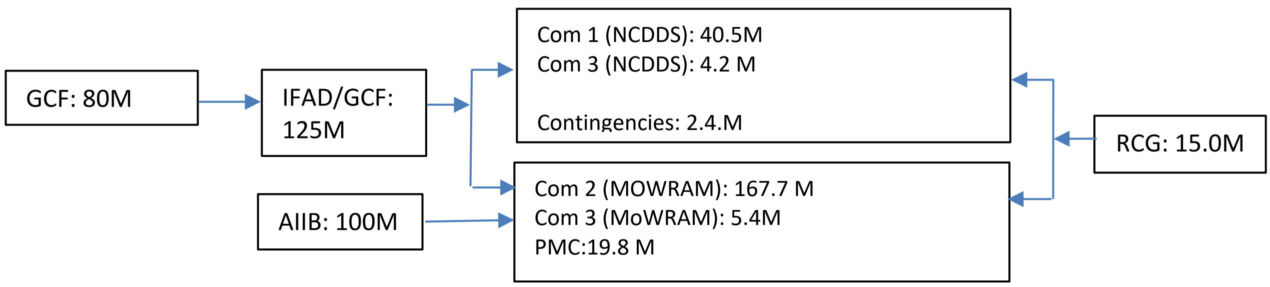
116. The project is attempting to transform farmer behaviour in their production decisions and beneficiaries will play a central role in all project activities and their participation and ownership will be critical to the success and sustainability of project interventions. Beneficiaries are expected to participate in the Farmer Field Schools and lead farmers are expected to disseminate the information to other households not directly participating in project activities. Similarly, farmers and other players along the value chain are expected to play an active role in the Multi-Stakeholder Platforms for greater market integration. In the irrigation sector, the water users will play an active role in scheme supervision and oversight and the FWUCs will play a critical role in the operation and maintenance of the schemes and will sustain the systems through payment of O&M charges as stipulated by the regulation.

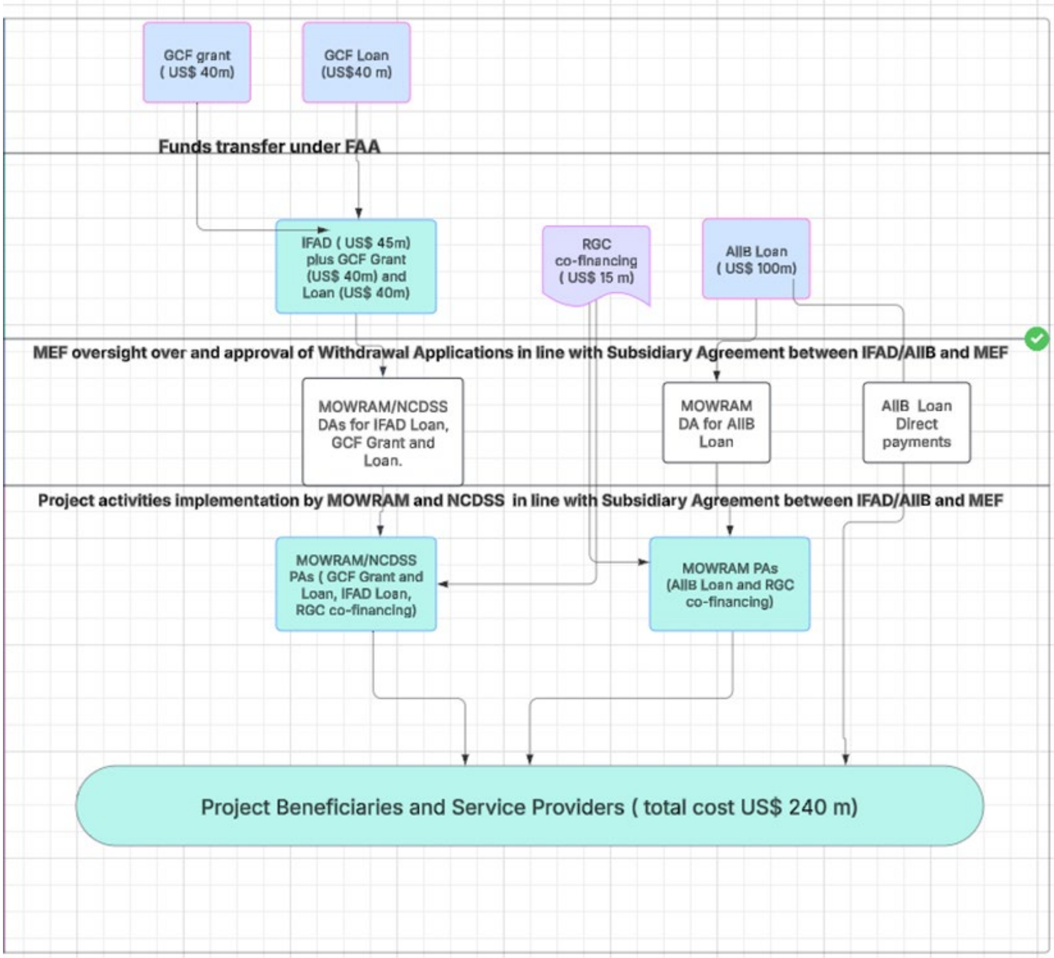
The Flow of Funds

117. The Asian Infrastructure Investment Bank (AIIB) and IFAD/GCF are the two main co-financiers of the project, and the RCG will provide contributions to the Project. AIIB and IFAD have provided grant funds and technical assistance for project preparation activities. An agreement will also be signed between GCF and IFAD as the AE for GCF funds. IFAD will also sign a Subsidiary Agreement with the RGC, represented by the MEF, for IFAD and GCF proceeds and RGC's financial support to the project. This Subsidiary Agreement will also serve as the co-financing agreement for the RGC's contributions. The GCF funds will be provided to IFAD and then transmitted directly to the Ministry of Water Resources and Meteorology and NCDD-S. The funds of the Asian Infrastructure Investment Bank (AIIB) will also be provided directly to MOWRAM for investment in the climate resilience and modernization of irrigation infrastructure. The Royal Cambodian Government will provide cash support directly during the implementation of the project. For financing the Project, IFAD will enter into a Funded Activity Agreement (trust account) with the GCF. For AIIB co-financing, AIIB will enter into a separate financing agreement with the RGC.

118. MEF will authorize MOWRAM PMU and the NCDDS PIU to open a designated accounts (DAs) at the National Bank of Cambodia for funding from IFAD, AIIB and GCF. MOWRAM PMU and NCDDS PIU shall open project accounts at a commercial bank to receive the fund transfer from the IFAD, AIIB and GCF DAs to pay for project activities.

Figure 6: Funds Flow Diagram (USD)





B.5. Justification for GCF funding request

119. The current project requires GCF funding to address both mitigation and adaptation measures which are beyond the ability of the Government to finance. There is need for significant financial resources for investments for climate change mitigation and adaptation in Cambodia. The total funding required for all mitigation actions is estimated to be above US$ 5.8 billion and US $ 2 billion for all adaptation actions.[54] The highest funding requirement is for infrastructure, water, and agriculture sectors. Cambodia is an LDC with many development priorities and investment needs. One of these priorities is accelerating the current effort to adapt its agriculture sector to climate change. The country's economy and its rural population heavily depend on this sector, and rural agrarian communities are highly vulnerable to the observed and anticipated impacts of climate change due to widespread poverty. The country is considered one of the most vulnerable countries to the impacts of climate change which is already impacting its GDP growth and could reduce its annual average GDP growth by 2.5% in 2030, and up to 9.8% in 2050.[55] This could delay the country's aspirations to reach upper middle-income status.

120. Cambodia's economy was badly affected by the COVID-19 pandemic, with GDP contracting by 3.1 percent in 2020, the country's first recession in 30 years. Simulations show that the poverty rate increased by 2.8 percentage points, pushing about 460,000 individuals into poverty.[56] The government responded quickly and effectively to the pandemic, assisting business and vulnerable families, but the associated

[54] Nationally Determined Contribution Update. December 2020.
[55] Nationally Determined Contribution Update. December 2020.
[56] World Bank. December 2022.

rise in spending meant that fiscal consolidation will be required in the period ahead. While the economy has recovered in 2021 and 2022, the weakening external environment means that returning to the strong rates of growth seen prior to the pandemic will prove challenging. The ongoing economic slowdown in the United States — Cambodia's largest export market. The Government ability to invest has been further eroded due to the focus on post COVID recovery. Cambodia is reluctant to borrow funds due to its rising indebtedness. The ratio of national debt to gross domestic product of Cambodia was at 38% in 2023 and was forecast to increase to 41.6% by 2028[57].

121. It is critical for the country to invest in its aging irrigation infrastructure and in making its infrastructure investments resilient to climate threats. The public sector's financial resources are limited, and the country is unable to do so on its own. The private sector has been reluctant to invest in the public irrigation system for several reasons. These include the absence of attractive financial incentives or government support, such as tax breaks or subsidies, which discourage private companies from undertaking large-scale irrigation projects. Furthermore, there are significant financial risks in investing in public infrastructure due to the high cost of developing and maintaining irrigation infrastructure which requires significant upfront investment and long-term commitment. Access to affordable financing can be challenging for private companies looking to undertake large infrastructure projects, including irrigation. The private sector perceives these investments as financially risky due to uncertain returns and extended payback periods. The regulatory and legal framework in Cambodia is not sufficiently clear or stable, causing hesitation among private investors. There are ambiguities in property rights, land tenure, and water usage rights which deter private entities from investing in irrigation projects. The CAISAR project is designed to build capacity of key institutions in the sector to better understand how to encourage private sector and local community participation through supportive frameworks.

122. The current system of payment for water services is not sufficiently developed at this stage to ensure the payment of services. Private companies are also often concerned about political instability and lack of transparency which can increase risks and raise doubts about the ability to protect investments and ensure a fair business environment. Building and managing irrigation infrastructure require specific technical expertise. The private sector does not always have the necessary knowledge or experience in this area. There are several social and environmental risks associated with irrigation projects, such as displacing local communities or harming ecosystems. These concerns can act as a deterrent for investment by the private sector which is not always in a strong position to deal with these concerns. Given the current lack of regulatory and incentive mechanisms, the RGC recognizes the need for public sector efforts to establish the necessary enabling conditions to leverage and unlock private sector investment in the medium to long term. In this context, the current project will also synergize with the GCF financed Public-Social-Private Partnerships for Ecologically-Sound Agriculture and Resilient Livelihood in Northern Tonle Sap Basin (PEARL-FP199) to encourage private-sector actors, particularly agricultural suppliers, traders/exporters, retailers, and local financial institutions, to work in partnership with the project's target beneficiaries to adopt and further invest in climate-resilient and high-value practices.

123. GCF financing will address the paucity of public sector funding and the lack of private sector financing due to the current stage of development of the regulatory and fiscal incentives that inhibit private sector investments due to the current market failure. The Government of Cambodia is trying to generate a reasonable mix of national and international funds, in addition to developing market mechanisms for the future which could encourage private sector investments. For now, it will use a mix of loan and grant funds from GCF, IFAD, AIIB, and the Royal Government of Cambodia. The project will use USD 40 million of the GCF grants and USD 40 million of loans from GCF and an additional USD 100 million of loans from AIIB, USD 45 million of loans from IFAD and equivalent of USD 15 million in-kind counterpart financing from the RGC. The project has proposed only 17% of the total estimated cost as a grant from GCF. This is the minimum level of concession required to make the investment viable for smallholders who are not in a financial position to finance the improvement in the modernization of the irrigation systems, the early warning systems or solar pumping of water from the tertiary canals to farmer's fields. The primary beneficiaries from the concessional funding will be smallholder farmers, particularly rice farmers. The GCF guiding principles which have been kept in mind while determining the level of

[57] Statista. April 2023.

concessional financing[58] is the balance between the levels of indebtedness capacity of the recipient so as not to encourage excessive indebtedness, tailoring the grant element of concessional finance to provide the appropriate incentive to facilitate the implementation of mitigation and adaptation activities and minimize market distortions and potential disincentives to private investment.[59]

B.6. Exit strategy (max. 500 words, approximately 1 page)

124. There is a clear exit strategy of the project from the various components that it is planning to execute and the investments that it is planning to undertake. The activities related to strengthening farm-level capacities will be designed in a manner that will enable the farmers to adopt and implement the improved production techniques, reduce emissions from rice cultivation, adopt techniques such as AWD for rice, enhance soil conservation, improve water management, diversify their cropping pattern, undertake and enhance other options such as vegetable production, poultry and aquaculture. The farmers will be provided opportunities for refresher trainings throughout the project period for enhanced retention and facilitated to establish links with both the public extension services and the private sector to provide the required adaptive inputs and technologies for enhanced adoption. Within Cambodia's pluralistic extension system, NGOs, CBOs, social enterprises, and private companies play crucial roles together with the public extension system in supporting smallholder farmers. Input suppliers, including social enterprises, are responsible for ensuring that services enhance production and productivity improvements, and that technical assistance is provided at the field level.[60]

125. The project will strengthen links with the private sector which generally provides technical assistance in the use of modern technologies and inputs, such as solar pump, mDSR and laser land levelling mechanization service and good quality seeds of short duration and climate adaptive rice varieties. This scaling up approach which relies on the private sector to strengthen their links with smallholders for extension advice and inputs is expected to provide a good exit strategy by shifting project assistance to the growing private sector. Opportunities will also be provided to farmers to link with those private sector entities in Cambodia that have tried several business models for irrigation using Information Technology. These include the Pay-per-use irrigation services using mobile technology, arrangements for technology leasing and financing, public-private partnerships to promote the adoption of smart water solutions and farming technologies. Some private companies in Cambodia have offered bundled solutions that combine smart water technologies with value-added services. These packages include not only the hardware but also training, maintenance, and support services. These efforts are on-going and being promoted by the MAFF. Where appropriate links will be established with these companies to promote these models by the private sector. NCDDS will also establish links with other institutions to ensure that the public extension systems and private companies continue to provide services and introduction of new adaptive technologies to farmers.

126. The irrigation and flood control investments of CAISAR which form the most significant investments by the project will be the responsibility of the Provincial Departments of Water Resources and Meteorology (PDWRAMs). These departments will participate during the design and implementation stages to ensure enhanced ownership of the design and are mandated to assume the responsibility for operation and maintenance upon scheme completion. Specifically, PDWRAMs are expected to undertake the following tasks; (i) ensure that the main canal infrastructure is properly maintained and repaired; (ii) ensure that the seasonal water management plan is endorsed and approved; (iii) ensure that in the dry season, the main canal system provides sufficient water; (iv) provide technical assistance to Farmer Water Use Communities (FWUC) and its members in the operation and maintenance of the irrigation systems/scheme; (v) assume leadership in the resolution of conflicts on water disputes; (vi) operation and maintenance of main canal systems with funding provided by the Government; (vii) Implement routine maintenance of the main canal system including all the associated structures; (viii) provide updated report to MOWRAM on the deferred maintenance needs of the system including an estimate of

58 The concessionality analysis using the grant element methodology employed by the IMF indicates that the GCF contribution would allow the credit operation to be highly concessional, with a grant element level close to 34.38% (https://www.imf.org/en/GECalculator)

59 Guiding principles and factors determining terms of financial instruments. B.05/07. October 2013.

60 Agricultural Extension in Cambodia: An Assessment and Options for Reform. IFPRI. January 2018.

the cost implications; (ix) monitor the volume of water in the main canal and the level released to downstream water users for proper scheme function.

127. The project will capitalize on the existing Cambodian water law under which the farmers have to contribute to the O&M costs of irrigation systems and this requirement is well accepted although at present O&M contributions are far below the required levels (usually 8-10 US/ha/crop). Experience from some pumped irrigation schemes developed under the Department of Foreign Affairs and Trade (DFAT) funded by the Cambodia Agriculture Value Chain (CAVAC) project shows that farmers are paying up to US 80/ha/crop for the O&M of the irrigation scheme, for routine maintenance in case of pumping schemes. CAISAR will pilot an irrigation fee that will be levied by the FWUCs who have the authority to collect the payment.

128. Well-functioning schemes enhance ownership and participation of local beneficiary communities which play their role in scheme cleaning and maintenance. To ensure the long-term institutional sustainability of the arrangements put in place, local communities and water user communities will also be expected to play their role and take over some key functions during project design and completion of each sub-project. The CAISAR project will invest in the development and training of Farmer Water Use Communities (FWUCs) to ensure efficient project management. The project team has conducted a thorough survey of FWUCs in the CAISAR command areas, which were included during project design and will be trained to lead the management of the selected schemes. The project will work with the FWUC Department at MOWRAM and PDWRAM to set up and train an appropriate number of FWUCs for each command area. Investment will be directed towards FWUC establishment and capacity building to ensure they have the capacity and capability to manage the operation and management of the scheme in an effective and sustainable manner. By investing in FWUC establishment and capacity building, strengthening technical capacity, and involving project stakeholders and local communities, the project team will enhance the potential for long-term financial and operational sustainability of the irrigation schemes.

129. During its institutional strengthening, MOWRAM will be assisted in developing a post-project strategy for sustainability and engaging the private sector in its design and implementation which can help develop innovative and sustainable models for management and operation and maintenance of the irrigation system in the country in the future. The private sector can be incentivized to participate when there are clear financial incentives with clear and transparent arrangements in place. It is expected that improved institutional strengthening of the sector institutions will make them more aware of these opportunities and open the way for innovation in the long-term management of the sector. However, for the foreseeable future, the scaling up of the modernization of the irrigation schemes can only take place once additional resources are made available for the purpose. It is expected that MOWRAM will finance infrastructure upgrades based on the experience from CAISAR. It will access its own resources and financing from the development partners for the purpose. The private sector can be incentivized to participate when there are clear financial incentives with clear and transparent arrangements in place. MOWRAM capacity will also be strengthened to introduce models that can incentivize the private sector for greater participation.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount		Currency	
	80.0		million USD ($)	
GCF financial instrument	**Amount**	**Tenor**	**Grace period**	**Pricing**
(i) Senior loans	40.0	20 years	5 years	0.75% interest + 0.5% service fee
(ii) Subordinated	Enter amount	Enter years	Enter years	Enter %

loans				
(iii) Equity	Enter amount			Enter % equity return
(iv) Guarantees	Enter amount	Enter years		
(v) Reimbursable grants	Enter amount			
(vi) Grants	40.0			
(vii) Results-based payments	Enter amount			

(b) Co-financing information	**Total amount**	**Currency**
	160	million USD ($)

Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
IFAD	Senior Loans	45.0	million USD ($)	25 years 5 years	1.28%-2.88%	senior
AIIB	Senior Loans	100.0	million USD ($)	35 years, max AMP 20 years	USD: SOFR+0.94%/1.44% RMB: 3 month RMB SHIBOR+0.43%/0.93%	senior
Government of Cambodia	Grant	15.0	million USD ($)	N/A	N/A	
Click here to enter text.	Options	Enter amount	Options	Enter years Enter years	Enter%	Options

(c) Total financing (c) = (a)+(b)	**Amount**	**Currency**
	240.0	million USD ($)

(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	Overall, 33.3% of the project will be financed through GCF, 18.7% through an IFAD loan, 41.6% through the AIIB, and the remaining 6.3% through an contribution from the government. The resources provided by the GCF funds will be primarily directed towards investments to improve the climate change adaptation capacities of the beneficiaries, flood proofing of the irrigation infrastructure and institutional strengthening, and project M&E. In addition, during implementation of some of the project activities related mainly to the development of value chains under component 1, efforts will be made to involve the private sector to contribute resources for activities related to storage, processing, transport and providing access to wholesale and retail markets.

C.2. Financing by component

Please provide an estimate of the total cost per component and output as outlined in section B.3. above and disaggregate by source of financing. More than one co-financing institution can fund a single component or output. Provide the summarised cost estimates in the table below and the detailed budget plan as annex 4.

Component	**Output**	**Indicative cost** million USD ($)	**GCF financing**		**Co-financing**		
			Amount million USD ($)	**Financial Instrument**	**Amount** million USD ($)	**Financial Instrument**	**Name of Institutions**
Component 1 Farm level climate	Output 1.1 Increased farmer	18.3	4.9	Grant	12.9	Senior loan	IFAD

adaptation and resilience	capacity in climate resilient agriculture.		0	Grant	0.5	Grant contribution	Government of Cambodia
	Output 1.2 Climate adaptive value chains developed by 4Ps and increased access to finance	4.0	3.8	Grant			
					0.2	Grant contribution	Government of Cambodia
	Output 1.3 Increased access to and use of climate information and advisory services for climate responsive water-use and crop planning	2.9	2.7	Grant	0.		
					0.2	Grant Contribution	Government of Cambodia
	Output 1.4 Increased resilience of farm road infrastructure to climate change.	15.2	15.0	Grant	0.2	Grant contribution	Government of Cambodia
Component 2. Upgrading and climate-proofing water infrastructure for increased resilience	Output 2.1 Irrigation schemes and ponds modernized	102.2	0		96.0	Senior Loan	AIIB
					6.2	Grant contribution	Government of Cambodia
	Output 2.2 Flood-proofed and improved drainage.	56.8	Enter amount34.4	Senior loan	18.3	Senior Loan	IFAD
					4.1	Grant Contribution	Government of Cambodia
	Output 2.3 Farmers Water User Communities (FWUC) established and trained.	5.0	4.6	Senior loan	0.4	Grant Contribution	Government of Cambodia
	Output 2.4 Water information system (CIEWS and SCADA) established and operational in three subprojects.	3.7	3.3	Grant	0.4	Grant Contribution	Government of Cambodia

Component 3. Strengthened institutional and regulatory capacity for low-emission climate-resilient development pathways.	Output 3.1 MOWRAM capacity development supported.	5.4	4.8	Grant	0.6	Grant Contribution	Government of Cambodia
	Output 3.2 Strengthening of NCDDS	4.2	3.8	Grant	0.4	Grant Contribution	Government of Cambodia
Project management cost	Project Management Unit	11.9			7.8	Senior Loan	IFAD
					2.7	Senior Loan	AIIB
					1.4	Grant contribution	Government of Cambodia
	Project Monitoring and Evaluation	8.0	1.7	Grant	3.6	Senior Loan	IFAD
					1.3	Senior Loan	AIIB
			1.0	Senior Loan	0.4	Grant contribution	Government of Cambodia
Contingency		2.4			2.4	Senior Loan	IFAD
Indicative total cost (USD)		240.0	80.0		160.0		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

If the project/programme is expected to support capacity building and technology development/transfer, please provide a brief description of these activities and quantify the total requested GCF funding amount for these activities, to the extent possible.

130. CAISAR is expected to invest in capacity building at various levels starting with the faming households through the FFS approach on the ground to help in the adaptation to climate change and techniques for rice, vegetable, poultry and aquaculture production and reduction of GHG emissions from rice crops. In addition, the institutional capacity of FWUCs will be strengthened to enable them to improve the governance of the irrigation system and collection of water user charges. The project also expects to invest in the technical capacity of key institutions such as MOWRAM and NCDDS for better formulation of the policy and regulatory framework for encouraging transformation in the agriculture and irrigation sector. In addition, the project will also invest in the introduction of new technologies such as solar pumps and the investment in the dissemination of climate information and Early Warning systems and smart solutions for operating and monitoring the irrigation systems. The total GCF financing which is expected to be used for capacity building and institutional strengthening is around USD 16.3 million.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential

131. The project is expected to lead to (i) improved farm-level climate adaptation, resilience, and water use efficiency; (ii) rehabilitated/upgraded, climate proofed, and modernized irrigation infrastructure for increased resilience and (iii) strengthened institutional and regulatory capacity for low-emission climate-resilient development pathways. Training farmers on climate change adaptation and mitigation, the Project will enhance their knowledge and skills and improve the environment for enhancing climate resiliency in irrigated agriculture. The CAISAR investments are expected to improve yields and provide better drought and flood resilience over the 25-year asset life, with the

project expecting significant yield and crop production improvements in all targeted irrigation sub-project areas.[61] The CAISAR project will be novel in Cambodia, with its focus on reducing the vulnerability of rural populations through the creation of a climate-resilient IWRM based project design that incorporates flood protection, sustainable water resource development, climate resilient solutions, environmentally aware design and gender equality.[62] The CAISAR project will promote economic growth and improve the livelihoods of those affected by enhancing agricultural production, reducing farmers' exposure to variability in crop yields, and increasing monthly income for families.

132. The use of innovative climate-resilient solutions will reduce the overall climate impact of the scheme and endeavor to reduce the emission of greenhouse gases from the existing and proposed agricultural practices with the introduction of solar pumps and improved water management and agronomic practices in rice cultivation (Land Laser Levelling, Direct seeding, Alternate Wetting and Drying techniques together with improved agronomic practices like IPM). The project will support the RGC's effort to improve and strengthen the disaster resilience of the population of four provinces in Cambodia through improved water information system and agriculture advisory services. Through combining irrigation rehabilitation with climate resilient design, the project will build a low carbon and climate resilient future, enabling rural populations to adequately adapt to current and future climate impacts and increase their food security and livelihoods whilst also enabling the RGC to meet its climate targets.

133. CAISAR is expected to impact 562,757 people directly and 1,151,618 people indirectly (Annex 23). The direct beneficiaries are expected to benefit from adopting improved and climate resilient cropping and diversification patterns, improved access to water, enhanced water governance, greater market integration for improved access to adaptive inputs and early warning systems. The combined impact of these investments will be to enhance water and food security. Improved irrigation infrastructure has the potential to increase Cambodia's potential rice yields. It is expected that changing from rain-fed irrigation or using dilapidated irrigation infrastructure to proper systems for irrigated rice cultivation with improved water management could result in significant production increase. Water security will be enhanced as a result of modernizing the irrigation systems, making them more resilient to climate risks and reducing conveyance losses. The project will enhance water use efficiency for an area of 32,056 ha out of a gross area of 80,000 in the target areas during the wet season for the rice crop. From this area, a second irrigated dry season rice crop can be established on 22,051 Ha (Table 2). In the remaining 10,005 Ha a non-rice crop will be established in the dry season, mainly homestead gardening including vegetables, chicken production and aquaculture production. The project will assist farmers gradually diversify their cropping patterns for additional income in the dry season. The project will develop the required irrigation infrastructure to allow for a second irrigated rice crop during the dry season using AIIB resources. GCF resources will be used to provide investments for flood control/drainage for the entire gross project area of 80,000 ha.

134. Reduction of GHG emissions from rice production will be achieved by shifting away from traditional flooded rice methods to improved water management with required drainage facilities diversifying to other crops/livestock/fisheries, promoting carbon sequestration activities, and use of renewable energy sources for pumping water from canals to fields instead of the diesel pumps currently used. The project will reduce and avoid GHG emissions as a result of better production practices for rice cultivation, improved soil management and replacement of diesel pumps with solar pumps for pumping water from the canals to the tertiary canals and farmer's fields. EX-ACT analysis has been undertaken to estimate the reduction in GHG emissions and baseline data sources are described in Table 1A and additionality assumptions are outlined in each sub-section (Annex 22). Briefly, baseline data is primarily from stakeholder consultations (e.g., with project commune-level representatives) and field surveys, including agronomy survey—conducted during feasibility and engineering design assessments undertaken by WAPCOS and MOWRAM's consultants, with support from IFAD and AIIB. The key factors contributing to the decrease is the reduction in the area under flooded rice, improved irrigation practices and greater use of Alternate Wetting and Drying

[61] Annex 2: Feasibility Study. WAPCOS Report. 2022.
[62] Ibid. 2023.

(intermittent flooding) practices for rice production and the replacement of diesel pumps with solar pumps, assuming a 75% adoption rate for the switch. Based on these factors it is expected that - the total mitigation potential of the project over 20 years is -1,006,507.43 tCO2-eq. This would amount to a carbon balance of -2.3 tCO2-eq per hectare per year (Annex 22).

135. The indirect beneficiaries will be generated through wider dissemination of the adaptation practices and technologies through lead farmers, private sector and agriculture extension workers and by enhanced availability of inputs in the market once the practices are adopted and generate a demand to which private suppliers respond. Benefits are also expected to accrue as a result of the more sustainable production technologies and increase in yields along the selected value chains through greater resilience and enhanced access to water. One of the most widely spread benefits will result from the use of early warning systems that will have a coverage beyond the project area. The early warning systems are expected to help prepare the farmers in case of changes in precipitation and temperature, droughts, floods, to help prepare them against crop loss through pest infestation and disease.

136. The design and construction of the irrigation systems are expected to reduce the risk of waterlogging and planned in a way that does not negatively impact the natural floodplain, which is important for the maintenance of ecosystem and the livelihoods of local communities. Another aspect which will add to water security and sustainability is through improved water governance. The project will strengthen local level institutions such as the FWUCs which have the mandate to operate and maintain irrigation schemes at the tertiary level. The project will strengthen their capacity for collection of water fee and plans to pilot mechanisms for collection that will be up scaled by MOWRAM using the existing Cambodian water law under which the farmers have to contribute to the O&M costs of irrigation systems. The project expects that farmers will enhance their contribution to O&M based on a more reliable and efficient system that will make the water governance system more financially sustainable.

D.2. Paradigm shift potential

137. There are some key activities that are expected to facilitate a paradigm shift and catalyze impact beyond the one-off project investments made through CAISAR. In line with CAISAR’s Theory of Change, the project is designed to take action at three levels; farm level; irrigation scheme level and at the national level for creating a strong institutional base and an enabling environment for water management. The project departs from a sectorial approach in water or agriculture management and integrates water management with agriculture innovation at all levels to address the growing climate challenges in irrigated agriculture. The integrated actions combining the various elements of the CAISAR strategy can have a transformative impact on reducing vulnerability of water and agriculture systems to climate change impacts while also reducing GHG emissions and enhancing the livelihood of rural populations who primarily depend on agriculture.

138. At the farm level, it is expected that the awareness and training to the farmers will facilitate their understanding of climate adaptive practices and encourage their adoption of climate resilient practices. Increased adoption of new and improved climate resilient agriculture technologies and practices will be increased through working closely with smallholders and training them in the use of these technologies, linking them with suppliers of the technologies, providing increased access to finance to enable them to continue to use these inputs and strengthening the value chain of key agriculture commodities. The project will promote a reduction in pesticide use and will include safeguards for farmers in the safe handling and use of chemicals. In addition, the project will promote efficient use of fertilizers ensuring that they match soil nutrient levels to avoid overuse and leakage of nutrients into water resources. The avoidance or reduction of production loss through better adaptation will help farmers reduce their losses from climate risks, increase production thereby increasing their incomes and food security and enable them to sustain farming as a key source of livelihood. Given the hydrometeorological observation infrastructure and capacity is insufficient to provide robust, basic data (including, in terms of spatial representation) and past efforts have been unable to scale up and sustain (e.g., stations are not maintained in good condition, sensors are non-functional or improperly calibrated), this project will shift how observation infrastructure is maintained

and put in place procedures for data quality assurance and data archiving through additional investments in equipment and capacity. Crucially, by illustrating financial returns to farmers from adopting more climate resilient and efficient crop management and water use practices, it will create bottom-up demand for CIEWS from both public (e.g., credible and timely data and analysis) and private sector (e.g., mobile apps). This will occur through investments in expanding agrometeorological advisory services that enable farmers to access tailored forecasts and advisories. The scale of potential impact is considerable: during the project, nearly all farming households in the provinces (375,171) will benefit from increased availability and accessibility to climate information and early warning on floods and droughts as well as crop-specific forecasts and advisories (which will leverage GCF-funded PEARL outputs) through traditional channels and ICT tools. Training of Trainers (TOT) for NGOs/PDAF/private entities on FFS under Activity 1.1.3, and the dedicated farmers and key stakeholders' trainings under Activity 1.3.4 will incorporate CIEWS in the curriculum and these trained individuals will raise awareness of CIEWS (availability of mobile apps, how to use crop advisories, etc.).Beyond the project, Farmer Field Schools (FFS) in Cambodia will integrate / mainstream this training to ensure scale and replication. These measures—CIEWS, farm-level actions to facilitate understanding of climate adaptive practices (Farmer Field Schools-FFS), investments in irrigation, strengthened water-use groups, access to markets for non-rice produce, etc.—will support the transformation towards low-emission and climate-resilient agriculture because farmers perceptions of risks will be informed by evidence and they have improved ability and willingness to translate such information (on climate risks) into actions, such as intermittent flooding management of rice. Cambodian agriculture will be on a more sustainable pathway by building resilience consistent with relevant national climate change adaptation strategies and plans.

139. The overall investment in modernizing irrigation systems and flood proofing drainage systems will protect the natural capital stocks, reduce the need for frequent rehabilitation and reconstruction and lower the cost of operation and maintenance. This will increase the efficiency of the systems, protect them from frequent breakdown and improve the overall reliability and sustainability of the irrigation system.

140. The more efficient delivery of irrigation water will increase production and encourage farmers to pay for irrigation services and maintain the systems. The ability to levy and collect higher charges will further enhance the operation and maintenance of the system and make it more sustainable. A more reliable and well-functioning system is likely to encourage the private sector to implement various business models that can recover costs and enhance the technical and financial sustainability of the system. Past investments in irrigation sector have found that innovative farmers are more willing to experiment with different varieties and cropping cycles, when any form of irrigated water is available.[63] This is also likely to bring about a sustainable shift in people's risk appetite for investment in the agriculture sector. This is likely to lead to a paradigm shift in the irrigation sector in Cambodia.

141. The adoption of solar water pumping systems is also expected to bring about a paradigm shift. The solar pumps are a cost-effective alternative to diesel-powered pumps, which were financially out of reach for many vulnerable smallholder farmers.[64] The inability to afford fuel for traditional pumping methods had previously led to decreased rice yields and even crop losses in some cases. Smallholder farmers see an opportunity to enhance their agricultural output and minimize losses through use of large-scale solar systems. These systems offer secure access to water for multiple cropping seasons, thereby increasing agricultural resilience. They also contribute to reducing environmental impacts and greenhouse gas emissions by replacing diesel pumps with off-grid solar electricity. This not only lowers energy costs for irrigation but also enhances crop resilience against climate change-related challenges like drought. Furthermore, the consistent water supply in canals created by solar water pumping systems fosters an environment suitable for fish and water birds thereby generating global public goods. This enables farmers to engage in subsistence fishing and gain an additional source of protein. By financing these systems, CAISAR can aid in combating

[63] Cambodia Agriculture Value Chain Project. Phase II. DFAT.
[64] The solar pumps are unlikely to lead to any type of mal-adaptation as the solar pumps will not be used for pumping ground water but simply to help convey the water from the main canals to farmer's fields.

climate change and mitigating its adverse effects on vulnerable farmers, thereby safeguarding their livelihoods.

142. The project will encourage the participation of private sector input suppliers and those supplying energy efficient technologies, improved water and soil management technologies which can assist with adaptation and mitigation of climate change. The project will also provide opportunities for knowledge management at the beneficiary level to farmers for exchange and dissemination of ideas through the action plans, through FFs and through the MSPs. The greater use of technology development, deployment and dissemination, can also be used through the provision of incentives to use these transformative technologies such as solar pumps, drip irrigation systems, soil conserving techniques, etc. The project will through its capacity building of key implementing partners also help them to identify how best to create an enabling environment for scaling up the adaptation and mitigation actions.

143. GCF investments are also expected to bring about a paradigm shift through strengthening some of the policy and regulatory institutions in the Irrigation sector such as MOWRAM. In addition, institutional strengthening of MOWRAM will contribute to transform it from a construction-oriented institution to a management oriented one that integrates both hard and soft approaches to irrigation sector management and enhance long term sustainability. The project will also provide opportunities for knowledge management and learning and policy advocacy through its annual review meetings and progress reports and special seminars and presentations. These efforts are expected to support Cambodia's Updated NDC.

D.3. Sustainable development

Greater Community Collaboration and Role of Women

144. There are several social co-benefits that are expected to be generated from the project. The farmers are expected to derive social dividends from their participation in the various forums that are established under the project. Farmers will be provided the opportunity to learn together through FFS and interact on a regular basis through the stakeholder and public private sector forums that are established. This is expected to generate additional benefits by sharing and discussing common problems and their solutions. The strengthening of FWUCs and their activation through a role in managing the O&M of improved irrigation schemes is also expected to bring communities closer together for a common purpose and build the spirit of cooperation and collaboration. Investments in the irrigation systems will involve community participation and capacity building of FWUCs. This empowerment can lead to improved decision-making and governance structures, which can have positive ripple effects on various aspects of community well-being. Previous experience of projects in Cambodia shows that where communities were given this role, they have subsequently taken charge and met the costs of O&M of the distribution canals.[65] This is expected to build community cohesion and spirit of collective action under CAISAR. CAISAR has established separate targets for participation of women in all project activities. By involving women and elaborating on why it is important to involve them in the decision-making regarding production decisions in the agriculture sector, the project will facilitate an improved understanding of how women's participation can positively impact on building climate resilience at the household and farm level and improve decision making regarding household food consumption, household health and nutrition. By encouraging greater participation of women in leadership positions and decision-making and promoting better access for women to agricultural information and capacity building, the project can increase their empowerment and capacity for enhancing their livelihoods.

Private Sector Integration in Rural Economy

[65] Cambodia Agriculture Value Chain Project. Phase II. DFAT.

145. Engaging the private sector will unlock significant co-benefits. The project explicitly aims to foster (and catalyze through grants) viable business models that enable private sector (e.g., input supplying MSMEs) to offer high-quality inputs (stress-tolerant seeds), affordable irrigation solutions (micro-irrigation, solar irrigation system), and other climate adaptive farming technologies, including through service delivery models (e.g., land leveling, direct seeding machines), which will be tailored to the agro-ecological context and assessed for climate mitigation/adaptation potential. Another key strategy involves engaging MSMEs to work directly with local farmer groups and cooperatives, nurturing their independent growth - e.g., through buyer-seller contracts. Cambodia also has successful private sector initiatives in the extension and CIEWS space - the project will leverage these existing initiatives (e.g., mobile apps) to accelerate CAISAR's impacts. Successful public-private partnerships will be showcased to representatives from other communes through demonstration sites or events at farmer cooperatives, where lead farmers will highlight gains in productivity, resilience, and income through their use of specific inputs or machinery as well as off-take contracts. These proven models will then be replicated across communes, driven by tangible evidence of increased private sector revenue and improved livelihoods—ensuring long-term sustainability and scalability.

D.4. Needs of recipient

146. **Country Vulnerability**: Due to a combination of political, geographic, and social factors, Cambodia is recognized as vulnerable to climate change impacts, ranked 140th out of 181 countries in the 2020 ND-GAIN Index.[66] According to the Notre Dame Global Adaptation Index, Cambodia has a high-vulnerability score and low-readiness which accounts for the high-risk ranking compared to other countries. Cambodia's vulnerability to climate change is largely a result of its geography, high reliance on the agriculture sector and low adaptive capacity (including limited financial, institutional and technical resources). Combined with a high exposure to climatic extremes such as floods and droughts, Cambodia is at high risk of natural disasters. Projected climate change trends indicate that Cambodia will experience more severe floods and droughts, which are expected to affect GDP by nearly 10% by 2050.[67] As a result, the Royal Government of Cambodia is committed to addressing the requirements of its population to ensure that they are adapted and resilient to current and future climate shocks.

147. **Water Resources:** Water resources in Cambodia are in a state of flux as a result of major human development interventions impacting on the Mekong River and Tonle Sap Lake.[68] Various studies have shown that lack of modern irrigation infrastructure is holding back agricultural production in Cambodia, and potentially enhancing the nation's vulnerability to climatic extremes.[69] Cambodia is simultaneously highly dependent on the resources provided by the natural river flow regime and the flood regime. In a context of dramatic changes to future water flows, likely exacerbated by climate change, a key focus area discussed in Cambodia's National Communication 2 (NC2) is on maintaining the flow levels necessary to sustain ecosystem services. It is hoped this will be achieved through management of irrigation, dams, and domestic water demand. The failure to maintain the necessary productivity of the ecosystem supporting Cambodia's inland fisheries would represent a major threat to the nation's primary source of protein.[70] Similarly, with a large proportion of the Cambodian population still dependent on natural water sources for domestic consumption, drought and other reductions to the natural water supply could have serious consequences for the population. The Tonle Sap Lake is a unique and vital natural resource in Cambodia. The lake's complex hydrological interactions with the Mekong River make it vulnerable to changes in the

[66] Climate Risk Country Profile. Cambodia. 2021. World Bank. Asian Development Bank.
[67] Ibid. 2021.
[68] ICEM (2020). Improving Climate-Smart Decision Making with the Cambodia Mekong Delta Digital Atlas tool. Summary Report for Decision Meeting. [4 June, 2020].
[69] Chun, J. A., Li, S., Wang, Q., Lee, W.-S., Lee, E.-J., Horstmann, N., . . . Vang, S. (2016). Assessing rice productivity and adaptation strategies for Southeast Asia under climate change through multi-scale crop modeling. Agricultural Systems, 143, 14–21. URL: https://koreauniv.pure.elsevier.com/en/publications/assessing-rice-productivity-and-adaptation-strategies-for-southea
[70] UNDP (2011). Climate change and water resources. Cambodia Human Development Report 2011. United Nations Development Program. URL: http://hdr.undp.org/sites/default/files/cambodia_2011_nhdr.pdf

Mekong River basin, including development-focused interventions taking place in upstream nations as well as climate change.[71] Any change in climate which can modify the flood pulse which feeds the Tonle Sap Lake during the peak monsoon season will have significant implications for its unique wetlands, forest and aquatic ecosystems.[72]

148. **Agriculture:** Climate change will influence food production via direct and indirect effects on crop growth processes. Direct effects include alterations to carbon dioxide availability, precipitation and temperatures. Indirect effects include impacts on water resource availability and seasonality, soil organic matter transformation, soil erosion, changes in pest and disease profiles, the arrival of invasive species, and decline in arable areas due to the submergence of coastal lands and desertification. These impacts are expected to damage key staple crop yields, even on lower emissions pathways. It is estimated that there could be a 5% and 6% decline in global wheat and maize yields respectively even if the Paris Climate Agreement is met and warming is limited to 1.5°C. Shifts in the optimal and viable spatial ranges of certain crops are also inevitable, though the extent and speed of those shifts remains dependent on the emissions pathway.[73] Rice is a staple crop in Cambodia, crucial to national food security and subsistence livelihoods. Cambodia faces some of the highest net rice yield losses in Southeast Asia, as a result of climate change.[74] Without adaptation, but with the benefits of increased atmospheric concentrations of CO2 included, the expected yield losses range from 10–15% by the 2040s under both RCPs 4.5 and RCP 8.5. These losses link primarily to increases in the average temperature during the growing season. The vulnerability of Cambodia's rice agriculture is linked particularly to the very high prevalence of rain-fed (rather than irrigated) systems, which are susceptible to damage from both lack and excess of water.[75]

149. **Poverty and Inequality:** Poverty reduction in Cambodia has been rapid, yet climate change threatens to slow progress. The current impacts from the COVID-19 pandemic are also likely to increase the country's poverty levels, with potential to move people back into poverty in the immediate to medium term. Many of the climate changes projected are likely to disproportionately affect the poorest groups in society. Poorer farmers and communities are least able to afford local water storage, irrigation infrastructure, and technologies for adaptation. The rate of poverty decline in Cambodia is broadly outpaced by the rate of GDP growth, indicating that growth is not inclusive and inequality may be growing, though data is scarce; second, poverty declines are strongly centered in urban areas, with changes considerably slower in rural areas.[76] Studies suggest that many households in Cambodia have a high probability of falling into extreme poverty even when exposed to relatively low intensity flood and drought events.[77] As subsistence agriculture remains prevalent in Cambodia, and rates of undernourishment high, the threat of yield reductions in staple crops and the potentially high cost of adaptation also represent major risks to livelihoods. While climate change is only one of the many pressures on livelihoods in the vicinity of the Tonle Sap Lake, the low adaptive capacity of rural communities, particularly their limited ability to diversify income sources presents major risks.[78]

150. **Financial Constraints:** The updated NDC has estimated that the total funding required for all mitigation actions in the country is over US $ 5.8 billion. The FOLU, waste, and energy sector actions

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[71] ICEM (2020). Improving Climate-Smart Decision Making with the Cambodia Mekong Delta Digital Atlas tool. Summary Report for Decision Meeting. [4 June, 2020].
[72] Climate Risk Country Profile. Cambodia. 2021. World Bank. Asian Development Bank.
[73] Tebaldi, C., & Lobell, D. (2018). Differences, or lack thereof, in wheat and maize yields under three low-warming scenarios. Environmental Research Letters: 13: 065001. URL: https://iopscience.iop.org/article/10.1088/1748-9326/aaba48
[74] Li, S., Wang, Q., & Chun, J. A. (2017). Impact assessment of climate change on rice productivity in the Indochinese Peninsula using a regional-scale crop model. International Journal of Climatology, 37(April), 1147–1160. URL: https://rmets.onlinelibrary.wiley.com/doi/full/10.1002/joc.5072
[75] Li, S., Wang, Q., & Chun, J. A. (2017). Impact assessment of climate change on rice productivity in the Indochinese Peninsula using a regional-scale crop model. International Journal of Climatology, 37(April), 1147–1160. URL: https://rmets.onlinelibrary.wiley.com/doi/full/10.1002/joc.5072
[76] ADB (2016). Measuring multidimensional poverty in three Southeast Asian countries using ordinal variables. ADBI Working Paper Series No. 618. Asian Development Bank. URL: https://www.adb.org/sites/default/files/publication/214871/adbi-wp618.pdf
[77] ADB (2017b). Risk financing for rural climate resilience in the greater Mekong subregion. Greater Mekong Subregion Core Environment Program. Asian Development Bank. P. 30. URL: https://www.adb.org/sites/default/files/publication/306796/risk-financing-ruralclimate-resilience-gms.pdf
[78] Climate Risk Country Profile. Cambodia. 2021. World Bank. Asian Development Bank.

require the highest funding. Total funding required for all adaptation actions is just over US $ 2 billion. Infrastructure, water, and agriculture require the highest funding.[79] The country needs assistance to mobilize the resources required to make some critical investments. Cambodia faces some key fiscal challenges in revenue generation due to a variety of reasons, including a narrow tax base, ineffective tax collection mechanisms, and a significant informal economy. Its tax-to-GDP ratio was 16.4% in 2021.[80] Cambodia has had a negative current account balance since the last 30 years and the situation sharply deteriorated over the last few years dropping to a -25.3% in 2022.[81] This deficit has strained its foreign exchange reserves, making it challenging to meet external obligations and finance essential imports. In such a situation, the government's ability to invest in development projects particularly irrigation infrastructure is constrained. While its debt to GDP ratio is not very high at 36.4% of GDP in 2022[82], The poor quality of its current infrastructure is reflected in its 2019 global ranking of 106 out of 140 countries. Cambodia investment requirements in infrastructure were estimated to be USD 28 bn between 2016 and 2040, based on the Global Infrastructure Hub's estimates.[83] The lack of depth and experience of Cambodia's local capital market is a significant impediment to alternative financing sources.

151. **Specific Needs in Project Areas**: The project sites were selected by the RGoC given their vulnerability to climate change in the water and irrigation sector and priority for the Government in the context of their SDG targets, especially given the moderate-to-high poverty incidence rates that prevail in the four provinces. The identification of the target areas is based on the need for upgrading the irrigation infrastructure, poverty status, food insecurity, vulnerability to climate change especially threat of high flood risks and low indicators of development. The targeting under the project will focus on agriculture cooperatives, farmer groups, MSMEs, smallholders, the poor, women and youth. CAISAR is designed to adopt a cluster approach which works with these diverse target groups to build their partnership and the synergy in selected value chains. Details of the targeting strategy is outlined in the PIM (Annex 21). MSMEs account for around 70% of employment in Cambodia and contribute around 58% of the country's GDP growth and are votal for agrciculture productivty and rural development. SMEs face several challenges, including concerns about financial access, technical skill and knowledge, insufficient facilitation and support. A survey to determine their specific constraints revealed that more than half of SMEs needed help with capacity development and marketing strategy. Almost half of them require financial assistance in the form of short- or long-term debt, as well as advice on their sale and operational strategies.[84] The Public Private Partnership Facility (4PF) under CAISAR is designed to strengthen micro, small, and medium enterprises (MSMEs) in the agricultural sector, particularly focusing on supporting young rural entrepreneurs.

152. **Smallholders**: Overall, there are 71,000 households in the target provinces from among whom, there are 47,093 farming households in the project area. The majority of the population are rural farmers (rice typically), whilst some also supplement their income via alternative jobs such as working in the service sector. Households typically do not have stable annual incomes or expenditures due to the annual cycle of the wet and dry season and the impact this has on agricultural yields and livelihoods. A majority of the population is at the edge of poverty despite most respondents having incomes above traditional poverty lines. There are few alternative income sources in rural areas, such as selling produce and groceries, business, or service providers, so most households are dependent on agriculture for their livelihoods.[85] Most rice farmers are smallholders who use low-input rainfed systems to produce rice for food and income. They have limited capacity to adapt to climate change and lack the technical and financial resources to access

[79] Cambodia's Updated Nationally Determined Contribution. Ministry of Environment. RGC. 2020.
[80] https://data.worldbank.org/indicator/GC.TAX.TOTL.GD.ZS?locations=KH. World Bank. 2021.
[81] https://data.worldbank.org/indicator/BN.CAB.XOKA.GD.ZS?locations=KH
[82] https://www.pwc.com/sg/en/publications/assets/page/cambodia-infrastructure-market-update-and-outlook.pdf
[83] Cambodia's Infrastructure Market Update and Outlook. 2022.
[84] Roles of SMEs in Cambodian Economic Development and Their Challenges. November 2021. 8th Annual NBC Macroeconomic Conference. Thy Sambath.
[85] Feasibility Study and Project Preparation AIIB. WAPCOS. July, 2023.

the technologies and adopt the practices that can build their resilience. CAISAR will directly target the smallholder farmers who are cultivating rice and help them adapt to climate risks and help them diversify and increase their sources of income from agriculture.

153. **Gender:** In Cambodia, women are the majority workforce in the agriculture sector and 20% of the households are women-headed. Key factors that account for the differences between women's and men's vulnerability to climate change risks include gender-based differences in time use; access to assets and credit, treatment by formal institutions, which can constrain women's opportunities, limited access to decision making, and a lack of sex-disaggregated data to influence policy change.[86] Under various led Government Programmes such as IWRM and the participatory irrigation management and development (PIMD) policies, women have gradually participated more in village and community water management forums and activities. However, due to the limited capacity and socio-cultural and physical constraints, only a few have been able to fully join farmer water user communities (FWUCs) in water and irrigation management as well as in disaster risk reduction activities. Compared with the past, women have become more active, committed and able to put forward their views. However, gender imbalance in FWUCs still exists, since women have many household chores to perform. Thus, only a few women have the time to join FWUCs and village and commune development activities, particularly in water and irrigation management. Compared with men, the number of women in FWUCs (and also in commune councils) is low at around 20%. Most of the important positions in FWUCs are dominated by men with women mostly holding positions as treasurers and accountants in the committee.[87] Women's participation in decision-making and productive activities needs to be proactively enhanced through special quotas and tailor-made interventions that are appropriate given their domestic, child-care and farm level responsibilities that involve both crops and livestock. Women also bear the largest brunt of the losses that households incur due to climate change as this directly impacts the household access to food, water and animal feed. CASIAR is designed to target women and deal with some of these key constraints.

154. **Youth:** Very few young people consider working in agriculture as a career aspiration; many see agriculture mainly as a backup option. They perceive agricultural work to be physically demanding, high risk and not very profitable.[88] It is expected that gradual structural transformation will lead to some of the poor gradually transitioning from agriculture-based to non-agricultural livelihoods. Project design recognizes this and places weight on creation of rural employment opportunities as an objective alongside improvements to farm-based livelihoods. Tailored capacity-building programs and incentives to retain youth in agricultural activities will be undertaken. CAISAR will through NCDDS collaborate with local NGOs to reach remote and marginalized populations. An effort will be made in CAISAR to target youth in activities which are of interest for them such as in the operation and maintenance of equipment and mechanized agriculture, work along supply chains in the transport, processing and trading activities. Youth will be trained as mechanics, drivers, processors and traders. Experience of programmes such as the Food and Agriculture Organization (FAO) which has developed a practitioner's guide for assessing skills and training needs of youth based on a case study in Cambodia will be used. Experience from ILOs project regarding the Cambodia Partnership for Sustainable Agriculture (CPSA) which has worked to expedite innovation, entrepreneurship and youth engagement in the agricultural sector will also be used. IRRI will be encouraged to work with local agencies, NGOs and research organizations and TVET to assist in the development of youth-tailored agricultural trainings which require improvement of training quality and easier access and appropriate training content.

D.5. Country ownership

155. **Relevance to and Alignment with Existing Policies such as Nationally Determined Contributions (NDCs), Nationally Appropriate Mitigation Actions (NAMAs), and National**

[86] World Bank Group (2016). Gender Equality, Poverty Reduction, and Inclusive Growth. URL: http://documents1.worldbank.org/curated/en/820851467992505410/pdf/102114-REVISED-PUBLIC-WBG-Gender-Strategy.pdf

[87] Gender and Water Governance: Women's Role in Irrigation Management and Development in the Context of Climate Change. Cambodia Development Resource. Institute. 2014

[88] Preparing and accessing decent work amongst rural youth in Cambodia. FAO, 2019.

Adaptation Plans (NAPs): The Government also updated its Nationally Determined Contributions (NDCs) in 2020. Cambodia's NDCs) update process utilized a bottom-up approach initiated by 16-line ministries. The effort was supported by the NDC Partnership and its Climate Action Enhancement Package (CAEP)[89]. This resulted in identification of 119 priority actions across all sectors. The implementation of these priority actions is being coordinated by the Department of Climate Change in close collaboration with the line ministries.[90] The updated NDC has been improved in several important ways. First, coverage has increased to include climate change mitigation targets in the agricultural and waste sectors, and also with more detailed actions in key sub-sectors, such as energy efficiency. It includes a stronger set of adaptation actions. It pays particular attention to gender and vulnerable groups, in order to ensure that the adaptation and mitigation actions contribute to a more inclusive society. Finally, a significant effort has been made to develop a solid framework for measurement, reporting and verification (MRV), so that the country is able to generate credible evidence on progress made and on challenges encountered.[91] CAISAR will directly support several NDC adaptation actions identified (e.g., Establish a national climate and flood warning system; Establish nationally standardized best-practice systems for irrigation; Establish a centralized and standardized approach to climate-resilient water management) through investments in irrigation infrastructure and CIEWS; improved institutional capacity for river basin management (including SCADA); effective irrigation management by rural water user groups and farmers; and so on. It will indirectly support MRV efforts through building PMU capacity on the use of EX-ACT tool in project M&E (which was already used in the NDC development for FOLU). GHG mitigation—to shift to low-emissions development pathways or carbon neutrality by 2050 targets 5 priority sectors: Agriculture; LULUCF; Energy; Transport; and Industry, while adaptation to climate change targets 5 vulnerable sectors: Agriculture; Water resources; Forest; Human health; and Coastal zone. Cambodia initiated the NAP process in 2014 to identify and address key vulnerabilities in key sectors such as agriculture and infrastructure, and is in the process of developing its NAP through GCF support. In its Second National Communication (SNC) to UNFCCC in 2015, Cambodia identified agriculture as the most vulnerable, followed by water resources, forests, coastal zones and health. SNC included a general vulnerability assessment and produced a vulnerability index to illustrate the geographic variation through layering variables such as socio-economic status, population and infrastructure. CAISAR, through support to farmers and water use groups on irrigation management, climate information and early warning services, design of resilient infrastructure, and institutional support will enable Cambodia to deliver on its mitigation and adaptation targets.

156. **Existing National Climate Change Strategy**: The country has in place a strong overarching policy structure for implementing its plans and strategies which include Cambodia's Sustainable Development Goals (CSDGs) 2016-30 (2018), and Cambodia Climate Change Strategic Plan (CCCSP)- 2014-2023. Each relevant ministry has developed associated action plans (Climate Change Action Plans - CCAPs), such as Climate Change Action Plan for Rural Development Sector (2021-2023). A new 10-year climate change strategy is under preparation with MOE leading the work. CCCSP 2016-2030 was the first ever comprehensive national policy document responding to climate change issues that occur in the country. It reflected the political will, commitment and readiness for reducing the negative impacts of climate change. CAISAR is in keeping with the emphasis of the CCCSP on building the climate change response of the country in the agriculture sector. In 2013, the Government developed a National Strategic Plan on Green Growth (NSPGG, 2013-2030). CAISAR will help to build mechanisms at the local and institutional level for green growth. At the local level, CAISAR will develop community action plans which will incorporate the

[89] CAEP support was delivered in part by UNDP and the World Bank.
[90] FAO, Germany, GGGI, IRENA, SNV, and UNDP.
[91] Cambodia's Updated Nationally Determined Contribution. Ministry of Environment. RGC. 2020.

principles of green growth. In addition, the capacity building of key institutions will incorporate aspects of green growth and embed them in the training and orientation of MOWRAM and NCDDS. The Long-Term Strategy for Climate Neutrality (LTS4CN 2021[92]) also focuses on the five NDC sectors, and prioritizes methane-intensive rice as one of the target areas for action. Cambodia's 2016 emissions were dominated by FOLU (60.9%) and agriculture (16.9%)[93]. In its first NDC update (2020), under a BAU scenario, Cambodia forecasts an increase in agriculture emissions from 125.2 MtCO2eq in 2016 to 155.0 MtCO2eq in 2030; agriculture will remain the second highest emitter sector. In the past it increased about 3% each year between 1994 and 2016 driven by methane from rice cultivation and enteric fermentation. In summary, CAISAR is designed to assist the country contribute to meeting its mitigation targets in the AFOLU/FOLU sector and its adaptation targets for the agriculture sector: for instance, out of the seven key mitigation actions identified in LTS4CN 2021 for agriculture, CAISAR will support at least 4-5 (namely, direct seeding practices, alternative wetting and drying practices, promotion of organic fertilizer, and possibly, less methane-intensive rice cultivars through advice on use of rice cultivars with maturity period most suitable to the local climatic and agroecological context).

157. **Existing GCF Country Programme:** GCF has been supporting Cambodia since 2014 on the green development initiatives. At present, GCF has already funded three single country projects worth over US$ 130 million. The development of this proposal's CIEWS activities was informed by PEARL (Public-Social-Private Partnerships for Ecologically-Sound Agriculture and Resilient Livelihood in Northern Tonle Sap Basin) analysis of needs and requirements, and IFAD has initiated discussions PEARL PMU on alignment of its CIEWS activities with that of the existing project. Based on the latest update, (12 Feb 2025) 12 Readiness grants (inclusive of 2 technical assistance) with the total funding amount of USD 5.63 million have been approved, and USD 3.13 million has been disbursed. GHG mitigation to reduce climate change impacts consists of 5 priority areas: Agriculture; LULUCF; Energy; Transport; and Industry, while adaptation to climate change consists of 5 priority areas: Agriculture; Water resources; Forest; Human health; and Coastal zone. In light of promoting a paradigm shift toward low emission and climate resilient development, this country programme aims to harmonize international climate finance and Cambodia fiscal priority and development policies, coordinating and allocating the GCF funding to areas with financial and development gap, ensuring programmes seeking GCF funding have incorporated environment and social safeguards as well as inclusive engagement process. A list of the pipeline projects developed until December 2021 are given in the country programme.[94]

158. The Royal Government of Cambodia (RGC) is committed to the sustainable development of Cambodia's water resources and strengthening resilience to unplanned shocks such as from climate change impacts. Establishing sustainable irrigation is a key priority for Cambodia to alleviate food poverty and ensure economic growth in the face of climate change. The RGC requested the Asian Infrastructure Investment Bank (AIIB) and the International Fund for Agriculture Development (IFAD) for assistance to finance the CAISAR project. AIIB approved the grant in February 2021 to assist the RGC with preparing the CAISAR project. IFAD had been leading the process with technical assistance from FAO and a team of independent consultants and the technical specialists from IFAD. The Ministry has been closely supervising and guiding the preparatory mission in the formulation of the project design.

159. **Commitment to Gender Aspects and Alignment to National Indigenous Policies:** In developing the Partnership Plan, Cambodia has adopted a rigorous approach to including gender considerations,

[92] LTS4CN https://unfccc.int/sites/default/files/resource/KHM_LTS_Dec2021.pdf
[93] Cambodia, Third National Communication, September 2022.
[94] GCF Country Programme Cambodia. December, 2021.

with every priority action examining gender implications. Gender equality is listed as a priority sector and Partnership Plan actions have an explicit focus on gender have already been budgeted. The government is also aware of the importance of strengthening institutional capacity to mainstream gender and has therefore brought the Ministry of Women's Affairs (MoWAs) into the process of climate action planning. The MoWAs is directing five of the six actions focused to gender in the plan. Furthermore, Cambodia's Partnership Plan pointedly identifies gender gaps and potential interventions to address these gaps to benefit women. The plan also contains a large number of statistics and targets related to gender equality, including indicators on participation in economic activity and the policymaking process, which will ultimately show differentials in how women and men benefit from the plan. Cambodia intends to strengthen the monitoring and evaluation systems of sectoral ministries to track the differential gender outcomes of climate change initiatives. The proposed project is aligned with Cambodia's national indigenous policies and strategies by adhering to the legal and institutional frameworks that recognize and protect Indigenous Peoples' rights, particularly those enshrined in the 2001 Land Law, which provides for the registration of collective land titles for Indigenous Communities, and the 2009 Sub-Decree on Procedures of Registration of Land of Indigenous Communities.

D.6. Efficiency and effectiveness

160. **Financial Structure**: The financial structure of the programme is based on the importance of investing in modernizing and climate proofing the irrigation and drainage infrastructure in the selected project areas. About 70% of the total project costs of USD 240 million will be allocated to the irrigation infrastructure. The soft activities included in component 1 will be allocated 16.9% (conducting FFS, organizing 4P and multi-stakeholder forums, providing early warning system), institutional strengthening at national level will be provided 4.0% while project management will be allocated 5% and the GCF contribution to project management costs will be mainly to assure the appropriate monitoring and evaluation of the entire operation.

161. For the use of GCF funding, both the grant and loan instruments, Table 4 indicates that 3% of the total GCF budget is allocated to the Project Management Costs (PMC), specifically under the M&E item. The remaining 97% is directed toward activities that benefit both direct and indirect beneficiaries. This includes 33% allocated to beneficiaries under Component 1 (funded by the GCF grant), 53% to beneficiaries under Component 2 (primarily financed through the GCF loan), and 11% to institutional beneficiaries under Component 3.

Table 4: Project Budget by Component and Funding Source.

Components	Total Cost (USD)	Funding Sources					%
		Green Climate Fund (Grant)	Green Climate Fund (Credit)	IFAD (Credit)	AIIB (credit)	Government (in kind)	
Component 1 - Farm Level Climate Adaptation and Resilience	40.4	26.3	-	12.9	-	1.2	16.9 %
Component 2 - Upgrading and climate-proofing water infrastructure for increased resilience	167.7	3.3	39.0	18.3	96.0	11.1	69.9 %
Component 3 - Strengthened institutional and regulatory capacity for low-emission climate-resilient development pathways	9.6	8.7	-	-	-	0.9	4.0%
Project Management and Coordination (PMC)a/	19.8	1.7	1.0	11.3	4.0	1.8	8.3%
Contingencies	2.4	-	-	2.4	-	-	1.0%
TOTAL	240.0	40.0	40.0	45.0	100.0	15.0	100.0 %

		16.7%	16.7%	18.8%	41.7%	6.3%	

a/ it includes M&E cost

162. **Local Communities and the Private Sector**: In order to make the investments viable, there is a strong role envisaged for local communities such as FWUCs, private farming households and private sector in the project area dealing with the four selected value chains. CAISAR hopes to generate demand for adaptive inputs, renewable energy equipment such as solar pumps and more efficient irrigation equipment which will be supplied by the private sector. Farming households are expected to finance all the production inputs on their land. The FWUCs will be strengthened to raise funds from user fees for the operation and maintenance of the irrigation schemes rehabilitated and modernised and climate proofed under the project. It is expected that CAISAR will help to establish effective market linkages so that the private sector is able to provide sustainable access to improved adaptive inputs and will also facilitate marketing of farm produce.

163. **Cost-Benefit Analysis of the Mitigation and Adaptation Outcomes:** GCF is being requested to provide total financing of USD 80 million for CAISAR. The programme is expected to benefit around 562,757 people directly and 1.15 million indirectly at a per capita cost of USD 46 which is above the average cost of GCF projects reported at USD 17 per capita.[95] The per capita cost within GCF and other programmes generally vary considerably depending upon the level of support provided to the beneficiaries. Projects which focus only on capacity building, awareness and information dissemination are much lower in cost while those which provide concrete support in terms of infrastructure investments are much higher in terms of the intensity of support. CAISAR provides the majority of its funding for infrastructure investments and thus its cost will be higher than the average GCF project. Based on preliminary estimates, it is expected that the programme will help to sequester a carbon balance of -2.3 tCO_2-eq per hectare per year. This would be achieved a unit cost of between USD 1.22 and USD 3 per metric tonnes. This is well below the benchmarks for many other GCF projects and the average cost of USD 4.75 per mt CO_2 equivalent reported in the GCF dashboard.[96]

164. **Economic and Financial Rate of Return.** The incremental economic and financial analysis of the project, considering all costs, indicates promising economic returns over a 30-year evaluation period and social discount rate of 9.47 percent and cost of capital of 11.36 percent[97]. The estimated Economic Internal Rate of Return (EIRR) is estimated at 16.3 percent, and the Net Present Value (NPV) is projected at USD 143.6 million, with a benefit-cost ratio of 1.26 and a payback period of nearly 12.0 years. Sensitivity and scenario analyses underscore the robustness of these evaluations, suggesting that the project would remain profitable even with a 20-year evaluation period. The sensitivity analysis shows that the project could become unprofitable if projected benefits decrease by more than 20.7 percent or if costs increase by more than 26.1 percent due to potential shocks during implementation, which is unlikely given the local economic context and macroeconomic trends. The results and scenarios explored are illustrated in the Table below.

[95] The per capita calculation uses the committed GCF funding of USD 16.7 billion and the 1 billion beneficiaries reported at the end of March 2024 by GCF in its dashboard.
[96] The unit cost uses the committed funding of USD 11.4 billion and the 2.4 bn CO_2 reported at the end of November 2022 by GCF in its dashboard.
[97] The social discount rate and the cost of capital in Cambodia were estimated using standard methodologies presented in the annexes and by comparing them with the rates used by other IFIs and commercial banks as part of the consistency analysis.

Table 5: Sensitivity and scenario Analysis

	Δ%		Overall project	Risk analysis
		EIRR (%)	**ENPV (USD M)**	
Base scenario (30 - year evaluation, @9.47%)		16.34%	143.57	Base scenario (conservative)
Project benefits	-10%	13.27%	74.25	Combination of risks affecting output prices, yields and adoption rates
	-20%	9.74%	4.94	
Switching value - benefits	-20.71%			
Project costs	10%	13.56%	88.61	Increase of construction material prices, input prices and services prices
	20%	10.98%	33.66	
Switching value - cost	26.12%			
Evaluation period (20 years)		14.27%	69.93	Uncertainty about the continuation of the activities supported by the project in the short, medium and/or long term
Evaluation period (10 years)		1.82%	-31.3	
Benefit delays	1 year	16.27%	142.8	Delays in project implementation
	3 year	15.88%	137.98	
Social discount rate (6 percent)		na	333.07	Linked with political and economic stability and macroeconomic context
Social discount rate (11 percent)		na	93.54	
Adoption rate - primary production (75%) - Base scenario		16.34%	143.57	Farmers' response to the implementation of agricultural practices
Adoption rate - primary production (60%)		15.64%	123.49	
Without considering the GHG emission reduction		14.51%	105.64	GHG emission reduction
Without considering avoided damage for floodings		10.42%	20.44	Floodings events
No expansion of new irrigated area in the OTP irrigation scheme after implementation of the project.		14.53%	93.06	Potential for irrigated area expansion after project implementation in the OTP irrigation scheme

165. In disaggregated terms, the implementation of activities under Components 1 and 2 is viable, with IRRs of 15.8 percent and 16.5 percent respectively. Moreover, all irrigation schemes demonstrate an IRR higher than the social discount rate used in this analysis, indicating the economic viability of each scheme. This viability is primarily due to the significant reduction in damages caused by annual flooding, both on and off the farms, and significant contribution of the benefits related to the GHG emission reductions.

166. The financial indicators for the entire operation were estimated from a private sector perspective incorporating the recovery costs of the irrigation infrastructure by estimating a tariff that users should pay per hectare per year[98]. This analysis was further complemented by an assessment of the operation's overall financial sustainability from a public finance perspective, considering both debt service requirements with financiers and the tariff estimated for irrigation system users. Both analyses demonstrate the overall financial viability of the project, with financial internal rates of return of 13.38 percent and 13.94 percent, respectively.

167. **Leverage and Co-financing:** The GCF grant will be used for climate proofing of infrastructure and strengthening the capacity of adaptation to climate risks for smallholders. In addition, CAISAR has already been able to leverage a USD 100 million loan from AIIB, US$ 25 million loan from IFAD 13 (33% Highly concessional and 67% Blend terms) and US$ 20 million from IFAD 14 (100% blend terms) and USD 15.0 million in-kind contribution from the RGC. There is a USD 20 million financing gap which IFAD will raise during implementation to further leverage GCF resources.

168. **Mobilization of Private and Commercial Funds:** CAISAR will attempt to secure debt capital from commercial financial institutions for financing smallholders and enterprises dealing with the four selected value chains. Some of the private sector enterprises will be expected to invest their own capital as well as borrow from the commercial financial institutions. There are several projects which

[98] Estimated annual fee (USD/ha/year): 208

have also put in place arrangements for enhanced access to financial services such as some of the IFAD supported projects in Cambodia. These will be used where appropriate.

E. LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

☒ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.

Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	The scale of practices and technologies that are currently being used by the farming communities and adaptation to climate risks are low. The private sector has limited role in the rural economy and does not have the incentives to engage with small fragmented rural producers.	<u>Low</u>	Enhanced awareness and access to climate adaptive practices and technologies that are adopted and disseminated in the project area and adjoining districts and eventually at the national level. Increased access to early warning systems which are used not just in the project area but are disseminated nationally through social media platforms. Increased presence of the private sector in the rural areas due to increased demand for adaptive practices and technologies.	The project will assist with training of 40,000 farmers for improved adaptation to climate risks and assist people with improved market integration, provide early warning systems to 170,000 people. The project will target 40% women in all training and support activities. The project will enhance water use efficiency for an area of 32,056 ha out of a gross area of 80,000 in the target areas during the wet season for the rice crop. The project will develop the required irrigation infrastructure to allow for a second irrigated rice crop during the dry season for an area of 22,051 ha using AIIB resources. GCF resources will be used to provide investments for flood control/drainage for the entire gross project

	Government has limited capacity to invest in modernizing of the irrigation system and confines itself to those for which it can raise funds through its own resources of through debt financing.		Growing role for private sector in the irrigation infrastructure through PPP partnerships due to increased cost recovery and a supportive policy environment for investment.	area of 80,000 ha. The project will assist farmers diversify their cropping patterns for additional income in the dry season. Close integration of the project into the National Systems being developed (such as Asset Management, flood forecasting and warning, water use monitoring as part of the National Water Resource Information System) will ensure upscaling to other systems in Cambodia.
Replicability	While there is good understanding of the need to enhance the capacity of the agriculture sector to adapt to climate change risks and capitalize on mitigation potential, there is limited financing and institutional capacity to undertake its replication in the agriculture sector.	<u>Medium</u>	Development of a model for climate adaptation and mitigation that can be upscaled through both public and private sector participation.	The project will invest in CR resilient agriculture technology and practices, modernize and climate proof irrigation infrastructure and strengthen institutional capacity of relevant institutions to enhance climate resilience. The lessons learned and experience gained will inform efforts to improve climate resilience in irrigated agriculture in Cambodia.
Sustainability	The sustainability of the current rice farming model is questionable. Increasing temperature and growing vulnerability of floods and droughts have exposed the vulnerability of the rice dominated agriculture	<u>Low</u>	Paradigm shift would see the development of a self-sustaining climate-resilient agriculture sector in which farmers will deploy climate resilient technologies and practices including water efficient adaption practices. Two factors will support this transformation. First, it provides them with better economic returns while adapting to CR methods. Second, they are provided with the technical, awareness, capacity support and modernized and climate	The project will facilitate and demonstrate the benefits of climate-resilient agricultural practices. It will mainstream the use of a range of climate-resilient farming technologies and practices that reduce the vulnerability of the evolving climate threats (flood, drought etc.) create financial benefits to the farmers ensuring increased production and protecting the production assets.

	sector in Cambodia. GHG emissions continue to increase and residue burning remains a significant problem despite repeated government attempts to reduce the practice.		proofed irrigation infrastructure in line with their service requirement to overcome the initial adoption barriers.	

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final[99]	
		Sources of information and methods used to collect and report data /information to measure progress against targets.	*The starting point or current value of the indicators before the implementation of the project*	*The estimated value of the indicator at the mid-point of the implementation*	*The estimated value of the indicator at the completion of the implementation*	*Externalities and factors outside project management's control that may impact the outcomes* *Data sources and methodologies applied for estimating baseline and targets*
<u>MRA4 Forests and Land Use</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Greenhouse Gas Emission Assessment using EX-ACT Tool. Baseline data (used to define Without-	0 tonnes of carbon dioxide equivalent (tCO2-eq)	Reduction of 45,750.34 tCO2-eq at the end of year 3	Reduction of 213,501.58 tCO2-eq at the end of year 7	Emissions reduction achieved in 20 years with 7 years project implementation period and 13 years capitalization phase.

[99] The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

		project scenario) and assumptions on additionality (With-project scenario) is described in Annex 22. The tool uses IPCC Tier 1 default values (or country-specific values, where available) to compute emissions and removals. Data from Mid-term and end-of-project socio-economic surveys will help refine with-project scenarios, based on cropping patterns and associated activities (e.g., use of intermittent flooding, fertiliser, residue practices)..			Reduction of 1,006,507 tCO2-eq over 20 years	**1. Methodological consistency:** This is achieved by employing the EX ACT Tool across all project phases - from initial ex-ante calculations through monitoring and final reporting (please refer *“CAISAR_Monitoring_EX-ANTE_20.02.2025.XSLX”*). This approach guarantees consistent GHG accounting methods throughout the project lifecycle. For detailed EX-ACT Tool methodology instead, please refer to the “Methodology for GHG accounting” section of Annex 22 or EX-ACT Guidelines (FAO. 2022. Ex-Ante Carbon-balance Tool | EX-ACT – Guidelines. Second edition – Tool version 9. Rome. https://doi.org/10.4060/cc0142en) **2. Alignment with National Systems:** The approach aligns with Cambodia's existing national frameworks, as the country's NDCs already mentions the EX-ACT Tool in conjunction with the Prospect+ model for Agriculture, Forestry and Other Land Use (AFOLU) sector calculations, as documented in Cambodia's 2023 NDC update (see Cambodia’s 2023 NDC update submission at : https://unfccc.int/sites/default/files NDC/2022 06/20201231_NDC_Update_Cabodia.pdf). Mid-term and endline surveys will collect information on crop cultivation and fish/livestock

						management practices that will impact GHGs: improved fertilizer management and use of organic fertilizers, changes in rice straw residue management, adoption of intermittent flooded rice with multiple drainage periods (alternate wetting and drying), uptake of solar irrigation and reduction in use of diesel pumps for rice, and land area under annual crops as well as fish pond area, number of poultry, livestock / fish feeding practices. Project progress reports on construction of irrigation canals, flood-protection embankments and rural roads will be used to update EX-ACT analysis on GHG emissions. There are several assumptions associated with these practices that are outlined in Annex 22. Without the project, there would be no reductions in GHG emission since current cropping patterns and management practices will continue. For instance, farmers will transition to an 80-day cultivation period for flooded rice, instead of 130-days with project thereby reducing the number of flooding days.
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	Project Management Unit (PMU) / Project Management and Implementation Consultant (PMIC)	Direct beneficiaries: 0	155,000 direct beneficiaries (62,000 female)	562,757 direct beneficiaries (225,103 female)	Based on the assumption that there will be political and economic stability that will enable the activities to be undertaken in a timely and efficient manner. See Annex 23B for methodology.

		progress/ completion reports.	Indirect beneficiaries: 0	0.5 M indirect beneficiaries (0.25M female)	1.15M indirect beneficiaries (575,809 female)	
<u>ARA2 Health, well-being, food and water security</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	- PMU/PMIC progress/ completion reports. - Mid-term and end-of-project household socioeconomic surveys	Direct beneficiaries: 0 Indirect beneficiaries: 0	155,000 direct beneficiaries (62,000 female) 0.5 M indirect beneficiaries (0.25M female)	562,757 direct beneficiaries (225,103 female) 1.15M indirect beneficiaries (575,809 female)	Based on the assumption that there will be political and economic stability that will enable the activities to be undertaken in a timely and efficient manner. See Annex 23B for methodology.
<u>ARA1 Most vulnerable people and communities</u>	<u>Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options</u>	- PMU/PMIC progress/ completion reports. - Mid-term and end-of-project household socioeconomic surveys	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 15,000 persons (6,000 female) Indirect beneficiaries: 60,000 persons (36,000 female)	Direct beneficiaries: 32,000 persons (12,800 female) Indirect beneficiaries: 160,000 persons (96,000 female)	• Economic and market trends remain favourable. • Effective implementation of FFS, FWUC, 4PF, infrastructure and CIEWS interventions. • National, regional or global disruptions to agriculture, particularly input markets (seeds, machinery, etc.), remain minimal. • Beneficiaries utilize improved access to CIEWS, improved market access (including inputs, mechanization), and other support via FFS and FWUCs for adoption of climate-resilient and low-emission practices and technologies for crops, fish and poultry to improve their productivity, reduce cost of production, improve farm

						income, reduce production loss, and/or diversify livelihoods. • Non-FFS participants are able to effectively learn from FFS participants. Based on the assumption that 80% of the 40,000 participants in the FFS will adopt the improved and or new climate resilient livelihood and crop diversification options. Each of those 40,000 FFS participants will influence one member in their household (typically, their spouse with whom they manage their plots and livestock), and an additional 2 households with 2 adult farmers.
ARA2 Health, well-being, food and water security	Supplementary 2.2: Beneficiaries (female/male) with improved food security	- PMU/PMIC progress/ completion reports. - Mid-term and end-of-project household socioeconomic surveys	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 56,000 persons (28,000 female) Indirect beneficiaries: 61,600 persons (30,800 female)	Direct beneficiaries: 134,400 persons (67,200 female) Indirect beneficiaries: 147,840 persons (73,920 female)	• Economic and market trends remain favourable. • Effective implementation of FFS, FWUC, 4PF, infrastructure and CIEWS interventions. • National, regional or global disruptions to agriculture and food supply remain minimal. • Beneficiaries utilize CIEWS, improved awareness of climate-resilient and low-emission practices, access to markets, and irrigation to improve food security. Based on the assumption that 96,000 households who adopt improved / climate-resilient livelihoods (40,000 FFS

						households and an additional 2 households influenced by each FFS participant household, with 80% adoption rate) will benefit from improved food security. Household size is 4.20 for the four provinces. 70% of households will improve their food security. Direct beneficiaries are two adult members, and indirect beneficiaries are other household members (2.2 average) of the same household. 50% of household members with improved food security are women. The baseline figure is left at 0 because the figures vary for each province. As of 2021, approximately **51.1% of Cambodia's population** experienced moderate or severe food insecurity, equating to about **8.6 million people**. Within this group, around **14.8%** faced severe food insecurity, affecting roughly **2.5 million individuals**. Source: CEIC
ARA4 Ecosystems and ecosystem services	Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice	- PMU/PMIC progress/ completion reports. - Mid-term and end-of-project household socioeconomic surveys	0 Hectares (Ha)	10,000 Ha	32,056 Ha	Beneficiaries are able to effectively use infrastructure, services, and technical support enabled by the project to improve water use efficiency / water management (including through FWUCs) and adopt other climate-resilient and low-emission practices

						(particularly for rice – retention of straw, intermittent flooding, organic fertilizer). Based on the area that will have access to improved irrigation by the 6 schemes which will be modernized under the project. 22,051 hectares of the command area, and an additional 10,001 hectares of the flood-protection area.
ARA4 Ecosystems and ecosystem services	Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under resoration and/or improved ecosystems	- PMU/PMIC progress/ completion reports. - Mid-term and end-of-project household socioeconomic surveys	0 Hectares (Ha)	10,000 Ha	32,056 Ha	Through FWUCs, lands will be brought under improved water management. Through FFS, lands will benefit from improved soil (and irrigation) management. Both actions contribute to better ecosystem services on production landscapes.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	Institutional and regulatory frameworks for sustainable low emission climate-resilient development in the irrigation sector remains low as manifested in: • Insufficient investment in systematic planning and long-term maintenance of infrastructure.	low	• Climate resilient and sustainable agricultural sector in the project areas. • Enhanced capacity of MOWRAM in strategic planning for modernized sustainable low emission climate-	The institutional and policy framework in Cambodia will be strengthened through strategic investments in developing MOWRAM, MoE, NCDDS, PDWRAM and FWUC staff capacity for: • improved designs, • use of Information Communication Technology,	National level (one country)

	• Technical capacities of relevant national institutions and FWUCs are rarely assessed in detail at the design stage. • In most irrigation schemes, there is largely no systematic cost-recovery practice leading to a lack of incentive for the FWUCs members to manage the infrastructure properly, making sure the voices of vulnerable groups are heard.		resilient irrigation sector.	• deploying water accounting and audit systems, • strengthening the early warning systems, • strengthening the policy and regulatory frameworks for climate action, • enhancing M&E capacity, and • encouraging greater collaboration between NDA and other stakeholders.	
Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation	• Limited extension services at the provincial level. • Agricultural extension has limited personnel, capacity and network to access remote communities. • Current extension personnel possess an inadequate understanding of climate change and climate-resilient practices and technologies that are suitable for local contexts.	low	• Increased capacity of farmers to deploy crop resilient practices at farm level. • Climate adaptive value chains developed by 4Ps and increased access to finance.	• Increased adoption of new and improved climate resilient agriculture technologies and practices will be enhanced through working closely with smallholders and training them in the use of these technologies, linking them with suppliers of the technologies, providing increased access to finance to enable them to continue to use these inputs and strengthening the value chain of key agriculture commodities.	Multiple sub-national areas within a country
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and	• Farmers have limited knowledge and tools to adapt their farming practices and production techniques to climate change.	low	• Increased climate adaptation and mitigated negative impact of extreme climate events.	• The project will introduce farmers with various climate resilient technologies and practices for both rice and non-rice activities such as	Multiple sub-national areas within a country

learning processes, and use of good practices, methodologies and standards	• They are largely unaware of climate resilient practices, stress-tolerant seed varieties and improved planting materials, efficient input management, or the benefits of applying these practices. • Limited access to credit on attractive and appropriate terms, and market intelligence.		• Improved livelihoods of smallholder farmers and vulnerable rural communities in four provinces of Cambodia.	vegetable production, poultry and aquaculture, assist in value addition and facilitate market linkages, introduce agro-meteorological information and services for climate informed agriculture planning and improve both physical and digital connectivity to facilitate improved production and marketing.	

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Outcome 1 Reduction in crop losses and improved (agricultural) production	Percentage increase in yield for rice and vegetables in targeted households benefiting directly from the project	Mid-term and end-of-project household socioeconomic surveys	0	Rice: 15% Vegetables: 10%	Rice: 30% Vegetables: 20%	There is significant scope for improving rice productivity, as many farmers currently do not use improved seed varieties. Vegetable farmers, on the other hand, are more likely to use improved varieties, but there remains room for improvement, particularly in the adoption of certified seeds
	Percentage decrease in post-harvest losses for rice and vegetables in targeted households benefiting directly from the project	Mid-term and end-of-project household socioeconomic surveys	0	Rice: 10% Vegetables: 15%	Rice: 20% Vegetables: 30%	Postharvest losses are particularly high, especially among vegetables. This indicates substantial room for improvement, particularly through better storage, processing, and handling practices.
Output 1.1	Total number of farmers (male and	Annual Progress Report	0	20,000 Male: 12,000	40,000 (Male: 24,000	It is assumed that each FFS (1600 in total) will have 25

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Increased farmer capacity in climate resilient agriculture.	female) trained in CR technologies through the FFS training programme			Female: 8,000)	Female: 16,000)	participants of whom 40% will be women. Each FFS participant represents one unique household. [40,000 FFS participant households hence represent 78.9% of the targeted commune farming households.
	Number of trainers (male and female) trained to demonstrate and propagate the CR technologies and practices to farmers	Annual Progress Report	0	100 (Male: 60 Female: 40)	200 (Male: 120 Female: 80)	Target values are number of trainers trained through ToT trainings conducted by the project
	Percentage of farmers adopting climate resilient farming practices promoted by the project	Mid-term and end-of-project household socioeconomic surveys.	0	50%	80%	Target values are percentage of farmers who implement the adaptation measures
	Average percentage of beneficiaries who are satisfied with the knowledge gained from the trainings provided	Training Surveys		70%	90%	A post-training survey will be conducted after each training session to understand the level of satisfaction among beneficiaries.
	Number of service providers who will provide mechanization services to farmers	Annual Progress Report -	0	10	20	These are service providers for LLLs (Laser Land Levelling) and direct rice seeder (mDSR)
Output 1.2 Climate adaptive value chains developed by 4Ps and increased access to finance	Number of farming households benefiting from the 4P model through increased access to finance	Annual Progress Report -	0	14,000	28 ,000	It is assumed that 70% of FFS farming households will have improved access to finance and will be able to invest it in their adoption of more sustainable agriculture practices.
	Number of MSMEs with increased access finance due to the linkages facilitated by the 4P model	Annual Progress Report - .	0	50	120	The target values are estimates from Output 1.2.

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Output 1.3 Increased access to and use of climate information and advisory services improved (for climate responsive water-use and crop planning)	Number of farmers (male and female) benefitting from the climate information and early warning services (CIEWS) delivered through the mobile app	Annual Progress Report	0	281,378 (Male: 168,827 Female: 112,551)	562,757 (Male: 337,654 Female: 225,103)	The target values are based on the assumption that 75% of farming households in the target provinces will benefit from climate information and early warning services (CIEWS) delivered through a mobile app. s
	Number of hydromet stations and agrometeorological stations rehabilitated	- Annual Progress Report	0	5	10	This includes the procurement and rehabilitation of six automatic hydromet stations and four agrometeorological stations, to enhance data collection capabilities in areas not covered by the Ou Ta Paong SCADA system
	Number of automatic weather stations established	Annual Progress Report	-	2	4	Four automatic weather stations will be established to generate reliable data to support decision-making across national, provincial, district, and local levels, while improving connectivity and data-sharing mechanisms among various departments.
Output 1.4 Increased resilience of farm road infrastructure to climate change	Total kilometers of roads upgraded in the project areas, taking into account current and future climate change risks	Annual Progress Report	0	5	90	The target values are based on the Preliminary Report on Road Survey.
	Number of farmers benefiting from improved road connectivity in the project areas	Annual Progress Report	0	5,000	10,683	There are a total of 9,417,017 people living in over 14,545 villages. This gives the average number of households in a Cambodian village. 66% of the population in each village is farming households.

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
						Assuming that 3 members (male and female farmer + youth or another adult farmer) from each household directly benefit from roads. 85 kms of road was mapped to 35 villages, based on the Preliminary Report on Road Survey.
Outcome 2: Increased water use conveyance and use efficiency	Number of Irrigation schemes and drainage facilities upgraded, rehabilitated and constructed.	Annual Progress Report	0	2	6	All 6 schemes developed, finalized and rehabilitated.
	Number of fully functional Water User Associations (WUAs)[100] sustainably maintaining irrigation systems	Annual Progress Report	0	6	11	
Output 2.1 Modernized irrigation schemes and ponds	Area provided with irrigation services - Improved (ha)	Annual Progress Report	0	10,000	22,051	The target value is a total area with rehabilitated irrigation systems under the 6 finalized projects.
Output 2.2 Flood-proofing and drainage improvement	Area provided with drainage services - Improved (ha)	Annual Progress Report	0	5,000	10,005	The target value is a total area with improved drainage systems under the 6 finalized projects
	Number of polder embankments designed and procured for climate-resilient water management and flood protection.	Annual Progress Report	0	1	2	This refers to the Ou Ta Paong and Steung Krang Bat polders

[100] A fully functioning WUA has the following characterstics: (i) officially registered or recognized by the relevant government authority; (ii) has a written constitution and/or bylaws outlining roles, responsibilities, and governance structure; (iii) Has an elected leadership committee and holds regular general meetings; (iv) Prepares and implements an annual operation and maintenance (O&M) plan for irrigation infrastructure; (v) Collects water user fees or other agreed contributions from members to fund O&M; and (vi) Maintains a basic financial record-keeping system, including annual budgets and financial reports.

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
Output 2.3 Farmers Water User Communities (FWUC) established and trained	Number of FWUC established and trained to ensure sustainable and effective operation and maintenance of irrigation schemes	Annual Progress Report	4	6	11	Four FWUC's are partially functioning. A total of 11 FWUC's are planned.
Output 2.4 Water information system established and operational in three subprojects	Percentage of farmer-households using Climate information and early warning (CIEWS) advisory services promoted by the project	Mid-term and end-of-project household socioeconomic surveys	0%	20%	75%	The target values are estimates of percentage of farmers using the CIEWS
	Number of people trained in using the Supervisory Control and Data Acquisition (SCADA) system	Annual Progress Report	0	20	40	It is assumed that staff with IT capacity will be available
Outcome 3: Enhanced environmental governance	Number of policies, regulations, and institutional frameworks strengthened or adopted to promote sustainable resource management and climate resilience	Annual Progress Report		5	10	It is expected that the policies would be developed through the strengthening of MOWRAM and NCDDs
Output 3.1 Strengthened MOWRAM	Number of MOWRAM staff (male and female) trained in river basin management, water accounting, SCADA system operation, and the design and implementation of climate-resilient green technologies	Annual Progress Report	0	125 (Male: 75 Female: 50)	250 (Male: 150 Female: 100)	The target values are as planned in Output 3.1.
Output 3.2 NCDDS strengthened at national and provincial level	Number NCDDS staff (male and female) trained on project management, policy formulation and	Annual Progress Report	0	175 (Male: 105 Female: 70)	350 (Male: 210 Female: 140)	The target values are as planned in Output 3.3.

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
	implementation and resource mobilization					
Project/programme co-benefit indicators						
Co-benefit 1 Increased decision-making power of women through greater community collaboration and participation.	Proportion of women actively participating in and influencing decision-making processes in community-based institutions (e.g., water user associations, farmer groups)	Mid-term and end-of-project household socioeconomic surveys	0	20%	40%	- Social norms and cultural practices allow space for women's participation - Capacity-building activities and facilitation are gender-sensitive - Women have time and resources to participate (e.g., childcare, transport) - Male champions and local leaders support inclusive governance - Institutional structures are receptive to inclusive reforms
Co-benefit 2 Greater private sector integration in rural economy.	Number of private sector actors (e.g., agri-businesses, cooperatives, SMEs) supported by the project and integrated in the rural value chain	Annual Progress Report	0	60	120	The target values are estimates from Output 1.2.
	Percentage increase in revenues of MSMEs supported by the project	Mid-term and end-of-project household socioeconomic surveys	0	15%	30%	4P interventions would lead to income increases for the beneficiaries

E.6. Project/programme activities and deliverables

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in annex 5 implementation timetable. Add rows as needed.
Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.

Activities	Description	Sub-activities	Deliverables
Output 1.1 Increased farmer capacity in climate resilient agriculture.			
Activity 1.1.1 Preparation of community-based action plans (AP) to transform agriculture with CR practices and technologies.	The AP will be prepared blending the scientific climate knowledge with the local knowledge and skills of farmers. It will be prepared at each commune level through interactive consultations facilitated by the project team focusing on linkages between climate change and agriculture and associated risk in the project areas and potential options for adaptation. Communities will identify and priorities potential adaptation options and prepare a plan to implement them. The plan will be revised annually, including the targets and crop choices and other adaptation technologies based on the new information and knowledge available from implementation of the plan each year as well as from the improved climate information services as they are available from implementation of Output 1.3	Sub-activity 1.1.1.1 Organize consultative workshops at commune level Sub-activity 1.1.1.2 apprise farmers on CC variability and impacts Sub-activity 1.1.1.3 compile existing adaptation approach and farmers perception Sub-activity 1.1.1.4 identify and prioritize future adaptation options based on above two Sub-activity 1.1.1.5. Prepare AP setting targets and with various adaptation options	2,800 farmer households trained/participated in the preparation of Action Plan to deploy CR technologies and practices at each commune level.

Activity 1.1.2 Preparation of training materials to support implementation of the AP	This activity will develop training curricula and compile, revise, and update available materials to be used to support farmer learning and adoption of CR technologies and practices as outlined in the action plans. Priority will be given to short visual learning materials combined with charts, leaflets and short notes that a user-friendly learning material are available to farmers who usually have lower literacy rate. The training materials will be reviewed regularly and modified and updated based on field experiences. The training material and the curricula will be mainstream in the MAFF extension department and will be used as key learning and training materials for scaling up application of CR technologies and practices in the country. Training materials will be prepared for rice (wet and dry season rice), vegetable production, poultry and aquaculture and other crops as prioritized by the farmers in the AP and will be prepared for the whole cropping season right from land preparation to final harvest.	Sub-activity 1.1.2.1 Compile and review all available materials on farm CR technology and practices (Cambodia and best practices worldwide) Sub-activity 1.1.2.2. Prepare training materials for Rice, non-rice crops, poultry and aquaculture, and design curricula to run the FFS (Activity 1.2.3) Sub-activity 1.1.2.3. Continuously modify and update the materials based on new information and lessons learned from implementation Sub-activity 1.1.2.4. Mainstream the materials and curricula to be used at national level	15 sets of training materials and curricula to deploy CR technologies at farm level
Activity 1.1.3 Conduct training of trainers to create a pool of expertise to demonstrate and propagate the CR technologies and practices.	TOTs will be organized to create a pool of expertise to propagate the CR technologies and practices, including the availability and use of early warning and climate-informed crop advisories. This will be organized to interested NGOs, Extension department of MAAF, universities or any external institutions and private entities. Altogether 200 champions will be trained, who will be able to work as Master Trainers (MTs). The training will be organized by the SMSs who will continue to provide the support to the MTs in testing and adapting to new technologies and practices and how CIEWS can complement on-farm management. The training will be conducted establishing 8 FFS (two in each province) to train on various aspects of the CR technologies and practices as identified in the AP.	1.1.3.1 Select potential candidates from provincial departments, extension, interested NGOs and other private entities 1.1.3.2 Establish 8 FFS for organizing the TOTs 1.1.3.3 Organize the TOTs for the 200 persons who would act as Master Trainers	200 Master trainers trained to propagate and expand the CR technologies and practices

Activity 1.1.4 Training and advisory on CR technologies and practices	Altogether, 40,000 farmers will be trained organizing 1600 FFS. Given the need to cover such huge numbers of farmers in a short time, a cascade approach will be followed. It will begin with a 200 FFS in the second first year and will be expanded utilizing newly trained farmers, who are young and innovative, but under the supervision of the MTs. The FFS will be used to link the farmers with markets and build their capacities in value chain development covered the Output 2. The last sessions of the FFS will focus on post-harvest management and market links. In addition, during the FFS, participating farmers will be demonstrated how the technologies and practices introduced are sustainable and provide higher returns to them as compared to conventional farming practices	1.1.4.1 Design a cascade approach to run 1600 FFS 1.1.4.2 Implement the FFS (20 members in each) based on designs above	40,000 farmers trained in CR technologies and practices through 1,600 FFS -
			-
			-
Activity 1.1.5 Strengthening and fostering tailored mechanization service providers for improved mechanization service delivery.	This activity focuses on improving the provision of high-quality mechanization services to farmers, especially to support them to shift towards the mechanized direct seeded rice mDSR. It will provide capacity support to service providers and assist them in bundling the services of new machinery in their existing business		Training materials on mDSR and other relevant knowledge products Service providers engaged in providing services for mDSR and LLL to the farming community

	models. The service providers will be linked with the machinery dealers and mechanics for smooth service delivery to farmers. There is already strong desire among farmers and willingness to pay to shift mDSR from existing manual broadcasting method. Access and knowledge are the key limiting factor for mDSR whereas business opportunity exist.	1.1.5.1 Contract a service provider to provide high level of knowledge inputs on mDSR and LLL through demonstration, training and knowledge products 1.1.5.2 Strengthen commune level extension works for strengthening and engaging private service providers for mDSR and LLL	
Activity 1.1.6. Community-based monitoring and evaluation (CBME) of the implementation	This grass root level CBME has particular importance in the context of climate change adaption measures as smallholder agriculture producers and supporting institutions needs to adapt not just once but constantly. Weather patterns and their effects will change continuously during and beyond the life of a project with considerable uncertainty regarding what the actual (versus the predicted) impacts of climate change in a given agricultural system. A set of community-based performance indicators will be identified in consultation with the farming communities and other relevant stakeholders to monitor and evaluate the outcomes of the interventions listed in the AP. Such indicators could be nos of farmers trained in applying specific technologies, nos of female farmers engaged in kitchen gardening, new varieties introduced etc. These indicators will be scientifically validated so that the data and information can be used in the project's M&E framework as well as in Climate Action Monitoring systems at national level.	1.1.6.1 select (3 to 5) active farmers in each commune level for monitoring 1.1.6.2 train the selected farmers on CBME process on how to monitor the implementation of AP 1.1.6.3 Produce CBME reports annually for each commune	Annual report by each commune level on status and progress of implementation
Output 1.2 Climate adaptive value chains developed through the adoption of the 4P model			
Activity 1.2.1. Value Chain study and planning	This activity, through a mix of theoretical and case-based training and resource	Sub-activity 1.2.1.1. Value chain studies (Rice, Vegetable, Poultry, Aquaculture)	1.2.1.1. In-depth studies on options (market potential, climate resilient potential, pro-

	allocation, capacitate provincial and district partners, is to prepare concise value chain action plans (VCAP) including mapping of commodity actors, processes, the added value at each link, and an initial identification of bottlenecks/challenges and opportunities. The VCAP will identify interventions that offer higher income potential, are inclusive of poor, women and youth, are resilient and can be adapted to climate change and that are or can be linked to private sector technical support and/or off-takers.	Sub-activity 1.2.1.2. Value chain action planning (VCAP)	poor potential) for the four selected value chains: rice, vegetable, chicken, and fish. 1.2.1.2. Four (4) VCAPs for rice, vegetable, chicken, fish developed and operationalized in each province.
Activity 1.2.2. Establish Multi-Stakeholder Platforms (MSPs)	Based on the VCAPs, this activity is to establish the district commodity-based Multi-Stakeholder Platforms (MSPs) with the purpose to improve value chain governance by enhancing coordination and strengthening relationships between actors within selected value chains. An MSP consist of relevant stakeholders within a value chain, including farmer representatives, farmers' and private sector organizations (including identified business partners), government representatives, traders, processing enterprises, input suppliers, consumer representatives and financial institutions.	Sub-activity 1.2.2.1. Establishment and operation of MSPs	1.2.2.1. Thirty-five (35) district MSPs established and implement value chain action plans and the Public Private Producer Partnership Facility (4PF); At least 350 MSP meetings organized with minimum 10 meetings per district over the project lifetime.
Activity 1.2.3. Public Private Producer Partnership Facility (4PF)	This activity is to establish a Public Private Producer Partnership Facility (4PF) that has the purpose to strengthen commercial actors – micro, small and medium agri-enterprises as well as commercial farms with strategic position in the value chain – so they can perform their role for their own benefit as well as for the benefit of related smallholder households. The facility has a special focus on promoting a new generation of young rural entrepreneurs. Grant beneficiaries receive technical and financial support to strengthen their business skills, and to carry out investments with good potential for	Sub-activity 1.2.3.1. Design and Establishment of 4PF Sub-activity 1.2.3.2. Capacity building and BDS provision to enterprises and farmers participated in 4PF Sub-activity 1.2.3.3. Evaluation and Delivery of 4PF grant finance to selected partnerships	1.2.3.1. 4PF manual developed and implemented 1.2.3.2. At least 120 MSMEs and 5000 farmers are capacitated/provided with BDS and technical services regarding business planning, accounting, marketing, certification, e-commerce, contract farming, and climate resilient agriculture. 1.2.3.3. At least 100 MSMEs engaged in 4PF; of these 30 young entrepreneurs aged 35 or less. MSME investment supported by the 4PF have a beneficial effect on 5000 farmers through farming contract, fair trade

	profitability and positive side-effects to target farmers	Sub-activity 1.2.3.4. Leveraging additional capital for 4PF	linkages, or by providing crucial services to target farmers 1.2.3.4. MOUs signed with at least three partner financial institutions (e.g. ARDB, SME, AMK…) on increasing provision of finance to project target beneficiaries (MSMEs, farmers); 50% of 4PF's MSMEs and farmers report increased access to finance.
Output 1.3. Access to climate information and advisory services improved for climate responsive water-use and crop planning			
Activity 1.3.1 Strengthening of agrometeorological stations in the project area	This activity will support the procurement and/or rehabilitation of 6 automatic hydromet stations in the areas not covered by the Ou Ta Paong SCADA system described in Output 2.4, the establishment of 4 agrometeorological stations in the project area increasing the coverage of data gathering capacity in the command area, and the rehabilitation/procurement of 4 automatic weather stations. The network of data will generate reliable information to influence decision making at national, provincial, district and local levels. It will also include improving connectivity between different levels and departments, along with mechanisms and procedures for data sharing.	Sub-activity 1.3.1.1 Identify locations for installation to the different stations Sub-activity 1.3.1.2 Prepare tender material Sub-activity 1.3.1.3 Procurement and supervision of installation of the equipment	14 stations installed and operational
Activity 1.3.2 Establish Provincial-Level Coordination Platform for climate-informed agricultural advisories	The platform will be formed at provincial level and will be responsible for providing agrometeorological services within the provinces. The platform will include provincial MOWRAM, MAFF, research institutes, key farmers, commune leaders and NGOs and relevant private sector. It will use data information available from Department of Meteorology and combine the necessary information from crop production sector including the crop related information from the farmers to produce advisory services to the farmers at local level. The platform will be operational within the first year of the project implementation	Sub-Activity 1.3.2.1 identify relevant stakeholders including key farmers, NGOs to represent the platform Sub-Activity 1.3.2.2 define scope and prepare TOR for the platform	Four (4) platforms formed at each project provinces

	together with its scope and detailed TORs. The provincial governors will provide required co-ordination for the functioning of the platform.		
Activity 1.3.3. Building the capacities to deliver services	The capacity development will focus on establishing the essential technical capacities of climate data collection and management, translating these data into agriculturally relevant information (agromet information) through modeling and statistical analyses, and holding simultaneous focus group discussions for implementing the co-creation process across multiple government entities and farming community. Most of these participants will be staffs/professionals responsible for either the production or management of climate and agriculture data, public services, or research and development institutes, and Cost recovery within these sectors representing the platforms. In addition, the staffs and other stakeholders will be also trained on operating Procedures (SOPs) as described in 1.3.2. Both structured training programs and on the job training approach will be used for capacity development. It is planned to train about 250 persons in different disciplines across various stakeholder groups. Visit to neighboring countries to learn on application of agromet services will be considered in the training programs. The upgraded agrometeorological data collection and processing capacities will enable an analysis of real-time weather and climate information, seasonal forecasts, historical records, crop parameters, soil moisture and temperatures, and pest and disease characteristics through relevant methodologies to provide crop-specific warnings and advisory services to the farmers	Sub-Activity 1.3.3.1 identify capacity needs and prepare a plan to address it Sub-Activity 1.3.3.2 Organize structured training programs to 250 participants Sub-Activity 1.3.3.3 Organize visit program to neighboring countries to learn about agrometeorological services	250 staffs trained 50 persons participate in the visit program to neighbouring countries

Activity 1.3.4 Establish the agromet information, systems and the awareness raising and capacity building of farmers and stakeholders in applying the services.	A standard operating system (SOP) will be developed for the production and dissemination of agrometeorological advisory services and data sharing needs and architecture targeting rice and vegetable crops aligning with other similar initiatives so as to have a common system of data collection, information and communication mechanism in the country. The services will be delivered in terms of weekly and monthly bulletin in websites, charts and leaflets, Audio and video materials (through local/regional medias and Farmer Field Schools) and SMS messages as appropriate. These will allow farmers to grow, harvest and store the target crops and other local value chain actors to anticipate yields and processing volumes with reliable information and consistent advisory services to make agriculture production, storage and value chains less vulnerable to climate change	1.3.4.1 Prepare advisory materials on weekly basis 1.3.4.2 Disseminate the materials using mobiles phones and other media as relevant 1.3.4.3 Obtain feedback from farmers regularly 1.3.4.4 continuously modify and updated the advisory services based on learning and feedback	Agromet advisory services are regularly prepared (once in a week) and disseminated to farmers
	Farmers' and other key stakeholders will be apprised of weather forecast process, challenges and limitations, reliability and how to use the information and advisory services in the decision-making process. The already available platform like the FFS (1.1.1) and the AP formulation process as well as other relevant platforms in communes, districts and provinces will be utilized for awareness raising programs. In addition, targeted trainings on "climate informed agriculture decisions and planning" will be organized. A total of 16 such training involving 320 key farmers and stakeholders are planned. These key people will act as resource persons to disseminate importance of climate informed agriculture planning during the FFS and AP process. Lessons and feedback from farmers will be continuously integrated in the advisory services to reflect their views, learning and advice. A specific platform using mobile applications (SMS, WhatsApp	1.3.4.5 Raise awareness using the existing platforms 1.3.4.6 conduct structured training on climate informed agriculture decisions and planning	Improved awareness raising to at least 50% of the farmers in the project target areas

	etc.) will be used to obtain feedback from farmers on the quality of services regularly.		
Output 1.4 Increased resilience of farm road infrastructure to climate change			
Activity 1.4.1 Identification, selection and initial planning	The initial planning shall be carried out organizing consultative meetings with the communities and other relevant stakeholders. This exercise will help the consulting firm (or experts engaged in design) to understand the local context and opportunities and challenges to shape the future road design. Lessons and experiences of the Community Participation Framework (ADB 2018) prepared under the Rural Roads Improvements Project III will be used in the community engagement process. At the end of the planning session relevant information on traffic numbers and type, impacts of increased precipitation or drought resulting from climate change, existing conditions etc. will be available for design purpose. Future maintenance and resources requirements will also be identified.	1.4.1.1 Organize consultative workshop with local communities 1.4.1.2 Conduct Walk through along roads 1.4.1.3 Asses current and future traffic movements 1.4.1.4 Finalize priorities areas of improvements based on above	Locally developed plan for farm road improvement
Activity 1.4.2 Technical survey, design considerations preparation of cost estimation	Survey and designs shall be carried out in accordance with the design principles mentioned earlier following the standard designs available in the country and available best practices. The design will integrate climate resilience features to ensure that the roads remain usable throughout the year. Key elements of the design approaches including for climate proofing will be: • Embankments safe from expected maximum precipitation intensity. • Drainage provisions capable enough for local runoff during the expected intense flooding • Pavement materials flexible to adjust both increased heat in line with the increase in temperature	1.4.2.1 conduct technical survey of road alignment and other structures 1.4.2.2 study hydrological impacts under the CC context 1.4.2.3 produce technical designs for road upgrading (including structures) and cost estimation with climate proofing 1.4.2.4. Prepare bidding documents	Design, cost estimations and bidding documents

	and increased precipitation intensity. • Application of bioengineering and NBS as appropriate and feasible especially for erosion control of embankments, increasing infiltration capacities, etc.		
Activity 1.4.3 Improve 90 Kilometers of farm roads	The farm roads will be improved based on the planning and design process identified earlier. In addition, bridges and culverts will be rehabilitated in line with the new hydrological requirements resulting from climate change. The consulting firm/experts engaged by the NCDDS will be responsible for producing the technical design and cost estimations and prepare the required bidding documents. The roads will be upgraded within their existing width, respecting the current land use pattern while avoiding impacts to any historical and cultural sites. While the construction of the road works will be executed through hiring suitable local contractors, it will be supervised by both technical experts and community members to ensure ownership of the road after completion.	1.4.3.1 select contractors following standard bidding process 1.4.3.2 implement construction works in line with designs of 1.4.2	90 Kilometers roads upgraded and climate proofed
Activity 1.4.4 Handing over of the completed roads	The completed roads will be handed over to MRD. The MRD will commit to the operation and maintenance of the roads in line with the sustainable road maintenance program. This will be agreed prior to construction in an MOU between the project and MRD. The costs of the annual O&M will be estimated and indicated in the MOU. The local communities will be engaged in monitoring and reporting of the construction works and will be trained for the purpose by the consultants.	1.4.4.1 prepare road specific operation and maintenance material 1.4.4.2 Train local communities for inspection and monitoring and minor repair 1.4.4.3 Handover completed roads to the MRD	Completed roads are handed over to MRD

Output 2.1 Irrigation schemes and ponds modernized			
Activity 2.1.1 Technical analysis, field surveys and preparation of plans for system upgrading.	This is a preparatory activity necessary for the implementation of infrastructure upgrading. This includes engineering surveys, preparation of designs/plans, and procurement.	2.1.1.1 Additional field surveys required 2.1.1.2 Detailed engineering design 2.1.1.3 Tender document preparation 2.1.1.4 Procurement process	Technical analysis, field surveys, plans/ design for system upgrade, & procurement completed
Activity 2.1.2 Implementation of infrastructure upgrading.	This activity involves the actual upgrading/ construction of infrastructures in accordance with the engineering design/plans prepared in Activity 2.1.1	2.1.2.1 Construction of 6 schemes	All schemes finalized by year 6.
Activity 2.1.3 Preparation of canal O&M plans including application of ICT and SCADA for operation	An O&M plan is prepared for the proper operation and maintenance of the upgraded infrastructures.	2.1.3.1 Preparation of detailed operation and maintenance plans produced for each of the six schemes. 2.1.3.2 Guidelines produced for use of ICT/SCADA	- Canal O&M plans including application of ICT and SCADA for operation completed
Output 2.2 Flood-proofed and improved drainage			

Activity 2.2.1 Technical analysis, field surveys and preparation of plans for system upgrading	This is a preparatory activity necessary for the implementation of infrastructure upgrading. This includes engineering surveys, preparation of designs/plans, and procurement for the flood proofing and drainage system.	2.2.1.1 Additional field surveys required 2.2.1.2 Detailed engineering design 2.2.1.3 Tender document preparation 2.2.1.4 Procurement process	- Technical analysis, field surveys, plans/ design for system upgrade, & procurement completed
Activity 2.2.2 Strengthening and construction of flood control and drainage infrastructures.	This activity involves the actual upgrading/ construction of infrastructures for flood proofing and drainage in accordance with the engineering design/plans prepared in Activity 2.2.1	2.2.2.1 Construction work for flood-proofing and drainage systems	Flood proofed and drainage systems installed in all six schemes.
Output 2.3 Farmers Water User Communities (FWUC) established and trained			
Activity 2.3.1 Formation and institutional strengthening of the FWUCs	The FWUC creation and strengthening methodology is participatory and based upon the Government of Cambodia PIMD strategy that is outlined in the 5 training modules prepared by the FWUC Department formulated in order to guide implementation required under Circular One and Prakas 306. The FWUC curriculum is provided in Annex2 and incorporates lessons learned under the Northwest Irrigation Sector Project, the Water Resources Management Sector Program, as well as work under-taken by ISC and also the FWN	2.3.1.1 Preparation of FWUCs for all 6 irrigation schemes 2.3.1.2 Preparation of documents for establishment	- 11 FWUCs (3 in OTP, 4 in LH & 4 in KP) formed -
Activity 2.3.2 Build technical capacities of FWUCs for canal O&M	Service for the establishment of the operational FWUC from the participatory approach of FWUC creation in the design, development and sustainable use of the rehabilitated irrigation schemes and Preks	2.3.2.1 Institutional: dialogue with MoWRAM, legal recognition, responsibility sharing agreement, 2.3.2.2 Organizational: governance, internal operating rules for water and scheme	- Functional and operational FWUCs on canal O&M completed

	to be rehabilitated by the Program to the signature of shared responsibility for the management and maintenance of irrigation infrastructure between FWUC and MOWRAM for all the irrigation schemes under the project area. The support to the FWUC shall address the following issues as listed under sub-activities	management, agricultural development and marketing, 2.3.2.3 Technical: participation to the design of the irrigation networks, execution of maintenance and servicing tasks, farming practices. 2.3.2.4 Financial: preparation of budgets, fundraising, accounting management, Irrigation Contribution Service (ISC) planning and collection, access to financial institutions, 2.3.2.5 Gender focus with strong action for women's empowerment and their social participation.	
Output 2.4 Water information system (CIEWS and SCADA) established and operational in three sub-projects.			
Activity 2.4.1 Preparing all processes for establishment of the SCADA and Water information system	One of the most important problems facing the agricultural sector, irrigation is indiscriminate and others regularly, causing wastage of large quantities of water. This problem can be considerably improved by adopting the SCADA model because of the advantages of real-time information on water levels and flow rates and the possibility to respond to fluctuations more accurately and timely. This is an emerging approach to the modernization of existing systems with basic low-cost to solve this problem using the SCADA system, which provides command and control and collect data on irrigation to rationalize and streamline the process of irrigation. This model is commonly adopted by the irrigation schemes around the world for the modernization of old manually gated systems. The development of irrigation systems strongly contributes to the socio-economic development, but there are still many limitations in these systems. For example, although operation and management of the systems play an important role, there is a little concern on this. There is a lack of training and higher education for human	2.4.1.1 Design of the system 2,4,1,2 Configuration and design of software 2,4,1.3 Installation and set-up 2.4.1.4 Performance assessment and acceptance	- Water information and SCADA system installed in all three sub-projects

	resources in the management and operation. Therefore, modernization for the management of irrigation systems is indispensable.		
Output 3.1 Training and capacity building support provided to MOWRAM			
Activity 3.1.1 Strengthening River basin management and Water Accounting Systems	Adequate water accounting by Cambodia's river basin managers is becoming more and more important with the increasing development of irrigation. CAISAR's investment in irrigation development envisages improved water availability to and throughout each of the schemes. This will require careful management of water resources storage, allocation, and delivery, and careful distribution of delivered water to the farmer/water users. Therefore, CAISAR will support capacity strengthening for water accounting in the river basins where CAISAR will rehabilitate and climate proof the irrigation infrastructure	3.1.1.1 Prepare and implement training courses in river basin management 3.1.1.2 Prepare and implement training courses in water accounting systems	- Trained staff in river basin management and water accounting and operational in implementation
Activity 3.1.2. Communicating with stakeholders	MOWRAM manages the water resources for the benefit of its stakeholders. The largest water users are the irrigation schemes, each managed by an FWUC. The Commune Leadership, the Village Leaders, and the FWUC Leadership (Committee) represent the interests of the citizens/farmers/water users. At a higher level, District and Provincial Leaderships seek for the public interest to be served as good as possible.	3.1.2.1 Training courses in stakeholder involvement 3.1.2.2 Training courses in communication processes and involvement	- Trained staff in communication processes and stakeholder involvement
Activity 3.1.3. Support to and guidance of Leaders of Commune, Village and FWUCs and Gender Awareness	The CAISAR activity will support MOWRAM and the four PDWRAMs in reviewing the current FWUC support system in the project-related schemes and to plan/implement improvements where desirable and feasible. This activity will need to see consultations with District, Commune, Village, and FWUC Leaderships. A next stage of project preparation will identify the capacity building needs in detail, using a participatory assessment approach.	3.1.3.1 General support to FWUC and gender awareness and inclusion	MOWRAM and FWUC members trained in gender awareness and inclusion

Output 3.2 Training and capacity building support provided to NCDDS at national and provincial levels			
Activity 3.2.1- Strengthening of NCDDS at the national level	The National Committee for Sub-National Democratic Development (NCDD) in Cambodia plays a crucial role in promoting democratic development through decentralization and deconcentration of reforms throughout the country. To effectively carry out this mission, the secretariat of NCDD (NCDDS) has been actively engaged in capacity development initiatives aimed at enhancing the skills and knowledge of sub-national government officials.	3.3.1.1 Training national level staff in relation to project management capacity, policy formulation and implementation, and institutional strengthening and coordination	Overall strengthening of NCDDS staff in policy coordination and implementation skills
Activity 3.2.2 – Strengthening of the provincial teams	NCDDS and provincial staff has the responsibility of overseeing and implementing policies and activities aimed at increasing agricultural productivity, better market access and gender smart activities and thereby improving livelihoods of farmers. However, it is often found to have weak links and ineffective due to a lack of resources, skilled personnel, and coordination mechanisms between different levels of government.	3.3.2.1 Conduct a detailed capacity development needs assessment 3.3.2.2 Training in technical concepts and cross-cutting subjects such as policy analysis, project management, market systems development and use of ICT tools. 3.3.2.3 Training provincial staff in sustainable agriculture, climate change adaptation and Information and communication technology (ICT)	Provincial better equipped to implement and manage climate change and resilience projects

E.7. Monitoring, reporting and evaluation arrangements

169. **Reporting from MOWRAM to GCF**: Programme-level monitoring and evaluation will be undertaken to meet the information needs of the key financing and implementing agencies and will be in compliance with IFAD, GCF, AIIB and RGC policies. As the main implementing agency, MOWRAM will ensure the existence of a well-designed, operational and effective supervision, monitoring and reporting system to analyze and quantify the causal and attributable change of the project. In particular, the monitoring and reporting system will cover all project components. MOWRAM will be responsible for coordinating and preparing a consolidated report for submission to all key stakeholders through the Project Steering Committee.

170. **Monitoring and Evaluation Framework/Plan**: The Logical Framework matrix, as elaborated in the Funding Proposal, shall be the main basis for the monitoring and evaluation of the project. The M&E plan will include a set of indicators that can be measured and monitored over time to track progress and identify areas for improvement. This will include monitoring of all project activities, outputs and outcomes. The monitoring of CAISAR will involve (i) process monitoring; (ii) monitoring of the implementation of Resettlement Plan; (iii) monitoring progress in construction of civil works; (iv) monitoring of the implementation of Environment Management Plan; (v) gender and social dimension monitoring, (vi) monitoring of the implementation of capacity building/training plan; and (vii) monitoring of compliance with loan covenants. M&E forms/matrices shall be developed for each of these aspects.

171. **Key Indicators**: Project performance monitoring and evaluation will aim to ascertain the success of the project in achieving the intended outputs and outcome through their respective performance indicators. These indicators will allow management and stakeholders to assess the extent to which the project is meeting its expectations and decide what actions ought to be taken to redress any failure to meet targets. The project team is considering additional digital tools to better monitor specific issues arising under the Project and use of such digital tools will be subject to Cambodia's data protection and data privacy regulations.[101]

172. Some of the key indicators for CAISAR will include training and adoption rates, changes in cropping pattern and diversification of crops, market integration and performance of the 4P forums and multi-stakeholder forums and the volume of produce and contracts negotiated, households reached through early warning systems and its reported benefits. Under component 2 the project will monitor the resilience of the irrigation infrastructure especially the flood defense and drainage systems, solar pump performance, durability and effectiveness of concrete irrigation canals, effectiveness of geotextiles in reducing erosion and improving water flow, improvement of transport links alongside canals, improvement of drought and extreme flood resilience, and impact on gender equality, etc. The performance of FWUCs will also be monitored and their success in putting in place mechanisms for improved governance and recovery of fees for irrigation system maintenance. The project monitoring system will track the greenhouse gas reduction benefits resulting from the adoption of alternate wetting and drying (AWD) in rice production. The monitoring system will use data-driven technologies such as Smart SCADA technology to monitor the system for leaks, flood damages, and other issues. Regular surveys and assessments will also be conducted to gather feedback from farmer water

[101] The augmented reality package constitutes of virtual 3D glasses connected to smart phone cameras onsite with a proprietary communication software to be purchased during Project preparation.

user communities (FWUCs) on the effectiveness of the irrigation system and its impact on their livelihoods.

173. **Types of Reports:** The M&E Specialist in the Programme Management Unit will coordinate and consolidate all reports from different implementing partners including a M&E Specialist in the NCDDS PIU and will be responsible for transmitting annual progress reports to IFAD, AIIB and the GCF Secretariat. GCF will be provided the following reports: i) an inception report; ii) Annual Performance Reports (APRs); iii) an interim evaluation; and iv) a terminal evaluation. The PMU, with the support of the PMIC, will provide six-monthly updates/reports on the project's implementation and performance which will be consolidated to provide APRs. The status of achievement of deliverables of activities/ sub-activities, and performance targets of the project's outputs will be provided in the six-monthly updates. AIIB/GCF/IFAD teams will coordinate with the PMU and the PIUs through regular progress meetings. The team will also conduct quarterly, six-monthly or annual review missions as required. The corrective actions will be agreed with the executing entity and recorded and subsequently implemented by the EE and monitored by AIIB/GCF/IFAD.

174. **CAISAR Safeguards and Gender Action Plan reporting**: The project places particular importance on ensuring the monitoring of differential impacts by gender especially monitoring components relevant to women and more vulnerable populations. Furthermore, the monitoring system will ensure monitoring of the action plans for gender and the social and environmental framework to safeguard their interests. This will ensure a comprehensive and holistic monitoring system.

175. **Interim and final evaluations:** IFAD has a well-structured system for undertaking annual supervision missions, a mid-term review and a project completion report. The project will undertake surveys at mid-term and at completion to assess the performance of the project, draw important lessons and incorporate beneficiary feedback. The external surveys will feed into these review reports. The interim or mid-term survey will incorporate key aspects of project performance up to that period and which will be incorporated in IFAD's Mid-Term Review Report. At project completion, a final evaluation will be undertaken to assess the overall performance and achievement of the outputs and outcomes. The mid-term and final evaluation will compare project results with the expected outreach, adoption of climate adaptation practices and assess the overall performance and the paradigm shifts outlined in the project log-frame and the indicators of resilience outlined. The project completion review will also assess the extent to which the intervention has contributed to the GCF's higher-level goal of achieving a paradigm shift in adaptation to climate change in the project districts.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Selected Risk Factor 1: Delay of Approval by GCF Board

Category	Probability	Impact
Technical and operational	Medium	Medium

Description

176. There is a risk related to the timing for approval of CAISAR by the GCF Board which is expected to be undertaken in 2025. Delay in this approval could impact the plans of MOWRAM, the financing schedule of IFAD and AIIB and impose additional costs and impact the financial returns from the project.

Mitigation Measure(s)

177. The NDA, IFAD and AIIB will try and expedite the process of design and approval to ensure that the FFP is submitted according to the agreed schedule and will endeavour to address all comments received from GCF as efficiently as possible.

Selected Risk Factor 2: Technical & Operational Capacity

Category	Probability	Impact
Technical and operational	Medium	Medium

Description

178. The institutional assessment report prepared by PPIC and WAPCOS, December 2022 noted that the weak and insufficient capacity of human resource at all levels of the irrigation management system is the primary cause of the slow rural development in Cambodia. The need for adequately trained professionals in water resources and irrigation management in Cambodia is a key issue. The capacity building can be performed through education, training, workshops, etc. MOWRAM will need enhanced capacity for improved water management and governance to develop a comprehensive water management plan that includes equitable water distribution and allocation and promotes efficient irrigation practices, such as drip irrigation and sprinkler systems, to optimize water use.

Mitigation Measure(s)

179. Based on the institutional assessments report of the Water Resources Management Sector Development Programme (WRMSDP), the training for MOWRAM and PDWRAM will include all personnel at the central and provincial levels that are involved in the CAISAR project schemes. CAISAR will also include an institutional strengthening component for MOWRAM. For improved implementation performance the training plan will cover topics such as (i) Project Orientation and Presentation of Project Plans; ii) FWUC Formation and Strengthening; iii) Gender Awareness and Gender Action Plan; iv) Data Management for Hydro and Meteorological Data and O& M for Meteorological Equipment; v) Environmental Awareness; vi) Construction Management and Supervision and vii) On-Farm Water Management and Operation and Maintenance.

Selected Risk Factor 3: Policy and Regulatory Framework

Category	Probability	Impact
Technical and operational	Medium	Medium
Description		
180. Inadequate policies and strategies to mainstream climate change agendas. While Cambodia has made impressive progress in formulation of climate change strategies and policies, mainstreaming of climate change across water and agriculture sector is still weak.		
Mitigation Measure(s)		
181. The policy efforts indicate that the public sector is aware of climate risks and the importance of mainstreaming climate agenda and is motivated to address those issues. The project aims to assist the line Ministries in revising its policy frameworks especially in water and agriculture sector mainstreaming from the climate change perspective / ensure climate change considerations are reflected in their projects and programmes. Capacity support to NDA and other relevant stakeholders have been proposed—through activities under Component 3—to improve NCDD-S's capacity (a GCF Direct Access Entity) to manage projects (particularly climate finance projects) and implement effective policies and programs at sub-national level, improve technical capacities of provincial teams on sustainable agriculture and climate change adaptation, etc.		
Selected Risk Factor 4: Weak coordination among institutions.		
Category	**Probability**	**Impact**
Technical and operational	High	Medium
Description		
182. Lack of coordination at the ministerial level is a common issue in the country and also impacts the working between the institutions in the water and agriculture sector, and between national and sub-national levels. This could slow down the project process with implications for cost and efficiency.		
Mitigation Measure(s)		
183. A robust coordination mechanism has been proposed establishing a Project Steering Committee. The PSC will have members from all key ministries involved in project implementation. The PSC will meet regularly and will address the issues related to coordination both at the central level and at field level to facilitate smooth project implementation. In addition, there is a clear division of responsibility of the agencies involved in the implementation between MOWRAM and NCDD-S. Initial orientation sessions, clear terms of reference and regular meetings between implementing partners at the provincial level will further help to reduce these risks.		
Selected Risk Factor 5: Procurement Delays.		
Category	**Probability**	**Impact**
Technical and operational	Medium	Medium
Description		
184. A principal reason for delay of projects during implementation is the delay in establishing procurement arrangements, prolonged procurement procedures or failure to sign contracts for delivery of the goods and services.		
Mitigation Measure(s)		
185. CAISAR will establish a dedicated team of national and international procurement experts to ensure procurement is timely and is approved as efficiently as possible. IFAD, AIIB and		

MOWRAM will closely monitor progress and provide guidance to ensure that IFAD, AIIBs and RGC's procedures are adopted. Project will adopt advanced procurement action for detailed engineering designs of irrigation schemes to ensure early project readiness. It will launch recruitment processes of staff and key service providers, timed so that key staff and key service providers will be contracted in a timely after the project effectiveness date. The project design includes a monitoring strategy and contingency for adjustments in the event either recession/inflation changes demand or important materials' cost inflate to a level that impacts the project stakeholders.

Selected Risk Factor 6: Social and Environmental

Category	Probability	Impact
Other	Medium	Medium

Description

186. Several social and environmental risks have been identified in the ESMF (Annex 6). These risks can negatively impact the project environment and the livelihoods of local population in various ways. There is a risk of insufficient consultation / engagement with stakeholders on key project interventions resulting in insufficient buy-in on project objectives and activities or dissatisfied stakeholders, which may eventually undermine project delivery and sustainability of project activities—for instance, operations and maintenance of project-supported infrastructure long term. The target populations and ecosystems are highly vulnerable to the impacts of climate change and environmental issues.

Mitigation Measure(s)

187. A SECAP mission was deployed to identify the potential social and environmental risks prior to project design which identified the frameworks and principles that would be adopted in case of any negative impact due to displacement or infringement of rights of any group or households. The project's ESMF (Annex 6) lays out the overall strategies to mitigate or minimize any harmful social and environmental risks, while also enhancing positive impact. A Stakeholder engagement plan (Annex 7) prescribes specific measures to ensure that project activities reflect their interests and needs, and to address any negative impact. A detailed gender action plan (Annex 8) highlights the inclusion and activities for women. To avoid the risk that effective grievance channels are not established for investigation and redress of stakeholder grievances, a GRM has been prepared. It includes plans for a multi-level GRM to be established and information made available to project stakeholders and affected persons through multiple channels. The project will not increase the exposure or vulnerability of target populations' livelihoods, ecosystems, economic assets or infrastructure to climate variability / hazards or result in ecosystem degradation. A rigorous climate risk analysis informed the selection of climate-adaptive investments in infrastructure, agricultural development and institutional strengthening, particularly to avoid risks of mal-adaptation; It will support resilience and adaptive capacity of smallholders by mitigating the negative impact of extreme climate events (reduced losses) and through improved incomes and livelihood diversification. Analysis of climate hazards, vulnerability and exposure will continue to inform planning processes during implementation, ensuring reduced vulnerability of irrigation and agricultural. These strategies and efforts will be further contextualized into ESMPs at project inception to implement targeted mitigation actions (e.g., occupational health and safety, risks associated with land acquisition, integrated fertilizer/pesticide management), as required.

Selected Risk Factor 7: Private Sector Engagement

Category	Probability	Impact
Other	Medium	Medium

Description

188. The private sector has limited incentives to engage with smallholder farmers due to the fragmented nature of their production and limited demand for their inputs.
Mitigation Measure(s)
189. CAISAR proposes to enhance demand for climate adaptive technologies and practices which will entice the private sector to engage more with the rural economy. The establishment of the PPP platforms, grants to MSMEs and multi-stakeholder forums is another mechanism that can be used to systematically engage the private sector for input supply and access to markets. In general, through policy engagement, IFAD will continue to support the government to improve ease of doing business in the agriculture sector.

Selected Risk Factor 8: Risk of money laundering, terrorist financing, or prohibited practices		
Category	**Probability**	**Impact**
Other	Medium	Medium
Description		
190. IFAD conducted financial management and procurement assessments during the project design phase. The assessments concluded that the executing entities (EEs) had the capacity to maintain financial management and procurement arrangements that meet IFAD"s standards. However, additional measures were agreed upon to further enhance the fiduciary capacity of the executing entities.		
Mitigation Measure(s)		
191. Proposed mitigation measures include (i) include IFAD's templates of self-certification forms for bidders and contractors on anticorruption, sexual harassment, sexual exploitation and abuse in the bidding document/RFP and contract documents; (ii) request bidders and contractors signing the Self-Certification Forms as a part of bids/proposals and contract documents; (iii) conducts screening of IFAD's counterparties against sanctions lists (using Frico Compliance Link) for all procurement activities; and (iv) The PIM will include detailed procedures for the prevention of money laundering and other prohibited activities. Compliance with these provisions will be monitored by IFAD. Further, no individual or entity listed on any UN Security Council sanctions list, including the UN Consolidated Sanctions list, will be involved in any manner with the project or its activities, either as a counterparty, implementer, beneficiary. The financing agreement will have provisions that "Any payment prohibited by a decision of the United Nations Security Council taken under Chapter VII of the Charter of the United Nations, shall not be eligible for financing by the Financing." IFAD's procurement regulations have relevant guidelines to monitor compliance with relevant laws and regulations. 192. Under IFAD"s Anti-Money Laundering and Countering the Financing of Terrorism (AML/CFT) Policy, IFAD has implemented a leading financial crime screening software solution, Firco Compliance Link (https://firco.ifad.org/). The software conducts screening of IFAD's counterparties against sanctions lists adopted by IFAD in addition to AML/CFT industry-standard databases that contain enforcement records and adverse media, to assist with Third Party Integrity Due Diligence (IDD). All IFAD personnel can run ad-hoc, or on demand, screening searches on potential counterparties IFAD is seeking to engage with, such as individual consultants, vendors, suppliers, private sector and other partners. The software has been integrated into IFAD's Online Procurement End-to-End system (OPEN). The screening will take place automatically for all procurement activities entered in OPEN by the borrower and covering any applicable exclusion and restrictive measures lists, adverse enforcements and assessment and recommendation based		

on IFAD internal policies and procedures. Results of the screening process will be displayed in OPEN and will support the project team and the IFAD country team to agree on the way forward.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

193. **Risk Category Assignment and Rationale:** The proposed environmental and social risk assignment for the project is Category A which was determined based on a comprehensive screening and assessment of E&S risks and impacts that may arise from proposed project activities. The risk assessment was conducted vis-a-vis the IFAD, AIIB, and GCF requirements. It is noted that risk assessment takes into account the followings: (i) nature, scope and scale of proposed project activities (particularly those under Component 1 and Component 2) – indicative through direct, indirect, and potential cumulative impacts; (ii) vulnerability of potential project affected people (both intended beneficiaries and adversely affected groups) as to project impacts, (iii) sensitivity of biodiversity present in the project's area of influence, and (iv) broader socioeconomic and environmental settings in the project sites (that bear inherent E&S risks). The risk assignment accounts for the likelihood of each risk materializing, and the severity of an impact on the achievement of the intended project results. It is noted that the risk assessment considers the effects of the E&S risk mitigation measures already utilized in the project's areas of influence and does not assume any future additional mitigation measures that are outside of the measures proposed in the above E&S instruments.

194. **Environmental and Social Assessment Instruments** - To manage the identified E&S risks and impacts effectively, a number of E&S instruments have been prepared in line with the requirements of AIIB, IFAD, and GCF. In line with that, at project level, three frameworks have been prepared (including Environmental, Social, Climate Management Framework (ESCMF), Land Acquisition and Resettlement Planning Framework, Stakeholder Engagement Plan). Two detailed assessments, were also conducted at project level, including, such as an Environmental, and Social, Climate Impact Assessment (ESIA) and Gender Analysis and Social Inclusion Plan. At the sub-scheme level, three Environmental, Social, Climate Management Plans have been prepared for three selected schemes: Ou Ta Paong, Krapeau Trom, and Yutasas,. These instruments have been developed based on a series of activities: : (i) **risk screening** (based on the proposed project activities against the IFAD's Social, Environmental, and Climate Assessment Procedures (SECAP), AIIB's Environmental and Social Framework (ESF) and the Green Climate Fund's (GCF's) Environmental & Social Safeguard Policy; (ii**) risk assessment** (through extensive field studies in 2022, 2023 and 2024 which involved consultation with more than 2,500 people at provincial, district, commune and village levels, (iii) **development of mitigation measures.** Risk assessments were conducted based on both primary data (those collected from field survey and stakeholder consultation for project purpose) and secondary data (that were obtained from desk research of E&S risks and impacts in similar projects in the region and other parts of the country.

195. **Main Outcomes ESCMF and Mitigation Measures** - The E&S instruments highlight environmental and social risks and impacts that are potentially associated with activities proposed under project component 1 and component 2. **Environmental risks and impacts** were identified by project stage: before construction, during construction, and during project operations. Before construction starts, environmental risks identified include unexploded ordnances (UXO). During construction stage, environmental pollutions that are caused by construction operations, include noise, air, water, soil pollution. These pollutions are anticipated to be temporary, localized, and as such manageable. During construction process, potential impacts on biodiversity is envisaged, particularly for species categorized as endangered or critical endangered by IUCN, and are exposed to impacts causes during construction, including physical and non-physical impacts from physical construction activities, and from human (e.g. project workers). The risks for biodiversity (e.g. hitting, hunting for trading and/or consumption) are anticipated in the form of direct and indirect impacts (during project construction), and in the form of cumulative impacts (over years

during project operation and due to combined effect caused by present and planned neighboring projects. Risk of greenhouse gas emission are identified as cumulative impact (during project operation) if proposed mitigation measures are not effectively taken. Health risk of farmers directly exposed to increased pesticide use is also envisaged as a cumulative environmental impact.

196. **Social risks and impacts** were also identified by project stage and are mostly associated with activities under project component 2. Before construction, permanent and temporary land acquisition is anticipated at certain places – such as at location where existing canals are widened for improved water intake (for feeder canals), and for improved water distribution (for secondary and tertiary canals). Based on the pre-feasibility studies, efforts are being made to avoid permanent land acquisition – through exploring engineering design options. So, minor permanent land acquisition (linear impact) is anticipated at canals or rivers subject to widening. Loss of permanent land (which is small scale) is anticipated to cause temporary economic displacement for households who lose part of their cropland permanently (if avoidance is not technically feasible). During construction, some risks are anticipated, including risks of disease transmission due interaction of project's labor force and local community members. The risk related to Sexual Exploitation and Abuse; Sexual Harassment is identified – due to presence of contractors' workers in the locality. The contractors' workers are also exposed to the risk related to occupational health and safety (during construction process). Risk related to low pay and unequal treatment on the part of workers hired locally is envisaged. Under component 1, project activities that support local farmers to enhance crop production through access to improved technologies may give rise to risk of elite capture and unequal water distribution during project operation stage may increase the risk of social inequality – particularly between water user groups located upstream vis a vise downstream. During operation stage, increased use of chemical inputs, and subsequent exposure to these chemicals during use, may increase health risk, particularly for those who expose themselves during pesticide spraying.

197. **Mitigation measures**. Mitigation hierarchy is applied where foreseen impacts are avoided through technical engineering design options. Where avoidance is not possible, risks and impacts are minimized through proposed mitigation measures, and further mitigated, or compensated for if residual impacts remain (See ESCIA for details). Mitigation measures are proposed in ESCIA, and ESCMPs for the three sub-schemes of Ou Ta Paong, Krapeau Trom and Yutasas. The SEP, to guide how project stakeholders are identified and engaged reiteratively in a meaningful consultation process during project implementation to receive timely feedback from those who think they are affected adversely by certain project activities. The project also has a Grievance Redress Mechanism (GRM) in place to enable affected people to raise their concerns and complaints to the project owner – through project's grievance procedures that are easily accessible and used.

198. **Due Diligence for Financial Intermediations** - The ESCMP will be implemented with safeguards specialists based at the central Project Management Unit, and E&S focal points to be appointed at the Provincial Project Implementation Units (PIU's). Contractors will adhere to relevant requirements of GCF, IFAD, AIIB and national laws and regulations. Capacity building, support will be provided to MOWRAM/ PMU and PDWRAM to strengthen their capacity to expedite project implementation and ensure effective project management. The capacity building will also cover (i) overall Project management, (ii) establishment of FWUCs and (iii) capacity building of FWUCs to ensure improved water use coordination. Where required international and national consultants will be engaged to fill and build capacity of MOWRAM/ PMU and PDWRAM.

199. **Capacity and Compliance Monitoring** - Capacity building is prioritized, and training is planned for all involved parties. Reporting and supervision mechanisms are outlined to ensure compliance. Topics include: (i) Project Orientation and Presentation of Project Plans; (ii) FWUC Formation and Strengthening; (iii) Gender Awareness and Gender Action Plan; (iv) Data Management for Hydro and Meteorological Data and O& M for Meteorological Equipment; (v) Environmental Awareness; vi) Construction Management and Supervision and (vii) On-Farm Water Management and Operation and Maintenance.

200. **Stakeholder Consultations -** Public consultations occurred at various project stages, including feasibility studies and during preparation of E&S instruments. Various rounds of stakeholder meetings were held as part of the preparation of Environmental, Social, and Climate Impact Assessment (project level) and Environmental, Social, Climate Management Plans (sub-scheme level. Continuous stakeholder engagement will be undertaken throughout CAISAR implementation and align with construction schedule – during which most impacts are anticipated.

201. **Safeguards Reports Disclosure -** Safeguards documents will be disclosed locally and on official websites of MOWRAM, GEF, IFAD and AIIB in Khmer and English as per disclosure requirements of the respective institutions

202. **Indigenous Peoples -** An Indigenous Peoples Planning Framework is in place, with mitigation measures for IP-related risks. Screening conducted during project preparation does not find any Indigenous Peoples who are present in the project's area of influence (for six sub-schemes). During project implementation, additional consultation will be carried out – based on the areas of influence to be confirmed when detailed engineering designs are final.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

203. **Gender assessment.** A gender assessment has been conducted in accordance with the principles and requirements of GCF's Gender Policy. The assessment examines the status quo of gender dimensions in agriculture and livelihood activities in the project area. It aims to (i) identify aspects of inequality between men and women and opportunities to promote gender equality through project investment; (ii) assess vulnerability of project population, with a focus on vulnerable and disadvantaged groups; (iii) propose measures to engage them in a meaningful consultation throughout the project cycle, particularly in investment and decision-makings that affect them; (iv) propose a gender action plan that ensures transparency, non-discriminatory access for vulnerable and disadvantaged groups while fostering accountability of project stakeholders during project implementation. The gender assessment was carried out based on the inputs gathered from key project stakeholders during project preparation. The consultation process also made use of a household survey which included 630 households for quantitative analysis (2022), and through focus group discussions and community meetings for qualitative review in the second half of 2023. Feedback was obtained on constraints and opportunities for promoting gender equality in the project area through project investments and how to enhance the resilience of the target group in addressing the climate change impact and propose priority intervention for the gender action plan. Annex 7 gives details of stakeholder consultations that were conducted during the preparation of the funding proposal.

204. **Key findings of the Gender Assessment.** The findings confirmed that in rice production, which is the project's major intervention, women participation was lower compared to men), particularly in production stages that are labor intensive (e.g. land preparation and harvesting). The more arduous tasks in rice farming have undergone mechanization. In project area, 50-70% of total farming households use machinery for land preparation (60%) and harvesting (70%). Women are involved mostly in unpaid household chores and small-scale animal husbandry. Yet they also play a key role in farming by contributing to seed preparation, seedling, crop care, weeding, and trading. It is estimated that male workers spend on average, 59.4% of their time on farming while female counterparts spend 40.5% of their time on farming-related tasks in addition to their household chores. This has implications for the adoption of climate-resilient technologies since socially determined gendered patterns of labour distribution in farming can influence preferences and adoption. With regard to household decision-making, many women in the project area participate actively in livelihood and budgeting decisions. Yet, major decisions concerning larger investments are still taken by men.

205. Addressing gender issues is essential in promoting and advancing the role and economic, social, political, legal and cultural status of women. The need to address this concern has been

increasingly acknowledged by the government of Cambodia since it would help to improve and sustain not only family economy but also national development and economic growth. Under IWRM and Participatory Irrigation Management and Development (PIMD) policies, women are being encouraged to gradually participate more in village and community water management issues. However, due to limited capacity and socio-cultural and physical constraints, only a few have been able to fully join farmer water user communities (FWUCs) as well as in disaster risk reduction activities. The study found that people prefer to have more women in management and development roles in local communities and in public work (such as FWUCs or commune councils) because women understand their issues and can solve some critical issues (water use conflict, money or irrigation service fee collection and management, women's issues and needs) better than men. Rural women have notably participated in meetings or training on women's rights, saving groups and agriculture. They are actively involved in their communities. Furthermore, they are well aware of their rights in society and are able to settle complicated issues of their families or communities. Compared with the past, women have become more active, committed and able to put forward their views. However, gender imbalance in FWUCs still exists, since women have many household chores to perform. Because of this, only a few women have the time to join FWUCs and village and commune development activities, particularly in water and irrigation management or in disaster risk reduction. At the same time, even though they are committed, women still lack capacities, experience and skills in this sector. These barriers have affected the ability of women to join in public affairs.[102] CAISAR will deal with some of these issues and set up the FWUCs in a manner that encourages their participation.

206. **Proposed intervention.** A gender assessment and a Gender Action Plan (Annex 8) has been developed for CAISAR. Key activities to support the implementation of the project's GAP include training of extension staff and implementing partners on gender and social inclusion, including the use of tools and approaches to promote normative and behavioral changes in gender dynamics at the HH and community level; exploring partnership with the Women Development Centers for delivering training to women; Developing gender-sensitive diagnostic tools and manuals for value-chain analysis and other diagnostic activities. Women's role in rice production is limited therefore project beneficiaries will be encouraged to diversify their income sources, and increase their household income by growing vegetables, raising poultry and aquaculture. A gender-sensitive study will be conducted at project start to identify opportunities for women along the value chains of interest to them. Interested beneficiaries will be trained and facilitated to enter into a business partnership with private sectors to supply the commodities to the market. The specific prongs of the gender strategy will include the following:

- **Encouraging women's Role in Irrigation Management and Development in the Context of Climate Change:** Under IWRM and participatory irrigation management and development (PIMD) policies, women will be encouraged to gradually participate more in village and community water management issues. CAISAR has established a quota for women's participation in the FWUCs.

- **Strengthening Women's Capacity in Adaptation to Climate Mitigation & Adaptation**: Women will be encouraged to participate in project trainings that help them address the potential impact of climate change which could include a) adoption of water-saving alternate wet and dry rice cultivation, b) adopt new seed varieties (which are more pest resistant and drought tolerant), c) adopt sustainable intensification packages that reduce seed rate, fertilizers and pesticide to save agricultural input costs and increase crop yield. Women will be invited to take part in relevant trainings jointly with their husbands to support adoption of climate-resilient technologies that also respond to their needs and preferences.

[102] Gender and Water Governance: Women's Role in Irrigation Management and Development in the Context of Climate Chang

- **Strengthening women's role in decision-making both at the household and community level.** It is expected that the role of women in decision making at household level will be enhanced as the outcome from Item 1 and 2 above. At the household level, gender specific tools will be used to encourage joint decision-making in technology adoption and investment decisions. Women will participate in consultation activities and planning of irrigation infrastructures, value-chain development and early warning system. The project will also promote women's participation and leadership in FWUCs, which will be trained on how to enhance equitable access to water for women and other vulnerable groups.

- **Alleviating women's workload.** This will be done by **raising awareness** on the importance of sharing household chore and farming workload to give women more time to learn new skills (climate-resilient farming technologies, irrigation system, financial management, business) that are essential for them to enhance their adaptation capacity.

- **Promoting women's economic empowerment.** This will be done by enhancing women's access to skills and business development training including farmer's field schools that are centered around project target value chains for commodities such as rice, poultry, vegetable and fish that are appropriate to them. In addition, project beneficiaries will be encouraged to diversity their existing sources of income, to improve the efficacy of existing income sources such as home-based gardening, animal husbandry (poultry and fish), promoting women's access to grant schemes as well as employment opportunities through value-chain development and PPPs.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

207. **Procurement Arrangements:** The procurement principles to be followed by the CAISAR project have been agreed between MOWRAM, AIIB and IFAD in a series of meetings during the preparation of the project concept and the funding proposal.[103] Public procurement in Cambodia is governed by the Law on Public Procurement (2023), and the Implementing Rules and Regulations Governing Public Procurement. For donor-funded projects, the Standard Procedures on Procurement for All Externally Financed Projects and Programs (SOP) was promulgated in February 2007 and subsequently amended in 2012 and 2019. The revised SOP document took into consideration the revised procurement policies of donors such as ADB and the World Bank. The MEF, through the General Department of Public Procurement (GDPP) is responsible for administration of the Law on Public Procurement. The GDPP oversees the activities for domestically financed procurement. The General Department for International Cooperation and Debt Management (GDICDM) oversees the externally financed procurement.

208. A procurement risk assessment for CAISAR was conducted according to the IFAD Procurement Manual. This analysis shows that the overall inherent procurement risk is moderate. The overall residual procurement risk after mitigation measures is taken is also rated as moderate. The main project procurement risks include the following: (i) The absence of national e-procurement system; (ii) There is no integrated system to monitor procurement values and records at the General Department of Public Procurement; (iii) Governance risks including corrupt practices in procurement; (iv) Procurement processes are often delayed; (v) Limited knowledge of IFAD and AIIB procurement and safeguard policies and requirements.

209. Mitigation measures for project procurement risks include the following: (i) Upload information on procurement plans, contract awards and resolution of procurement complaints on IAs/project websites; (ii) Using IFAD's Online Procurement End-to-End system (OPEN) and ICP Contract Monitoring Tool for monitoring project procurement activities; (iii) Request

[103] Following missions in December 2020, January 2021, January 2023 and September 2023.

bidders and contractors signing the Self-Certification Forms on anticorruption, sexual harassment, sexual exploitation and abuse as a part of bids/proposals and contract documents; Conduct prior and ex-post reviews to strengthen enforcement of the debarment system; (iv) Monitoring closely procurement and contract implementation progress with periodic progress reports and follow-up actions; and (v) Carry out coaching and on-the-job trainings on IFAD and AIIB procurement and safeguard policies.

210. Procurement using AIIB/GCF/IFAD funds will follow the Standard Operating Procedures on Procurement for All Externally Financed Projects/Programs in Cambodia (SOP) to the extent such are consistent with the IFAD Project Procurement Guidelines, and AIIB's Procurement Policy and AIIB's Interim Operational Directive on Procurement Instructions for Recipients. Procurement using AIIB financing will be undertaken with oversight by AIIB.

211. NCDDS PIU will be directly responsible for project procurement activities of Component 1 and Output 3.2 under component 3. MOWRAM PMU will be directly responsible for project procurement activities of Component 2 and Output 3.1 under component 3. The NCDDS PIU will have a procurement officer, a procurement assistant and a procurement specialist. MOWRAM PMU will have a procurement officer, procurement assistant(s) and a procurement specialist.

212. A detailed Project Procurement Arrangement Letter (PAL) will be issued to establish the detailed procurement requirements and thresholds for procurement methods and IFAD prior review. The initial 18-month Procurement Plan (based on the draft 18-month procurement plan at the project design stage), and subsequently the annual procurement plans in consistency with the AWPBs shall be prepared and IFAD no objection obtained before proceeding with procurement activities. IFAD's Online Procurement End-to-end System (OPEN) will be used for preparation and monitoring implementation progress of project procurement plans. Requests for IFAD prior review and no objection shall be routed through the OPEN system. Data and information of all contracts and related payments must be listed and updated regularly in the IFAD Client Portal - Contract Monitoring Tool (ICP-CMT). The register of contracts must be kept up to date on a quarterly basis.

213. **Financial Management:** MOWRAM PMU and NCDDS PIU shall follow the Royal Government of Cambodia (RGoC)'s Standard Operating Procedures (SOP) as the principal guidance on financial management, each unit shall also develop a Financial Management Manual for the specific guidance on project financial management to meet IFAD and GCF requirements. The computerized accounting software shall be used as the main tool for bookkeeping, budget monitoring and financial reporting. NCDDS will send to MoWRAM the AWPB, quarterly and financial reports for consolidation and submission to IFAD and AIIB.

214. The key financial management risks are (i) Potential low capacity of financial management staff at implementing agencies, (ii) Lack of prior experience of MOWRAM with IFAD-funded projects, (iii) Previous external audit reports did not meet IFAD compliance standards, and (iv) Complicated disbursement and fund flow arrangements resulting from multiple financiers. The mitigation measures to deal with these risks include the inclusion in the design the following measures to mitigate the risks. Recruitment of qualified consultants who possess relevant experience and educational qualifications, development of a Project Implementation Manual (PIM) and a Financial Management Manual (FMM) that outline clear roles for funds flow and disbursement arrangements, as well as guidelines for filing supporting documents and maintaining accounting records for all implementing partners, customization of accounting software to facilitate accurate record keeping and financial reporting and close coordination with the Ministry of Economy and Finance (MEF) to ensure adherence to IFAD audit requirements.

215. Financial Management arrangements will include (i) project-recruited staff and staff seconded by the government with adequate qualifications and experience for the FM team; (ii) AWPB following GCF budget format; (iii) report-based disbursement modality to request funds from GCF and IFAD sources; (iv) comprehensive SOP and FMM for adequate internal controls; (v) customized accounting software for financial accounting and reporting, financial reporting and monitoring

adjusted to accommodate GCF reporting requirements; and (vi) external audit tailored to meet GCF requirements.

209. **Staffing:** MOWRAM is the Lead Project Agency, and NCCD is an implementing partner; MOWRAM and NCCD already have a finance unit with experience in international donor-funded projects that can support CAISAR. PMU and PIU will be set up for MOWRAM and NCCD with a combination of seconded staff and contract staff. The seconded Finance Officer will oversee the project's financial management to ensure the effective FM system and internal control are maintained in accordance with the procedures in SOP and PIM. The seconded Finance Officer will have relevant experience working on other internationally funded projects. The contract Finance staff will be hired to support project FM functions daily including: a) Ensuring sufficient supporting documentation and approvals for payment; b) Posting transactions to the accounting software; c) Providing periodical and financial reports to the management; d) submitting of IFR and WA to IFAD, providing other support to Seconded Finance Officer as required. The Finance staff will be recruited with relevant experience and qualifications. FM functions will have clear roles and responsibilities to ensure the segregation of duty and payment authorization in accordance with SOP and PIM.

210. **Planning and budgeting**: The project budget is separate from the government budget; however, it will be integrated into the Public Investment Plan; the procedures for budget preparation are described in the SOP, and the AWPB will also include the projection of counterpart contributions. AWPB is subject to approval by MEF before submission to IFAD for no Objection. NCCD-PIU will prepare the AWPB and submit it to MOWRAM-PMU; PMU will consolidate the AWPB and submit it to IFAD for NOL. The AWPB and procurement plan will be prepared in IFAD format with details and a summary of activities and categories. The budget will be monitored in the accounting software. PMU/PIU will ensure to keep track of the progress reports against the budget and provide required information for interim financial reports and annual financial reports.

211. **Internal control**: According to the PEFA for Cambodia, the effectiveness of internal audits was rated C. PMU and PIU will be set up separately from the government system and will follow the internal control procedures from the Government's Standard Operating Procedures (SOP), which is deemed to be effective for internationally funded projects. The SOP articulates the responsibilities and critical elements of internal control, financial delegation, transaction processing, and computerized accounting system controls. PIM and Financial Management Manual will outline key FM arrangements and controls. The Internal Audit Department of MOWRAM and NCCD will conduct the internal audit of PMU and PIU annually to ensure compliance with the internal control procedures; the outcome of the internal audit will be shared with IFAD.

212. **Accounting systems, policies, procedures, and financial reporting**: PMU and PIU accounting procedures will follow Standard Operating Procedures (SOP) for internationally funded projects. According to SOP, CAISAR will apply the Cambodia Public Sector Accounting Standard (cash basis), which is based on the International Public Sector Accounting Standards (IPSAS). Standard accounting software shall be used for PMU/PIU that will be compatible with IFAD and other donor's financial reporting requirements. Standard Government's chart of accounts will be set up in the accounting software to capture all payment transactions; in addition, project components, categories, sources of funds, and other segments will be set up and recorded for the transactions in the software to ensure financial reports and budget reports produced in accordance with IFAD and Government requirements. The details of the accounting process and records are outlined in the SOP. All accounting records are required to be maintained in electronic format for a period of 10 years after project closure. MOWRAM PMU will be responsible for the consolidation of financial reports from NCCD-PIU for each quarter; within 30 days, NCCD PIU will send to MOWRAM PMU the interim financial report (IFR) in the agreed format that will finally be consolidated and submitted to IFAD for justification of advance and requesting new advances.

213. **External audits**: CAISAR project financial statements will be audited by a private audit firm appointed by MEF, with the TOR prepared by MEF taking into account the IFAD requirements. The audit will be conducted in accordance with Cambodian Auditing Standards, which is adapting the International Standards on Auditing. The unaudited annual Financial Statement will be submitted to IFAD within 4 months after the end of the fiscal year; the audited financial statement will be submitted to IFAD within 6 months after the end of the fiscal year. The management letter with an assessment of the project's internal control matter will be included as an integral part of the audit report.

214. There will be a single subsidiary agreement between IFAD and the RGC, represented by the MEF. In the Subsidiary Agreement (SA) with the Executing Entity (EE), it will be emphasized that the GCF has the right to conduct ad-hoc checks on the use of its funds and perform ex post evaluations. The budget allocation between MOWRAM and NCDD-S will be based on the prepared cost tables.. MOWRAM will create and manage four designated accounts for each funding source (AIIB, IFAD, GCF Loan, and GCF Grant). Each funding source will have its own designated account. Additionally, NCDD-S will maintain three designated accounts for the GCF proceeds (Loan and Grant) and IFAD co-financing. The Financial Management Manual (FMM), which will be attached to the Project Implementation Manual (PIM), will specify the financial data requirements and the corresponding timing for the submission of interim financial reports, annual financial statements, project audit reports, GCF semi-annual reports, and GCF annual performance reports. The MOWRAM Project Management Unit (PMU) will prepare withdrawal applications for the project (GCF and IFAD funds), which will then be approved by MEF officials.

G.4. Disclosure of funding proposal

Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.

Indicate below whether or not the funding proposal includes confidential information.

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.
☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

☐	Annex 1	NDA no-objection letter(s) **(template provided)**
☐	Annex 2	Feasibility study - and a market study, if applicable
☐	Annex 3	Economic and/or financial analyses in spreadsheet format
☐	Annex 4	Detailed budget plan **(template provided)**
☐	Annex 5	Implementation timetable including key project/programme milestones **(template provided)**
☐	Annex 6	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): **(ESS disclosure form provided)** ☐ Environmental and Social Impact Assessment (ESIA) or ☐ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☐	Annex 7	Summary of consultations and stakeholder engagement plan
☐	Annex 8	Gender assessment and project/programme-level action plan **(template provided)**
☐	Annex 9	Legal due diligence (regulation, taxation and insurance)
☐	Annex 10	Procurement plan **(template provided)**
☐	Annex 11	Monitoring and evaluation plan **(template provided)**
☐	Annex 12	AE fee request **(template provided)**
☐	Annex 13	Co-financing commitment letter, if applicable **(template provided) 0**
☐	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

☐	Annex 15	Evidence of internal approval **(template provided)**
☐	Annex 16	Map(s) indicating the location of proposed interventions
☐	Annex 17	Multi-country project/programme information **(template provided)**
☐	Annex 18	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☐	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☐	Annex 20	First level AML/CFT (KYC) assessment
☐	Annex 21	Operations manual (Operations and maintenance)
☐	Annex 22	Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects).
☐	Annex 23	Estimation of Beneficiary Calculation
☐	Annex 24	Exist Strategy Matrix.

☐ Annex 25 Scalability Matrix with Tool Kits

Annex 26 Knowledge Management

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

No-objection letter issued by the national designated authority(ies) or focal point(s)

KINGDOM OF CAMBODIA
Nation Religion King

Ministry of Environment
Nº : 1452/0225 MoE

Phnom Penh, 24-February-2025

Ms. Mafalda Duarte
Executive Director
Green Climate Fund Secretariat
Songdo Business District
175 Art Center-daero
Yeonsu-gu, Incheoun 22004
Republic of Korea

Re: Funding proposal for the GCF by IFAD regarding Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR)

Dear Ms. Duarte,

We refer to the project titled CAISAR in Cambodia as included in the funding proposal submitted by IFAD to us on 18 February 2025. The undersigned is the duly authorized representative of Ministry of Environment, the National Designated Authority of Cambodia. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Cambodia has no-objection to the project as included in the funding proposal;

(b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Cambodia;

(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Please, **Ms. Duarte**, accept the assurances of my consideration and regards.

Sincerely Yours,
Minister

EANG Sophalleth

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR)
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	International Fund for Agricultural Development (IFAD)
Environmental and social safeguards (ESS) category	Category A
Location – specific location(s) of project or target country or location(s) of programme	4 target provinces of the Kingdom of Cambodia (Kampong Speu, Kampong Chhnang, Kandal, and Pursat).
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Thursday, February 20, 2025
Language(s) of disclosure	English and Khmer
Explanation on language	Khmer is the official language of Cambodia.
Link to disclosure	**Environmental, Social, and Climate Impact Assessment (ESCIA) Report** English: https://www.ifad.org/documents/d/new-ifad.org/environmental-social-and-climate-impact-assessment-report Khmer: https://www.ifad.org/documents/d/new-ifad.org/caisar-escia-khmer **Executive Summary** Khmer: https://www.ifad.org/documents/d/new-ifad.org/environmental-social-and-climate-impact-assessment-report-executive-summary-khmer **Non-Technical Summary** English: https://www.ifad.org/documents/d/new-ifad.org/environmental-social-and-climate-impact-assessment-report-non-technical-summary Khmer: https://www.ifad.org/documents/d/new-ifad.org/environmental-social-and-climate-impact-assessment-report-non-technical-summary-khmer
Other link(s)	**Environmental, Social and Climate Management Framework (ESCMF)** English:

	https://www.ifad.org/documents/d/new-ifad.org/environmental-social-and-climate-management-framework Khmer: https://www.ifad.org/documents/d/new-ifad.org/caisar_-escmf-february-2025-khmer- **Executive Summary** English: https://www.ifad.org/documents/d/new-ifad.org/environmental-social-climate-management-framework-executive-summary Khmer: https://www.ifad.org/documents/d/new-ifad.org/environmental-social-and-climate-management-framework-executive-summary-khmer
Remarks	Confirmation that the Environmental, Social, and Climate Impact Assessment Report and the Environmental, Social and Climate Management Framework are consistent with the requirements for a Category A project is subject to further due diligence on the Funding Proposal.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Thursday, February 20, 2025
Language(s) of disclosure	English and Khmer
Explanation on language	Khmer is the official language of Cambodia.
Link to disclosure	**Environmental, Social, and Climate Management Plan (ESCMP) Yutasas Sub-Scheme** English: https://www.ifad.org/documents/d/new-ifad.org/esmp-yutasas Khmer: https://www.ifad.org/documents/d/new-ifad.org/caisar_-escmp-yutasas-february-2025-khmer- **Executive Summary** Khmer: https://www.ifad.org/documents/d/new-ifad.org/esmp-executive-summary-yutasas-khmer
Other link(s)	**ESCMP Ou Ta Paong Sub-Scheme** English: https://www.ifad.org/documents/d/new-ifad.org/esmp-ou-ta-paong Khmer: https://www.ifad.org/documents/d/new-ifad.org/caisar_-escmp-ou-ta-paong-february-2025-khmer- **Executive Summary** Khmer:

	https://www.ifad.org/documents/d/new-ifad.org/esmp-executive-summary-ou-ta-paong-khmer **ESCMP Krapeu Troum Sub-Scheme** English: https://www.ifad.org/documents/d/new-ifad.org/esmp-krapeu-troum Khmer: https://www.ifad.org/documents/d/new-ifad.org/caisar_-escmp-krapeau-trom-february-2025-khmer- **Executive Summary** Khmer: https://www.ifad.org/documents/d/new-ifad.org/esmp-executive-summary-krapeu-troum-khmer
Remarks	Confirmation that the Environmental, Social, and Climate Management Plans are consistent with the requirements for a Category A project is subject to further due diligence on the Funding Proposal.
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report	Land Acquisition and Resettlement Planning Framework; Indigenous Peoples Planning Framework; Stakeholder Engagement Plan
Date of disclosure on accredited entity's website	Thursday, February 20, 2025
Language(s) of disclosure	English and Khmer
Explanation on language	Khmer is the official language of Cambodia.
Link to disclosure	**Land Acquisition and Resettlement Planning Framework** English: https://www.ifad.org/documents/d/new-ifad.org/land-acquisition-and-resettlement-planning-framework Khmer (Included in the ESCMF): https://www.ifad.org/documents/d/new-ifad.org/caisar_-escmf-february-2025-khmer- **Executive Summary** English: https://www.ifad.org/documents/d/new-ifad.org/indigenous-peoples-planning-framework-executive-summary

	Khmer: https://www.ifad.org/documents/d/new-ifad.org/land-acquisition-and-resettlement-planning-framework-executive-summar-khmer **Stakeholder Engagement Plan (SEP)** English: https://www.ifad.org/documents/d/new-ifad.org/stakeholder-engagement-plan Khmer: https://www.ifad.org/documents/d/new-ifad.org/caisar-stakeholder-engagement-plan-khmer **Executive Summary** English: https://www.ifad.org/documents/d/new-ifad.org/stakeholder-engagement-plan-executive-summary Khmer: https://www.ifad.org/documents/d/new-ifad.org/stakeholder-engagement-plan-executive-summary-khmer **Indigenous Peoples Planning Framework** (Annex 14 of the ESCMF) English: https://www.ifad.org/documents/d/new-ifad.org/environmental-social-and-climate-management-framework Khmer: https://www.ifad.org/documents/d/new-ifad.org/caisar_-escmf-february-2025-khmer- **Executive Summary** English: https://www.ifad.org/documents/d/new-ifad.org/indigenous-peoples-planning-framework-executive-summary Khmer: https://www.ifad.org/documents/d/new-ifad.org/indigenous-peoples-planning-framework-executive-summary-khmer
Other link(s)	N/A
Remarks	Confirmation that the other ESS reports are consistent with the requirements for a Category A project is subject to further due diligence on the Funding Proposal
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Wednesday, December 25, 2024
Place	MOWRAM website (Executing Entity): https://doc.nwrdmc-mowram.gov.kh/public/doc_aiib)

	Local Offices of the communes in the 4 target provinces: 1) Pursat PDWRAM Chamka Chek, Sangkat Phtas Prey, Pursat City, Pursat province Phone: +855 52 951 772 2) Kampong Chhnaing PDWRAM Lortuk Trey village, Kampong Chhnaing commune, Kampong Chhnaing city, Kampong Chhnaing province 3) Kampong Speu PDWRAM Teng Village, Vor Sar commune, Samroung Tong district, Kampong Speu province 4) Kandal PDWRAM Street 21A, Prek Samroung village, TA Khmao commune, Ta Khmao City, Kandal province AIIB website (Co-Financier): https://www.aiib.org/en/projects/details/2023/proposed/Cambodia-Climate-Adaptive-Irrigation-and-Sustainable-Agriculture-for-Resilience-Project-CAISAR.html
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, June 30, 2025

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP270

Proposal name:	Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR) in Cambodia
Accredited entity:	International Fund for Agricultural Development
Country:	Cambodia
Project size:	Medium

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The project targets four high-risk provinces – Kampong Speu, Kampong Chhnang, Kandal and Pursat – where agricultural livelihoods are most threatened by extreme weather and water scarcity but also present opportunities. The selection of these areas reflects robust climate vulnerability mapping and socioeconomic targeting, including a focus on women-headed households and areas with high poverty density.	The project's success depends heavily on the capacity and coordination of multiple domestic agencies, such as Ministry of Water Resources and Meteorology and the National Committee for Sub-National Democratic Development secretariat. Weak institutional capacity, particularly at the subnational level, along with historically under resourced extension services, could pose risks to implementation efficiency and long-term sustainability.
The proposal effectively combines adaptation and mitigation goals. This is complemented by strong financing structure with a USD 240 million budget, integrating contributions from multiple sources, including GCF, International Fund for Agricultural Development (IFAD), Asian Infrastructure Investment Bank (AIIB) and the Royal Cambodian Government. Together, these contributions demonstrate robust multilateral and national support and long-term sustainability.	The financial viability of irrigation modernization depends heavily on cost recovery through Farmer Water User Committees (FWUCs), weak existing capacity and lack of a systematic fee collection mechanism could create risk. Without clearer strategies for long-term operation and maintenance, financing and capacity-building, there could be risk of infrastructure degradation and underutilization post-project.
With a clear focus on operations and maintenance (O&M), the project anticipates a return on investment within 20 years, which highlights its focus on sustainability and long-term success.	

2. The Board may wish to consider approving this funding proposal in accordance with the terms listed in the term sheet agreed between the Secretariat and the accredited entity (AE), and, if considered appropriate, subject to the conditions set out in annex II to document GCF/B.42/02.

II. Summary of the Secretariat's assessment

2.1 Project background

3. The objective of the Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR) project is to strengthen the resilience of Cambodia's rural communities and agricultural systems to climate change. Key components of the project include modernizing the irrigation infrastructure, promoting sustainable agricultural practices and reinforcing institutional capacities. Targeting four highly vulnerable provinces – Kampong Speu, Kampong Chhnang, Kandal and Pursat – the project seeks to address rising temperatures, erratic rainfall and increasing occurrences of climate-induced disasters such as floods and droughts. These provinces were strategically selected because of their high population density, prevalence of poor and women-headed households and exposure to worsening climate risks such as severe water shortages and infrastructure degradation. The project adopts a gender-responsive and socially inclusive approach, ensuring that women, youth and marginalized groups are integrated into resilience-building interventions.

4. The CAISAR project is structured around three interconnected components. The first component focuses on enhancing farm-level climate resilience by scaling up climate-smart agriculture practices, improving crop water management, strengthening value chains for rice, vegetables, poultry and aquaculture, and enhancing farmers' access to climate information and rural roads. The second component supports the climate-proofing and modernization of irrigation systems and drainage infrastructure, including the introduction of solar-powered irrigation pumps and training of Farmer Water User Committees (FWUCs) to promote equitable and sustainable water use. The third component strengthens the enabling institutional environment through capacity-building of key government agencies (i.e. the Ministry of Water Resources and Meteorology (MOWRAM) and the National Committee for Sub-National Democratic Development (NCDD) secretariat), policy development, digitization of climate tools and support for river basin management and planning systems.

5. CAISAR seeks to catalyse a paradigm shift towards climate-resilient, low-emission agriculture in Cambodia by promoting the adoption of proven techniques such as alternate wetting and drying (AWD), mechanized direct seeded rice (mDSR) and land laser leveling (LLL) through technical assistance provided by the International Rice Research Institute. The project also integrates market-based approaches, private sector engagement, such as public-private-producer partnerships (4Ps) and digital innovations for farmer training and service delivery to reinforce long-term sustainability.

6. CAISAR is expected to directly benefit 562,757 rural people and reach an additional 1,151,618 people indirectly through system-wide improvements. The project also anticipates a mitigation impact through a reduction in greenhouse gas (GHG) emissions of 1,006,507.43 tonnes of carbon dioxide equivalent (t CO_2 eq) over 20 years, contributing meaningfully to Cambodia's climate targets, with 7 years of implementation and a 30-year lifespan.

7. The proposal outlines a total budget of USD 240 million. This budget leverages contributions from GCF totalling USD 80 million (USD 40 million grant and USD 40 million loan),

alongside contributions from the International Fund for Agricultural Development (IFAD) with USD 45 million, the Asian Infrastructure Investment Bank (AIIB) with USD 100 million and the Royal Government of Cambodia providing USD 15 million.

8. The project's exit strategy emphasizes long-term financial and institutional sustainability through capacity-building, integration with national systems and clear cost recovery mechanisms. By empowering provincial departments and FWUCs to manage and maintain irrigation infrastructure, the project aims to institutionalize operations and maintenance (O&M) practices beyond the project timeline. The use of community-based action plans, digital advisory services and mobile-based tools for market and climate information strengthens local ownership while reducing dependence on external technical support. Furthermore, the alignment of CAISAR with other government-led initiatives such as the GCF project Public-Social-Private Partnerships for Ecologically-Sound Agriculture and Resilient Livelihood (PEARL) and Sustainable Assets for Agriculture Markets, Business and Trade (SAAMBAT) increases the likelihood of continued domestic financing and integration into public investment cycles.

9. Project risks, however, remain. A key concern is the weak baseline capacity of FWUCs to collect fees and manage funds for irrigation O&M, which threatens the long-term viability of the upgraded infrastructure. To mitigate this, CAISAR embeds targeted technical assistance and performance-based training to operationalize cost recovery models and improve financial governance at the local level. Similarly, the project's reliance on cascading training models – through Master Trainers and Farmer Field Schools – could face quality control and retention challenges, especially without ongoing funding post-project. To address this, CAISAR includes linkages to existing credit schemes and explores blended finance and private sector engagement to sustain service delivery. By anchoring financial sustainability into both institutional frameworks and farmer-level systems, the project lays a credible foundation for long-term impact and scalability.

10. From the GCF perspective, the identified risks are typical of complex climate adaptation projects in developing country contexts but are systematically addressed through the project's design, governance arrangements and risk mitigation measures. The project's emphasis on inclusive stakeholder engagement, capacity-building, financial inclusion and adherence to environmental and social safeguards (ESS) standards aligns well with GCF investment criteria for impact, sustainability and paradigm shift.

11. GCF will maintain active oversight through regular monitoring, technical support and adaptive management to ensure that risks are effectively managed throughout the project lifecycle. This proactive risk management approach is essential to safeguard the GCF investment and maximize the climate resilience benefits for Cambodia's vulnerable agricultural communities.

2.2 Component-by-component analysis

Component 1: Farm-level climate adaptation and resilience (total cost: USD 40.4 million; GCF cost: USD 26.3 million)

12. Component 1 focuses on strengthening farm-level climate adaptation and resilience by equipping smallholder farmers with the knowledge, tools and support necessary to adapt to climate change impacts. The project promotes climate-smart agricultural practices, such as improved crop varieties, soil and water management techniques and integrated pest management, to enhance productivity and reduce vulnerability to droughts, floods and shifting weather patterns. Through Farmer Field and Business Schools, capacity-building and

demonstration plots, farmers will gain hands-on experience in resilient farming methods tailored to local conditions. Special emphasis is placed on empowering women, youth and vulnerable groups to participate in decision-making and benefit from project interventions. The component also supports the development of community-based organizations and local extension services to ensure sustainability and knowledge transfer.

13. The project operates in a highly climate-sensitive environment where extreme weather events such as floods, droughts and erratic rainfall patterns pose direct threats to project outcomes. These climate risks could undermine irrigation infrastructure and agricultural productivity gains. However, residual climate risks remain, necessitating continuous monitoring and adaptive management. Aside from climate risk, cultural risks in the project may impede the adoption of climate-resilient practices if traditional agricultural methods, local decision-making norms or scepticism towards external expertise limit community acceptance. To address this, the project emphasizes culturally sensitive engagement strategies, including collaboration with community leaders and adapting communication approaches to align with local values and knowledge systems.

Component 2: Upgrading and climate-proofing water infrastructure for increased resilience (total cost: USD 167.7 million; GCF cost: USD 42.3 million)

14. Component 2 focuses on climate-proofing water infrastructure to enhance the resilience of Cambodia's irrigation systems and agricultural productivity in the face of climate change. The project will support the rehabilitation, modernization and construction of climate-resilient irrigation schemes, including canals, water storage facilities and control structures, with a strong emphasis on integrating climate risk considerations into design and operation. Special attention is given to ensuring that infrastructure investments are robust against extreme weather events, such as floods and droughts, and that they contribute to sustainable water management at the watershed level. The project will also strengthen the capacity of FWUCs and local authorities in the operation and maintenance of these systems, ensuring long-term functionality and community ownership.

15. Specifically for components 1 and 2, financial and market risks for the project include fluctuations in market prices for agricultural inputs and outputs, limited access to finance for smallholder farmers and the potential for cost overruns during implementation. These challenges are compounded by operational and implementation risks stemming from the complexity of integrated landscape and irrigation interventions across diverse geographic and socioeconomic contexts.

16. Procurement delay, logistical constraints and the need for effective coordination among multiple stakeholders could impact project timelines and quality. A significant additional risk relates to infrastructure: the successful delivery of climate-resilient irrigation and landscape management depends on the construction and ongoing maintenance of critical infrastructure such as irrigation canals, water storage facilities and related assets. This infrastructure is vulnerable to extreme weather events, flooding and potential design or construction flaws, which could lead to service disruptions, increased maintenance costs or even asset failure. Operational risks are mitigated via robust monitoring and evaluation (M&E) systems within the Project Management Unit (PMU).

Component 3: Strengthened institutional and regulatory capacity for low-emission climate-resilient development pathways (total cost: USD 9.06 million; GCF cost: USD 8.07 million)

17. The project supports a range of innovative and sustainable activities designed to strengthen institutional capacity and climate governance for resilient water and landscape management. Emphasizing the integration of climate-resilient irrigation and ecosystem-based adaptation approaches into national and subnational policies, the project enhances the sustainability and effectiveness of climate adaptation efforts across sectoral planning and financing. Capacity development is also a cornerstone, with training provided to key institutions. These institutions are critical in coordinating climate finance, overseeing project implementation and ensuring the sustainability and replication of climate-resilient investments.

18. The success of CAISAR heavily depends on the capacity and coordination of multiple government agencies, including MOWRAM, the Ministry of Economy and Finance (MEF), and the NCDD secretariat. Potential risks include institutional fragmentation, limited technical capacity at the provincial and local level and delays in decision-making or fund disbursement. The project addresses these risks through capacity-building, clear roles and responsibilities and the establishment of a Central Project Implementation Unit with strong technical expertise and monitoring capabilities.

Gender, environmental and social management framework, monitoring and evaluation components

19. The project, classified as category A, anticipates significant positive environmental and social (E&S) benefits, including enhanced climate resilience and reduced GHG emissions, while recognizing the need for careful management of any potential adverse impacts. Institutional arrangements involve MOWRAM as the main executing entity, working in close coordination with MEF and the NCDD secretariat, and supported by international technical assistance to ensure compliance with ESS standards. The project's environmental and social management framework includes a zero-tolerance policy for sexual exploitation, abuse and harassment (SEAH), with specific risk mitigation procedures integrated into project management and a grievance redress mechanism (GRM) to address concerns during implementation. Project risks and further information are noted in the relevant sections below.

Project management (total cost: USD 19.8 million; GCF cost: USD 2.7 million) and contingencies (total cost: USD 2.4 million; GCF cost: USD 0 million)

20. This project will be managed by the PMU, supported by technical specialists in areas like irrigation, climate resilience and community development. The PMU includes finance, operations and M&E officers to oversee project progress and ensure that goals are met effectively. Project risks and further information are noted in the relevant sections below.

III. Assessment of performance against investment criteria

3.1 Impact potential *Scale: High*

21. The project is considered to have significant impact potential, aiming to transform the vulnerability landscape of Cambodia's rural agricultural communities through integrated climate adaptation and water management interventions. It targets four highly vulnerable provinces – Kampong Speu, Kampong Chhnang, Kandal and Pursat – which face increasing climate risks, including rising temperatures, erratic rainfall, floods and droughts.

22. The project will directly benefit 562,757 people (225,103 female) and indirectly reach an additional 1,151,618 people (575,809 female). This represents approximately 9.6 per cent of

Cambodia's total population, with direct beneficiaries accounting for 3.16 per cent and indirect beneficiaries for 6.47 per cent.

23. The project will target approximately 25,604 hectares across six subschemes in the four provinces over its 7-year implementation period, with an expected lifespan of 30 years. It anticipates a significant mitigation impact through a reduction in GHG emissions of 1,006,507.43 t CO_2 eq over 20 years through the adoption of climate-smart agricultural practices such as AWD, mDSR and LLL.

3.2 Paradigm shift potential *Scale: High*

24. The project aims to catalyse a significant shift towards climate resilience and sustainable agriculture in Cambodia's vulnerable rural provinces by transitioning from a state of high exposure to climate risks to one where irrigation systems are modernized, agricultural practices are climate smart and institutional capacities are strengthened. This transition supports resilient farming communities and sustainable water management, enabling adaptation to climate change impacts such as droughts, floods and erratic rainfall.

25. To achieve this, the project promotes a comprehensive approach combining climate-smart agriculture with integrated water resource management. It focuses on rehabilitating and climate-proofing irrigation infrastructure while empowering FWUCs and provincial departments to manage water resources equitably and sustainably. These interventions are integrated into commune and district development plans, ensuring the participation of marginalized groups, including women and youth, and incorporating local knowledge and practices.

26. The project implements a range of climate-adaptive interventions such as AWD, mDSR and LLL, alongside strengthening value chains and improving farmers' access to climate information and rural infrastructure. It also enhances the enabling environment by building institutional capacity, digitizing climate tools and supporting policy development to facilitate replication and scaling up.

27. Overall, the project seeks to create a sustainable and resilient future for Cambodia by embedding climate adaptation into local governance, agricultural practices and water management systems. It ensures meaningful stakeholder participation and knowledge-sharing across multiple levels, thereby fostering long-term sustainability and resilience in the face of climate change.

3.3 Sustainable development potential *Scale: High*

28. The project is strategically poised to contribute significantly to sustainable development in Cambodia, particularly aligning with Sustainable Development Goals 1 (No poverty), 2 (Zero hunger), 6 (Clean water and sanitation), 12 (Responsible consumption and production) and 13 (Climate action).

29. By modernizing irrigation infrastructure, scaling climate-smart agriculture and strengthening institutional systems, CAISAR aims to enhance agricultural resilience, improve water resource efficiency, reduce GHG emissions and strengthen climate adaptation capacities among vulnerable rural communities. This integrated approach supports the development of resilient food systems, the sustainable management of water and land resources and inclusive economic opportunities in climate-affected provinces. Through its focus on gender inclusion, institutional capacity-building and the introduction of financially sustainable mechanisms for

service delivery and infrastructure maintenance, the project is expected to lead to improved food security, income stability and long-term resilience to climate shocks – contributing meaningfully to Cambodia's achievement of the Sustainable Development Goals.

3.4 Needs of the recipient *Scale: High*

30. The project targets the acute needs of Cambodia's rural population and agriculture-dependent economy, which are increasingly threatened by climate change, poverty and food insecurity. Focusing on climate-vulnerable provinces and communities, particularly smallholder farmers and women-headed households, CAISAR aims to strengthen resilience through sustainable irrigation development, climate-smart agriculture and institutional reform.

31. CAISAR directly addresses challenges such as prolonged droughts, erratic rainfall, declining agricultural yields and flood-prone infrastructure. By equipping communities with adaptive farming techniques, improving water management systems and enhancing institutional coordination and financial sustainability, the project seeks to build long-term resilience to climate shocks.

3.5 Country ownership *Scale: High*

32. The project exemplifies strong country ownership, aligning closely with Cambodia's national strategies on climate resilience, sustainable agriculture and water resource management. CAISAR is firmly embedded within the Royal Government of Cambodia's priorities, including its nationally determined contributions, the Rectangular Strategy, and the National Water Resources Management and Sustainable Irrigation Road Map. By working collaboratively with key ministries such as MOWRAM, the Ministry of Environment and the NCDD secretariat, the project ensures alignment with national plans and builds on local institutional knowledge and systems.

33. Country ownership is further reinforced through active engagement with provincial departments and community-level structures like FWUCs, enabling localized planning and implementation. This participatory approach ensures that interventions are context specific and community driven, enhancing both the effectiveness and the long-term sustainability of the project while fostering a shared national responsibility in addressing climate change and agricultural resilience.

3.6 Efficiency and effectiveness *Scale: Medium to high*

34. The project is expected to deliver significant and quantifiable climate benefits. It is expected to reach mostly smallholder farmers, who are expected to benefit from improved water security, reduced flood risk and enhanced agricultural productivity. The project strengthens early warning systems, institutional capacity and the adoption of climate-resilient practices. On the mitigation side, the project is expected to reduce GHG emissions by over 1 million t CO_2 eq, primarily through reduced methane emissions from improved rice cultivation and the replacement of diesel-powered pumps with solar photovoltaic systems. Climate co-benefits represent over 23 per cent of total economic benefits, with avoided flood damage (17.8 per cent) and GHG emission reduction (5.5 per cent) quantified.

35. The CAISAR project demonstrates strong economic and financial efficiency. The economic internal rate of return is 16.34 per cent, exceeding the applied social discount rate of 9.47 per cent, and the benefit–cost ratio is 1.26. Component-level analysis confirms that both

farm-level adaptation and irrigation infrastructure investments are economically viable, with economic internal rates of return ranging from 13.3 to 37.1 per cent across the six schemes. The project demonstrates resilience to shocks, remaining economically viable under scenarios involving cost increases or benefit reductions of up to 20 per cent.

36. From a financial standpoint, the project is marginally viable under conservative conditions: financial internal rates of return from both the private (13.38 per cent) and the government (13.95 per cent) perspective are positive, but returns are moderate. From the government perspective, incorporated loan repayment obligations and cost recovery via irrigation tariffs, the project breaks even from the government perspective at year 20 if tariffs are enforced and the system well maintained. This supports the requested GCF concessionality as sensitivity-tested financial returns show limited headroom. There is also clear differentiation between GCF and other lenders, with GCF grants targeting non-revenue-generating, high climate impact components (e.g. flood protection, GHG emission reduction, institutional capacity) and the use of GCF concessional loans reduce the financing burden on the government for the irrigation investments. GCF funding de-risks the projects, catalyses co-financing from AIIB and IFAD and supports adaptation measures. Increasing the GCF loan rate or reducing the grant would affect project affordability and concessionality, potentially making the package unviable for the government, therefore increasing its fiscal stress.

37. The GCF grant and concessional loan are used in a targeted and efficient manner. Grants are directed to non-revenue-generating, public good components, while concessional loans support a partially cost-recoverable infrastructure. The overall financing structure achieves a grant element of 33 per cent, aligning with minimum concessionality principles while enabling co-financing from AIIB and IFAD.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

38. **Project background**. The project aims to increase climate adaptation, mitigate the negative impact of extreme climate events and improve the livelihoods of smallholder farmers and vulnerable rural communities in four provinces of Cambodia (Pursat, Kampong Chhnang, Kampong Speu and Kendal). The E&S risks and impacts of the project will potentially come from (a) possible changes to, and more effective implementation of, policies and regulatory frameworks on land use, land classification and restriction of access; (b) infrastructure development, specifically the modernization of irrigation schemes and ponds, flood-proofing, drainage and road improvements; and (c) from the enterprises to be supported by the project. The E&S co-benefits include improved habitats, reduction in pesticide use and increased agricultural production and food diversity through increased yields in vegetables, fish and poultry products. The improved irrigation systems are also expected to increase household incomes, leading to better diets and nutrition and access to better healthcare services, which could result in overall improved health conditions.

39. **Environmental and social risk category and safeguards instruments.** The project has been assigned an E&S risk category A classification. This was determined on the basis of a comprehensive screening and assessment of E&S risks and/or impacts that may arise from the proposed activities under the project. The assessment considered the nature, scope and scale of the project activities, the vulnerability of potential project-affected people, the sensitivity of biodiversity in the project's area of influence, and the broader socioeconomic and environmental settings in the project sites. The Secretariat concurs with the assigned risk

category and notes that it is within the AE accreditation level. Since the specific designs and locations of the various interventions (subprojects) will still be developed or finalized during implementation, the framework approach has been adopted for this project. The project has an environmental, social and climate management framework (ESCMF), a land acquisition and resettlement planning framework, an Indigenous Peoples planning framework and a stakeholder engagement plan.

40. **Compliance with GCF's environmental and social safeguards standards**. The paragraphs below describe the project's compliance with GCF's ESS standards requirements.

41. **ESS 1: Assessment and management of environmental and social risks and impacts**. The ESCMF was developed on the basis of a review of relevant government laws, regulations, ordinances and other legal documents; various background papers and reports pertaining to E&S conditions of the proposed project areas and potential subproject sites; and numerous field visits, interviews through livelihood household surveys, focus group discussions, key informant interviews and other community meetings in those areas. An environmental, social and climate impact assessment has also been prepared, including environmental, social and climate management plans prepared for the three selected subschemes of Ou Ta Paong, Krapeu Truom and Yutasas.

42. The project is expected to have positive environmental impacts, which include improved ecosystem services, soil fertility/soil health, forest health and natural resources management. Key potential negative environmental impacts include those that are related to small-scale infrastructure (e.g. construction/repair of small-scale rural roads, irrigation channels and small-scale reservoirs). These impacts may include erosion and run-off to water bodies during excavation and earthwork activities; traffic safety hazards during construction and operation; construction-related air, water and land pollution issues such as the generation of dust, nuisance noise, vibration, and solid waste and wastewater generation from workers' campsites. Inherent area-related risks include the potential presence of unexploded ordnances (UXOs) at project sites and the potential for increased use of pesticides due to crop intensification/improved crop yields. The main social impacts of the project include the potential for workers and host communities to acquire sexually transmitted and/or other communicable diseases, conflicts arising with local communities and gender-based violence (GBV) and/or SEAH due to migratory labour influx; occupational and labour-related accidents during project construction; risks related to forced labour and/or child labour or age-inappropriate work or abuse of ethnic minority workers; and the possibility of social exclusion due to vulnerability, land tenure and disability issues.

43. **ESS 2: Labour and working conditions.** The labour and working conditions standards apply to both direct project workers (PMU employees) and contracted workers in subprojects (employees of contractors). The ESCMF highlights key labour issues in the country, including child labour, forced labour and gender-based discrimination, despite existing laws aligned with international standards. Other issues include inadequate provisions for occupational health and safety, exposure to hazardous materials and waste, transmission of diseases, absence of contract for seasonal ethnic minority workers, SEAH in the workplace and exposure to UXO hazard. Management measures include adoption of a workers' code of conduct and camp guidelines, adoption of a labour management procedure, conduct of adequate due diligence on labour and working conditions on primary suppliers and sources of aggregate materials, establishment of monitoring procedures, inclusion of labour issues in the conduct of site-specific environmental, social and climate impact assessments, compliance with national regulations, provision of access to a workers' GRM, compliance with occupational health and safety procedures and inclusion of labour-related measures in agreements with workers and contractors/subcontractors, as well as conduct of training and capacity-building.

44. **ESS 3: Resource efficiency and pollution prevention.** The project contributes to resource use efficiency by targeting the optimization of water and energy resources. The project aims to improve water use efficiency at the farm level by implementing climate-resilient water management practices; rehabilitation and modernization of irrigation systems to ensure efficient water distribution and reduce water wastage; and promoting the use of water-saving technologies such as drip irrigation and AWD methods in rice cultivation. The project will also incorporate the use of renewable energy sources, such as solar pumps, to reduce reliance on conventional energy sources and lower GHG emissions and will promote energy-efficient practices in the O&M of irrigation and flood control infrastructure. The ESCMF, however, identifies potential environmental pollution that may arise during project implementation. These include noise, air and water pollution from construction activities, and possible contamination of the environment from pesticides and agrochemicals during project operations. Mitigation measures have also been identified to address these risks, including erosion control to minimize sedimentation, noise and air emissions, and waste management. Other approaches to mitigate pollution issues include the adoption of an E&S code of practice for construction and the adoption of guidelines for a workers' camp. The project will also adopt the integrated pest management approach to minimize the use of pesticides in crop production.

45. **ESS 4: Community health, safety and security.** Project activities on the ground may affect community health and safety; these include construction-related hazards that increase the likelihood of traffic accidents, the spread of communicable diseases such as human immunodeficiency virus/acquired immunodeficiency syndrome as a result of labour influx, a heightened risk of GBV/SEAH and potential exposure to UXO hazards from land clearing and excavation. To mitigate these risks, management measures include strict enforcement of health and safety regulations, implementation of traffic and waste management plans, employment of disease prevention initiatives such as health screenings and medical facilities, GBV/SEAH prevention through the adoption of worker codes of conduct and the adoption of a standard UXO safety protocol in accordance with local regulations.

46. **ESS 5: Land acquisition and involuntary resettlement.** The project seeks to minimize land acquisition and involuntary resettlement by exploring alternative designs and sites. However, some infrastructure subprojects such as irrigation, flood protection and road rehabilitation may require land acquisition and/or necessitate the physical relocation of people. Additionally, some project activities may support changes in land-use policies, which could restrict people's access to land or restrict land to some productive uses, affecting individuals and communities that depend on agriculture and natural resources. A land acquisition and resettlement planning framework has been developed to address this requirement. The land acquisition and resettlement planning establishes principles, procedures and measures to ensure fair compensation and proper resettlement for affected individuals.

47. **ESS 6: Biodiversity conservation and sustainable management of living natural resources**. The project may negatively impact biodiversity and forest sustainability, particularly through habitat destruction, soil erosion and water pollution. Construction activities for irrigation and flood protection could lead to the loss of wetlands, forests and aquatic ecosystems. Land clearing and excavation may degrade vegetation and wildlife habitats, while soil erosion and sedimentation could reduce water quality and harm aquatic life. Additionally, run-off from construction sites, along with pollutants like oil and chemicals, may further threaten ecosystem functions. Poor water management could also facilitate the spread of invasive species, disrupting native biodiversity. To mitigate biodiversity risks and impacts, the project will conduct a detailed habitat assessment and implement measures to preserve critical habitats, such as wetlands and forests, by avoiding or minimizing construction within these areas. The project will likewise implement reforestation and revegetation programmes in areas where land clearance is necessary and only use native species to restore these habitats. To

manage soil erosion, control measures such as silt fences, erosion control blankets and sediment basins will be established to prevent soil erosion. These will complement the soft measures such as working with local authorities and communities to develop land-use plans that minimize negative impacts on agriculture and forestry. Water and waste management plans to minimize waterbody alteration will also be developed and implemented. The project will establish these measures following the mitigation hierarchy approach (i.e. avoidance, minimization, restoration or rehabilitation, or offset) consistent with the requirements of ESS 6.

48. **GCF Indigenous Peoples Policy & ESS7 (Indigenous Peoples)**. The AE has confirmed that Indigenous Peoples or communities will be affected by the project. A detailed and exemplary Indigenous Peoples planning framework is included in the ESCMF, which contains mitigation measures for risks related to Indigenous Peoples. Initial screening conducted during project preparation does not find any Indigenous Peoples present in the project's area of influence (for six subschemes). During project implementation, additional consultation will be carried out – based on the areas of influence to be confirmed when detailed engineering designs are final. Indigenous Peoples could benefit from the project through measures such as (i) Indigenous Peoples with farmland in command areas will benefit from improved irrigation, reliable water supply, agricultural extension services and support from local farmer cooperatives, designed to be culturally appropriate and gender responsive, aligning with Indigenous Peoples' needs and traditions; (ii) if minor land acquisition is unavoidable, compensation will be provided, along with additional developmental support based on meaningful consultation with Indigenous Peoples to restore or improve pre-project income levels; (iii) for subprojects affecting Indigenous Peoples, a social assessment will be conducted to evaluate cultural, socioeconomic and livelihood conditions. This assessment will inform the preparation of site-specific Indigenous Peoples plans that include mitigation and development measures tailored to each affected Indigenous Peoples group and include culturally sensitive consultations, recommend benefit-sharing strategies, detail mitigation and compensation measures, outline monitoring and GRMs; (v) Indigenous Peoples will be involved in the M&E of the implementation of an Indigenous Peoples plan to ensure ownership and accountability; and (vi) training and support will be provided to Indigenous Peoples organizations to enhance their ability to engage effectively in project activities. In line with its roles and functions, the Indigenous Peoples Advisory Group is available to provide advice to the AE and executing entities. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples focal point will be available for assistance at any stage, including before a claim has been made.

49. **ESS8: Cultural heritage.** The project could potentially have an impact on cultural heritage sites. The ESCMF has identified strategies to minimize any adverse impacts, namely the conduct of archaeological surveys before construction in culturally sensitive areas; the development of mitigation strategies to protect cultural heritage sites and involve local communities in preservation efforts; and the adoption and implementation of a chance finds procedure in the case of discoveries of cultural heritage sites during construction.

50. **Implementation arrangements.** The implementation of the CAISAR project will be coordinated across multiple agencies. A PMU, composed of personnel from MOWRAM, the Department of Hydrology and River Works and provincial departments of water resources and meteorology, will handle daily project execution. Oversight will be provided by the Project Steering Committee, chaired by the Minister of Water Resources and Meteorology and the Inter-Ministerial Resettlement Committee, led by MEF, which will manage land acquisition and resettlement. A dedicated team will ensure compliance with the ESCMF. The Department of Farmer Water User Communities will coordinate agricultural support activities, ESS and stakeholder engagement. Provincial departments of water resources and meteorology will assist with resettlement coordination, information dissemination and local compliance. The project will also establish FWUCs to enhance community involvement in irrigation scheme

management. These local groups will help to monitor infrastructure conditions, resolve water disputes and ensure sustainable operation and maintenance. Regular assessments will evaluate material quality, infrastructure resilience and the social impact of project activities. The project will also deploy key personnel, including an Environmental and Climate Safeguards Specialist, who will oversee E&S documentation, compliance monitoring, training and adherence to management plans. A Gender and Social Safeguards Specialist will ensure implementation of social management measures, while a GRM Focal Point will handle grievance resolution, and a Resettlement Plan Specialist will ensure fair compensation and relocation for affected households. Additionally, MOWRAM will appoint at least one Environmental Specialist, one Social Specialist and one GRM Focal Point for full-time project support. These specialists will play a crucial role in ensuring compliance, conducting training and maintaining ESCMF standards throughout implementation.

51. **Stakeholder engagement and information disclosure**. The project underwent extensive consultations, including several rounds of stakeholder meetings. The ESCMF outlines a comprehensive process of stakeholder engagement and consultation, which included multiple missions, meetings and public consultations at various levels. Continuous stakeholder engagement will be undertaken throughout the project implementation to receive timely feedback from those who might be affected by certain project activities. A Stakeholder Engagement Plan has been developed to outline the strategies and processes for engaging stakeholders throughout the life of the project. The ESCMF indicates that all safeguards-related documents and project information will be disclosed locally in an accessible place and in a form and language understandable to key stakeholders (Khmer and English).

52. **Grievance redress mechanism (GRM).** The project will establish a GRM that is accessible, transparent, free and user-friendly. The ESCMF also outlines the procedures for lodging grievances and the resolution processes for general complaints, as well a those related to land acquisition, labour and working conditions, and SEAH/GBV. As part of overall implementation of the subproject, the GRM will be established by the implementing agencies, which will have procedures, responsible persons and contact information and will be readily accessible to the public and be able to resolve issues at the lowest level as quickly as possible. The GRM will be put in place before the commencement of subproject activities.

53. **Sexual exploitation, abuse and harassment (SEAH) safeguarding.** The revised GCF Environmental and Social Policy adopted by decision B.BM-2021/18 requires safeguarding from SEAH in GCF-financed activities. The AE provided SEAH safeguarding in its submission of the funding proposal. The project includes activities that involve engaging with community members, and capacity-building activities will be undertaken for authorities at the national and provincial level. Risks factors encompass gender inequalities that tend to be greater in rural areas of Cambodia where the farming communities are located; construction of roads and irrigation facilities which will require immigration of labour or an influx of workers, likely to be mostly male, from outside the farming communities or project locations who will need to be accommodated in the communities; and land acquisition and associated livelihoods disruption which may result in physical and/or economic displacement. SEAH risks related to interventions that will necessitate community interactions have been identified and included in the ESCMF. Further assessment of SEAH risks will be undertaken at the subproject level. PMU members will receive SEAH training. Contractors will also be required to have financial resources set aside for recruiting a SEAH consultant to conduct public awareness-raising on SEAH at the subproject level, and to undertake SEAH management measures to minimize SEAH risks. A workers' code of conduct is annexed to the ESCMF, which all construction workers will be required to sign. The project-level GRM will receive and record complaints related to SEAH. In the case of reported incidents, SEAH survivors will be referred to a local GBV service provider for immediate support. In addition, confidential reporting will be ensured. Procedures for SEAH

complaints are included in the ESCMF. Additionally, the GCF Independent Redress Mechanism will also be available to stakeholders. SEAH risks will be monitored and reported on the basis of risks and impacts contained in site-specific environmental and social management plans, which will be implemented by contractors.

4.2 Gender policy

54. The AE has provided a gender assessment and action plan and therefore complies with the requirements of the GCF Gender Policy. The legal and institutional framework that demonstrates the existence of an enabling environment for addressing gender issues and women's empowerment is outlined in the assessment. Identified issues in the national framework for gender include lack of provision of equal remuneration for work of equal value, scope to make the training provided by the Ministry of Women's Affairs at its development centres more responsive to the labour market, and the need to mainstream gender into existing climate change strategies and policies. Women are identified as vulnerable to climate change. Root causes of the challenges and sources of vulnerabilities consist of traditional social norms that perpetuate and reinforce the traditional division of labour between women and men, particularly the burden of household duties, time poverty, limited access to resources, and lower decision-making autonomy in their households and at the community level. In addition, female-headed households, those living in poverty and those with disabilities are also identified as vulnerable in the gender assessment. Gender dimensions in the agriculture sector, as well as the roles and responsibilities of women and men on targeted value chains such as rice, vegetables and poultry, are analysed, as are the contextual details of gender-related matters from the consultations with community members in the provinces where the project will operate. In addition, access to agricultural extension services and markets where women face disadvantages are also analysed in the assessment.

55. The gender action plan covers activities, performance indicators, timelines and responsibilities. The baseline and sex-disaggregated targets for project mid- and end-term as well as female-headed households have been included in the gender action plan. It includes activities to support women in accessing finance and establishing agribusinesses and rural enterprises and to improve access to agrometeorological advisories. Training will be offered to communities on climate-resilient practices and O&M of irrigation canals as a way of empowering them in their communities. Women will be engaged in the preparation of community-based action plans for the adoption of climate-resilient crop water management practices. The project will promote the inclusion of women in leadership positions of FWUCs to influencing decision-making. A gender consultant and officer will be recruited during the project's implementation to support gender mainstreaming. The AE is recommended to undertake further gender analyses at the subproject level and identify any additional actions that can be implemented to address the needs of women and men.

4.3 Risks

4.3.1. Overall project assessment (medium risk)

56. The Secretariat considers the overall project as medium risk.

4.3.2. Accredited entity/executing entity capability to execute the current project (medium risk)

57. IFAD will be the AE for this project. Since the late 1990s, IFAD has implemented 12 projects with a total project cost of ~USD 1 billion for agricultural and rural development projects in Cambodia. The AE will work with other executing entities, namely, MEF, MOWRAM and the NCDD secretariat.

4.3.3. Project-specific execution risks (medium risk)

58. **Credit Risk:** The borrower of the GCF loan will be the Government of Cambodia (rated B2 by Moody's); therefore, GCF is exposed to the sovereign credit risk. The country is currently facing several challenges given the uncertainties around trade policies. However, the concessional terms from GCF will be advantageous for the country's debt affordability. The GCF loan will be ranked pari passu with other co-financiers such as IFAD and AIIB.

59. **Co-financing risk:** AIIB provides a sizable funding to the programme; however, the agreement will be made between the government and AIIB directly. In case of delay or reduction of the co-financing, the impact might not be materialized. To partially mitigate the risk, the term sheet includes provisions that AE to ensure that co-financing is proportionally disbursed in accordance with the budget and the AE/EEs have to inform GCF as soon as possible in case of cancellation or delay in the co-financing.

4.3.4. Compliance risk (medium risk)

60. The project is being implemented in regions with developing regulatory frameworks, which may present challenges in terms of financial oversight. Additionally, the focus on small-scale agricultural finance introduces risks related to fund allocation and potential misuse due to weaker local institutions. Considering IFAD's engagement with local government agencies and financial intermediaries, there is also the possibility of encountering risks related to sanctions compliance and political influence.

61. IFAD has established control measures to mitigate these risks, including a thorough due diligence process for all project partners. This includes KYC checks and regular sanctions screening to ensure compliance with AML/CFT requirements. The project also benefits from ongoing financial monitoring and auditing, which helps detect any irregularities early on. Furthermore, IFAD's on-the-ground presence in the project's areas of operation supports oversight, including the provision of technical assistance to enhance local governance and financial integrity. These measures help maintain transparency and reduce the likelihood of non-compliance.

62. While controls such as due diligence, sanctions screening, and continuous monitoring manage most risks, local political dynamics and enforcement limitations may still present challenges. Considering these factors, the residual compliance risk rating for the project is medium, reflecting both the project's intended safeguards and ongoing external vulnerabilities inherent in the operating environment.

4.3.5. GCF portfolio concentration risk

63. In the case of approval, the impact of this proposal on the GCF portfolio risk remains non-material and within the risk appetite in terms of concentration levels, results area, single proposal or AE concentration.

4.3.6. Recommendation

64. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall project	Medium
Accredited entity/executing entity capability	Medium
Project-specific execution	Medium
Compliance	Medium
GCF portfolio concentration	Within the monitoring threshold

4.4 Fiduciary

65. MOWRAM will establish a PMU for the implementation of the CAISAR project and will be responsible for the overall management of the project. The PMU will be led by the Secretary of State, and will include an Operations Officer, Gender Specialist, Finance Officer, Human Resources and Administration Officer, Procurement Officer and Monitoring and Evaluation Officer. Consulting firms and individual consultants will be recruited to support the project implementation and undertake detailed design services and construction supervision.

66. The project highlights number of risks in relation to financial management, including low capacity of financial management staff at implementing agencies with risks such as lack of prior experience of MOWRAM with IFAD-funded projects; previous external audit reports that did not meet IFAD compliance standards; and complicated disbursement and fund flow arrangements resulting from multiple financiers.

67. The mitigation measures to address these risks include recruitment of qualified consultants who possess relevant experience and educational qualifications, development of a Project Implementation Manual and a Financial Management Manual that outline clear roles for funds flow and disbursement arrangements, as well as guidelines for filing supporting documents and maintaining accounting records for all implementing partners, customization of accounting software to facilitate accurate record keeping and financial reporting and close coordination with MEF to ensure adherence to IFAD audit requirements.

68. CAISAR project financial statements will be audited annually by a private audit firm.

4.5 Results monitoring and reporting

69. This is a cross-cutting project targeting mitigation result area 1 (energy generation and access), adaptation result area 1 (most vulnerable people and communities), and adaptation result area 2 (health, well-being, food and water security). The project is expected to contribute to a reduction in emissions of 213,500 t CO2 eq by the completion date of the programme (7 years from the effective date) and 1 million t CO2 eq for the whole lifetime of the programme's proposed technologies (20 years) and contributes to the agricultural resilience of more than 1.5 million beneficiaries (about half of them female) in the targeted areas of Cambodia that are prone to climate change vulnerabilities by the end of the project implementation period.

70. The AE has produced a fairly comprehensive annex 22 explaining how the GHG emission reduction was calculated using the EX-ACT tool (Environmental eXternalities ACcounting Tool) of the Food and Agriculture Organization of the United Nations, incorporating elaboration on how the GHG emission baseline scenario was calculated as well as additionality for each proposed activity. The activities include other land use; annual cropping system; flooded rice cropping systems; livestock (poultry); livestock (fish); and inputs management and infrastructure management.

71. The project's beneficiaries were accurately calculated against each adaptation benefit resulting from improved agricultural practices and better access to irrigation for farming, as well as along the whole agricultural value chain, including the engagement of private sector actors and policymakers. In total, about 1.5 million people will benefit from the project.

72. The results management arrangements proposed under the project were refined over multiple rounds of review and engagement with the AE. Specifically, changes were made to the project theory of change to ensure that it was clear, logically coherent and linked to the climate rationale and activities proposed under the project. The project logical framework was also refined over multiple rounds to ensure that the proposed indicators and targets were realistic and linked to the proposed theory of change.

73. The AE was also provided with advice for refining the supporting annexes, including the implementation timetable, monitoring and evaluation plan and GHG emission reduction calculation, as well as adaptation beneficiary calculation. However, it should be noted that the AE is required to pay more attention to the elaboration of the M&E plan to ensure that it will be able to guide effective monitoring, reporting and evaluation of the project's results, particularly those indicators proposed in the logical framework.

74. The AE has responded positively to guidance provided and refined the funding proposal accordingly.

4.6 Legal assessment

75. The Accreditation Master Agreement was signed with the Accredited Entity on 24 September 2018 (the "AMA"), and it became effective on 9 November 2018. The Accredited Entity's five-year accreditation term lapsed on 8 November 2023 but was extended by Decision B.37/18, paragraph (q) until the earlier of three years from the date of lapse of the five-year accreditation term or the date on which a revised accreditation framework is adopted by the Board. The Board approved the re-accreditation of the Accredited Entity pursuant to Decision B.37/18, and negotiation of the Amended and Restated Accreditation Master Agreement is ongoing.

76. The Accredited Entity has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

77. The proposed project will be implemented in the Kingdom of Cambodia, country in which GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. Moreover, the ability of GCF to undertake redress activities and/or investigations in such a country may be hindered due to the absence of privileges and immunities for relevant GCF personnel.

78. Therefore, it is recommended that the Board considers whether disbursements of GCF proceeds should only be made after GCF has obtained satisfactory protection against litigation and expropriation in this country, or has been provided with appropriate privileges and immunities for GCF and its personnel.

79. Since the Accredited Entity's direct counterparty is a sovereign state (as the executing entity), the Secretariat has agreed to disapply the provisions of the AMA allowing GCF to step in to the Accredited Entity's contractual arrangements with the Executing Entity, i.e., Royal Government of Cambodia as, due to the sovereign nature of the counterparty and the contracts, it is not feasible for the Accredited Entity to allow GCF or any other party to take over its contractual relationship with the Executing Entity under the subsidiary agreement. Accordingly,

and similar to the arrangements for FP143, FP162, FP183, FP236, SAP012 and SAP017, as a condition for execution of the FAA, the Accredited Entity is required to inform the National Designated Authority (NDA) of the host country of this disapplication and its implications. An express provision has been included in the term sheet to provide for such disapplication.

80. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

(a) Submission by the Accredited Entity to the Fund of a certificate or legal opinion, in form and substance satisfactory to the GCF Secretariat, within 120 days after Board approval, confirming that the Accredited Entity has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed project;

(b) Signature of the FAA in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval, or the date the Accredited Entity has provided a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later; and

(c) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP270

Proposal name:	Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR) in Cambodia
Accredited entity:	International Fund for Agricultural Development
Country:	Cambodia
Project size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. **Persistent vulnerability of Cambodia's agriculture sector**. As a cornerstone of rural livelihoods and national food security, Cambodia's agriculture sector is constrained by systemic development challenges that undermine its productivity and resilience. Most of Cambodia's smallholder farmers rely on rain-fed, low-input subsistence farming systems, characterized by inefficient water use, degraded soil health, limited access to technology and underdeveloped market linkages. Institutional capacity for extension services remains weak, while investments in climate-smart practices and infrastructure (e.g. irrigation, post-harvest management and sustainable land management) are insufficient. Thus, farming communities face persistent poverty, food insecurity and limited opportunities to improve their incomes or adapt to emerging risks. These underlying vulnerabilities make Cambodia's rural economy particularly susceptible to shocks and reduce its ability to transition towards a more sustainable and inclusive growth path.[1]

2. **Climate change as a threat multiplier.** The impacts of climate change are exacerbating Cambodia's existing agricultural challenges, acting as a threat multiplier across ecological, economic and social dimensions. Increasing erratic rainfall, prolonged droughts, frequent storms and rising temperatures are disrupting traditional cropping calendars, lowering yields and accelerating land degradation. Extreme weather events linked to climate change have increased in frequency and intensity, damaging irrigation infrastructure, washing away topsoil and increasing the prevalence of pests and diseases. These climate shocks disproportionately affect women, land-poor households and other marginalized groups, deepening rural poverty and inequality. Without climate-resilient agricultural systems and early warning mechanisms, the sector's exposure to climate variability will continue to grow, thus threatening national food security and undermining Cambodia's broader development goals.[2]

3. The Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR) project is designed to strengthen Cambodia's agricultural resilience, water security and institutional capacity in response to increasing climate risks. The project intends to implement across three strategic levels each corresponding to expected outcomes, namely (1) reduction in crop losses and improved production; (2) increased efficiency in water use conveyance and use; and (3) enhanced environmental governance. Farm-level climate adaptation seeks to enable farmers to implement climate-smart agricultural practices through improved water

[1] See funding proposal, section B.1.
[2] As footnote 1 above.

management, early warning systems and value chain integration. The irrigation scheme modernization is designed to support flood-proofing, enhance drainage and upgrade irrigation efficiency to mitigate drought and flood impacts. Lastly, through institutional strengthening on the national level, the project is expected to reinforce government agencies and Farmer Water User Communities (FWUCs) to ensure the long-term sustainability of interventions.

4. The proposed project has a total financing package of USD 240.1 million: USD 80 million from GCF and USD 160.1 million in co-financing contributions from the International Fund for Agricultural Development (IFAD) (USD 45 million), the Asian Infrastructure Investment Bank (AIIB) (USD 100 million) and the Royal Cambodian Government (USD 15.1 million). The GCF contribution is evenly split between a USD 40 million grant and a USD 40 million concessional loan, enabling a balance between immediate support and long-term financial sustainability. Categorized as a medium-sized project by GCF standards, the intervention will be implemented over 7 years, with a total operational lifespan of 30 years to ensure lasting climate resilience outcomes. Given its scope and potential environmental and social risks, the project has been classified under environmental and social safeguards category A, requiring strong safeguards and risk management systems throughout implementation.[3]

1.2 Impact potential *Scale: Low to medium*

5. **Adaptation impact.** The CAISAR project is aimed at strengthening the climate resilience of Cambodia's agriculture sector, with wide-reaching benefits for communities, ecosystems, livelihoods and infrastructure. By rehabilitating and modernizing irrigation systems and canals, the project seeks to improve access to water for productive use, reduce vulnerability to floods and droughts, and ensure more stable agricultural production in the face of erratic weather patterns. Climate-resilient infrastructure, including flood-proofed systems, are designed to protect farms and enhance the reliability of water supply throughout the year. The integration of climate-smart agricultural practices (e.g. efficient irrigation methods, sustainable land management and pest control) are planned to increase productivity, food security and farm incomes. These improvements aim to support diversified cropping and multiple harvests per year, helping rural households to adapt to a changing climate. To ensure long-term impact, CAISAR is designed to strengthen institutional and community capacity through training for government staff, extension workers and FWUCs, equipping them with the knowledge and tools to manage irrigation systems and climate services effectively. The project seeks to promote social inclusion and gender equity by involving women, youth and Indigenous groups in decision-making and providing targeted livelihood support to the most vulnerable households. It is also geared to enhance financial inclusion by expanding access to rural credit, supporting financial literacy and working with local financial institutions to develop climate-resilient financial products. Through these combined efforts, CAISAR is expected to build stronger, more adaptive communities, promoting sustainable resource use and laying the foundation for a climate-resilient rural economy. The CAISAR project is expected to benefit a total of 1,714,375 people, comprising 562,757 direct beneficiaries (including 225,103 women), representing approximately 3.16 per cent of Cambodia's population, and 1,151,618 indirect beneficiaries (including 575,809 women), or 6.47 per cent of the national population.[4]

6. **Mitigation impact.** The funding proposal states that CAISAR is projected to reduce and/or sequester approximately 1,006,507 tonnes of carbon dioxide equivalent (tCO_2 eq) over a 20-year period, averaging around 50,325 tCO_2 eq per year. These reductions are set to be achieved through the adoption of climate-smart agricultural practices such as alternate wetting and drying for rice cultivation, crop diversification, improved soil and fertilizer management,

[3] See funding proposal, sections A.7 to A.14.
[4] See funding proposal, section A.6.

moving from diesel to solar-powered irrigation systems,[5] and improved fertilizer and residue management. Such practices aim to reduce methane and nitrous oxide emissions from rice cultivation and CO_2 emissions from energy use. Potential emissions from road transport, food processing and waste are acknowledged and these are expected to be negligible compared with the substantial reductions from improved agricultural practices, according to the project accredited entity (AE). The project focuses on production-level interventions, targeting minimally processed commodities such as rice, vegetables, poultry and fish, which have limited energy demands. Hence, any increase in post-harvest emissions will be marginal and offset by reductions from cleaner energy use and improved field practices.

1.2.1. Risks to impact.

7. **The rehabilitation of a 100 km road[6] may lead to greenhouse gas (GHG) emissions.** The AE indicates that the all-weather road within the command area is expected to have several key impacts, as outlined in the environmental, social and climate impact assessment (ESCIA).

(a) An influx of up to 735 skilled and unskilled workers is anticipated, primarily sourced from local communities during construction; the ESCIA states that this is a “short-term” influx and, because of the project’s limited scope, no significant post-construction population increase is expected.

(b) Deforestation is not considered a risk as the project focuses on knowledge improvement and irrigation access rather than large-scale land clearance, with no direct or indirect deforestation impacts identified.

(c) On biodiversity, some species may be temporarily affected during construction but mitigation measures are in place, as detailed in the biodiversity assessment.[7]

8. In response, the view of the independent Technical Advisory Panel (iTAP) is that the rehabilitation of the 100 km all-weather road within the command area is expected to have several impacts and the following key concerns remain unaddressed:

9. Influx of people. The anticipation of a short-term influx of 735 workers during construction noted in the ESCIA does not adequately assess the long-term migration pressures resulting from improved road access; improved connectivity could attract new settlers or investors, leading to unplanned population growth, a risk that requires further evaluation. It could also lead to land-use changes (e.g. conversion of land use from agriculture to residential and/or commercial uses).

10. Road length and mapping. The ESCIA repeatedly refers to short road sections, yet the total project involves 80–100 km of roads,[8] a significant length that should not be dismissed as negligible. A clear map delineating rehabilitated segments would improve transparency and help to assess cumulative impacts.

11. Deforestation exclusion. The ESCIA terms of reference explicitly exclude deforestation risks, arguing that the project is limited to "knowledge improvement and irrigation access". However, if rural road financing is included, this omission becomes a critical gap. Road rehabilitation, especially over 80–100 km, can indirectly drive deforestation by facilitating land encroachment or illegal logging; either unintended consequences (e.g. deforestation risks

[5] The emission estimates were calculated using the Food and Agriculture Organization of the United Nations (FAO) Environmental eXternalities ACcounting Tool (EX-ACT Tool) based on Intergovernmental Panel on Climate Change tier 1 methodology and are aligned with Cambodia’s nationally determined contributions. Full details of the calculations can be found in annex 22 to the funding proposal.

[6] One of the major construction activities under component 1, output 1.4, representing 37 per cent of the funding allocation for component 1.

[7] See funding proposal, ESCIA, annex 1 and section 7.4.2, and mitigation strategies in section 9.2.1.

[8] See funding proposal, annex 4, activity 1.4.3.

and/or maladaptation as well as malmitigation impacts) should be properly assessed, or the road rehabilitation component should be reconsidered.

12. Impact on GHG emissions. The road rehabilitation may or may not have a potentially significant impact on climate change mitigation. This is not clearly addressed in the funding proposal. This requires an assessment to explain and provide clear evidence regarding whether or not the road rehabilitation component is immaterial.[9]

13. Limited scope of assessments on road projects. As per the ESCIA tables on "project activities that may cause potential impacts on the environment and social aspects" at the Ou Tapaong, Lum Hach, Brambei Mom, Krapeu Trom, Yutasas, Stueng Krang Bat schemes, the assessment of road impacts is limited in scope.[10] The focus of the project's planned construction activities (such as the rehabilitation of canals and farm roads along with changes in land use and water management practices which present significant environmental risks) is limited to dredging, excavation and alteration of the existing water infrastructure. The report states that these all have the potential to cause habitat loss and fragmentation, affecting wildlife movement and species survival, disturbance to nesting and foraging areas, along with disruption to natural water flow and hydrological patterns, and could further degrade both terrestrial and aquatic ecosystems. The report, however, does not assess the potential impacts (e.g. increased traffic, urbanization, GHG emissions) of a rehabilitated all-weather road in these areas – areas that still form part of the Southern Tonle Sap Floodplains watershed given that this encompasses portions of Pursat, Kampong Chhnang, Kandal and Kampong Speu provinces.

14. For a robust assessment, the ESCIA should (i) evaluate post-construction migration risks; (ii) provide detailed road mapping; and (iii) either provide an analysis on unintended consequences (e.g. deforestation analysis, land-use change) or simply remove sub-activity 1.4.3 from the project's scope.

15. Monitoring and evaluation. The monitoring and evaluation framework should be used to reassess and update GHG emission reduction estimates if the actual adoption rates of the introduced practices or technologies vary from those used in the original (ex ante) GHG projections. This ensures that the reported emission reductions reflect real-world implementation outcomes and can be communicated accurately.

16. Subject to the resolution of this issue, the funding proposal's impact potential is assessed as low to medium.

1.3 Paradigm shift potential *Scale: Medium to high*

17. Advancing a paradigm shift in climate-resilient agriculture and water management, the CAISAR project seeks to make a significant departure from conventional, siloed approaches by integrating climate-resilient infrastructure, sustainable agriculture and institutional reform. By modernizing irrigation schemes and canals, the project is expected to improve year-round water availability, support diversified cropping and reduce communities' exposure to floods and droughts. This integration is designed to ensure that water infrastructure directly supports climate-smart agriculture, ecosystem-based adaptation and resilient rural livelihoods.

18. CAISAR is designed to contribute to low-emission development by replacing diesel pumps with solar-powered irrigation systems, reducing GHG emissions and enhancing energy security. The planned use of renewable energy reflects a forward-looking commitment to low-carbon, climate-resilient agriculture.

[9] The iTAP has already raised this similar issue in FP227, "Increase Resilience to Climate Change of Smallholders Receiving the Services of the Inclusive Agricultural Value Chains Programme", where IFAD was the AE. Lessons learned should have been applied to this proposed project.
[10] See funding proposal biodiversity assessment report for CAISAR.

19. The institutional capacity and mandate of the Ministry of Water Resources and Meteorology (MOWRAM) is envisioned to shift from infrastructure delivery to strategic water governance and integrated planning (see para. 37 below). The capacity-building at both the national and the subnational level, including support for provincial departments and FWUCs, are planned to improve their ability to manage water equitably and adaptively over the long term.

20. CAISAR plans to promote inclusive access to rural finance via Village Savings and Loan Associations and FWUCs, linking smallholders to credit, insurance and input markets. The public–private–producer partnerships (4Ps) initiative is designed to unlock concessional financing and crowd in private investment, thus reducing dependency on subsidies and enabling climate-resilient investments by smallholder farmers and agri-enterprises.

21. Through support for policy reform, the project is geared to help to embed climate resilience into national irrigation planning and governance. MOWRAM, in its planned enhanced role, is expected to help to support the replication and scale-up of climate-smart water-agriculture systems across the country.

22. CAISAR is designed to prioritize knowledge exchange and farmer training**,** facilitating the uptake of climate-smart agriculture technologies and promoting behavioural change. This is intended to enhance adaptive capacity not only during project implementation but also into the future.

23. Overall, CAISAR represents a strong paradigm shift by combining infrastructure modernization, low-carbon innovation, institutional transformation and financial inclusion. It strengthens the resilience of communities, ecosystems, investments and infrastructure, while laying the foundation for sustainable, scalable impact across Cambodia's agriculture and water sectors.

24. The paradigm shift potential is assessed as medium to high.

1.4 Sustainable development potential *Scale: Medium to high*

25. **Sustainable Development Goals (SDGs)**. The proposal aims to contribute to the achievements of the following: SDG 1 (No poverty) via helping farmers to earn more and become more resilient to climate impacts; SDG 2 (Zero hunger) through better farming methods and improved food production; SDG 5 (Gender equality) by giving women more access to finance, training and livelihood opportunities; SDG 6 (Clean water and sanitation) by protecting water sources and using water-saving farming techniques; SDG 8 (Decent work and economic growth) by supporting small businesses and rural jobs in farming and beyond; SDG 9 (Industry, innovation and infrastructure) by improving roads, storage, irrigation and access to digital tools; SDG 10 (Reduced inequalities) by focusing on poor, remote and vulnerable communities; SDG 12 (Responsible consumption and production) by promoting sustainable farming that protects the land; SDG 13 (Climate action) by helping people to adapt to climate change and reduce emissions; SDG 15 (Life on land) by restoring forests, protecting soil and improving biodiversity; and SDG 17 (Partnerships for the goals) by working with government, communities and international partners.

26. **Advancing rural socioeconomic development.** The CAISAR project is designed to enhance agricultural productivity and income security for smallholder farmers, thus playing a vital role in strengthening rural livelihoods in Cambodia. The planned improved irrigation infrastructure, formation of FWUCs and integration of climate-resilient practices empowers farmers to sustainably manage water resources and increase yields. Capacity-building, the 4Ps

initiative and improved access to finance and markets aim to generate inclusive economic opportunities, particularly for vulnerable communities.[11]

27. **Promoting environmental sustainability.** Environmental preservation is central to the CAISAR design given that the project prioritizes nature-based solutions such as agroforestry, soil and water conservation, and efficient solar-powered irrigation systems; nature-based solutions help to restore degraded ecosystems and reduce GHG emissions. These interventions are aligned with Cambodia's climate goals under the Paris Agreement and are expected to improve biodiversity, enhance soil health and mitigate flood and drought risks while ensuring climate-resilient agricultural landscapes.[12]

28. **Empowering women and promoting gender equity.** CAISAR is designed to incorporate a robust gender action plan to advance women's empowerment in agriculture and water governance. It targets increased female participation in decision-making bodies like FWUCs, provides training in climate-smart agriculture and financial literacy, and supports women-led livelihood diversification. The CAISAR project seeks to address traditional gender roles and to ensure equal access to project benefits, to strengthen women's economic agency, and to contribute to gender-responsive climate adaptation.[13]

29. **Risk to sustainable development potential.** The issue on rehabilitation of a 100 km road which may lead to GHG emissions and could adversely impact other SDGs such as SDG 13, and 15.

30. The sustainable development potential is assessed as medium to high.

1.5 Needs of the recipient *Scale: Medium to high*

31. **Need to reduce vulnerability of target beneficiaries.** Over 75 per cent of Cambodia's rural population relies on rain-fed agriculture, making them highly susceptible to changing precipitation patterns and water shortages. The project addresses this by establishing community-specific early warning systems, providing drought, flood and extreme weather advisories to safeguard vulnerable groups.[14]

32. **Need for equitable access to climate-smart irrigation.** Smallholder farmers (especially those with limited land), women, Indigenous Peoples and marginalized communities often lack access to irrigation infrastructure. These groups remain more exposed to droughts and water scarcity in the absence of equitable distribution of irrigation. Ensuring fair access helps to reduce rural poverty and food insecurity by enabling all farmers to benefit from productivity improvements. Under CAISAR component 2[15] affordable water infrastructure is expanded, ensuring that low-income and marginalized farmers have access to efficient irrigation systems. Indigenous farming communities in flood-prone regions will gain resilient water management solutions tailored to local climate risks.

33. **Need for access to finance.** Many farmers lack access to concessional financing or formal lending mechanisms, limiting their ability to invest in climate-adaptive technologies. Financially constrained farmers will receive support through concessional finance, unlocking long-term adaptation investments. The financial barriers faced by vulnerable farmers will be

[11] See funding proposal, section D.3.
[12] As footnote 11 above.
[13] See funding proposal, sections D3 and D.4, and annex 8, "Gender Assessment and Social Inclusion Plan".
[14] See funding proposal, output 1.3, "Increased access to climate information and advisory services for climate responsive water-use and crop planning".
[15] See funding proposal, section B.3, component 2.

addressed via the 4Ps initiative,[16] which promotes microfinance, concessional loans and climate-smart credit lines.

34. **Need to reduce gender disparities.** Women farmers represent nearly 50 per cent of the agricultural workforce, yet face structural barriers in land ownership, credit access and participation in irrigation governance. CAISAR aims to ensure that women-led cooperatives and smallholder farmers actively participate in agricultural decision-making, promoting gender-equitable climate resilience strategies. Women farmers will benefit from improved irrigation and financial inclusion.

35. Removing barriers to climate adaptation, strengthening financial accessibility and embedding social equity into agricultural resilience strategies ensures that the needs of Cambodia's most vulnerable populations are prioritized in national climate adaptation efforts.

36. The needs of the recipient are assessed as medium to high.

1.6 Country ownership *Scale: Medium to high*

37. **No-objection letter secured**. The Ministry of Environment issued a no-objection letter on 24 February 2025.

38. **Alignment with national climate and development priorities.** CAISAR is strategically aligned with Cambodia's National Adaptation Plan (NAP), Climate Change Strategic Plan and nationally determined contributions, with a strong focus on the agriculture and water sectors. It supports key national adaptation targets, including the development of climate and flood warning systems, standardized best practices for irrigation and strengthened institutional capacity for climate-resilient water management. The project advances the objectives of the Climate Change Strategic Plan by enhancing food, water and energy security; safeguarding critical ecosystems; promoting low-carbon technologies; and strengthening climate governance through inclusive, coordinated approaches. Leveraging climate finance, concessional funding and private sector partnerships enables CAISAR to embed long-term sustainability through policy alignment and blended financing mechanisms.

39. **Strategic partnership with MOWRAM.** MOWRAM is the designated national authority responsible for the management of Cambodia's water resources. Its extensive mandate includes river basin planning, irrigation development and management, flood and drought monitoring, and coordination of climate-resilient water infrastructure. The ministry's leadership in the CAISAR project is institutional and strategic given its mandate, operational scope and capacity to implement complex, climate-responsive irrigation programmes. MOWRAM will lead CAISAR as an executing entity and host the Project Management Unit (PMU), which is tasked with the coordination of all components and consolidation of planning and reporting. The PMU, staffed by experienced government and project-recruited personnel, will oversee technical quality, financial management, procurement, and monitoring and evaluation. The PMU will also manage component 2 (modernization of irrigation systems) and output 3.1 (institutional capacity-building), which are central to the project's transformational ambition. MOWRAM has a strong track record in project execution and interministerial coordination; it has chaired a Project Steering Committee, which includes high-level representation from the Ministries of Economy and Finance, Interior, Agriculture and Environment, as well as provincial governments. MOWRAM is positioned to effectively manage cross-sectoral integration and risk mitigation. The ministry is actively strengthening its institutional capacity through targeted training in climate-resilient irrigation design, Supervisory Control and Data Acquisition (SCADA) systems, river basin management and gender and social inclusion. These capacity-building efforts are embedded within output 3.1, making MOWRAM the leader and a direct beneficiary of

[16] See funding proposal, output 1.2, on climate adaptive value chains developed by 4Ps and increased access to finance.

strengthened operational capability. This embedded institutional strengthening, along with its centralized role in water governance, justifies the ministry's leadership and ensures that the project will be anchored in national systems, increasing the likelihood of long-term sustainability and scale-up across Cambodia's water and agriculture sectors.

40. **Access to local implementation partners supporting MOWRAM.** To ensure effective, inclusive and sustainable implementation of the CAISAR project, MOWRAM collaborates with several key national and subnational institutions, each of which brings specialized capabilities and mandates.

41. **The National Committee for Sub-National Democratic Development**, which plays a pivotal role in implementing component 1 and component 3, output 3.2. The committee has extensive experience in facilitating decentralized service delivery and works through provincial- and district-level agencies. It ensures strong coordination with complementary rural development and climate adaptation programmes such as Ministry of Agriculture, Forestry and Fisheries/Agricultural Services Programme for Innovation, Resilience and Extension, Ministry of Commerce/Accelerating Inclusive Markets for Smallholders, Ministry of Rural Development/Sustainable Assets for Agriculture Markets, Business and Trade and the GCF project Public-Social-Private Partnerships for Ecologically-Sound Agriculture and Resilient Livelihood (PEARL) to maximize synergy in delivering extension services, capacity-building and climate information.

42. **Provincial departments of water resources and meteorology**, which serve as key technical arms of MOWRAM at the provincial level. They support the PMU in coordinating field activities, stakeholder engagement, water resource planning and dissemination of project-related information and advisories to local communities.

43. **The Department of Farmer Water User Committees**, which oversees the establishment, training and operational guidance of FWUCs, ensuring that these community-level institutions are equipped to manage irrigation infrastructure sustainably.

44. **FWUCs**, which are the direct interface with farming communities, responsible for managing water distribution, maintaining irrigation systems and making local decisions that promote efficient, equitable water use. Their involvement is essential to build ownership, ensure sustainability and promote climate-resilient farming at the grassroots level.

45. These institutions complement MOWRAM's leadership and technical expertise, ensuring a coordinated, decentralized approach to climate-resilient irrigation development across Cambodia.

1.6.1. Risk to country ownership.

46. **Limited technical capacities of local partners.** Given the central role played by MOWRAM and its relevance in Cambodia's water resources sector, several institutional limitations may affect the effective and efficient implementation of the CAISAR project. The Institutional Assessment conducted by PPIC (Project Preparation Implementation Consultant (PPIC) and Water and Power Consultancy Services (WAPCOS) in December 2022 reports that MOWRAM has limited technical and human resource capacity, particularly in the areas of modern irrigation management and governance systems, which remain critical bottlenecks to rural development outcomes.[17]

47. **Recommendation.** While the funding proposal outlines relevant mitigation measures,[18] several capacity-building gaps remain that should be addressed to ensure long-term impact:

[17] See funding proposal, section F, para. 170.
[18] See funding proposal, section F, para. 179.

48. **Sustainability of training.** The current training approach is limited to the project timeline, with no mechanism to sustain capacity development. The AE should work with centres of excellence like the Institute of Technology of Cambodia to establish a permanent training unit or curriculum within MOWRAM to support ongoing professional development.

49. **Lack of certification and assessment.** There are no defined competency benchmarks or systems to assess training effectiveness. The AE should introduce a certification framework linked to job roles and implement pre- and post-training assessments to track skill development and application.

50. **Limited focus on digital tools.** Digital technologies such as SCADA, geographic information system based planning and mobile data systems are not clearly and/or fully integrated into the training. The AE should expand training modules to include these tools and support MOWRAM in adopting cloud-based or geographic information system integrated platforms for real-time water management.

51. **Weak stakeholder linkages.** The training plan does not extend to FWUCs beyond their formation or include the private sector. The AE should design joint training for leaders from MOWRAM, the provincial departments of water resources and meteorology and FWUCs and engage private service providers in technical training to strengthen local support systems.

52. **Gender inclusion.** While gender awareness is noted, there is no clear strategy to involve women in technical training or irrigation leadership. A gender-responsive training plan should be developed to encourage women's participation, with tailored content and incentives.

53. Country ownership is assessed as medium to high.

1.7 Efficiency and effectiveness *Scale: Medium to high*

54. The GCF commitment of USD 80 million (USD 40 million grants, USD 40 million concessional loan) is seen to play a pivotal role in mobilizing an additional USD 160.1 million in co-financing, resulting in total project financing of USD 240.1 million. This reflects a leverage ratio of 3:1, indicating that every United States dollar of GCF funding attracts two additional United States dollars from other financiers. AIIB is providing USD 100 million in senior loans, representing the largest share (41.6 per cent) of total financing; AIIB has indicated that this loan is likely to be approved within the second half of 2025.[19] The concessional nature of GCF funding improves the project's risk-return profile and encourages AIIB and IFAD to co-finance the CAISAR project. GCF funding is strategically targeted towards climate adaptation measures; the lowering of investment risks and increasing financial viability enables the mobilization of large-scale capital for climate-resilient development, demonstrating its critical role in unlocking transformational finance and fostering long-term partnerships in climate-vulnerable contexts.

55. **Complementarity and synergy with PEARL.** CAISAR and the PEARL project are highly complementary, both supporting climate-resilient, market-driven agriculture in Cambodia's Northern Tonle Sap Basin. PEARL focuses on fostering public–social–private partnerships to engage agricultural suppliers, traders, retailers and financial institutions with local communities. Key areas of synergy include the following: (i) CAISAR will build on PEARL agrometeorological advisory services to deliver timely, tailored climate information to farmers and value chain actors; (ii) joint efforts will strengthen market access, enhance value chains and improve the business environment for smallholders; (iii) CAISAR targets public institutions like MOWRAM and the National Committee for Sub-National Democratic Development, while PEARL engages private sector actors, together fostering a holistic, cross-sector capacity-building approach; (iv) CAISAR climate-smart irrigation and farming methods complement PEARL ecologically sound practices, advancing agricultural sustainability and resilience; and (v) both

[19] AE/iTAP video conference held on 12 May 2025.

projects promote private investment in climate-resilient agriculture, improving access to finance and market integration. Together, CAISAR and PEARL reinforce each other's objectives, enabling a more integrated and impactful approach to building climate resilience in Cambodia's agriculture sector. The synergy between CAISAR and PEARL enhances efficiency by leveraging shared systems and minimizing redundancies, and improves effectiveness by reinforcing and integrating interventions across the public and private sector, resulting in stronger, more sustainable outcomes.

56. **Robust economic and financial analysis.** The CAISAR project demonstrates strong economic and financial viability, underscoring its potential to deliver impactful, cost-effective and sustainable climate-resilient development outcomes in Cambodia. The project is expected to achieve high returns under both economic and financial scenarios, validating its efficiency and effectiveness in resource allocation and outcome delivery.[20] The detailed risk assessment of CAISAR shows the analysis on how different scenarios impact their financial viability through key indicators (i.e. internal rate of return (IRR) and present net value (NPV)). The base scenario, evaluated over 30 years at a 9.47 per cent social discount rate, yields a strong IRR of 16.34 per cent and an NPV of USD 143.57 million, indicating robust profitability under conservative assumptions.

57. **Project benefits reduction**. A 10 per cent decline in benefits (due to lower output prices, yields or adoption rates) decreases the IRR to 13.27 per cent and the NPV to USD 74.25 million. A 20 per cent drop pushes the IRR close to the discount rate at 9.74 per cent, with the NPV at nearly breakeven point at USD 4.94 million. The switching value shows that benefits can fall by 20.71 per cent before the project becomes unviable.

58. **Cost increases.** A 10 per cent rise in costs (e.g. higher construction, farm inputs or service prices) reduces the IRR to 13.56 per cent and the NPV to USD 88.61 million. A 20 per cent increase lowers the IRR to 10.98 per cent. The project can withstand up to a 26.12 per cent cost escalation before losing economic feasibility.

59. **Shorter evaluation periods.** Reducing the assessment to 20 years drops the IRR to 14.27 per cent and the NPV to USD 69.93 million. A 10-year horizon results to a near-zero IRR (1.82 per cent) and a negative NPV (–USD 31.3 million). This highlights the importance of long-term sustainability.

60. **Benefit delays.** A one-year delay has minimal impact, but a three-year delay slightly reduces the IRR to 15.88 per cent and the NPV to USD 137.98 million; this suggests that the project is resilient to implementation lags.

61. **Social discount rates.** A lower 6 per cent rate increases the NPV to USD 333.07 million, reflecting a higher value in stable economic conditions. A higher 11 per cent rate reduces the NPV to USD 93.54 million, indicating sensitivity to macroeconomic risks or increases in the cost of doing business.

62. **Adoption rate changes.** If farmer adoption of new practices falls from 75 to 60 per cent, the IRR declines to 15.64 per cent, and the NPV drops to USD 123.49 million, underscoring the importance of farmer commitment and constant stakeholder engagement.

63. **Exclusion of environmental benefits.** Ignoring GHG emission reductions lowers the IRR to 14.51 per cent, and NPV to USD 105.64 million. Omitting flood mitigation benefits drastically reduces the IRR to 10.42 per cent and the NPV to USD 20.44 million, emphasizing the project's low carbon and climate resilience impacts.

64. **Limited irrigation expansion.** If no new irrigated areas are added post-project, the IRR falls to 14.53 per cent and the NPV to USD 93.06 million, highlighting the need for scalability.

[20] See funding proposal, annex 3 (economic and financial analysis).

65. This analysis demonstrates the project's financial resilience while identifying critical vulnerabilities, particularly benefit reductions, cost overruns and shortened timelines, which could undermine its success. It also underscores the significance of environmental co-benefits and long-term planning in ensuring economic sustainability.

1.7.1. Risk to efficiency and effectiveness.

66. **Increase in GHG emissions.** The potential risk of increased GHG emissions due to the road rehabilitation project will also impact the economic and financial viability of the CAISAR project. The reduction in GHG emissions accounts for 5.47 per cent of benefits,[21] monetized using AIIB social carbon price projections through 2056.[22] Without the reduction in GHG emissions, the economic internal rate of return (EIRR) falls to 14.51 per cent.[23]

67. **No road rehabilitation project**. If the road rehabilitation project is excluded as one of the funded components, the following scenarios may happen:

(a) The effectiveness of CAISAR will be affected (i.e. post-harvest losses remain at 10–15 per cent instead of being reduced to 5 per cent, product loss during transport stays at 10 per cent, rather than being halved, market commercialization rates drop from 90 per cent (with project) to 50–70 per cent (without project), travel time to markets would remain at five hours instead of being reduced to two hours).[24]

(b) The cost–benefit advantage of component 1 will be affected (i.e. EIRR may fall below the social discount rate of 9.47 per cent, especially in rural areas where road conditions are critical for economic activity). The economic viability assessments for component 1, which includes road rehabilitation (i.e. NPV of USD 36.5 million, EIRR = 15.79 per cent, benefit–cost ratio = 1.11) may be affected.[25] However, this impact remains to be assessed considering that the sensitivity and risk analysis does not explicitly mention how the road rehabilitation component will impact the IRRs and NPVs.

68. **Limited assessment of climate extreme events.** The risks of climate variability (e.g. extreme weather wiping out a season of crops produced, disrupting yields or damaging irrigation and/or road infrastructures) is not explicitly mentioned. It may, however, be indirectly captured in the benefit reduction scenarios (–10 per cent/–20 per cent). The potential missing climate risks impact assessments that could be materially relevant (but are not mentioned) include drought resilience (impact on water availability for irrigation), temperature variability (effect on crop yields), long-term climate change projections (e.g. shifting rainfall patterns beyond the 30-year horizon) and climate extreme events (e.g. tropical cyclones). The AE should clearly integrate an assessment of how climate extreme events impact investments under difference scenarios.

69. **Unclear alignment with other GCF-funded programmes.** The iTAP notes that the CAISAR project shares significant thematic and operational overlap with the Cambodian Climate Financing Facility (CCFF), an existing GCF-funded programme with the Korea Development Bank as AE. CCFF aims to mobilize finance for low-carbon and climate-resilient investments in Cambodia through a dual-track approach: a lending facility offering concessional debt to projects and financial institutions, and a technical assistance facility supporting capacity-building for market actors, including local financial institutions. CCFF is managed by the Agricultural and Rural Development Bank, a Cambodian State-owned bank, and is capitalized with a USD 100 million lending facility and a USD 9 million grant-funded technical assistance facility. Given the shared objectives, particularly around promoting climate-resilient agriculture,

[21] GHG emissions are quantified using the FAO EX-ACT Tool (v.9.4).
[22] See funding proposal, annex 3, section II.f (greenhouse gas emission reduction).
[23] See funding proposal, annex 3, section III (results, sensitivity analysis and risk analysis).
[24] See funding proposal, annex 3, section II.c (component 1: assumptions on the post-harvest development).
[25] See funding proposal, annex 3, section III.c (return on investment for each component).

strengthening rural financial systems and building institutional capacity, the apparent lack of coordination between CAISAR and CCFF is concerning. The current funding proposal does not provide any evidence of alignment or collaboration with CCFF implementers, raising questions about the potential for duplicative efforts or suboptimal use of GCF concessional resources. At a minimum, this represents a missed opportunity to enhance synergies and increase the overall effectiveness of GCF interventions in Cambodia. At worst, it risks inefficiencies in programming and fragmentation of impact. The iTAP encourages greater diligence in ensuring coordination among GCF-supported initiatives operating in the same sectoral and geographic space. The no-objection process should serve as a safeguard to promote strategic alignment and avoid siloed implementation, particularly in countries like Cambodia with relatively limited institutional capacity.

70. Efficiency and effectiveness is assessed as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

71. The iTAP recommends that the Board approve this funding proposal. Given the key environmental and institutional risks identified in the CAISAR project, specific disbursement conditions for IFAD are recommended.

72. To address the potential GHG emissions, maladaptation risks or unintended consequences associated with the 100 km road rehabilitation component (see paras. 8–9, 25 and 45– 46 above), IFAD should be required to submit the following as the conditions precedent to the first disbursement of the GCF Proceeds under the funded activity agreement:

Conditions precedent to the first disbursement of the GCF Proceeds:

73. Delivery to the GCF by the Accredited Entity, in a form and substance satisfactory to the Secretariat, of:

(a) An updated Environmental, Social, and Climate Impact Assessment (ESCIA), along with a road-specific Environmental and Social Action Plan (ESAP) under Annex 6 (E&S Documents) to the funding proposal, to be submitted prior to any construction activities. This updated assessment must address cumulative GHG emissions, potential land-use change, post construction migration pressures, detailed road mapping, and deforestation risks;

(b) An updated Annex 22 (Assessment of GHG Emission Reductions and their Monitoring and Reporting) to the funding proposal and supporting documents enclosed with Annex 22, to recompute the GHG calculations, including indirect impacts related to road rehabilitation and construction in terms of road usage, potential land-use change and deforestation;

(c) An updated Annex 3 (Economic and Financial Analyses) to the funding proposal to align with revised GHG data findings and reflect their impact on the economic internal rate of return and other investment metrics;

(d) An updated Annex 11 (Monitoring and Evaluation Plan) to the funding proposal, which should be used to reassess and update GHG emission reduction estimates if the actual adoption rates of the introduced practices or technologies differ from those used in the original (ex-ante) GHG projections, with a view to ensuring that the reported emission reductions reflect actual implementation outcomes and are presented accurately; and

(e) Other relevant annexes, as needed, to reflect the changes and outcomes of the ESCIA, ESAP, Annex 22, Annex 3 and Annex 11 to the funding proposal.

74. To mitigate institutional risks (See paragraphs 41 and 42), IFAD should be required to provide the following covenants under the funded activity agreement:

Covenants to be included in the funded activity agreement:

(a) The Accredited Entity shall ensure that the Project Implementation Manual includes the time-bound institutional capacity strengthening plan, prepared by the Accredited Entity in support of MOWRAM, covering training in climate-smart irrigation and financial management, supervisory control and data acquisition (SCADA) system upgrades, and third-party technical support in relation to Output 3.1 of Component 3.

(b) The Accredited Entity shall monitor the implementation of the institutional capacity strengthening plan on a quarterly basis, and inform the outcomes to the Fund in the APR.

75. These conditions are essential to ensure effective risk management, safeguard compliance, and alignment with GCF's environmental and fiduciary standards.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP270)

Proposal name:	Climate Adaptive Irrigation and Sustainable Agriculture for Resilience (CAISAR) in Cambodia
Accredited entity:	International Fund for Agricultural Development
Country:	Cambodia
Project size:	Medium

Impact potential
IFAD takes note of the assessment and recommendations provided. IFAD would like to reassure GCF that its feasibility analysis of the roads showed that it is unlikely to have any significant impact on the environment as there will not be any direct clearing of forest during construction of the roads; the roads will be rehabilitated to enable market access and while some short sections might be widened, there is no new road construction. In the full-blown Biodiversity Impact Assessment, the area where the roads are to be constructed do not coincide with known forest stands, and it is unlikely to trigger externalities such as deforestation in the period after construction. Furthermore, many of the roads will be built on canal embankments and will enhance flood control and drainage systems. These types of roads do not entail any major post construction migration risks, and they were not assessed as significantly contributing to more GHGs owing to forest loss or increased traffic volumes. Nevertheless, the AE commits to review the ESCIA to address and report on some of the concerns identified by GCF.
Paradigm shift potential
IFAD takes note of the positive assessment.
Sustainable development potential
IFAD takes note of the positive assessment.
Needs of the recipient
IFAD takes note of the positive assessment.
Country ownership

IFAD takes note of the positive assessment.
Efficiency and effectiveness
IFAD takes note of the positive assessment.
Overall remarks from the independent Technical Advisory Panel: IFAD thanks the iTAP for its positive review of the proposal and takes note of the iTAP conditions precedent to the first disbursement of the GCF Proceeds under the funded activity agreement to address the potential GHG emissions, maladaptation risks or unintended consequences associated with the 100 km road rehabilitation component to ensure effective risk management, safeguard compliance, and alignment with GCF's environmental and fiduciary standards. IFAD also confirm to monitor the implementation of the institutional capacity strengthening plan on a quarterly basis, and inform the outcomes to the Fund in the APR.

KINGDOM OF CAMBODIA

Nation – Religion – King

MINISTRY OF WATER RESOURCES AND METEOROLOGY

CLIMATE ADAPTIVE IRRIGATION AND SUSTAINABLE AGRICULTURE FOR RESILIENCE (CAISAR)

GENDER ASSESSMENT &
GENDER ACTION & SOCIAL INCLUSION PLAN

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Abbreviations

CAISAR	Climate Adaptive Irrigation and Sustainable Agriculture for Resilience
EA/IA	Executing Agency/Implementing Agency
ESCP	Environmental and Social Commitment Plan
ESF	Environmental and Social Framework
ESMF	Environmental and Social Management Framework
ESMP	Environmental and Social Management Plan
ESO	Environmental and Social Officers
ESS	Environmental and Social Standards
FPICon	Free Prior and Informed Consultation
GASIP	Gender Assessment and Social Inclusion Plan
IDA	International Development Association
ILO	International Labour Organization
IP	Indigenous People
IPP	Indigenous Peoples' Plan
IPPF	Indigenous Peoples Planning Framework
MoWRAM	Ministry of Water Resources and Meteorology
MoSALVY	Ministry of Social Affairs, Labor, Vocational Training and Youth Rehabilitation
NCDD	National Committee for Sub-National Democratic Development
NGO	Non-Government Organization
PDWRAM	Provincial Department of Water Resources and Meteorology
PMU	Project Management Unit
PIU	Project Implementation Unit
PPC	Project Preparation Consultants
RGC	Royal Government of Cambodia
RP	Resettlement Plan
RPF	Resettlement Planning Framework
SA	Social Assessment
SEA	Sexual Exploitation and Abuse
SEP	Stakeholder Engagement Plan
SEP	Stakeholder Engagement Plan
SH	Sexual Harassment
SIB	Subproject Information Booklet
VAC	Violence Against Children

1. INTRODUCTION

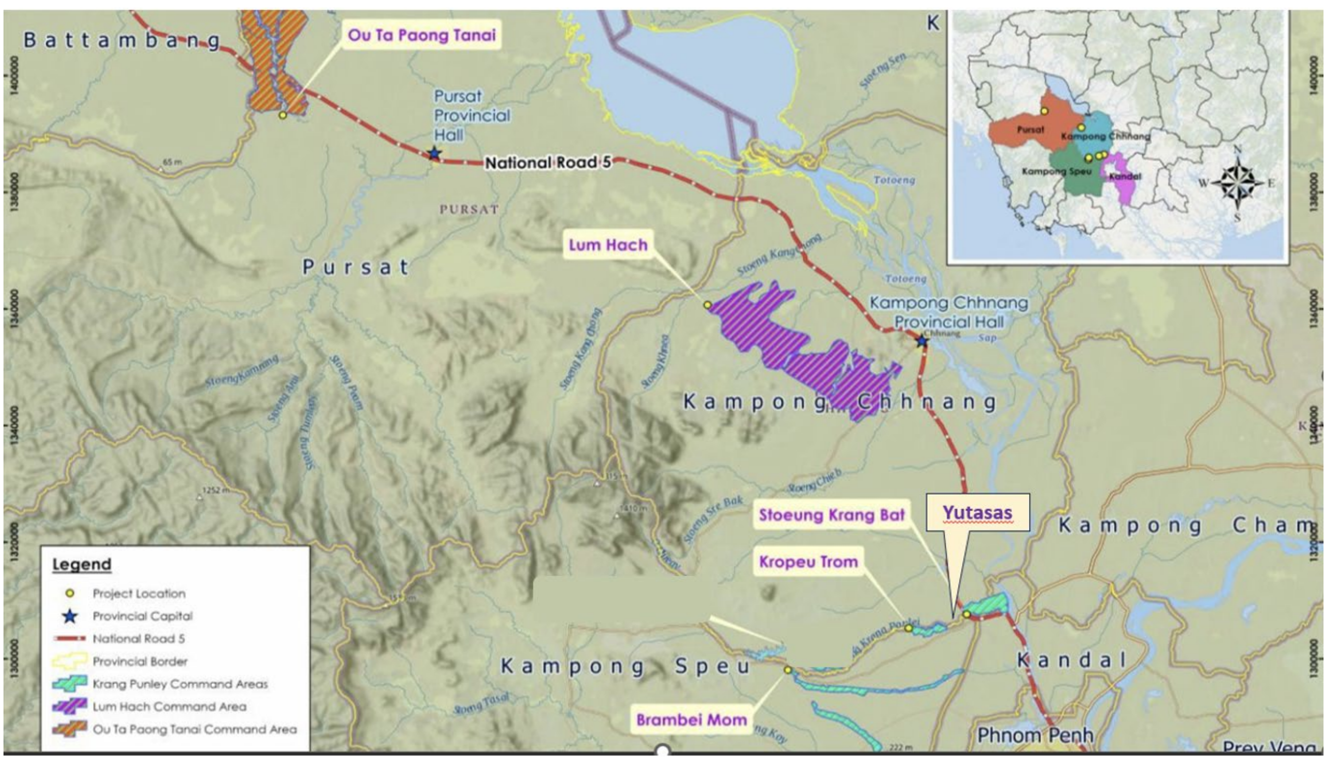
1.1 Project Background

1.1.1 Project objective

The project objective is to increase climate adaptation, mitigate the negative impact of extreme climate events, and improve livelihoods of smallholder farmers and vulnerable rural communities in four provinces of Cambodia. Mitigation is a co-benefit of this adaptation project, as it will also contribute to reducing GHG emissions, including methane emissions from rice fields. These objectives will be achieved by implementing three components that aim at addressing climate change vulnerabilities, increasing agriculture productivity, and developing institutional capacities. The investments to be implemented will (i) combine robust climate-resilient water management and agricultural practices at the farm level, (ii) establish climate-proofed irrigation and flood control infrastructure and (iii) develop institutional capacity to plan, maintain and operate irrigation and flood control infrastructure in a changing climate context.

The project objective is to increase climate adaptation, mitigate the negative impact of extreme climate events, and improve livelihoods of smallholder farmers and vulnerable rural communities in four provinces of Cambodia, including Pursat, Kampong Chhnang, Kampong Speu, and Kandal provinces (see map below).

Figure 1 − Map showing six irrigation sub-schemes located in four provinces



1.1.2 Beneficiaries

The CAISAR project aims to bring about benefits to 100,000 households (500,000 people, 3% of the population). The project will be executed through the Ministry of Water Resources and Meteorology (MOWRAM) and the National Committee for Sub-National Democratic Development Secretariat (NCDD-S), a Direct Access Accredited Entity to the GCF.

1.1.3 Project activities

The Project will carry out a wide range of activities in the four project provinces. These activities are organized into three project components:

Component 1. *Improving farm-level climate adaptation, resilience, and water use efficiency*

The objective of this component is to build climate resilience (CR) of smallholder farmers and enhance sustainable production through evidence-based planning and context-relevant climate resilient practices at the farm level. This component is designed to address the lack of knowledge and skills to deploy technologies and practices at farm level by farmers and the lack of appropriate extension services to propagate them. It will introduce farmers with various climate resilient technologies and practices for both rice and non-rice activities such as vegetable production, poultry and aquaculture.

Output 1.1: Increased farmer capacity in climate resilient agriculture

This output will focus on developing farmer's capacity in deploying CR technologies and practices to transform the agricultural production system to adapt to the changing climate context. Farmer's will be trained to first develop Action Plans (AP) to re-orient farmer behaviour and assist them in transforming the agriculture production system in a manner that is better adapted to factoring in the agro-ecological context and expected climate change impacts.

- Activity 1.1.1 Preparation of community-based action plans (AP) to transform agriculture with CR practices.
- Activity 1.1.2 Preparation of training materials to support implementation of the AP.
- Activity 1.1.3 Conduct training to create a pool of expertise to demonstrate and propagate the CR technologies and practices.
- Activity 1.1.4 Training and advisory on CR technologies and practices
- Activity 1.1.5 Strengthening and fostering tailored mechanization service providers for improved mechanization service delivery.
- Activity 1.1.6 community-based monitoring and evaluation (CBME) of implementation

Output 1.2 Climate adaptive value chains developed by 4Ps and increased access to finance

This output involves improving and enhancing some value chains that are key for the project area and include rice, vegetable, chicken and aquaculture value chains, through the use of Public Private Producer Partnerships (4Ps) and increased access to finance, which will improve market access, climate adaptability, and ensure increased income for smallholders in the value chains.

- Activity 1.2.1 Value chain study and planning
- Activity 1.2.2 Establish District Multi-Stakeholder Platforms (MSPs)
- Activity 1.2.3 Public Private Producer Partnership Facility (4PF)

Output 1.3. Increased access to and use of climate information and advisory services for climate responsive agriculture planning

This sub-component will strengthen the production and dissemination of tailored agro-meteorological information to inform climate responsive management and planning of agriculture in the project target areas through ICT technologies. The aim is to ensure that agro-meteorological services are accessible and useful to farmers to manage climate risks, access to and use of water and efficient cropping systems.

- Activity 1.3.1 Strengthening of agrometeorological stations in the project area
- Activity 1.3.2 Establish Provincial-Level Coordination Platform for climate-informed agricultural advisories
- Activity 1.3.3. Building the capacities to deliver services
- Activity 1.3.4 Establish the agromet information, systems and the awareness raising and capacity building of farmers and stakeholders in applying the services.

Output 1.4: Increased resilience of farm road infrastructure to climate change

- Activity 1.4.1 Identification, selection and initial planning
- Activity 1.4.2 Technical survey and design considerations, preparation of cost estimation
- Activity 1.4.3 Improve 90 Kilometers of farm roads.
- Activity 1.4.4 Handing over of the completed works.

Component 2: Irrigation Infrastructure for increased resilience

Component 2 is linked with Component 1 such that it facilitates the implementation of CR on farm crop and water management practices through improved field level water supply delivery and drainage. It will focus on rehabilitating and modernizing irrigation and flood protection/drainage infrastructure in the six sub-projects, including irrigation and drainage canals, flood control embankments, and ponds, to provide high-efficiency climate-resilient irrigated agriculture systems for adapting to both increasing flood and drought conditions.

Output 2.1 Modernized irrigation schemes and ponds

- Activity 2.1.1 Technical analysis, field surveys and preparation of plans for system upgrading
- Activity 2.1.2 Implementation of infrastructure upgrading
- Activity 2.1.3 Preparation of canal O&M plans including application of ICT and SCADA for operation

Output 2.2: Flood-proofed and Improved Drainage

- Activity 2.2.1 Technical analysis, field surveys and preparation of plans for system upgrading
- Activity 2.2.2 Strengthening and construction of flood control and drainage infrastructures.

Output 2.3: Farmers Water User Communities (FWUC) established and trained

- Activity 2.3.1 Formation of institutional strengthening of the FWUC
- Activity 2.3.2 Build technical capacities of FWCU for canal structure O&M

Output 2.4: Water information system (CIEWS and SCADA) established and operational in sub-projects

- Activity 2.4.1 Preparing all processes for establishment of the SCADA and Water information system

Component 3. Institutional Strengthening

Output 3.1 Strengthened MOWRAM Capacity

- Activity 3.1.1 Strengthening River basin management and Water Accounting Systems
- **Activity 3.1.2. Communicating with stakeholders**
- **Activity 3.1.3. Support to and guidance of Leaders of Commune, Village and FWUCs and Gender Awareness**
-

Output 3.2 Strengthened NCCDS at national and provincial level

- **Activity 3.2.1 Strengthening of NCDDS at the national level**
- **Activity 3.**2 **.2 Strengthening of the provincial teams**

1.1.4 Project's Environmental and Social Risks and Impacts

While the project will bring about an overall positive, long-term social and environmental impacts, some project activities, particularly those under Component 2 (Upgrading and Climate-Proofing Water Infrastructure for Increased Resilience) are anticipated to generate the following environmental and social risks and impacts:

During project implementation, the following environmental risks and impacts are anticipated – at specific construction sites, due to implementing some of the above mentioned project activities:

Environmental Risks and Impacts

- Generation of noise and vibration due to construction operation
- Pollution of air (dust), water (disturbance), soil (leakage of oil, chemicals…)
- Generation of solid, hazardous, domestic waste (temporary during construction stage)
- Potential impact on biodiversity at farming ecosystem and health of farmers due to crop intensification in the target command area (during scheme's operational stage)
- Occupational health and safety (for project workers, particularly contractors' workforce)
- Disease transmission (for project workers, particularly contractors' workforce)
- Road and traffic safety (for project workers, particularly contractors' workforce)
- Unexploded ordnance (associated with physical construction activities that involve ground-breaking, excavation…).

Most of these risks and impacts are temporary, localized and small-scaled. Mitigation measures for these environmental risks and impacts are being proposed to avoid these impacts, including measures through construction measures. Where avoidance is not possible, these environmental risks and impacts will be minimized during construction phase – based on the following principles:

- Technical standards and regulations must be abided by.
- Disturbance to the livelihoods of the local people must be minimized.
- The proposed measures must be environmentally and socio-economically feasible.
- Construction equipment and methods must be environmentally friendly.
- Monitoring activities must be conducted on a regular basis.

Social Risks and Impacts

During project implementation, the following social risks and impacts are anticipated – at specific construction sites.

- Potential land acquisition (mostly linear impact for rehabilitating/construction of irrigation canals)
- Temporary restricted water access for farmers in target command area during construction phase
- Gender inequality (e.g., during the process of crop intensification)
- Sexual Exploitation and Abuse (SEA), Sexual Harassment (SH), Gender Base Violence (GBV) due to labor influx
- Exclusion of vulnerable groups (e.g. ethnic minorities, poor, people with disabilities, female-headedness (e.g. because of their restricted access to project information, language, and other personal disadvantages…)
- Disease transmission (due to labor influx and locations that is specific for certain diseases)
- Road and traffic safety (during to increased transportation activities during construction)
- Hunting, trading, and consumption of animals from the wild (due to labor influx).

1.1.5 Mitigation Measures

To ensure project's risks and impacts are avoided/minimized/mitigated, the project has prepared the following documents:

- Resettlement Policy Framework (RPF)
- Indigenous People Planning Framework (IPPF)
- Stakeholder Engagement Plan (SEP)
- Gender Assessment (including Gender Action Plan)
- Environmental, Social, Climate Management Framework (ESCMF)
- Environmental, Social and Climate Impact Assessment (ESCIA)

These documents have been developed in accordance with the national laws and regulations, the safeguards policy of the AIIB (Environmental and Social Framework), IFAD (Social, Environmental and Climate Assessment Procedures), and relevant policies of GEF (Policy on Environmental and Social Safeguards, Policy on Gender Equality, Principles and Guidelines for Engagement with Indigenous Peoples.

The above documents are prepared to ensure anticipated environmental and social risks and impacts are identified as proportional to the significance of the project's environmental and social risks and impacts that are present during project design, pre-construction, during construction, and operational stages.

1.2 Rationale of Gender Assessment

The purpose of the Gender Assessment (GA) is to understand the status-quo of key gender related issues in the context of agricultural production/irrigation management in Cambodia, particularly at project level. Based on these understandings, actions for gender mainstreaming will be proposed, and monitoring and evaluation plans will be developed through a proposed action plan. The plan will ensure the project enhances the participation of beneficiaries (women and men) in project planning, implementation, monitoring and evaluation.

The GA will identify gender gaps– through examining the situation of women and examining gender stereotypes, gender roles and responsibilities in farming practices, constraints to and opportunities for gender mainstreaming. Based on the gender gaps, which are identified between the current gender practices, national policies, and gender policies of the AIIB, IFAD and GCF. Specific actions are being proposed to ensure gender mainstreaming actions will be implemented in an integral manner with appropriate project investments activities.

The gender action plan aim to:

1) Support climate change interventions and innovations through a holistic gender mainstreaming approach which means that 40 % of the total beneficiary will be women across these activities.
2) Promote gender equity through climate change mitigation and adaptation interventions whereas minimizing social, gender-related and climate-related risks in all project activities; and
3) Contribute to reducing the gender gap caused by inherent social, economic and environmental vulnerabilities and exclusions as well as potential impacts induced by the climate change in four project provinces

2. METHODS

2.1 Research Design

2.1.1 Data Collection

Most quantitative data collection described below were conducted in 2022 by the Livelihoods and Vulnerability Survey Team. Information and charts that are used in this report are acknowledged.

Focus Group Discussions (FGD)**.** FGDs were conducted in small groups (7-8 persons each). FGD gathered people from similar backgrounds (e.g. ethnic groups, natural resource dependent) and experiences to discuss pre-defined topics. This information was collected to support the gender analysis and gender assessment exercise, and the preparation of the required social and environmental safeguards documents. All FGDs were guided by moderators who are trained local governmental staff. The group facilitators introduced topics for discussion and helped groups to participate and facilitate a lively and natural discussion. FGD aimed to obtain the insight of local people on specific issues, particularly the range of their opinion and ideas, and any inconsistencies and variation that exists in their community with regards to their beliefs, knowledge, attitude and practices with regards to their agricultural productions and their access to land and water resources to support their subsistence and long-term livelihoods.

No.	Time	Number of Community meeting	Number of participants		
			Male	Female	Total
1	26-39 June 2023	7	270 (82%)	58 (18%)	328 (100%)

Key Informant Interview (KII). KII aims to collect information about select pressing issues and problem in the target community – through individuals who are locally known as experts in the topics under question, and who have a broad social network with individuals and parties who play certain roles (e.g. production, consumption…) in a value chain that are likely to be supported by the project. About one-hundred key informant interviews have been conducted by the Livelihood and Vulnerability Survey Team.

Household Survey. Household surveys were administered to invited representatives of local households using questionnaires uploaded through personal mobile devices of local government staff who conducted the survey.

2.1.2 Research Participants

Sampling and Sample size for household survey

Province	Irrigation Schemes	Between from Canal Up to 100 M from a Canal (Close to Canal)	Between 100 M to 2.5 KM	Beyond 2.5 KM	Total
		N	N	N	n
Pursat	O Ta Paong Tanay Scheme	74	64	14	152
	Sub-total 1	**74**	**64**	**14**	**152**
Kampong Chhnang	Lum Hach Scheme	65	43	23	131
	Krang Ponley Kandal Scheme	6	3	5	14
	Anlong Chrey Scheme	37	14	2	53
	Sub-total 2	**108**	**60**	**30**	**198**
Kampong Speu	Brambei Mom Scheme	25	42	20	87
	Kra Peu Troum Scheme	22	17	19	58
	Krang Ponley Kandal Scheme	11	7	16	34
	Sub-total 3	**58**	**66**	**55**	**179**
Kandal	Krang Ponley Kandal Scheme	34	51	16	101
	Sub-total 4	**34**	**51**	**16**	**101**
	Grand Total	**274**	**241**	**115**	**630**

2.3 Data analysis

2.3.1 Qualitative analysis

Qualitative data obtained from focus group discussion and key informant interviews were analyzed using content analysis.

2.3.2 Quantitative analysis

Household survey data was collected through questionnaires prepared by the Livelihood and Vulnerability Survey Team using KoboToolbox platform. KoboToolbox was selected because it supports offline data collection, and it offers functionality appropriate to the needs of this survey. Data collected on the field were submitted to KoboToolbox platform after each day of household interview. Data were cleaned using Microsoft Excel and were analyzed by the author using the statistical package IBM SPSS Statistics for Windows (version 28.0, Armonk, NY, USA).

Descriptive statistics were collected for select questions in the questionnaire to describe the sample distribution. Various other statistical methods were used as appropriate for additional analysis.[1] For this study, all significance levels were set to $p < 0.05$.

[1] For descriptive analysis, Person's correlation and Spearman's correlation were applied. For inferential analysis, Chi-square χ^2 test statistics was used to examine associations between categorical variables.

3. RESULTS OF GENDER ASSESSMENT

3.1 Overview at Country Level

3.1.1 Background on gender issues

Social norms for gender

Cambodia is ranked 92th on the The Global Gender Gap Index 2023 rankings. This is an improvement compared to the The Global Gender Gap Index 2013 where it was on 104th rank. Gender norms are often considered a key underlying cause of gender inequalities in Cambodia. Underlying gender inequalities are cultural and traditional norms that limit girls' and women's choices and options[2]. Back in history, traditional codes (Chbab Srey/ Proh) have been adopted for several centuries. The code specifies the role of men in a family as 'head of the household' and advise women "to maintain peace within the home, walk and talk softly, and obey and respect her husband" [3]. The codes, however, posited men as heads of the family, bread winners, and are responsible for protecting women and making decisions[4]. For women and men who have never heard of Chbab Srey/ Proh, they hold the views that are consistent with those codes – quoting common proverbs that reflect the same beliefs[5]. Even for students today who are not explicitly taught to know and follow Chbab Srey /Proh, the expectations for a 'virtuous Khmer woman' are still communicated which affect how girls perceive about themselves that shape the gender norms in the next generation. The ideals described in Chbab Srey/Proh influence public policies – in a manner that shape the operation of public and private organizations thereby contributing to maintaining women in a subordinate position[6]. Although these codes have been no longer taught at schools, these teachings are still disseminated through families and other social institutions.[7] In fact, social expectations for Khmer women are reinforced and negotiated in day-to-day interactions in households, communities and workplaces. Women entrepreneurs face negative judgements by their family (54%), and community (71%) when starting their businesses[8].

Women are traditionally responsible for household expenditure including expense control. When it comes to loan, they also take on the responsibility of seeking out locals' credit and assume greater responsibility for loan repayments than their husbands. However, limited credit access and high interest rate is commonly known[9].

Gender discrimination is deeply embedded in, and reinforced by, social attitudes[10]. According to the United Nation (2022), women, gender norms are related to vastly unequal distribution of unpaid domestic and care work in which women undertale, on average, 90% of the work. During COVID-19 pandemic, a substantial

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[2] UNIFEM, WB, ADB, UNDP and DFID/UK. 2004. A Fair Share for Women: Cambodia Gender Assessment. UNIFEM, WB, ADB, UNDP, DFID/UK. Phnom Penh, ISBN:1-932827-00-5

[3] Anderson, E., & Grace, K. (2018). From Schoolgirls to "Virtuous" Khmer Women: Interrogating Chbab Srey and Gender in Cambodian Education Policy. Studies in Social Justice, 12(2), 215–234. https://doi.org/10.26522/ssj.v12i2.1626

[4] Lamb, V., Schoenberger, L., Middleton, C., & Un, B. (2017). Gendered eviction, protest and recovery: A feminist political ecology engagement with land grabbing in rural Cambodia. The Journal of Peasant Studies, 44(6), 1215–1234. https://doi.org/10.1080/03066150.2017.1311868

[5] Brickell, K. (2011a). "We don't forget the old rice pot when we get the new one": Discourses on Ideals and Practices of Women in Contemporary Cambodia. Signs: Journal of Women in Culture and Society, 36(2), 437–462. https://doi.org/10.1086/655915

[6] United Nation 2022. Gender Equality Deep-Dive for Cambodia – Common Country Analysis.

[7] United Nation 2022. Gender Equality Deep-Dive for Cambodia – Common Country Analysis.

[8] UNIDO & UN Women 2021. Policy Assessment for the Economic Empowerment of Women in Green Industry – Country Report: Cambodia.

[9] ADB - Gender Analysis for Climate-Friendly Agribusiness Value Chains Sector Project. Accessed September 13, 2023 (https://www.adb.org/sites/default/files/linked-documents/48409-002-sd-06.pdf).

[10] Ledgerwood, J. 1996. Women in Development: Cambodia. Asian Development Bank. Manila.

proportion of men and women spent increased time on domestic work whereas about 30% of employed women reported that their partners did not provide any additional assistance. It is noted under such circumstances, elderly women and girls assumed extra unpaid work in the absence of other support. The need to maintain livelihoods by carrying on more unpaid responsibilities keeps many women in vulnerable work, and is a constraint to women's participation and expansion of business, or advancing in their careers, or taking leadership roles[11]. However, in practice, the household division of labor has been changing: women engage in a broader range of tasks, including those normally associated with men[12]. Prevailing cultural norms encourage young men to engage in paid work, but discourage women. Although it has been increasingly recognized that there are more paid job opportunities for women than for men, such as in garment industry, people still feel that young women should remain in their villages[13].

Women assume a wider range of domestic and nondomestic roles than men. They also wake up before men and go to sleep after men. Women are generally responsible for 90% of the household work, including care of dependents and the sick. People in households, mostly women and girls, spend from one to two hours each day collecting water for the family's needs. Women and girls work longer hours than men and boys. Despite that unpaid domestic work is one of the biggest obstacles to promoting gender equality in Cambodia, it is rarely addressed by public policies or development efforts[14].

Labor Force in Key Sectors

The Cambodian labor force participation (LFP) rate for both the male and female population (aged 15 to 64) is high compared to other countries in the EAP region. Cambodian LFP rates are similar to those of Vietnam, perhaps Cambodia's closest regional peer in terms of structure of their economies[15]. Agriculture plays an important role in Cambodia's employment landscape. About 40% of all employed men and women work in the sector. This is on par with agricultural employment in Vietnam and Myanmar, though higher than Indonesia and Thailand. It is noted that during the period 2011-2016, there was a rapid shift out of agriculture because women moved into manufacturing while men moved into construction (at a similar rate). Men also shifted out of agriculture with a similar pattern albeit they moved out of agriculture at a faster rate – 17.2% compared to 11.5 percent of women – and moving into construction (from 7% to 14 percent) and services (from 20.7 to 26.2%)[16] (See also "Internal Migration" below).

Figure 2 – Education attainment of survey participants

[11] United Nation. 2022. Gender Equality Deep-Dive for Cambodia – Common Country Analysis.

[12] UNIFEM, WB, ADB, UNDP and DFID/UK. 2004. A Fair Share for Women: Cambodia Gender Assessment. UNIFEM, WB, ADB, UNDP, DFID/UK. Phnom Penh, ISBN:1-932827-00-5.

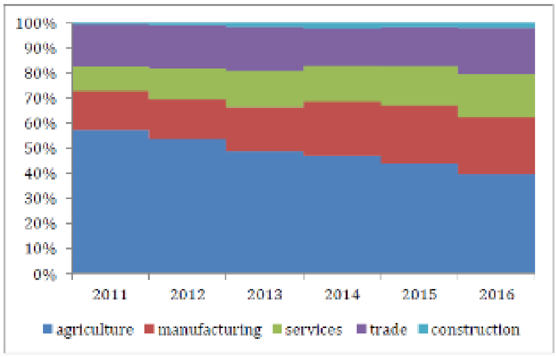
[13] Asian Development Bank (ADB), 2001. Participatory Poverty Assessment: Cambodia. Manila.

[14] United Nation. 2022. Gender Equality Deep-Dive for Cambodia – Common Country Analysis.

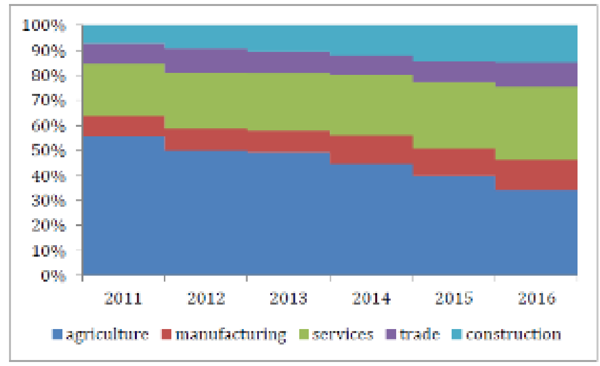
[15] Dimitria Gavalyugova & Wendy Cunningham. 2020. Gender Analysis of the Cambodian Labor Market. The World Bank.

[16] Dimitria Gavalyugova & Wendy Cunningham. 2020. Gender Analysis of the Cambodian Labor Market. The World Bank.

a. Female



b. Male



Source: Author's calculations based on the CSES 2011-2016

b. Male

Source: *Dimitria Gavalyugova & Wendy Cunningham (2020)*

Internal Migration

By 2013, there are about 25 percent of the Cambodian population (or 3.7 million) who migrated. Of the total internal migration, rural-rural accounts for 58.4%, followed by rural-urban (24.5%, mostly to Phnom Penh), and urban-urban (12%). Men dominated rural-rural migration (60%) whereas women dominated rural-urban migration. A large percentage of men migrants were engaged in construction while 58.5% of women migrants migrated to Phnom Penh for work in garment factories (32.2%), small business (23.4 percent), domestic work (11.1 percent), and the service/entertainment sector (10.3 percent)[17]. By 2023, there are around 1,326 garment factories in Cambodia. In Kampong Speu, there are 189 factories, including textile, garment, and footwear sectors with a total of 145,399 workers (110,843 women). Kampong Chhnang has 18 factories with 30,207 employees.

Migration has a strong influence on gender-based division of labor in rural Cambodia. This affects the level of participation of women in agriculture value chains, particularly in the face of increasing migration of young women and men to garment factories, construction work, and service industry, etc. Low income from farming activities and debts are key driving factors for rural people to migrate, leaving elderly, women, and children home[18].

Gender Based Violence

One in five Cambodian women have experienced physical or sexual violence from an intimate partner, and more than one third of men report having perpetrated partner violence. Common understanding that men are entitled to sex regardless of consent, directly contributes to gender-based violence. These harmful gender norms are a root cause of gender-based violence. During the COVID-19 surge, the progress made in reducing gender-based violence was slowed, or reversed. There is no national data on sexual harassment. However, small studies

Indicated that sexual harassment affects even more women than domestic or partner violence. Public spaces and public transport are common sites for sexual harassment[19]. Studies on sexual harassment found that 89% of all respondents (not just those who reported harassment) felt unsafe working or studying at night and 24%

[17] Ministry of Planning (2013) A Crump series study: Women and migration

[18] MAFF 2022. Gender Mainstreaming Policy and Strategic Framework in Agriculture (2022-2026).

[19] United Nation. 2022. Gender Equality Deep-Dive for Cambodia – Common Country Analysis.

felt unsafe when using public spaces at any time[20]. Significant gender inequalities continue with particular implications for women who face intersecting forms of discrimination such as women from ethnic minority group, people with disabilities, LGBTQI, and women living in rural areas of Cambodia where gender inequalities tend to be larger[21]. There is no national data on sexual harassment. However, small studies suggested that sexual harassment affects even more women than domestic or partner violence[22] (See ESCIA for more).

3.1.2 Background on impacts of Climate Change on Women

In Cambodia, women in rural areas are especially dependent on local natural resources for their livelihood, because of their domestic responsibilities to secure water, food and energy for cooking and other household activities. The effects of climate change, including drought, uncertain rainfall and deforestation, make it harder for them to secure these resources. Compared with men, women face historical disadvantages, including limited access to decision-making and economic assets that compound the challenges of climate change[23].

High dependence on rainfall for agriculture makes Cambodian agriculture vulnerable to weather shocks. Male and female farmers reported the negative impacts of shorter rainy seasons, floods, and more frequent drought spells[24]. Climate change is leading to more variable growing seasons and water deficits. Gender inequality intersects with climate risks and vulnerabilities. Poor women have particularly limited access to resources, restricted rights, limited mobility and muted voice in shaping decisions, making them highly vulnerable to climate change. The nature of that vulnerability varies widely and climate change will magnify existing patterns of inequality, including gender inequality[25].

Though rural women play an equal (and somehow influencing role) in intra-household decision making such as keeping money, deciding selling prices (jointly with husband), spending money for daily living activities (See Section 3.3.13 below), they are less involved in day-to-day farming in the field (compared to men). Thus, they are not up-to-date as to new farming techniques (e.g. how to select appropriate pesticide and fertilizers formula), new technology. Women's involvement in direct farming activities of poorer for the householding doing one crop per year (vis-à-vis two or more crops per year) due to lack of irrigation access during dry season.

There is ample evidence that women have less access to irrigation water than men and have less say in how their local water districts allocate the water. Several sources report[26] that the number of women in FWUCs is lower than that of men. Most of the important positions in FWUCs, such as chair and first and second vice-chair, are dominated by men, who tend to take overall management control, including water allocation, irrigation system operation and maintenance and water use conflict resolution, as well as in rice planting. Men also

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[20] Ministry of Women's Affairs. (2014). Violence Against Women and Girls: Cambodia Gender Assessment [Policy Brief]. Ministry of Women's Affairs.

[21] Evans, A. 2019. How Cities Erode Gender Inequality: A New Theory and Evidence from Cambodia. Gender & Society, 33(6), 961–984.

[22] United Nation. 2022. Gender Equality Deep-Dive for Cambodia – Common Country Analysis.

[23] 52nd session of the Commission on the Status of Women (2008) Gender perspectives on climate change. http://www.un.org/womenwatch/daw/csw/csw52/issuespapers/ Gender%20and%20climate%20 change%20paper%20final.pdf

[24] Thomas, T., T. Ponlok, R. Bansok, T. De Lopez, C. Chiang, N. Phirun, and C. Chhun. 2013. 'Cambodian Agriculture: Adaptation to Climate Change Impact.' IFPRI Discussion Paper 1285, Washington, DC.

[25] Ministry of Women Affairs 'Standardized Guideline Mainstreaming Gender and Climate Change in Sectoral Ministries' Planning, Budgeting and Implementation' December 2015.

[26] CDRI, 2014 Gender and Water Governance: Women's Role in Irrigation Management and Development in the Context of Climate Change.

provide technical input and final (household and community) decision-making. In meetings where water allocation is discussed are dominated by men and women do not speak up as much as men do to voice their needs. There is also a gender imbalance in receiving information on water supply.

Women farmers currently have limited capacity and opportunities to diversify agricultural practices and lessen dependency on climate sensitive and stressed natural resources; limited access to knowledge regarding new agricultural production and post-production techniques and technologies; and limited mobility to avoid disasters stemming from their domestic and agricultural responsibilities.[27]

Vulnerability is further heightened for women in agriculture, as they are often unpaid family workers with few options for coping with disasters. Climate change vulnerability of both the agriculture and fisheries sectors heightens the level of risk for women's livelihoods, as they have significant involvement in post-harvest activities.[28]

Recommendations for climate change mitigation and women's economic empowerment

The 4th East Asia Gender Equality Ministerial Meeting entitled "Building Resilience to Economic Crisis and Moving Forward", held in Siem Reap in November 2011[29], made the following specific recommendations for Cambodia on a green economy:

- Women entrepreneurs: Mechanisms (such as women's business associations) should be strengthened to promote entrepreneurship, access to business-related information, green technology and financial services.
- Organic and sustainable agriculture: the Ministry of Agriculture, Forestry and Fisheries should develop an organic agriculture policy framework and program. Extension services should proactively target female farmers and women's groups.
- Green procurement: Public sector procurement should prioritise wherever possible green products and services, especially sustainably produced (organic and chemical free) food and other products for government offices, schools, hospitals and embassies. Priority should be given to women-led enterprises.
- Public Private Partnerships (PPP): Promotion of PPPs will be a key strategy for promoting green growth. Regulatory frameworks need to be adjusted to provide appropriate incentives. Sector-based policy research is needed to identify the most important opportunities.
- Women's economic empowerment bears largely untapped potential for the development of green products and services, including finance, skills, and especially knowledge and information on how to start, manage and grow a competitive, sustainable green business. Women should be widely encouraged to start and improve their own businesses as entrepreneurs, where urgently needed productivity increases, innovation and employment could be generated by and for them.
- **Progress and Challenges**

[27] Ministry of Women Affairs 'Standardized Guideline Mainstreaming Gender and Climate Change in Sectoral Ministries' Planning, Budgeting and Implementation' December 2015.

[28] Gender Assessment. Accessed June 28, 2023 (https://www.greenclimate.fund/sites/default/files/document/gender-assessment-fp076-adb-cambodia.pdf)

[29] Ministry of Women Affairs 'Standardized Guideline Mainstreaming Gender and Climate Change in Sectoral Ministries' Planning, Budgeting and Implementation' December 2015.

MOWA's Gender Mainstreaming Guidance Manual (2017) has reviewed the progress of national gender mainstreaming methods. As part of this exercise, challenges that remain are identified. Both progress and challenges are summarized by MOWA in table below.

Topic	***Progress***	***Challenges***
Awareness on gender & gender mainstreaming issues	There is a certain level of awareness and initiatives have been taking on gender & gender mainstreaming in different planning fields e.g. CC and DRR	Gender mainstreaming in CC investments is new to the sector ministries, civil society organizations and other sectors.
Institutional and community capacity and cross-sectoral coordination with a focus on women's role in CC adaptation and mitigation	The NSDP and national level Rectangular Strategy, the Gender Master Plan on Gender and CC, the Cambodia CC Strategic Plan (CCCSP), and the CC Action Plans are sector and theme specific policy documents. The Pilot Program on Climate Resilience (PPCR) Coordination Team, TWGs, etc. are strengthening institutional capacity and cross- sectoral coordination measures. The Gender and CC Committee (GCCC) was established to facilitate gender mainstreaming in CC investment initiatives.	Limited technical and financial resources limit the planned initiatives. Gender mainstreaming in CC investment initiatives have yet to become part of the regular agenda. Lack of defined roles in CC adaptation, DRR and mitigation.
Gender mainstreaming in CC investment become a regular part of the development agenda	The RGC recognized gender concerns as a part of all development initiatives.	No specific initiative has yet been taken to incorporate gender mainstreaming in CC investments as part of the regular agenda.
The Government, CSO and private sector dialogue platform	There are different networks, gender and development networks etc. have been working.	Absence of the platform that brings national policy makers from the Government, DPs, CSOs and private sectors and individual experts for the discourse on gender mainstreaming in CC and economics of gender mainstreaming in CC etc. fields.

3.1.3 National Policies related to Gender & Climate Change

National Legal frameworks

Cambodia has adopted a legal framework that aims to promote the role of women as part of national effort towards economic empowerment for women and gender equality. At the constitutional level, anti-discrimination are prohibited and are included in the Constitution. Article 36 of the Constitution recognizes the value of women's work in the home as equal to that performed outside of the home. The Labor Law, which was adopted in 1997, specifies provisions that protect female laborer.

Ratification of international convention

In terms of international convention, the Royal Government of Cambodia has ratified 13 International Labor Organization conventions (See table below). Cambodia is also a signatory to various United Nations' human rights covenants and conventions. These include the Convention for Elimination of All Forms of Discrimination against Women, which was ratified in 1992. However, it is noted that the C006 - Night Work of Young Persons (Industry) Convention, 1919 (No. 6) has not been yet in force.

Convention	Date	Status
C087 - Freedom of Association and Protection of the Right to Organise Convention, 1948 (No. 87)	23 Aug 1999	In Force
C098 - Right to Organise and Collective Bargaining Convention, 1949 (No. 98)	23 Aug 1999	In Force
C100 - Equal Remuneration Convention, 1951 (No. 100)	23 Aug 1999	In Force
C111 - Discrimination (Employment and Occupation) Convention, 1958 (No. 111)	23 Aug 1999	In Force
C138 - Minimum Age Convention, 1973 (No. 138)*Minimum age specified: 14 years*	23 Aug 1999	In Force
C122 - Employment Policy Convention, 1964 (No. 122)	28 Sep 1971	Not IF
C006 - Night Work of Young Persons (Industry) Convention, 1919 (No. 6)	24 Feb 1969	In Force
C013 - White Lead (Painting) Convention, 1921 (No. 13)	24 Feb 1969	In Force
C150 - Labour Administration Convention, 1978 (No. 150)	23 Aug 1999	In Force
C122 - Employment Policy Convention, 1964 (No. 122)	28 Sep 1971	In Force

Despite the presence of anti-discrimination legislation, there is no provision for 'equal remuneration for work of equal value', although 'equal pay for the same work' is included in the Constitution[30]. Provision for minimum wage is in place and has been applied to the garment, textile, and shoe industries, the industries where a large number of female workers are employed. However, the adequacy of the minimum wage is still open to question. Many women working in the domestic, tourism, and entertainment sectors are not covered with regards to minimum wage.

In terms of land ownership, Land Law (2001) stipulates that women have equal rights to land ownership. Overall, 70% of all land titles are issued jointly between husband and wife. In practice, however, there are examples of gendered land ownership in which men take over the legal title to land even when women have an equal rights to land ownership as per Land Law (2001). It seems that women are not aware of their rights and have little recourse to legal advice[31].

Cambodia's Rectangular Strategy (Phase IV):

The Government of Cambodia's Rectangular Strategy Phase IV (2018-2023) provides a framework for policies and strategies addressing poverty reduction and promoting the economic empowerment of women. The objectives of RS III relate to economic growth, employment particularly for youth, and strengthening institutional capacity and governance. The four strategic rectangles of the RS IV are:

[30] ADB & ILO 2013 Quarterly Briefing Note GMS TRIANGLE Project Oct-Dec Geneva.

[31] GCF, Gender Assessment, 2018, Climate-Friendly Agribusiness Value Chains Sector Project.

- **Rectangle 1** - Human resource development: 1). Improving the quality of education, science and technology; 2). Vocational training; 3). Improving public healthcare and nutrition; and 4). Strengthening gender equality and social protection.
- **Rectangle 2** - Economic Diversifications: 1). Improving logistics system and enhancing transport, energy and digital connectivity; 2). Developing key and new sources of economic growth; 3). Readiness for digital economy and industrial revolution 4.0; and 4). Promoting financial and banking sector development.
- **Rectangle 3** – Promotion of private sector development and employment: 1). Job market development; 2). Promotion of SME and entrepreneurship; 3). Public-private partnership; and 4). Enhanced competitiveness.
- **Rectangle 4** – Inclusive and sustainable development: 1). Promotion of agricultural and rural development; 2). Strengthening sustainable management of natural and cultural resources; 3). Strengthening management of urbanization; and 4). Ensuring environment sustainability and readiness for climate change.
- Master Plan On Gender And Climate Change (2018-2030)

Cambodia Climate Change Strategic Plan 2014-2023

The threat of climate change, manifested by the increase of droughts, storms or floods, has been recognized as a key global development challenge. Not only does climate change have broad impacts on the natural environment, it also impacts the economy and social development. The gravity of these impacts varies among regions, income groups and occupations, as well as between women and men. In Cambodia, the impacts of climate change are crucial for agriculture and the lives of mostly rural communities. As the majority of Cambodian people still rely on rain-fed rice farming as a main source of household income, a change of climate directly threatens people's livelihoods, in both the short and long term. Poor management of natural resources has had serious impacts on the local economy, as most communities rely on natural forest cover; a majority of these people are women and children.

The RGC recognizes that the rural poor of Cambodia, the majority of whom are women, are most vulnerable to climate change impacts due to their high dependence on agriculture and natural resources. This vulnerable group is susceptible to diseases due to their limited resources and capacity to adapt to climate change impacts, including a lack of preparedness to cope with climate risks and hazards. There is a need to mainstream gender into climate change response measures, such as into existing policies and laws and sector climate change strategic plans (SCCSPs) in order for this cross-cutting issue to be supported by all government agencies, especially at national and sub-national levels, by development partners, NGOs, civil society organizations (CSOs), research and academic institutions and the private sector. In response, the Government has put in place 'Strategic Objective 2: Reduce sectoral, regional, gender vulnerability and health risks to climate change impacts'.

Generally, all policies, strategies and projects dealing with climate change are either about climate change adaptation or climate change mitigation. It is important to draw a distinction between the two. Climate change adaptation aims to reduce vulnerabilities and build resilience to the impacts of climate change. This is done by strengthening national institutional capacity for vulnerability assessment and adaptation planning. It includes national efforts to integrate climate change adaptation measures into development planning and ecosystem management practices. The overall work on climate change adaptation is guided by and contributes to the Nairobi Work Program on Impacts, Vulnerability and Adaptation, a program developed by the UN-Framework

Convention for Climate Change (UNFCCC)5 to help countries understand climate change impacts and adapt to climate change.

Climate change is expected to compound and amplify development challenges, stresses and problems in Cambodia, further affecting poor and marginalized people, particularly women and children. Women have disproportionate access to financial resources, land, natural resources, education, health, rights and other development services that are essential for effective adaptation to climate change. For the vast majority of women working in the informal sector and in small enterprises who lack capital and access to credit, information or knowledge, recovering from the devastating effects of environmental disasters is nearly impossible[32].

3.1.4 Existing implementation arrangements

The institutions to support women's economic empowerment include the Ministry of Women's Affairs (MOWA) and gender mainstreaming action groups (GMAGs) in each line ministry that prepare and implement sectoral gender mainstreaming action plans (GMAPs). In 2013, MOWA launched a Millennium Development Goal (MDG) Acceleration Framework Cambodia Action Plan focused on women's economic empowerment to contribute to the achievement of other MDGs in poverty reduction, health, and education. The plan prioritizes three areas of intervention: (i) providing training for jobs for women that are consistent with market demands; (ii) ensuring that women have the capacity to lead and grow their micro, small, and medium-sized enterprises and can move from the informal to the formal sector; and (iii) improving livelihoods in rural communities, especially for poor women. In 2014 MoWA launched the Cambodia Gender Assessment and 5 Year Strategic Plan for Gender Equality and Women's Empowerment (Neary Ratanak IV), which includes policy recommendations on Women's Economic Empowerment (including Agriculture), education, health, political participation, and climate change.

MOWA also manages a network of 13 women's development centers (WDCs) nationwide, which are vocational centers offering training programs in areas such as handicraft production, hairdressing, tailoring, and food processing. WDCs face a host of challenges and are not reaching their full potential as centers that promote women's economic empowerment. There is scope to improve training to be more responsive to the labor market and include entrepreneurial skills training, business development services, and current market information. There is considerable interest in MOWA to introduce public–private partnerships at WDCs as a way of increasing their market and entrepreneurial orientation, and to ensure sustainable financing.

The Ministry of Agriculture, Forestry and Fisheries (MAFF) has prepared a Gender Mainstreaming Policy and Strategy in Agriculture 2016–2020 which includes objectives relating to greater participation of women in the civil service; enhanced capacity to integrate gender; increased ability of rural women to access and manage resources; and building and promoting gender equality in access to extension services. With regard to the latter, specific mention is given to assistance with social land concessions, participation in the private sector, participation in village and community groups, and access to credit, and extension services. The CFAVCP GAP is designed to align to and support the operationalization of the MAFF Gender Policy.

3.2 Profiles of Project's Target Groups

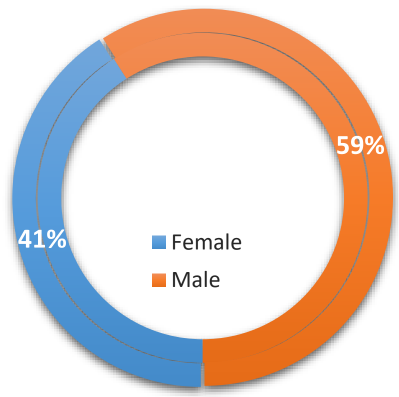
3.2.1 Demographic Information

Of the total 630 households who were interviewed in the household survey, 40.95% of respondents (n=258) are female and 59.05% (372) are male. There are 20.8% (n=131) of surveyed households who are female headed households. Most households in the survey are Khmer (98.9%, n=623). Only 7 households are from the Khmer

[32] Oxfam America (2010) Rural Women, Gender and Climate Change. Cambodia.

Islam group. These households are located in the O Ta Paong Tanay Scheme which is located in three villages of Prasat, Rumlech, Koun Tnaot villages of Rumlech commune, Bakan district, Pursat province. All Khmer people in the survey are Buddhist and all 7 Khmer Islam households follow Muslim. There are 71 people with disability (2.6%) out of total 2,687 people of 630 households in the survey).

Figure 3 – Gender of Survey Respondents



The household size is 4.27 people (n=2,688), of which 1,399 (52%) are female and 1,289 (48%) are male. Average age of household head is 50.9 years.

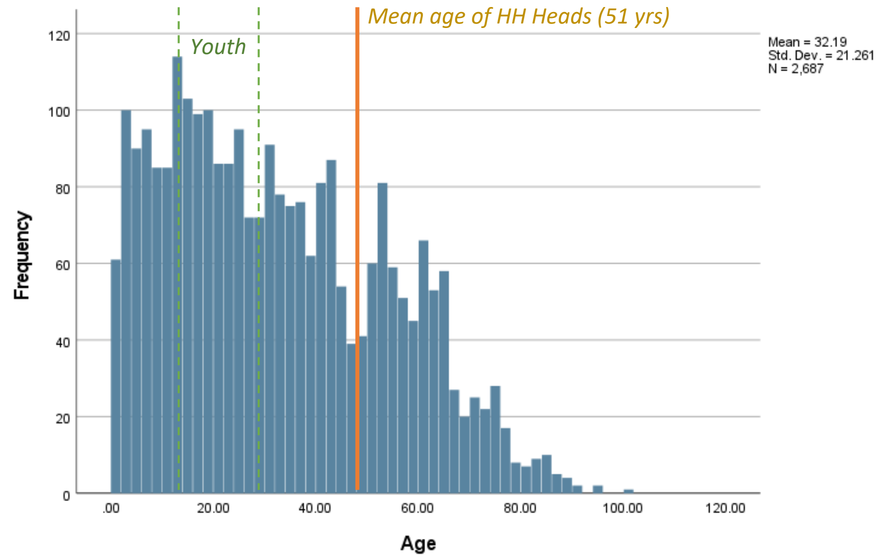
Table 1 – Descriptive Statistic of Age of Household Head and Household member

Statistics	Household Head	Household members
Mean	51.0	32.2
SD	13.3	21.3
Min	18.0	1.0
Max	85.0	100.0
Median	52.0	30.0
Mode	53.0	18.0

Survey participants live in the villages for an average of 33 years (min=1, max=85, SD=18.7).

Youth. It is noted that the youth (members from 18 to under 30 years of age) takes up 23.8% (n=572) in the sample. If extended to cover those between 18 and 50, the percentage would go up to 44.1% of the survey population (with average age of 32).

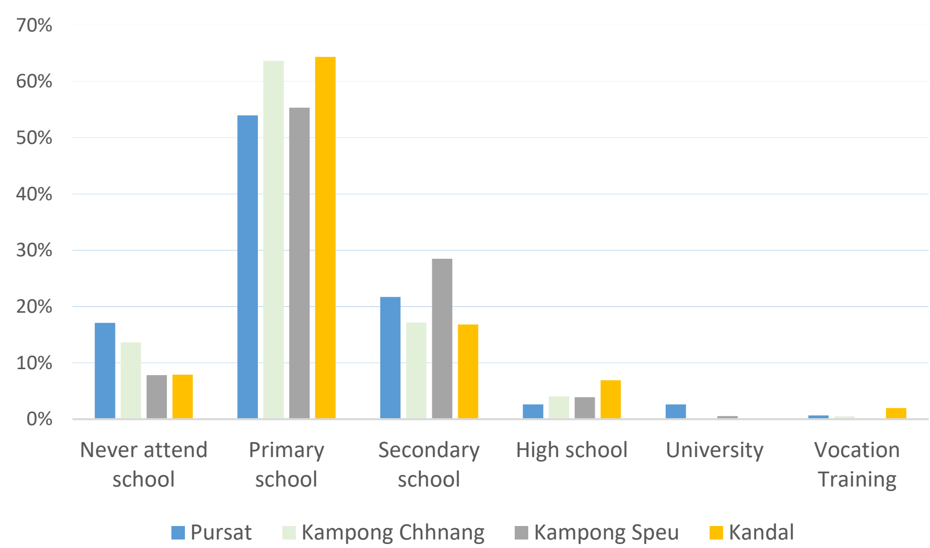
Figure 4 – Age distribution of all household members (n=2,787)



3.2.2 Education

Nearly half of respondents (46.04%) completed primary school, followed by secondary level (21.70%), high school (10.33%). Higher education attainment is 2.36%. About 9.1% of respondents dropped out of school or never attended. According to a focus group discussion that was carried out under the Livelihood and Vulnerability survey, elderly people typically had very limited/no access to education due to the history of Cambodia, and they have not been able to find relevant education to catch up. Younger generation is different. They took the opportunities to learn and have higher education. Some households have sent their children to Phnom Penh or abroad for education.

Figure 5 – Education of survey participants



3.2.3 Assets

- Houses

Of the 630 HHs, 96% own a house. Only 4% said they are living in their children's, or relatives' house. For those who own a house, house status is as follows:

Table 2 – Types of House

No.	House type	Percentage
1	Wooden/iron on walls (corrugate iron roof/roofing cement)	29%
2	Semi- permanent houses (roof of brick or tiles)	28%
3	Simple house (corrugated wall wood and roofing sheets)	28%
4	Permanent house with one or more floors (15%),	15%
5	temporary houses	0.48%

In terms of land status for the house, 48% of respondents have land title registered. 49% are still under process of registration with local authorities but their land are recognized as legal. Some people are using land that are not under their ownership for temporary farming.

- Land access

Most survey participants (95.7%, n=603) own the land that they currently use. Only 2.4% (n=15) use lands of their relatives. The remaining (1.6%) use land from their children (n=5), parents (n=3), state (n=3), and renting (n=1). The average land area owned/used by surveyed households is 5,668.5 m (SD=8,720.1).

Table 3 – Descriptive statistic of land use owned/used by survey households

Statistics	Agricultural Land (m^2)	Residential Land (m^2)
Mean	12,824.9	5,668.5
SD	14,398.7	8,720.1
Min	354.0	66.0
Max	130,000.0	100,000.0
Median	8,900.0	2,200.0
Mode	10,000.0	5,000.0

Below is land distribution in four project provinces.

Table 4 – Land Ownership

Province	Subproject	Ownership		Renting		Relatives		Children		Parents		State land		Total	
		N	%	N	%	N	%	N	%	N	%	N	%	N	%
Pursat	O Ta Paong Tanay Scheme	320	94.67%	4	1.18%	10	2.96%	2	0.59%	2	0.59%	0	0.00%	338	100.00%
	Sub-total 1	**320**	**94.67%**	**4**	**1.18%**	**10**	**2.96%**	**2**	**0.59%**	**2**	**0.59%**	**0**	**0.00%**	**338**	**100.00%**
Kampong Chhnang	Lum Hach Scheme	300	67.11%	3	0.67%	4	0.89%	5	1.12%	0	0.00%	0	0.00%	312	69.80%
	Krang Ponley Kandal Scheme	25	5.59%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	25	5.59%
	Anlong Chrey Scheme	108	24.16%	1	0.22%	1	0.22%	0	0.00%	0	0.00%	0	0.00%	110	24.61%
	Sub-total 2	**433**	**96.87%**	**4**	**0.89%**	**5**	**1.12%**	**5**	**1.12%**	**0**	**0.00%**	**0**	**0.00%**	**447**	**100.00%**
Kampong Speu	Brambei Mom Scheme	174	51.03%	0	0.00%	6	1.76%	0	0.00%	2	0.59%	0	0.00%	182	53.37%
	Kra Peu Troum Scheme	120	35.19%	0	0.00%	0	0.00%	1	0.29%	0	0.00%	0	0.00%	121	35.48%
	Krang Ponley Kandal Scheme	37	10.85%	0	0.00%	1	0.29%	0	0.00%	0	0.00%	0	0.00%	38	11.14%
	Sub-total 3	**331**	**97.07%**	**0**	**0.00%**	**7**	**2.05%**	**1**	**0.29%**	**2**	**0.59%**	**0**	**0.00%**	**341**	**100.00%**
Kandal	Krang Ponley Kandal Scheme	147	91.88%	3	1.88%	5	3.13%	1	0.63%	0	0.00%	4	2.50%	160	100.00%
	Sub-total 4	**147**	**91.88%**	**3**	**1.88%**	**5**	**3.13%**	**1**	**0.63%**	**0**	**0.00%**	**4**	**2.50%**	**160**	**100.00%**
	Grand Total	**1231**	**95.72%**	**11**	**0.86%**	**27**	**2.10%**	**9**	**0.70%**	**4**	**0.31%**	**4**	**0.31%**	**1286**	**100.00%**

Source: CAISAR Livelihood and Vulnerability Report 2022

Table 5 – Land Title

Province	Subproject	Land Title		Commune/village Title		illegal Possession		Don't Know		Total	
		N	%	N	%	N	%	N	%	N	%
Pursat	O Ta Paong Tanay Scheme	239	70.71%	90	26.63%	3	0.89%	6	1.78%	338	100.00%
	Sub-total 1	**239**	**70.71%**	**90**	**26.63%**	**3**	**0.89%**	**6**	**1.78%**	**338**	**100.00%**
Kampong Chhnang	Lum Hach Scheme	225	50.34%	77	17.23%	8	1.79%	2	0.45%	312	69.80%
	Krang Ponley Kandal Scheme	11	2.46%	13	2.91%	0	0.00%	1	0.22%	25	5.59%
	Anlong Chrey Scheme	19	4.25%	80	17.90%	4	0.89%	7	1.57%	110	24.61%
	Sub-total 2	**255**	**57.05%**	**170**	**38.03%**	**12**	**2.68%**	**10**	**2.24%**	**447**	**100.00%**
Kampong Speu	Brambei Mom Scheme	78	22.87%	104	30.50%	0	0.00%	0	0.00%	182	53.37%
	Kra Peu Troum Scheme	14	4.11%	104	30.50%	0	0.00%	3	0.88%	121	35.48%
	Krang Ponley Kandal Scheme	1	0.29%	37	10.85%	0	0.00%	0	0.00%	38	11.14%
	Sub-total 3	**93**	**27.27%**	**245**	**71.85%**	**0**	**0.00%**	**3**	**0.88%**	**341**	**100.00%**
Kandal	Krang Ponley Kandal Scheme	26	16.25%	129	80.63%	0	0.00%	5	3.13%	160	100.00%
	Sub-total 4	**26**	**16.25%**	**129**	**80.63%**	**0**	**0.00%**	**5**	**3.13%**	**160**	**100.00%**
	Grand Total	**613**	**47.67%**	**634**	**49.30%**	**15**	**1.17%**	**24**	**1.87%**	**1286**	**100.00%**

Source: CAISAR Livelihood and Vulnerability Report 2022

3.2.4 Occupation

Overall job pattern.

72.86% of respondents (n=459) are farmers. 99% of farmer respondents said they grow rice and no other crop. Main crop takes place during the wet season albeit slight differences between project provinces. It is noted that main crop in Kandal is in dry season (see graph below).

Job of households heads.

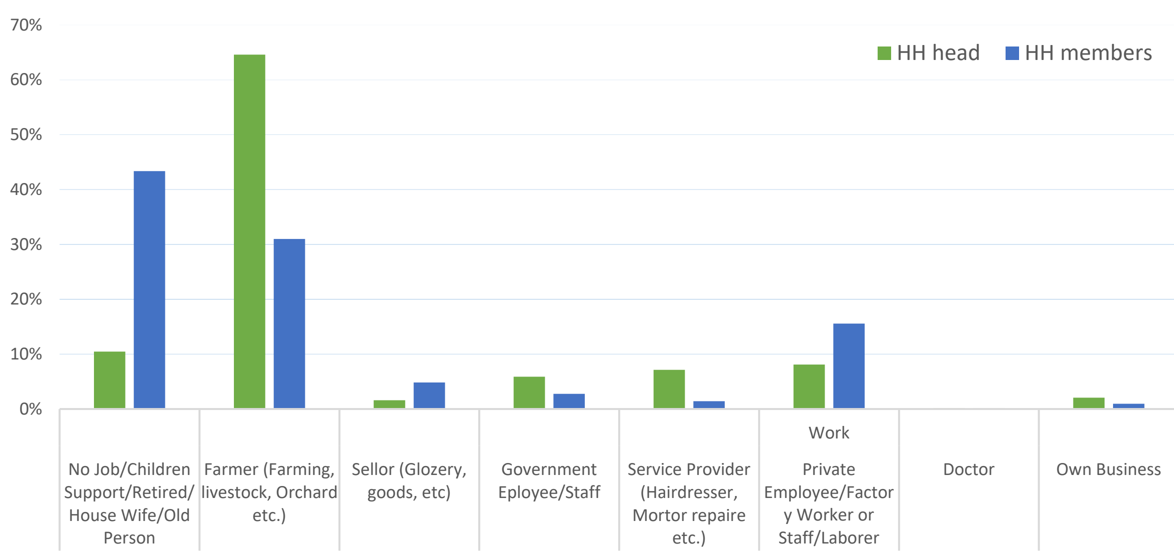
64% of households head in the survey (n=407) do farming, followed by unpaid work (10.5%, n=66), paid workers (8.1%, n=51), retailed service (7.1%, n=45), government employee (5.9%, n=37), etc. For those who are involved in farming. It is noted that in addition to cultivating rice, 73% indicated they are involved in animal husbandry to earn additional income. Children are most popular (50%), followed by cows (34%), ducks (9%), buffalos (2%), and other livestock. 48% of respondents do animal husbandry for income. About 23% raise livestock for home consumption.

It is noted that 32% of household heads (n=201) do two jobs compared to 68% who only do 1 job. The reason why they do the second jobs because they want to earn more income[33] to meet family's expenditure[34].

Job of household members.

While 64% of households do farming to earn income, and half of them (32%) do the second job, only 30% of household members are involved in farming. These members are usually children and tend to work in non-farm work such as work for the private sector (15.6%), services (4.8%), and government (2.8%). It is noted that 43.4% of household members are not income earners. They are retired, doing household chores, or jobless.

Figure 6 – Job of household heads and household members



Source: Livelihood and Vulnerability Data 2022. Graph by author

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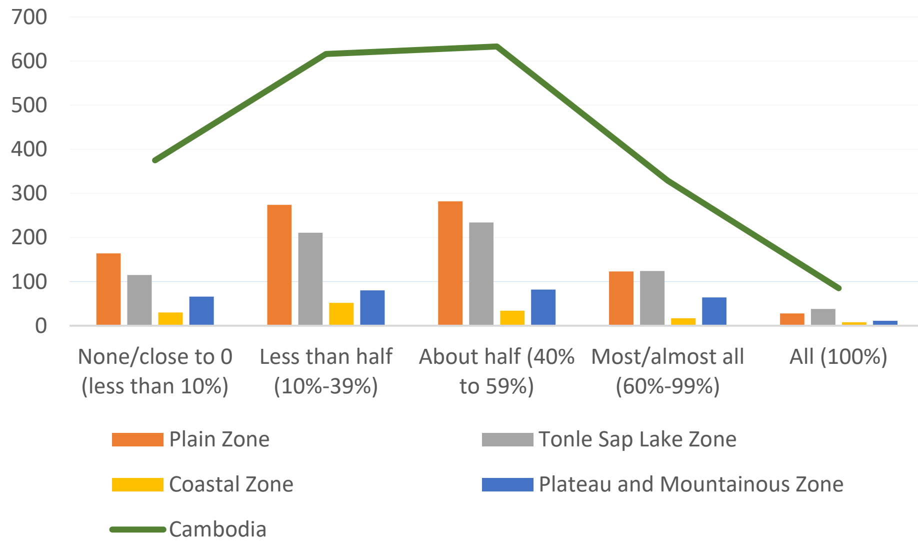
[33] $\chi^2(202)=238.085$, $p<0.05$

[34] $\chi^2(55)=77.458$, $p<0.01$

The level of main key income (64%) of households matches the findings from the national Agriculture Survey 2020.

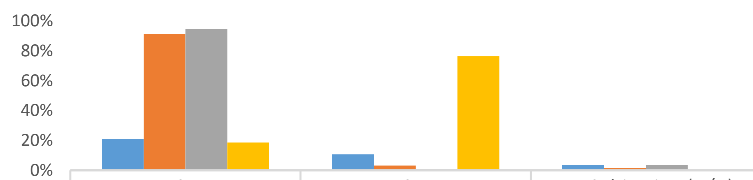
Figure 7 – Share of household income accounted for by agricultural income



Source: Data from Cambodia Agriculture Survey 2020. Graph by author

3.2.5 Agricultural production

Figure 8 – Cropping Pattern by project provinces



	Wet Season	Dry Season	No Cultivation (N/A)
Pursat	20.86%	10.70%	3.74%
Kampong Chhnang	91.13%	3.23%	1.61%
Kampong Speu	94.44%	0.62%	3.70%
Kandal	18.64%	76.27%	0.00%

Source: Data from Livelihood and Vulnerability Survey. Graphed by author

Table 6 – Crop Type

Province	Subproject	Banana , Corn		Cashew Nuts		Mengo		No Cultivation (N/A)		Orang		Rice		Vegetable crops		Total	
		N	%	N	%	N	%	N	%	N	%	N	%	N	%	N	%
Pursat	O Ta Paong Tanay Scheme	0	0.00%	0	0.00%	0	0.00%	7	3.74%	0	0.00%	177	94.65%	3	1.60%	187	100.00%
	Sub-total 1	**0**	**0.00%**	**0**	**0.00%**	**0**	**0.00%**	**7**	**3.74%**	**0**	**0.00%**	**177**	**94.65%**	**3**	**1.60%**	**187**	**100.00%**
Kampong Chhnang	Lum Hach Scheme	0	0.00%	1	0.40%	0	0.00%	3	1.21%	0	0.00%	176	70.97%	0	0.00%	180	72.58%
	Krang Ponley Kandal Scheme	0	0.00%	0	0.00%	0	0.00%	1	0.40%	0	0.00%	10	4.03%	0	0.00%	11	4.44%
	Anlong Chrey Scheme	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	57	22.98%	0	0.00%	57	22.98%
	Sub-total 2	**0**	**0.00%**	**1**	**0.40%**	**0**	**0.00%**	**4**	**1.61%**	**0**	**0.00%**	**243**	**97.98%**	**0**	**0.00%**	**248**	**100.00%**
Kampong Speu	Brambei Mom Scheme	0	0.00%	0	0.00%	1	0.62%	4	2.47%	0	0.00%	89	54.94%	1	0.62%	95	58.64%
	Kra Peu Troum Scheme	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	62	38.27%	1	0.62%	63	38.89%
	Krang Ponley Kandal Scheme	0	0.00%	0	0.00%	0	0.00%	2	1.23%	0	0.00%	2	1.23%	0	0.00%	4	2.47%
	Sub-total 3	**0**	**0.00%**	**0**	**0.00%**	**1**	**0.62%**	**6**	**3.70%**	**0**	**0.00%**	**153**	**94.44%**	**2**	**1.23%**	**162**	**100.00%**
Kandal	Krang Ponley Kandal Scheme	1	1.69%	0	0.00%	0	0.00%	0	0.00%	1	1.69%	56	94.92%	1	1.69%	59	100.00%
	Sub-total 4	**1**	**1.69%**	**0**	**0.00%**	**0**	**0.00%**	**0**	**0.00%**	**1**	**1.69%**	**56**	**94.92%**	**1**	**1.69%**	**59**	**100.00%**
	Grand Total	**1**	**0.15%**	**1**	**0.15%**	**1**	**0.15%**	**17**	**2.59%**	**1**	**0.15%**	**629**	**95.88%**	**6**	**0.91%**	**656**	**100.00%**

Source: Livelihood and Vulnerability Report

Table 7 – Cropping Season

Province	Subproject	Wet Season		Dry Season		Both Season		No Cultivation (N/A)		Total		Total of harvest (Ton/Kg)
		N	%	N	%	N	%	N	%	N	%	Average
Pursat	O Ta Paong Tanay Scheme	39	20.86%	20	10.70%	121	64.71%	7	3.74%	187	100.00%	5558
	Sub-total 1	**39**	**20.86%**	**20**	**10.70%**	**121**	**64.71%**	**7**	**3.74%**	**187**	**100.00%**	**5558**
Kampong Chhnang	Lum Hach Scheme	168	67.74%	0	0.00%	9	3.63%	3	1.21%	180	72.58%	2036
	Krang Ponley Kandal Scheme	2	0.81%	8	3.23%	0	0.00%	1	0.40%	11	4.44%	1790
	Anlong Chrey Scheme	56	22.58%	0	0.00%	1	0.40%	0	0.00%	57	22.98%	1850
	Sub-total 2	**226**	**91.13%**	**8**	**3.23%**	**10**	**4.03%**	**4**	**1.61%**	**248**	**100.00%**	**1983**
Kampong Speu	Brambei Mom Scheme	89	54.94%	0	0.00%	2	1.23%	4	2.47%	95	58.64%	1401
	Kra Peu Troum Scheme	62	38.27%	1	0.62%	0	0.00%	0	0.00%	63	38.89%	1096
	Krang Ponley Kandal Scheme	2	1.23%	0	0.00%	0	0.00%	2	1.23%	4	2.47%	1100
	Sub-total 3	**153**	**94.44%**	**1**	**0.62%**	**2**	**1.23%**	**6**	**3.70%**	**162**	**100.00%**	**1274**
Kandal	Krang Ponley Kandal Scheme	11	18.64%	45	76.27%	3	5.08%	0	0.00%	59	100.00%	1189
	Sub-total 4	**11**	**18.64%**	**45**	**76.27%**	**3**	**5.08%**	**0**	**0.00%**	**59**	**100.00%**	**1189**
	Grand Total	**429**	**65.40%**	**74**	**11.28%**	**136**	**20.73%**	**17**	**2.59%**	**656**	**100.00%**	**2750**

Source: Livelihood and Vulnerability Report

Table 8 – livestock production (household level)

Province	Subproject	Cow		Buffalos		Pig		Duck		Chicken		Goose		Total		b. Number
		N	%	N	%	N	%	N	%	N	%	N	%	N	%	Mean
Pursat	**O Ta Paong Tanay Scheme**	**71**	31.56%	**5**	2.22%	**8**	3.56%	**28**	12.44%	**113**	50.22%	**0**	0.00%	**225**	100.00%	**18**
	Sub-total 1	**71**	**31.56%**	**5**	**2.22%**	**8**	**3.56%**	**28**	**12.44%**	**113**	**50.22%**	**0**	**0.00%**	**225**	**100.00%**	**18**
Kampong Chhnang	**Lum Hach Scheme**	**83**	26.18%	**11**	3.47%	**20**	6.31%	**19**	5.99%	**99**	31.23%	**2**	0.63%	**234**	73.82%	**9**
	Krang Ponley Kandal Scheme	**3**	0.95%	**0**	0.00%	**0**	0.00%	**0**	0.00%	**6**	1.89%	**0**	0.00%	**9**	2.84%	**8**
	Anlong Chrey Scheme	**31**	9.78%	**0**	0.00%	**2**	0.63%	**4**	1.26%	**36**	11.36%	**1**	0.32%	**74**	23.34%	**7**
	Sub-total 2	**117**	**36.91%**	**11**	**3.47%**	**22**	**6.94%**	**23**	**7.26%**	**141**	**44.48%**	**3**	**0.95%**	**317**	**100.00%**	**9**
Kampong Speu	**Brambei Mom Scheme**	**50**	24.63%	**0**	0.00%	**2**	0.99%	**8**	3.94%	**55**	27.09%	**0**	0.00%	**115**	56.65%	**17**
	Kra Peu Troum Scheme	**27**	13.30%	**0**	0.00%	**1**	0.49%	**7**	3.45%	**38**	18.72%	**0**	0.00%	**73**	35.96%	**8**
	Krang Ponley Kandal Scheme	**2**	0.99%	**0**	0.00%	**0**	0.00%	**3**	1.48%	**9**	4.43%	**1**	0.49%	**15**	7.39%	**9**
	Sub-total 3	**79**	**38.92%**	**0**	**0.00%**	**3**	**1.48%**	**18**	**8.87%**	**102**	**50.25%**	**1**	**0.49%**	**203**	**100.00%**	**13**
Kandal	**Krang Ponley Kandal Scheme**	**11**	16.92%	**0**	0.00%	**0**	0.00%	**3**	4.62%	**50**	76.92%	**1**	1.54%	**65**	100.00%	**9**
	Sub-total 4	**11**	**16.92%**	**0**	**0.00%**	**0**	**0.00%**	**3**	**4.62%**	**50**	**76.92%**	**1**	**1.54%**	**65**	**100.00%**	**9**
	Grand Total	**278**	**34.32%**	**16**	**1.98%**	**33**	**4.07%**	**72**	**8.89%**	**406**	**50.12%**	**5**	**0.62%**	**810**	**100.00%**	**12**

Source: Livelihood and Vulnerability Report

Table 9 – Use of livestock production (household level)

Province	Subproject	Yes		No		Total	
		N	%	N	%	N	%
Pursat	O Ta Paong Tanay Scheme	129	84.87%	23	15.13%	152	100.00%
	Sub-total 1	**129**	**84.87%**	**23**	**15.13%**	**152**	**100.00%**
Kampong Chhnang	Lum Hach Scheme	114	57.58%	17	8.59%	131	66.16%
	Krang Ponley Kandal Scheme	7	3.54%	7	3.54%	14	7.07%
	Anlong Chrey Scheme	42	21.21%	11	5.56%	53	26.77%
	Sub-total 2	**163**	**82.32%**	**35**	**17.68%**	**198**	**100.00%**
Kampong Speu	Brambei Mom Scheme	67	37.43%	20	11.17%	87	48.60%
	Kra Peu Troum Scheme	41	22.91%	17	9.50%	58	32.40%
	Krang Ponley Kandal Scheme	10	5.59%	24	13.41%	34	18.99%
	Sub-total 3	**118**	**65.92%**	**61**	**34.08%**	**179**	**100.00%**
Kandal	Krang Ponley Kandal Scheme	53	52.48%	48	47.52%	101	100.00%
	Sub-total 4	**53**	**52.48%**	**48**	**47.52%**	**101**	**100.00%**
	Grand Total	**463**	**73.49%**	**167**	**26.51%**	**630**	**100.00%**

Source: Livelihood and Vulnerability Report

3.2.6 Current practices

Mechanization. Due to internal migration over the past few decades for better income, the rural labor force has been reduced, leaving room for increased use of machinery to compensate for traditional human labor. In the project provinces, an average 66% of sample (n=630) think that mechanization is very frequently used while 32% think it is used frequently. Harvesting needs more machinery use (75%) compared to land preparation (65%). It is noted that while 60% of surveyed households use machinery for heavy works such as land preparation and harvesting, the remainder (40%) still use buffalo and manpower for the heavy works. Where needed, families reach out to neighbors to ask for labor help and return to them when they need (labor exchange).

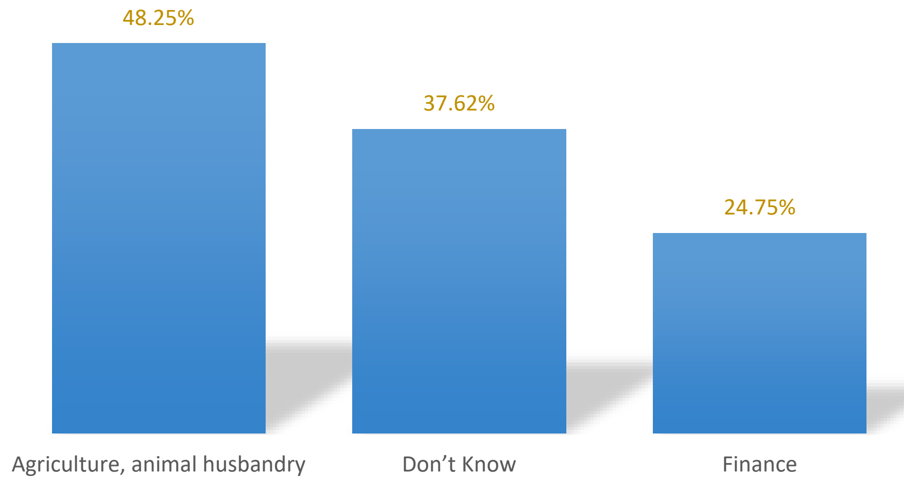
Use of new farming technology. 65.3% of surveyed households (n=319) have been using traditional rice cultivation practices. With regards to applying new farming techniques, 21.7% (n=106) use occasionally, 11.6% (n=57) use frequently, and 1.2% used very frequently. 61.2% (n=299) think mechanization is better whereas 38.7% (n=189) think new farming techniques are better. Data suggested that those who have already applied improved farming practices find it better. They think that it crop is better because of appropriate use of fertilizers (33.8%, n=408), pesticides (29.3%, n=354), use of machinery (12.2%, n=147), training (10%, n=121), and use of organic farming (10%, n=122). On the contrary, crop is less productive when land is less fertile (26.6%, n=303), lack of irrigation (24.3%, n=277), overuse of fertilizers and pesticides (15.5%, n=177), use the same farming practice (14.2%, n=162), and same crop (10%, n=116). It is noted that 65.5% (n=426) used pesticides and fertilizers whereas the remainder (32.3%) don't. Those who use pesticides and fertilizers think these boost growth of crop (42.8%, n=330), improve crop yield (41.6%, n=321), and improve soil fertility (14.6%, n=113).

Climate change and new technologies

Most surveyed households (90%, n=568) are not aware of the potential impact of climate change on their agricultural production. However, 40% (n=256) think they have experienced the impact of climate change and the remainder have not experienced it. 63% (n=397) have never heard of improved farming technology that helps increase their income.

Training Needs

When asked about the support/training that women need, half (48%) preferred training in agriculture/animal husbandry and finance (24.7). However, 37% of households don't know what they need in terms of knowledge to support their livelihood development.



3.2.7 Income level

In this survey, monthly income of the surveyed households were collected. Monthly household income refers to the total income that all members of the household could earn each month, on average, excluding children. It is noted that all family members who can make an income contribute their income to the family's common expenditure. The level of individual contribution varies depending on family. Below is the distribution of monthly household income across four project provinces. The average monthly household income is 152.8 USD (SD=126.7).

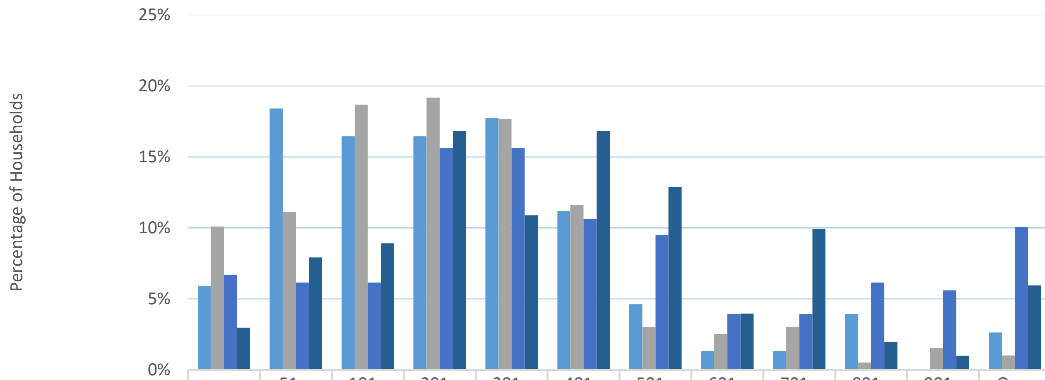
Table 10 – Descriptive Statistic of Monthly Income of Household

Monthly household income	USD
Mean	393.1
SD	327.6
Min	10.0
Max	2,150. 0
Median	322.5
Mode	100.0

Source: Analyzed by author using dataset from Livelihood & Vulnerability survey

Below is the distribution of various levels of monthly income of households across 4 provinces. It is noted that of 630 households interviewed, about one-third (34.6%) said their income is stable. The remaining (65.4%) do not think their income, which is already low, is stable. Distribution of monthly income of the survey households are summarized in the graph below.

Figure 9 – Distribution of monthly household income by project provinces

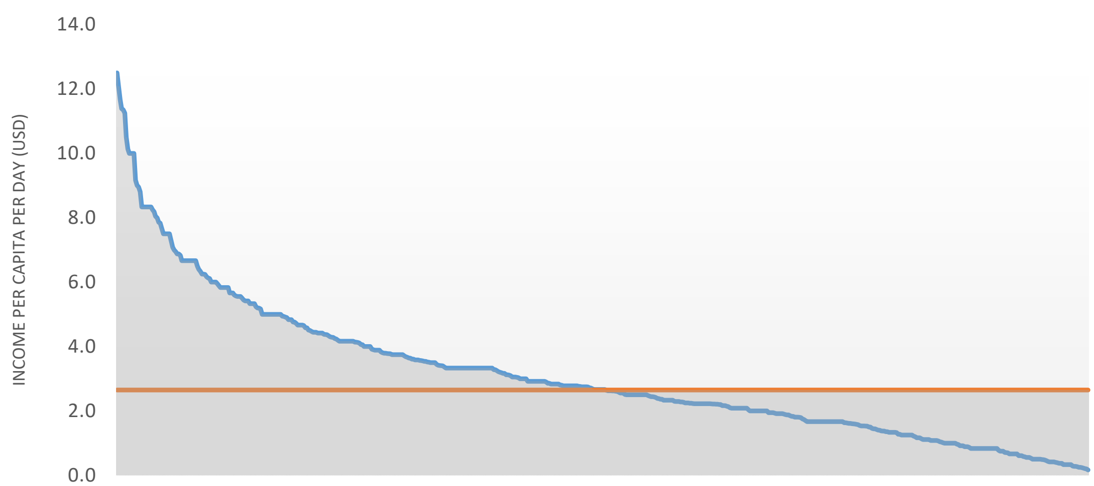


	< 50$	51 - 100$	101 - 200$	201 - 300$	301 - 400$	401 - 500%	501 - 600$	601 - 700$	701 - 800%	801 - 900$	901 - 1,000$	Over 100$
Pursat	5.92%	18.42%	16.45%	16.45%	17.76%	11.18%	4.61%	1.32%	1.32%	3.95%	0.00%	2.63%
Kampong Chhnang	10.10%	11.11%	18.69%	19.19%	17.68%	11.62%	3.03%	2.53%	3.03%	0.51%	1.52%	1.01%
Kampong Speu	6.70%	6.15%	6.15%	15.64%	15.64%	10.61%	9.50%	3.91%	3.91%	6.15%	5.59%	10.06%
Kandal	2.97%	7.92%	8.91%	16.83%	10.89%	16.83%	12.87%	3.96%	9.90%	1.98%	0.99%	5.94%

Source: Data from Livelihood and Vulnerability Survey. Graphed by author

It is noted that given the mean income of 152.8 USD per household per month, the mean income per capita per month is estimated at 35.87 USD (or 148.161.75 Riel) per capita per month. This is equivalent to 1.19 USD (4,938.72 KHR) per day which is below the national poverty line – as established by the World Bank for Cambodia (which is KHR 10,951 per person per day using 2019/2020 prices.

Figure 10 – Distribution of income per capita per day vs poverty (orange) line (about 2.65 USD, or KHR 10,951, per day)



Source: Data from Livelihood and Vulnerability Survey. Calculated & graphed by author

It is noted that although 49.7% (n=313) are under the poverty line (based on self-reported income), only 14% of the survey households (n=88) have ID poor cards. It is also noted that the no relation is found between the level of household income and the area of agricultural owned by the households[35]. This suggests that the

[35] *r*=.059, p>.05

income currently earned by households is not from land. In others, land has not been made full use of because of lack of irrigation water.

3.2.8 Expenditure

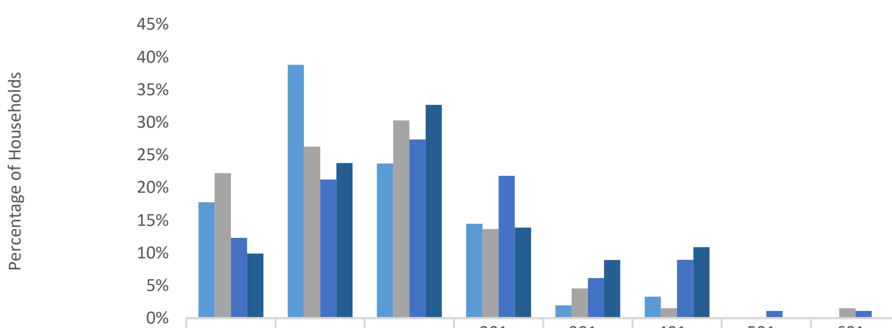
The average monthly expenditure of the surveyed participants is 176.6 USD. This is lower than the average monthly income which is 393.1USD.

Table 11 –Descriptive Statistic of Monthly Household Expenditure

Monthly household expenditure	USD
Mean	176.6
SD	131.8
Min	10.0
Max	700.0
Median	150.0
Mode	100.0

Below is the distribution of various level of monthly expenditure of sampled households (n=630) from four provinces.

Figure 11 – Distribution of monthly expenditure at household level (by project province)



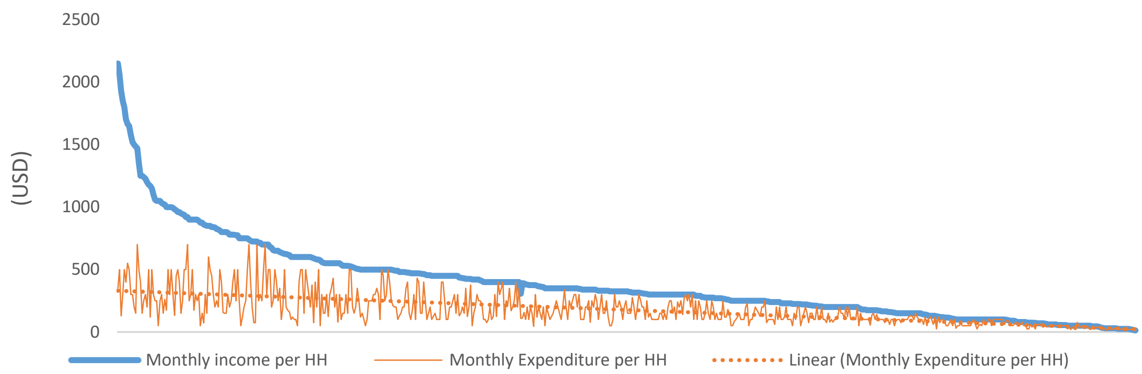
	< 50$	51 - 100$	101 -200$	201 - 300$	301 - 400$	401 - 500%	501 - 600$	601 - 700$
Pursat	17.76%	38.82%	23.68%	14.47%	1.97%	3.29%	0.00%	0.00%
Kampong Chhnang	22.22%	26.26%	30.30%	13.64%	4.55%	1.52%	0.00%	1.52%
Kampong Speu	12.29%	21.23%	27.37%	21.79%	6.15%	8.94%	1.12%	1.12%
Kandal	9.90%	23.76%	32.67%	13.86%	8.91%	10.89%	0.00%	0.00%

Source: Data from Livelihood and Vulnerability Survey. Graphed by author

There is a fairly strong positive relationship between household size and income[36], and a strong positive relationship between income–expenditure[37]. However, the data (graph below) suggests that surveyed households appear to confine their expenditure within the income they are capable of earning. Households that earn more tend to spend only basic costs to save money.

Figure 12 – Expenditure vs Income (Monthly at Household level)

[36] r_s=.51, p<.001
[37] r_s=.74, p<.001



Source: Data from Livelihood and Vulnerability Survey. Graphed by author

The reasons why both income and expenditure are unstable are due to the variation between wet season and dry season, according to the explanation from focus group discussion and key informant interviews conducted by the Livelihood and Vulnerability Survey Team. In particular, during wet season, key expense include income lower than expenditure (24.3%), medical costs (19.9%)family expenses (18.7%), children's education (13.4%), loan repayment (10.4%), investment in agricultural production (9.5%), flooding (2.4%), and, less cultivation (1.5%).

During dry season reasons given, key reasons include income lower than expenditure (19%), children's education (14.8%), social expenses (13.6%), medical care (10.4%), rice drought damages (9.2%), debt payment (8%), lack of irrigation water (6.8%), payment for fertilizer cost (1.8%), , and low product income (1.5%).

3.2.9 Social networking

61.1% of respondents (n=385) have had no experience working in a community farmer group. 68.3% of them (n=430) are not involved in any group activities at village/commune level. Most women in the country use their own resources to start their business, or rely on informal sources and savings to expand their businesses. They have no other choice since women are underserved by banks[38].

3.2.10 Loans

At the country level, according to the Cambodia Microfinance Association (CMA), 80% of their clients are women. However, loan size offered by these institutions are typically small whereas interest rate is higher than commercial banks[39].

In this survey, 28.9% of surveyed households (n=189) got a loan over the past one year. The purpose of the loan (in descending order) is for house repair, investment in business, living expenses, purpose of consumables, and to buy land (See graph below). Most of them will try to take a loan from a legitimate bank, and not the local lender because of the high interest.

[38] International Finance Cooperation 2021. Exploring the Opportunities for Women-owned SMEs in Cambodia (page 6).
[39] MAFF 2022.

There is an association between borrowing loan and family's income[40], expenses[41], household size[42]. No association can be found between loan borrowing and area of riceland[43], suggesting that surveyed households have not borrowed loan for investment in rice production.

Figure 13 – Expenditure vs Income (Monthly at Household level)

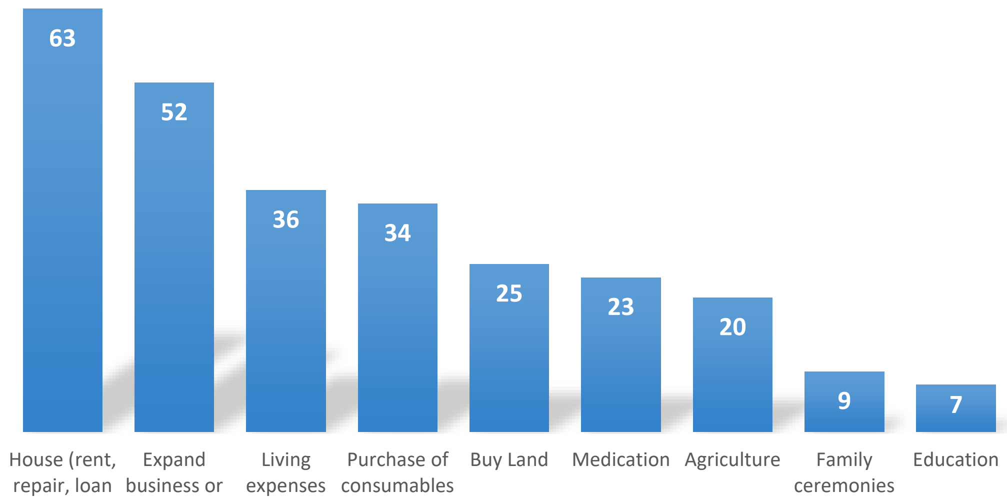


Table 12 – Descriptive Statistic of Loan size

Statistics	Loand size (in Riel)
Mean	6549.0
SD	7189.9
Min	2000.0
Max	50,000.0
Median	4000.0
Mode	10,000.0

It appears that participants prefer taking a loan from a private lender or money lender (67%), followed by relatives (9.3%). The average interest rate (per month) is 1.36%. 33.5% of loan borrowers (n=61) indicated they faced challenges repaying the loans. Most households in the survey (94.7%, n=597) are not members of any microcredit/self help group. 93.9% (n=592) of women in surveyed households don't have any personal savings although 80.4% (n=507) say that women in their family have a freedom of determining how money is used in their family and 17.3% (n=109) don't have.

[40] $\chi^2(202) = 246.337$, $p<0.05$
[41] $\chi^2(55) = 89.307$, $p<0.05$
[42] $\chi^2(11) = 52.649$, $p<0.01$
[43] $\chi^2(107) = 110.593$, $p>0.05$

Province	Subproject	Private lender or money lender		Community lender		Team lender		Relative lender with interest rate	
		N	%	N	%	N	%	N	%
Pursat	O Ta Paong Tanay Scheme	68	68.69%	6	6.06%	14	14.14%	4	4.04%
	Sub-total 1	**68**	**68.69%**	**6**	**6.06%**	**14**	**14.14%**	**4**	**4.04%**
Kampong Chhnang	Lum Hach Scheme	42	35.00%	1	0.83%	2	1.67%	8	6.67%
	Krang Ponley Kandal Scheme	8	6.67%	0	0.00%	1	0.83%	3	2.50%
	Anlong Chrey Scheme	29	24.17%	10	8.33%	6	5.00%	2	1.67%
	Sub-total 2	**79**	**65.83%**	**11**	**9.17%**	**9**	**7.50%**	**13**	**10.83%**
Kampong Speu	Brambei Mom Scheme	41	33.61%	4	3.28%	8	6.56%	6	4.92%
	Kra Peu Troum Scheme	26	21.31%9.3	0	0.00%	2	1.64%	7	5.74%
	Krang Ponley Kandal Scheme	15	12.30%	0	0.00%	1	0.82%	0	0.00%
	Sub-total 3	**82**	**67.21%**	**4**	**3.28%**	**11**	**9.02%**	**13**	**10.66%**
Kandal	Krang Ponley Kandal Scheme	55	71.43%	3	3.90%	5	6.49%	9	11.69%
	Sub-total 4	**55**	**71.43%**	**3**	**3.90%**	**5**	**6.49%**	**9**	**11.69%**
	Grand Total	**284**	**67.94%**	**24**	**5.74%**	**39**	**9.33%**	**39**	**9.33%**

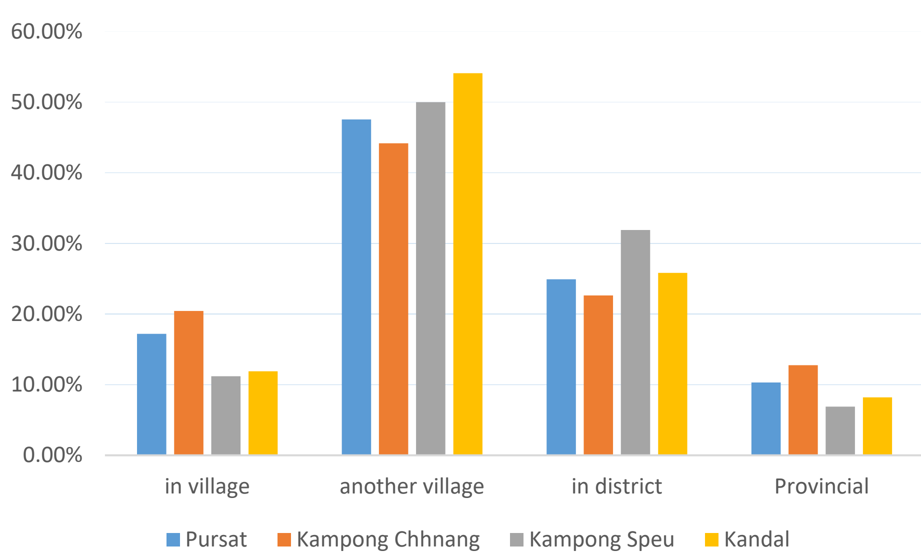
Source: CAISAR Livelihood and Vulnerability Survey (2022)

3.2.11 Public services

- Health services

Based on the survey results, health services do exist in many villages, but respondents told enumerators that often people would travel to neighboring villages (1-2 KM) or to a district (more advanced health service) further away (around 5 KM) where there is a more advanced health service (Figure 5-26).

Figure 14 – Access to local or provincial health care facilities

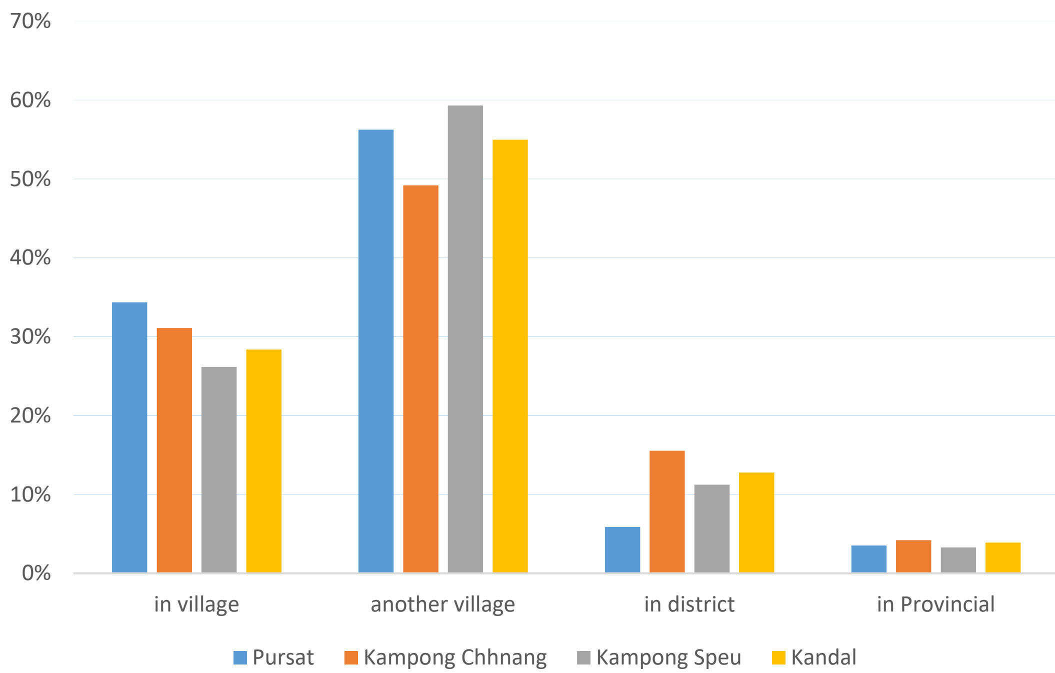


Source: data from CAISAR Livelihood & Vulnerability report. Graphed by author.

- Education services

Based on discussions held with the FGDs and KIIs, enumerators found that students usually go to study in another village and for upper secondary school (High School) students must go to the district center. As discussed above, this poses a difficulty as it costs money, as students may have to stay away from home, which means they cannot participate in household chores. Students can travel 1-2 KM to a neighboring village school or 4-5 KM if travelling to a district center.

Figure 15 – Access to local or provincial educational facilities by province



Source: data from CAISAR Livelihood & Vulnerability report. Graphed by author.

3.2.12 Vulnerability of Households with less than 1ha of riceland

In the survey sample (n=630), there are more than half of households (62%) who have less than 1ha of riceland (as shown in table below).

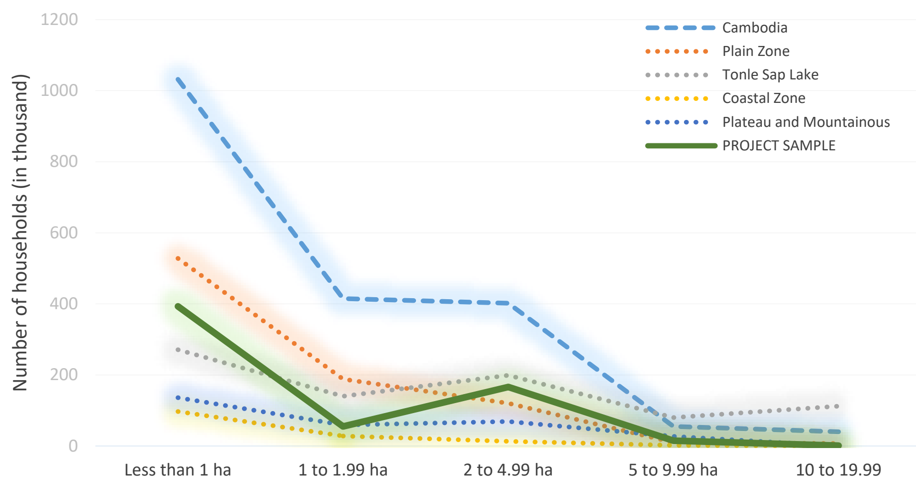
Table 13 – Distribution of agri-land size of the sample

Groups	n	Percentage
Poor (less than 1ha)	393.0	62%
Middle (≥1 ha, <2ha)	55.0	9%
Better-off (≥2 ha)	182.0	29%
TOTAL		100%

Source: Dataset of Livelihood Survey. Calculated by author.

Compared to the regional, and the national land distribution data, distribution of agri-land size from the project's sample appears to follow the same distribution at regional and national levels. It is noted that the proportion of households who have less than 1ha of agricultural landholding makes up the largest percentage, followed by the group with landholding from 1 to 4.99ha per household (See chart below).

Figure 16 – Distribution of agricultural land – project data vs national data



Source: Data from CAISAR Livelihood Survey and 2020 National Agriculture Census. Calculated by author. Project data is timed by 1,000 to show its patten vis-à-vis national data.

The households with less than 1ha of riceland is characterized by the followings:

- Average HH size is **lower** than that of those having more than 1 ha of riceland[44]
- Average size of residential land is **lower** than that of those having more than 1 ha of riceland[45]
- Average age of HH head is **higher** than that of those having more than 1 ha of riceland[46]
- Average percentage of monthly food expenditure is **higher**[47]
- Have **one job** while those having more than 1 ha have two jobs[48]
- **More likely to borrow loan** compared to those having more than 1 ha [49]
- Average HH income is lower that of the non-youth[50]

3.3 Gender Analysis

3.3.1 Labor Division

For those who are involved in farming as their main livelihood activities, virtually all of them (97.6%, n=451) dedicate their full-time farming. The male workers spend, on average, 59.4% of their time farming vis-à-vis their female counterparts who spend 40.5% of their time farming.

In rice farming, traditionally, the first stages of rice cultivation are male-designated and the latter stages female-designated. Men generally perform land preparation tasks, while seedling preparation and weeding are commonly assigned to women[51]. All other jobs such as transplanting, uprooting, harvesting, and marketing are

[44] $t(628)=2.168$, $p<0.01$, mean difference: 0.47
[45] $t(628)=4.726$, $p<0.01$, mean difference: 535.9
[46] $t(628)=3.270$, $p<0.01$, mean difference: 3.5
[47] $t(628)=3.542$, $p<0.01$, mean difference: 3.7
[48] Chi-square test is statistically significant: $\chi^2(5.578)=$, $p<0.05$
[49] Chi-square test is statistically significant: $\chi^2(5.578)=$, $p<0.05$
[50] Chi-square test is statistically significant: $\chi^2(5.172)=$, $p<0.05$
[51] MAFF 2015. Gender Mainstreaming Policy and Strategic Framework in Agriculture (2016-2020).

generally shared tasks[52]. However, given the migration of men for better income, mechanization has become more popular. As a result, in households with men migrating, women undertake all farming tasks: land preparation, irrigation, threshing, and recruitment of labor, farm management, harvesting, and selling. Growing secondary crops such as vegetables also increases. These secondary crops are mainly grown in home gardens, increasing nutritional status for households as well as households' food security. [53].

In livestock, most rural women (70%) are involved in raising small livestock such as chicken and ducks, and pigs. These are important sources of food and supplementary household's income. For most poor women, livestock serves as a primary form of savings, as well as insurance against economic shocks due to accidents, illness, death and natural disasters. However, as livestock in Cambodia faces rapid market restructuring, poor livestock producers, particularly women, experience constraints to meeting a growing number of regulations with regards to sanitary standards, animal transportation, storage standards, etc) required by more structured markets.[54] In general, women raise poultry, men do cattle and pigs.

For household chores, women still do 90% of the work. During COVID-19 breakout, women spent even increased time on domestic and care work[55]. Elders carry full responsibility for raising grandchildren whose mothers have migrated[56]. In the project area, similarly, in families where female labor has to work in nearby garment factories, or migrants for paid work, domestic works are transferred to young girls and old people. According to the National Institute of Statistics (2021), unpaid household workers are more than twice as likely to be women and are in a particularly precarious situation, depending on other members of their household for any income[57].

3.3.2 Income

At the individual level, the average income of all female household members is 121.4 USD per month (n=238). This is lower than the income of male members which is 166.6 USD per month (n=366). The difference in income between male and female members is 45.2 USD, and is statistically significant[58]. However, at household level, there is no statistically significant difference in the mean income between female-headed household and male-headed households[59]. The average income between two groups is small (mean difference=17.3 USD).

In term of use of income, according to the national Demographic and Health Survey (2020-2021), when men were asked about who makes decisions regarding how their earnings are used, 48% of married men (aged 15–49) reported that they jointly make decisions with their wives, 46% said that their wives mainly make decisions, and 6% said they make these own decisions. In fact, 97% of married women (aged 15–49) who can make cash earnings for their employment participate in making decisions about the use of their earnings whereas 63% make these decisions mainly alone.

3.3.3 Decision Making

- **Children & Livelihood**

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[52] ADB 2012.Cambodian Climate Change Resilient Rice Commercialization: Socio Economic Assessment and Gender Analysis.

[53] MAFF 2015. Gender Mainstreaming Policy and Strategic Framework in Agriculture (2016-2020).

[54] MAFF 2015. Gender Mainstreaming Policy and Strategic Framework in Agriculture (2016-2020).

[55] United Nations 2022. Gender Equality Deep-Dive for Cambodia – Common Country Analysis.

[56] Green, W. N., & Estes, J. 2019. Precarious Debt: Microfinance Subjects and Intergenerational Dependency in Cambodia. Antipode, 51(1), 129–147. https://doi.org/10.1111/anti.12413.

[57] MAFF 2022. Gender Mainstreaming Policy and Strategic Framework in Agriculture (2022-2026).

[58] $t(602)=3.996$, $p<0.01$, Cohen's $d=0.333$

[59] $t(628)=0.653$, $p>0.05$, Cohen's $d=0.053$

Data from the Livelihood Survey indicated that issues around children (education, childcare, marriage) are decided mainly by both husband and wife albeit there is a remarkable proportion of households where women are the only one who make the decisions related to their children. In terms of livelihoods, while decision related to daily livelihoods (e.g. expense for households, same of home-based produce, visit friend and family are decided by both husband and wife, there is a considerable part of respondents who indicate decisions related to livelihood of women, family expense, attending trainings, etc. are decided solely by women (See table below).

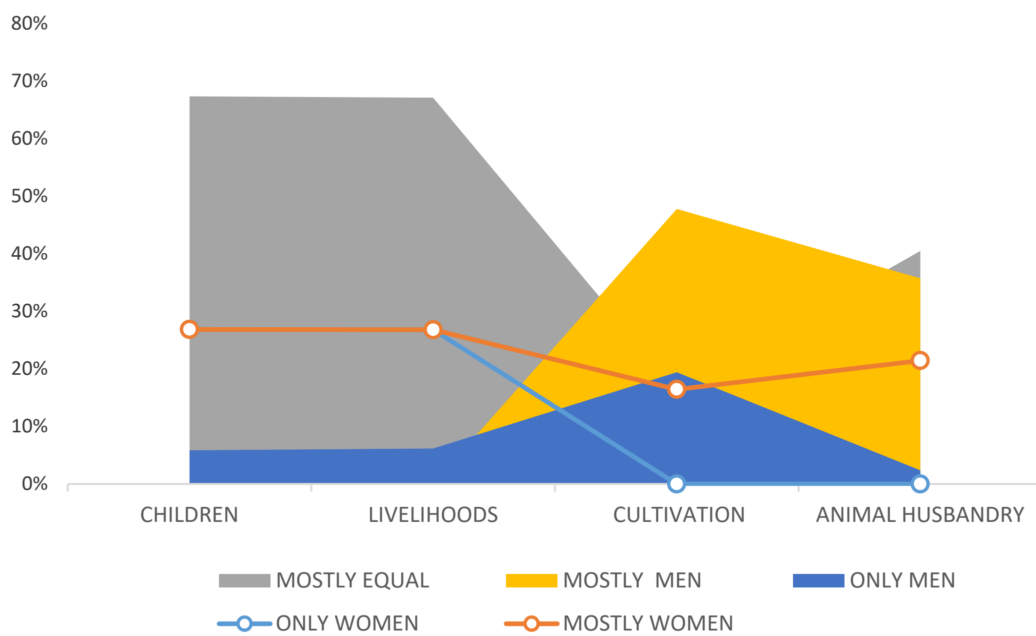
	ONLY WOMEN	MOSTLY WOMEN	MOSTLY EQUAL	MOSTLY MEN	ONLY MEN
CHILDREN	**1352**	**0**	**3393**	**0**	**295**
School enrolment of children, particularly for girls	189		426		15
Continuing education for girls beyond primary	191		416		23
Girls' age at marriage	144		464		22
Selection of match for girls	158		447		25
Place of delivery	140		379		111
No. of child to be born	186		416		28
No. of male child preferred	165		431		34
Child immunization	179		414		37
LIVELIHOODS	**1181**	**0**	**2958**	**0**	**271**
Type of livelihood for women	191		414		25
Spending on Household necessities/items	215		387		28
Sale of home-based products	214		389		27
Visits to parents' family/friends	136		477		17
Attend skill training	138		438		54
Forming/joining commune FWUCs/FWUSG	143		429		58
Participating in commune meetings	144		424		62

Source: Data from Livelihood Survey. Analyzed by author.

- **Cultivation & Animal husbandry**

While Children & Livelihood issues tend to gravitate to women, issues related to agricultural production (cultivation and animal husbandry) tends to be mostly with men. This is explained by the fact that agricultural works are physically heavy and poisonous in some instances such as pesticide application (See chart below). It is interesting to learn from the consultation from 7 communes in 4 provinces in June 2023, consultation participants indicated that most trainings are attended by women.

Farm works	ONLY WOMEN	MOSTLY WOMEN	MOSTLY EQUAL	MOSTLY MEN	ONLY MEN
CULTIVATION	0	11	11	32	13
Buy seeds, fertilizers, pesticides					
Prepare land					
Daily crop care					
Hire additional labor					
Irrigate					
Spray pesticides					
Apply fertilizers					
Harvest					
Decide selling prices					
Attend trainings					
ANIMAL HUSBANDRY	0	9	17	15	1
Select/buy varieties					
Buy feed					
Daily care (e.g. feeding, bathing)					
Choose veterinary services					
Decide selling prices					
Attend training					

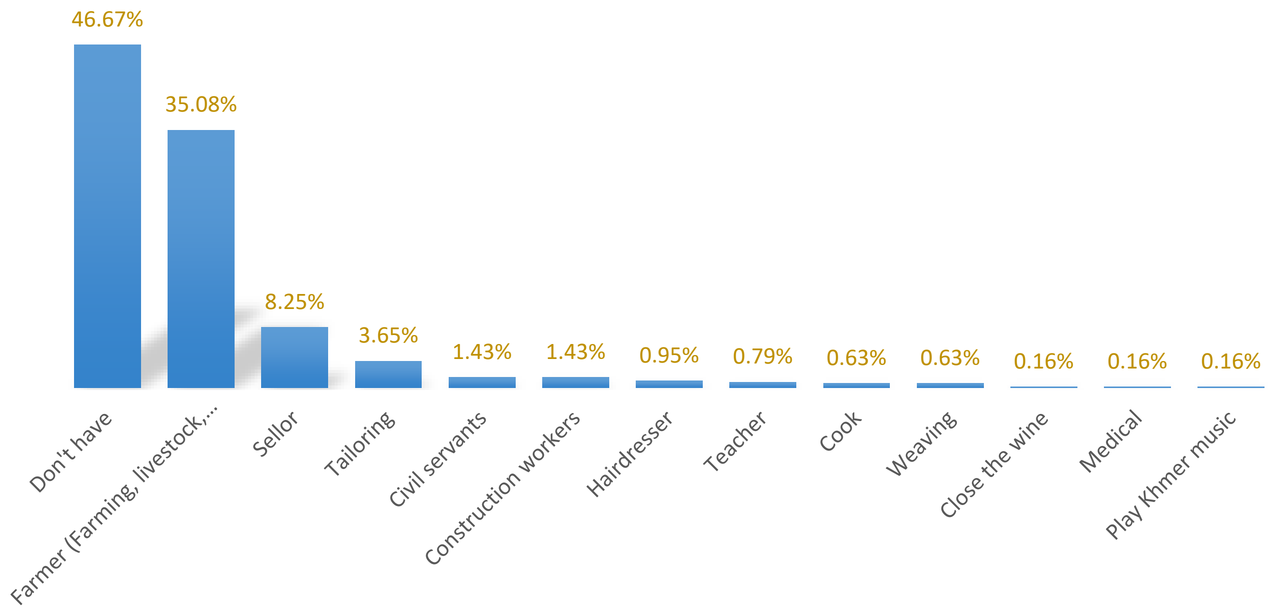


Source: Data on Children and Livelihood from CAISAR Livelihood Survey report. Analyzed by author.

At national level, when it comes to spending money for women's health care, purchasing a major household, and visiting their family and relatives, 88% of currently married women aged 15–49 can decide on their own or jointly with their spouse (Demographic and Health Survey 2020-2021).

- ***Skills of female household member***

Of the total 630 households in the survey, in 46% of households (n=294), female members don't have working skills. 35% (n=221) have farming skills such as farming, livestock, orchard.



In the opinion of the surveyed households, women should have the following education, followed by freedom of decision making, good husband, income opportunity, and own saving.

- ***Access to Agricultural Extension Service***

It is estimated that only 10 percent of the Cambodian farmers with access to agricultural extension services are women[60]. The constraints affecting rural Cambodian women's ability to have adequate access to agricultural extension services include distance to the point of service provision; insensitivity to the level of literacy; time; lack of childcare options; household responsibilities; mobility; and socio-cultural characteristics[61]. Men are still traditionally regarded as household head and are often automatically the recipient of new information[62].

- ***Market access***

Women face a number of disadvantages: lower mobility, lower level of literacy, less access to training, less access to market information, and less access to productive resources. Lower financial literacy of women and travel safety are identified as the main gender gaps in Cambodia to access to markets for women[63].

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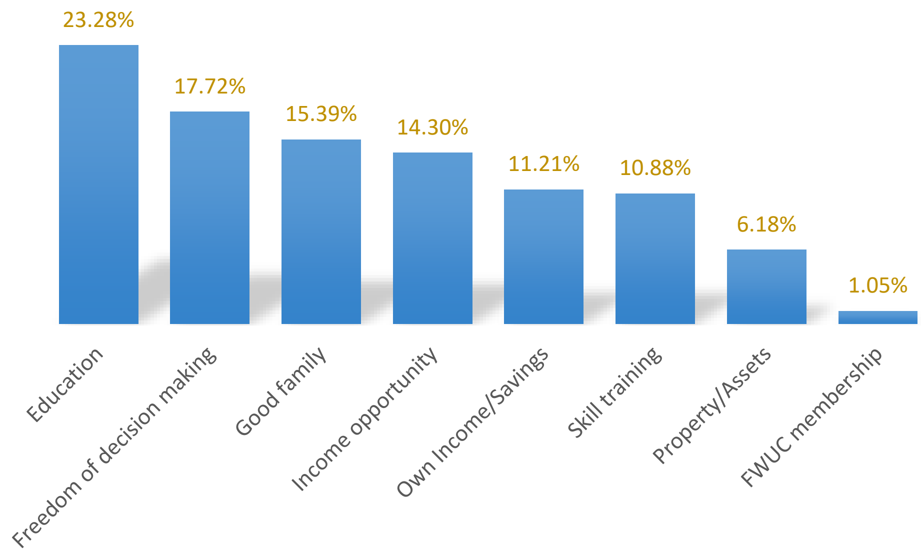
[60] https://www.phnompenhpost.com/national-post-depth/cavac-empowering-women-farmers-through-tech (accessed September 17, 2023).

[61] MAFF 2015. Gender Mainstreaming Policy and Strategic Framework in Agriculture (2016-2020).

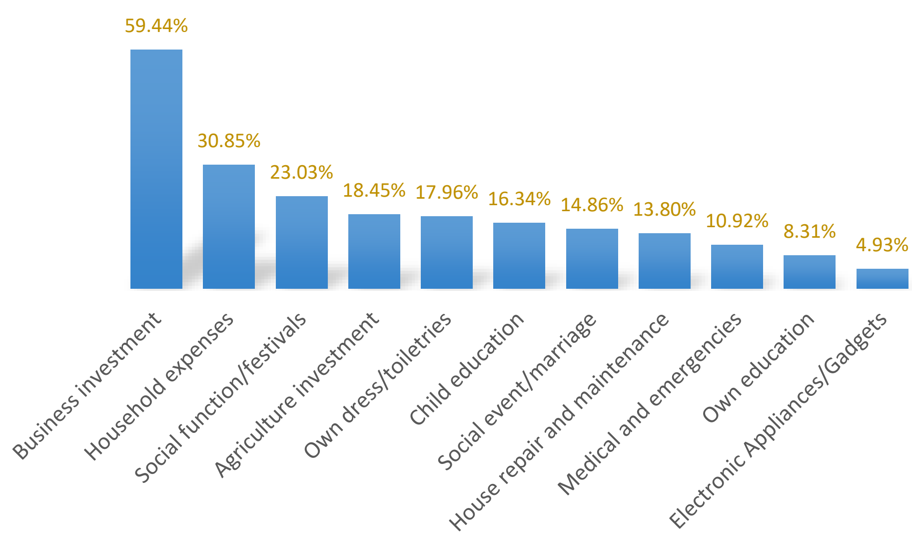
[62] MAFF 2022. Gender Mainstreaming Policy and Strategic Framework in Agriculture (2022-2026).

[63] MAFF 2022. Gender Mainstreaming Policy and Strategic Framework in Agriculture (2022-2026).



- ***Potential financial investment***

80.48% of survey respondents (n=507) said female members have freedom of spending household income. If they have financial resources, they plan to spend on the followings:



3.4.4 Key gaps

This section summarized the key **gaps in working skills** among female members:

- **64%** of surveyed households (n=407) **do farming as the household's main income generation activity**. However, only **half of them** (35.08%, n=221) have female members with farming skills.
- **In 46%** of surveyed households (n=294), female members (working age) **don't have working skills**.

- There is a **big dismatch** between **current income generation activities** (64% in farming) and **desired investment** (59% in non-farming).

4. GENDER ACTION & SOCIAL INCLUSION PLAN

From the above analysis, key gaps with regards to participation of women in farming activities and household chores have been identified. To reduce the gender gaps, it is important that barriers to reducing the gaps be pinpointed, opportunities that facilitate the gender mainstreaming effort be identified, and priorities set forth to address the barriers while maximizing the opportunities. The section below identified key barriers, opportunities, and priorities in promoting participation of women in the economic empowerment process. The following section is a guidance to integrate women in the project level activities. The sub-projects activities will adopt the same guidance and incorporate the women folk in the subproject activities.

The section below identified key barriers, opportunities, and priorities in promoting participation of women in the economic empowerment process.

4.1 Key Barriers to Participation of Women

This section highlighted five key barriers that may potentially affect the participation of women in project activities. The barriers are inter-related thus need to be addressed together to reduce the constraints.

Barrier 1 – Gender norms

In Cambodia, gender stereotypes, which are based on social norms, have long shaped how rural men and women divide their responsibilities and time for household chores and income generation activities. This affects their mobility, arrangements in decision making, and their wellbeing. This gender norm was originally based on labor division between women and men based on their physical and reproductive health. While men are generally responsible for physically heavy works in farming, including works that are considered harmful to health (e.g spraying pesticide), women's role focuses on child bearing, childcare, household chores. Because of these functions, women are involved in farm work that is physically suitable for them at home such as home gardening, animal husbandry (e.g. raising poultry). Given this responsibility pattern, men generally represent his family for works that happen outside the family (e.g. community works) and have more social interaction compared to their female counterparts (See opportunities and priority for project to overcome this barrier in the next sections).

Barrier 2 – Household chore

The gender norm defines that domestic works are key women's responsibility. Traditional codes (Chbab Srey/ Proh) have been adopted for several centuries. The code specifies men as 'head of the household' and advise a woman to maintain peace at home and respect her husband" [64]. As such, women spend most of their time at home, taking care of domestic works that are unpaid. Chore work does not only take them most of their time of the day but also prevents them from taking part in activities outside their home: attending community meetings, agricultural training, visiting friends or relatives, or keeping up with their peers in the neighborhood. Because of their work burden, only a few women have the time to join FWUCs and village and commune development activities, particularly in water and irrigation management or in disaster risk reduction. This can negatively affect women's adaptive capacity, jeopardizing the resilience of poor rural households.

[64] Anderson, E., & Grace, K. (2018). From Schoolgirls to "Virtuous" Khmer Women: Interrogating Chbab Srey and Gender in Cambodian Education Policy. Studies in Social Justice, 12(2), 215–234. https://doi.org/10.26522/ssj.v12i2.1626

For families that are involved in home gardening and animal husbandry, household chore and home farming reinforces the gender stereotype. This keeps women busy at home, depriving them of opportunities to update their general knowledge, social trends, or even refreshing the skills that they use daily in their domestic work and animal husbandry. This forms a vicious cycle that maintains the existence of gender norms (See opportunities and priority for projects to overcome this barrier in the next sections).

Barrier 3 – Human capital

Education, adult literacy and skills are low for both men and women, and particularly lower among women. Income generation in rural areas is difficult. Thus children start involving in farming and household chores from a very young age. Therefore, they tend to drop out from school as they move through primary, secondary and high school. According to the 2021 national socioeconomic survey, main reasons why persons aged 6-17 years did not attend school are: a) work to contribute to family income (27%), b) don't want to study (16.2%), c) do not do well in school (13.5%), d) too poor to continue education (8.3%), and e) Must help with household chores (7.3%). Surprisingly, the rates are higher among females when it comes to work to contribute to family income (29.7% and 25.0% for female and male, respectively), and household chores (10.3% and 4.9% for female and male, respectively). Early marriage in rural areas is still common. This reduces the opportunities of young females to take on further education because of child bearing and household chores (mentioned in Barrier 1 and 2). Adult literacy is overall 85% at country level. However, adult literacy among women is lower than men – 80% and 90%, respectively. In rural area, adult literacy is even lower − 75.3% and 86.7% for women and men, respectively, and tend to decrease among both men and women who are 25 years and above[65]. More than two-thirds of households in the lowest income quintile depend on agriculture. However, agriculture is mainly held back by low level of mechanization, inadequate irrigation, little use of fertilizer, and predominant use of low-value rice strains [66]. Farming skills that are associated with old farming practices are also limited (See opportunities and priority for projects to overcome this barrier in the next sections).

Barrier 4 – Wage, Job, Migration

Although the gender wage gap has gone down − from 24% in 2017 to 19% in 2020, the wage gap between women and men remains a challenge, representing an enduring barrier for women jobseekers, and for both men and women under the unfavorable impacts on productivity and economic growth. Also, average women are found to work less hours than men in wage employment. But this does not take into account the time women typically spend for their household chores[67]. In the rural area, in general, and in four project provinces, in particular, the job market is not active. This is especially evident in remote, mountainous areas where farm works are dominant. Given this, when searching for jobs, women tend to look for jobs near home while men can travel far to work as migrant workers. In the context of this project, it is likely that women, including women in ethnic minority communities, are likely to focus on their domestic farming, or farming activities (for those who have access to productive land). There is a high level of out-migration of young people (60-80% in each project commune) (See opportunities and priority for project intervention in the next sections).

Barrier 5 – Adoption of New Farming Technologies

Traditional farming practices − characterized by dominant use of old varieties, low mechanization, old crop caring techniques, monocrop, and local market, are among potential constraints to project's effort in promoting adoption of new technologies that take into account factors related to crop intensification (e.g. more crops per year, short crop duration, better crop care), environment protection, loan, and market for new rice strains. It is also important to note that since 30-50% of rice are retained for home use, farmers may encounter challenges

[65] National Institute of Statistics, Ministry of Planning, 2021. Cambodia Socio-Economic Survey 2021.

[66] World Bank. 2012. Matching Aspirations - Skills For Implementing Cambodia's Growth Strategy

[67] UNDP. 2021. The Gender Wage Gap in Cambodia. Phnom Penh, Cambodia.

in adoption of new varieties that they may not be interested in eating new rice varieties. They may also be concerned about how they sell the new strains (market), and how to learn quickly to apply successfully new technology packages that will come alongside new varieties, shorter crop duration, labor shortage, market demand and how to ensure better selling prices. Trial and adoption of new technology package may take a few years: from Knowledge, Persuasion, Decision, Implementation, and Confirmation[68]. To achieve the project goal, project farmers need to trial and apply new farming techniques that the project will introduce. Since most farmers don't change the current farming practice overnight, it is important that the project walk the farmers through a well designed learning and adoption process, which needs to be facilitated with project interventions, to promote full adoption of introduced technologies and achieve the one of the project goal – enhancing income and livelihoods of rural people in the context of climate change (See opportunities and priority for project intervention in the next sections).

4.1.1 Potential Risks to achieving gender related Impacts are

- Gender based violence (GBV) at the HH level and during outside activities where women will participate.
- Gender inequality and knowledge gaps in the decision making activities where both men and women will be participating and where women will be participating for the first time due to the project activities
- Increase of sexual abuse for women and children living near the project site due to the influx of outside workers/ labourers.
- Losing of jobs for women due to increased mechanization

These potential risks will be mitigated through strong gender inclusion in all project related activities, continuous awareness generation activities, training and capacity development of both men and women farmers. The GBV will be addressed through special training and awareness to the women farmers along with their children. All other risks will be mitigated through the targeted project activities mentioned in 4.2.

4.2 Opportunities for Gender Mainstreaming into Project Activities

The project offers a unique opportunity where participation of women in project design and implementation is subject to monitoring and evaluation. This ensures the target on gender mainstreamings as to different project activities are set (as baseline) and monitored for changes (mid-term and endterm). Although 73% of surveyed households do animal husbandry, this activities serve as family's saving as income is low. There is still a chance for developing more in animal husbandry and other farming activities such as cash crop, income from post-harvest (e.g. straw, mushroom production).

Opportunity 1 – Increasing sharing of household chore between men and women

While household chores remain a key responsibility of women, at national level, there has been an increased share of household chores, particularly among households where demand for job/income necessitates adjustment to traditional intra-household labor division. This is indicated during COVID-19 pandemic when men participated more in household works, and in households where women work full-time in garment/footwear factories (including women who migrate and those who work in nearby communes). This suggests that there has been a gentle shift in labor division in situations where adjusted workload between men and women in a family is essential to maintaining/improving family income. The willingness of men to share household chores is fundamental to reducing time and workload a woman typically takes on a daily basis. This fosters women with

[68] Le Anh Tuan & Grant R. Singleton. 2018. Promoting Adoption of Agricultural Technologies - A Guidance Note.

additional time for self-care, child development, learning opportunities, and new income opportunities such as attending project training on new farming technology, financial management, and taking up new jobs created in the value chain development process promoted under the project.

Opportunity 2 – Female are increasingly involved in household decision makings

Though gender norms traditionally limit women's participation in decision making at household and community levels, empirical evidence suggests a new opportunity: while women's participation in public decisions (community level) are still rare, women have been playing an increasing role in decision making at household level, particularly those related to income/financial management and children development. This suggests that men have increasingly recognized the role of women in their family well-being, especially in child care/ development, management of family expenditure, loans, and trading of farm produce. Most women have participated in deciding selling prices of their farm produce – jointly with their husbands albeit their participation in agricultural production process as a labor has decreased due to increased mechanization which frees up their time and participation in labor intensive tasks such as soil preparation and harvest. Improved water access (thanks to project investment) will generally double income from rice production and reinforce the existing roles of women but will require more participation of women in the decision making process, particularly for households that join the project's value chain for long-term and sustainable rice production.

Opportunity 3 – Increasing female participation in waged jobs and migration

Poverty (exacerbated by natural disasters) have left rural people with no choice but to migrate. This is especially true to migrant who don't have sufficient land for farming and monthly income cannot cover expenditures[69]. Data has suggested that migration of women to Phnom Penh and other provinces has been increasing, and more remarkably over the past decade. Of 58.5% of rural women who moved to Phnom Penh, 32.2% work as garment workers, 23.4% as small business owners, 11.1% as domestic workers, and 10.3% as service/entertainment workers. Most migrant women (82.9%) sent remittance home (compared to men - 75.9%[70]). In project area where a high level of women participation in garment factories already exists (e.g. at Krapeu Troum scheme of Krang Ponley subproject in Oudong district, Kampong Speu), there would be a significant lack of women's participation in rice production under the project. However, the opportunity for farming households to increase the number of rice crops (from one to two crops a year) will increase income for all farming households, including households in which women worked full time in nearby factories. Increased mechanization also frees up labor intensive farm workers and reduces physical works of women in rice farming.

Opportunity 4 – Project opportunities for gender awareness raising among youth farmers

Various studies suggested that income gaps are not common among youth households (compared to income gaps in households headed by persons more than 40 years of age). The openness of youth facilitates sharing of household chores and other farming works. Targeting farming youth households may gain low bearing fruit with regards to gender equality. This will be a great opportunity to maintain this momentum since intervention among youth to minimize gender gaps, and effectives. This does not only serve as a good example at the local level but also prepare for transgenerational transmission of gender equality.

Opportunity 5 – Project opportunities for increased women participation in project activities

The project offers a unique opportunity where participation of women in project design and implementation is subject to monitoring and evaluation. This ensures the target on gender mainstreamings as to different project

[69] World Food Programme 2019. Vulnerability and Migration in Cambodia

[70] Ministry of Planning 2013. A CRUMP series study: Women and Migration.

activities are set (as baseline) and monitored for changes (mid-term and endterm). Although 73% of surveyed households do animal husbandry, these activities serve as family's saving as income is low. There is still a chance for developing more in animal husbandry and other farming activities such as cash crop, income from post-harvest (e.g. straw, mushroom production).

4.3 Priorities for Gender Mainstreaming

Priority 1 – Alleviating women's workload and taking into account their labour needs and constraints.

Women are traditionally responsible for most household chores. This unpaid work takes them not only most of their time in a day but also restricts them from learning a chance to upgrade their farming skills which are essential for improved crop yield, and for being accepted into a value chain. To free up women gradually from traditional household chores, it is essential that their husbands share the household chores, such as childcare, cooking, home gardening, and animal husbandry. When women have more time to participate in training, community meetings, and farming activities, they contribute to increasing family income – through optimized labor and farm yield. Hence, the use of gender tools and approaches that tackle discriminatory gender norms and promote mindset and behavioral changes among rural households, towards greater sharing of household chores and responsibilities is essential. Promoting women's access to water for irrigation, livelihood activities and domestic uses coupled with the adoption of labour-saving climate-smart technologies will also contribute to alleviate women's workload in farming and household activities. It is important that the technologies delivered are informed by considerations for women's labour and productive needs.

In addition, training and extension activities, particularly those targeting women, should be delivered during off-farm season, or during time when farm work is low. It is also important to be sensitive in arranging the timing of such meetings and training during the day. Women's workloads have diurnal variations and it is important to be sensitive to this aspect. Meetings and training should be organized at a time during the day when it does not clash with the time of the day when demands on the women for domestic chores is highest. This is to facilitate sharing of household chores on the part of the male, of work shifting, which enable women to go outside their home for training and meetings. Training content and capacity building events need to be designed based on women's needs and aspirations. As such, these need to be gender sensitive, e.g. type/focus of technologies/techniques, training events, proximity to villages/communes.

Priority 2 – Improve women's skills, and access to financial resources for Improved Income and resilience

Improving women's skills that are essential to their family's income is important to economic empowerment that the project intended to foster. Skills to be selected for training, development and usage should be based on the needs of the target households, particularly the female members of the target groups. Nonetheless, it is important that women farmers are trained alongside men on new agricultural practices and technologies, including promoting access to irrigation and mechanization. Involvement of women in agricultural training will vary depending on the context and household livelihood characteristics. It is clear that women from farming households who are not involved in other non-farm jobs will have different needs for skill development compared to women from farming households who work in nearby garment factories. It is also equally important to support women from farming households to develop non-farm skills to diversify their income sources, such as work in small-scaled agribusiness which deal with food processing and trading rather than production itself. Access to loans and grants are equally important to facilitate application of new skills for income generation and small enterprise development Local financial service provider could be supported/encouraged to ensure women participation is a sufficient and necessary condition to quality for a pro-poor loan. Grant schemes should also include special conditions and requirements to facilitate women's participation.

To increase income from rice cultivation, women and men farmers need to adopt new rice varieties that are high-yielding (See Barrier No. 5 above), and increase the number of crops from existing one crop per year to at least two crop per year. Gender work is needed to ensure that both women and men are trained on new agricultural practices, that their preferences and labour constraints are taken into account and that joint decisions on adoption of new rice varieties, crops and mechanization are made within the household. Increased use of machinery by both women and men is important. This is to ensure increased rice production at commune level does not face the existing labor shortage (due to increasing migration for non-farm works). Home gardening (vegetable) and animal husbandry (e.g. chicken, duck raising) should be promoted since these works can be done at home which is suitable for women and increase additional income and source of food for children's nutrition. The economic empowerment of women should also be promoted by supporting women's involvement in off-farm activities, such as processing and value-addition as well as investments in green products and women-led green enterprises.

Priority 3 – Promoting women's participation and leadership in community-based planning, decision-making and water governance

Women's poor decision-making capacities and participation in community affairs, which is often driven by discriminatory social and cultural norms, are still challenges to be overcome. FWUCs are still heavily male dominated although relevant institutions are very open-minded about women's participation in the community. Yet women's experiences in water and water governance and in community-based climate change adaptation activities remain limited compared with those of men. Therefore it is important that the project strengthen the capacity of women through FWUCs and other local social groups and motivate men and women to work towards gender equality; This will require strengthening water governance by addressing its social and gender dimensions to ensure that water resources and irrigation systems are managed in a transparent, participatory and equitable manner, including handing over responsibilities to women in FWUCs such as planning and decision making in water management and development, early warning system and disaster risk reduction activities;

Overall project activities should require women participation to enable the household to benefit from project activities. This aims to ensure family and husband consistently support the women to participate in project activities in a longer time and women participation is not as ticking the box. Investments that benefit groups or larger communities should require at 30% of women participation for the planning meeting to proceed. If this target is compromised, there is no impetus for later meetings that have a higher level of women participation. Thus, the target of at least 40% women participation in all project planning and implementation will be compromised. Long-term support for women's participation will facilitate change in the gender norms and behaviours.

Priority 4 – Promote Collective Farming among Smallholder Farmers

Contract farming is a model where farmers and private sectors (e.g. collector, agri-businesses) work together in the long run – through legally bonded relationships, for mutual economic benefits. When farmers change to new farming practices, it is important that they work closely with each other to ensure the crops (despite individually cultivated) are consistent in terms of variety, technology adoption. Collective farming and contract farming should be facilitated through local Farmers' Cooperatives, or a local agri-business. It is important to ensure that women producers are actively involved in those institutions and agri-business initiatives.

Opportunities for developing women-led agri-enterprises should also be explored, particularly in vegetable production, which is one of the target crops.

Priority 5 – Targeting Youth Farmers as Change Agents

Youth under the age of 30 represents 2/3 of the country's population and continues to grow at around 1.6% per year[71]. In the project area, the skill of youth and access to land resources are limited. The average income of male and female among youth is not different statistically compared to that between male and female in the group older than youth age (more than 30 years of age). Targeting youth farmers in the project area is one effective way to promote improved income at household level. This could be achieved as youth is more active in learning and applying new knowledge, they are more capable physically and thus can spend more effective time in farming activities and later on value chains. Their initial results achieved from project support can serve as champion in the social change process within their community and thus can create a good spill-over effect in terms of learning for youth outside the project area, including groups that are just above youth age, such as the group between 30-50 years of age who are quite active in farming. As there is no income gap between male and female in youth, supporting youth groups, including the group just above youth age (i.e. 30-50), can create a long-term impact which may last beyond project life in terms of income equality. Investment for youth should focus on skills, finance access, business development.

4.4 Anticipated Trends during Project Life and Beyond

The project interventions are anticipated to give rise to the following trends. These trends would become clear when construction of irrigation sub projects are completed and put back into operation:

- **Increased labor market** is anticipated to meet increasing demand of intensified rice production, particularly in increasing need in crop care such as fertilizers and pesticide application more job (hired labor) created, including jobs for women such as crop care.
- **Increased mechanization,** particularly at land preparation and harvesting stages, thanks to increased production scale and road access more job (hired labor) created and female is freed up from heavy jobs such as harvesting, trawling.
- **Opportunities to target market segments that bring farmers premium** (thanks to selection of rice variety and production technologies that target high-end markets, particularly for farmers with large farm size (e.g. more than 4ha) increasing opportunities for women to participate in trading and value chain.
- **Increased income** (double or more) thanks to a) increased number of crops per year and b) increased crop yield thanks to better crop production based on improved variety and crop management technologies (e.g. IPM, integrated crop management, alternative wet dry…) increasing decision making for women, particularly in nutrition and education for children.
- **Increased land concentration** (e.g. farmers with small farms leave farming and sell to better-off), particularly when water access becomes more reliable thanks to project investment.
- **Increased individualized farming** if contract farming (collective farming) is not promoted successfully
- **Reduced labor exchange** because of a) more synchronized irrigation and b) increased labor market reduced social networking at farm neighborhood level.
- **Reduced selling prices** due to increased farm output for the same market.

[71] UNDP 2023 (https://www.undp.org/cambodia/projects/promoting-decent-youth-employment-cambodia). Accessed on 23 September 2023.

5. GENDER ACTION & SOCIAL INCLUSION PLAN

5.1 Approach to Developing Gender Action & Social Inclusion Plan (GASIP)

The GASIP is aligned with the logical framework of the project. This aims to ensure the activities proposed in the GASIP are aligned up with overall project design and promote gender mainstreaming in all investment areas that are feasible. In addition, GASIP are proposed based on empirical evidence gathered from the project area, in-country and international experience in gender mainstreaming. Gender actions are proposed on the basis of a) analysis of gender gaps, b) constraints to gender mainstreaming, and b) opportunities to overcome such constraints. Key priorities (for promoting gender mainstreaming) are identified for the project. This paves the way to proposing various gender activities that are integrated into project investment.

5.2 Mainstreaming GASIP into Project Investment Plan

The following plan will guide the project to mainstream gender, youth and social inclusion in all project's components and activities.

With regard to gender the plan intends to strengthen women's adaptive capacity so as to contribute to broader household and community resilience. In line with the three SOs of IFAD's Gender Policy, this will be done by:

1) **Alleviating women's workload.** This will be done by a) **raising awareness** on the importance of sharing household chore and farming workload to give women more time to learn new skills (climate-smart farming technologies, irrigation system, financial management, business) that are essential for them to enhance their adaptation capacity.
2) **Promoting women's economic empowerment.** This will be done by enhancing women's access to skills and business development training that are centered around project target value chains for commodities such as rice, poultry, vegetable and fish that are appropriate to them. Women will be encouraged to participate in project trainings that help them address the potential impact of climate changes which include a) adoption of water-saving alternate wet and dry, b) adopt new seed varieties (which are more pest resistant and drought tolerant), c) adopt sustainable intensification packages that reduce seed rate, fertilizers and pesticide to save agricultural input costs and increase crop yield. Women are invited to take part in relevant training jointly with their husbands to support adoption of climate-smart technologies that also respond to their needs and preferences. In addition, project beneficiaries will be encouraged to diversity their existing sources of income, to improve the efficacy of existing income sources such as home-based gardening, animal husbandry (poultry and fish); promoting women's access to grant schemes as well as employment opportunities through value-chain development and PPP
3) **Strengthening women's role in decision-making both at the household and community level.** It is expected that the role of women in decision making at household level will be enhanced as the outcome from Item 1 and 2 above. At the household level, gender specific tools will be used to encourage joint decision-making in technology adoption and investment decisions. Women will participate in consultation activities and planning of irrigation infrastructures, value-chain development and early warning systems. The project will also promote women's participation and leadership in FWUCs, which will be trained on how to enhance equitable access to water for women and other vulnerable groups.

Key activities to support the implementation of the project's GASIP include training of extension staff and implementing partners on gender and social inclusion, including the use of tools and approaches to promote normative and behavioural changes in gender dynamics at the HH and community level; establishing partnership with the Women Development Centers and other specialized service-providers for delivering training to women; Developing gender-sensitive tools for value-chain analysis and other diagnostic activities.

6. IMPLEMENTATION OF GASIP

6.1 Implementation Arrangements

6.1.1 PMU at Central Level

- Oversee all operational aspects of provincial PMU for the project, including the GAP implementation.
- Provide overall guidance to provincial PMU with regards to GAP implementation, particularly implementing the GAP as an integral part of provincial annual socioeconomic development program and gender mainstreaming in agro- and agricultural sector.
- Monitor PMU's project activities and ensure that activities set forth in the GAP are implemented timely and effectively.
- Undertake specific studies focusing on gender and social inclusion for documenting and disseminating success stories/good practices in a range of relevant topics focusing on the nexus between gender and climate change (e.g. women adopting climate-smart labour-saving technologies; women using irrigation system; how joint decision-making contribute to strengthening farmers' resilience etc.)
- Organise consultation activities for climate-related policy-dialogues with government institutions to ensure that the voices of rural women are integrated in national strategic frameworks.
- Support and monitor training activities on gender and social inclusion with local implementing partners and extension staff.

6.1.2 PMU at Provincial Level

- Responsible for overall implementation of the project, including the GAP.
- Ensure funding is allocated sufficiently and timely for implementation of the planned activities
- Work as lead agency for GAP implementation and coordinate with other governmental agencies, such as Provincial Hall and functional agencies – at provincial, district and commune levels, to carry out planned activities.
- Establish partnership with Women Development Centers (WDCs) and other women's organizations and service-providers to deliver training to women.
- Train local implementing partners on gender and social inclusion tools and approaches – in collaboration with PMU Gender and Social Inclusion officer
- Maintain regular monitoring of GAP implementation, at quarterly, bi–annually, and annually
- Conduct periodic review and evaluation of GAP implementation to ensure the overall progress and quality of GAP implementation is on track with the planned activities.
- Where necessary, update the GAP based on development needs and feedback from IP communities in the project area. These development needs should be in line with the logical framework of the project.
- Ensure that women and other vulnerable groups have dedicated access points to feedback mechanisms and GRM

6.1.3 District Hall

- Support PMU in selecting communes that are potential for planning and carrying out activities proposed in the GAP.

- Ensure the activities set forth in GAP are carried in a manner that is integrated with the overall socioeconomic development plan of the district, and in connection with other annual district development programs.
- Provide overall guidance to commune hall within the district in GAP implementation.
- Support information campaigns in the project's area with a focus on gender and social inclusion

6.1.4 Commune Governments

- Collaborate and support PMU in implementing GAP.
- Support PMU in carrying out consultation/ training/ workshops/ with effective participation of female representatives of target households and of female headed households.
- Provide regular support and encourage the participation of women and female headed households in all project related activities, particularly consultation meetings, trainings, workshops.

6.1.5 Project Target groups

- At household level, arrangements should be made to enable women to participate in project activities, including consultation and planning meetings, training and workshops. This will empower women to participate and make decisions in planning and economic activities to support household's income and well-being.
- Gender transformative approaches, such as GALS should be used to promote joint decision-making at the household level and ensure that climate-smart technologies and farm infrastructure benefit all household members and alleviate women's workload.
- Women and women's groups and enterprises are encouraged to participate in workshops, trainings and grant schemes that are designed to promote their participation and economic empowerment.
- Women are encouraged to participate and lead FWUAs
- Provide gender-sensitive comments/suggestions/feedback, including raising questions for issues they are concerned about.

6.2 Monitoring and Evaluation Arrangement

The implementation of the GASIP will be monitored annually vis-à-vis achievement using suggestive performance indicators (See Annex 1 – Section A. Summary of GASIP Performance Indicators). Evaluation of the GASIP will be conducted at midline and end-line. It is noted that the midline review will serve the opportunities for the project to update the original plan based of the progress that has been made over the first three years and will be adjusted based on the overall project performance progress

6.3 Costs and Budget

Budget for gender mainstreaming will be from project financing.

No.	Items	Time	Cost	**Total** (USD)	Notes
1	**Recruit a gender consultant**	Year 1	2,500 x 12	30,000	
	● Update GASIP to reflect detailed project implementation plan ● Updated Indicators, Baseline & Target based on IFAD's Gender Monitoring template				

	● Developing data collection monitoring forms ● Develop guidelines for mainstreaming gender aspect into awareness raising campaign and relevant training materials ● Build capacity for PMU's Gender Officer ● Build capacity of extension staff and other implementing partners on the use of participatory gender tools and approaches in diagnostic and extension activities (GALS?)				
3	**Recruit and maintain a gender officer**	6 years	1,200 x 12 x 6	86,400	
2	**Conduct gender awareness raising campaign in Year 1 and repeated at Midline**	Year 1 & 3	2,000 x 6 x 2 schemes	24,000	
4	**Gender mainstreaming** ● Training materials ● Training delivery			0	Costs integrated into respective consulting services
5	**Monitoring & Evaluation by PMU**			0	Costs integrated into project financing (Component 3)
			TOTAL	**140,400**	

Annex 1 – Template of Updating Annual GASIP

- Prepared By: PMU of _________________ province
- Submitted to:
- Fiscal Year: _________________

HOW TO USE THE TEMPLATE

- This template has two parts, including Part A (GASIP Indicators and Part B (Annual GASIP)
- Provincial PMUs needs to prepare and submit this Plan to MoWRAM before the start of each Fiscal Year to request budget allocation, especially for activities related to Project's GASIP
- Provincial PMUs will discuss with provincial project's stakeholder and fill up targets in Part A before the beginning of each fiscal year such that targets tentatively set for Mid-line and End-line are achieved.

PLEASE NOTE

FOR PART A

- The percentage of IP households receiving project benefits are calculated based on the List of Project Beneficiary Households (not the count of any total IP households that receive project benefit) to avoid duplication of counting.

FOR PART B

- Activities proposed in the Annual GASIP should be aligned with respective Component and Output (based on project's Logical Framework) to allow annual monitoring, and evaluative review by mid-term and project-end.
- Each proposed activity should come with a proposed budget, indicating the period (in months) from activity planning to completion. Each activity should indicate the lead stakeholder who is responsible for the expected result of the activity, and relevant stakeholders who need to participate to realize the activity.

A. SUMMARY OF GASIP PERFORMANCE INDICATORS

Project Component				Indicators	TARGETS					
	Budget Allocation (in USD)				Year 1	Year 2	Year 3 (Mid-line)	Year 4	Year 5	Year 6 (Endline)
			Ethnicity	**% of IP households** (in total beneficiary households) receiving benefited from project			**20%**			**40%**
				% of Khmer households (in total beneficiary households) receiving benefited from project			**30%**			**60%**
			Gender	**% of FEMALE people** who benefit from project			**25%**			**50%**
				Of which % of IP female						**40%**
				Of which % of Khmer female						**60%**
				Of which % of female from vulnerable households (e.g. female headed household…)						
				Of which % below the poverty line						
			Youth	**% of YOUTH people receiving <u>services promoted or supported</u> by project**			**15%**			**30%**
				Of which % of IP youth						
				Of which % of Khmer youth						
				Of which % of female person						
				Of which % of male person						
				Of which % below the poverty line						
				% of YOUTH people <u>with new jobs/employment opportunities</u>			**15%**			**30%**
				Of which % of IP youth						
				Of which % of Khmer youth						
				Of which % of female person						
				Of which % of male person						

B. ANNUAL GASIP

PROJECT ACTIVITIES	BUDGET (in USD)	MONTHS												STAKEHOLDERS IN CHARGE			REMARKS
		1	2	3	4	5	6	7	8	9	10	11	12	PMU STAFF INVOLVED	LEAD AGENCY IN CHARGE	RELEVANT STAKEHOLDERS	
Component 1: IMPROVING FARM-LEVEL CLIMATE ADAPTATION, RESILIENCE, AND WATER USE EFFICIENCY																	
Output 1.1 – Deployment of farm-level climate adaptation and water use efficiency measures																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
Output 1.2 – Climate adapted, value added, and market led agricultural investment																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
Output 1.3 – Improve enabling conditions, capacities and disaster risk management strategies																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
Sub-component 1.4 Rural roads.																	

ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
Component 2: UPGRADING AND CLIMATE-PROOFING WATER INFRASTRUCTURE FOR INCREASED RESILIENCE																	
Output 2.1 – Modernization of irrigation scheme and ponds																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
Output 2.2. – Flood-proofing and Drainage improvements																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
Output 2.3 – Establishments and training of Farmers Water User Communities (FWUC)																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
Output 2.4 – SCADA																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	
Component 3: INSTITUTIONAL STRENGTHENING																	
Output 3.1 – Strenthening MOWRAM capacity																	
ACTIVITY																	

ACTIVITY																	
ACTIVITY																	
Output 3.2 – Strengthening of NDA and NCDD																	
ACTIVITY																	
ACTIVITY																	
ACTIVITY																	

Annex 2 – Gender Action and Social Inclusion Plan (GASIP)

All targets were set in this table as a tentative one. The Gender Action Plan is expected to update annually to engage the participation of female beneficiaries and the Department of Women's Affairs. This is to promote the ownership and buy-in from local people and the Department of Women's Affairs.

Project activities	Responsive Activities	Indicators		Base-line	Mid-line	End-line	Responsible institutions	Timeline	Notes
		•							
							•		
COMPONENT 1: IMPROVING FARM-LEVEL CLIMATE ADAPTATION, RESILIENCE AND WATER EFFICIENCY OUTPUT: • Climate resilient crop water management practices at farm level enabled • Climate resilient value added, and market led agriculture investment secured • Enabling conditions facilitated and capacities for climate resilient on farm water management and agriculture practices improved									
Sub-Component 1.1 Deployment of farm-level climate adaptation and water use efficiency measures									
Activity 1.1.1 Preparing community-based action plans for adapting to climate resilient crop-water management practices and their monitoring **Activity 1.1.2** Preparation of training materials to support implementation of the AP	1.1.1.Women farmers from both couple and WHHs are actively involved in community-based planning for adoption of climate resilience crop-water management practices. 1.1.2. training women in climate-smart technologies.	• % female PARTICIPATING in each of the planning meetings. • Number of action plans integrating specific women's preferences. • Percentage of female-headed households participating in community-based action plans 1.1.2 • % female ATTENDING each awareness raising activities on new technologies		10% 0% 0% 0%	20% All 20 % 30%	40% All 40 % 50 %	PMU PMU PMU PMU		

Project activities	Responsive Activities	Indicators		Base-line	Mid-line	End-line	Responsible institutions	Timeline	Notes
Activity 1.1.3 Conduct trainings to create a pool of expertise to demonstrate and propagate the CR technologies and practices **Activity 1.1.4** Training and advisory on CR technologies and practices **Activity 1.1.5** Strengthening and fostering tailored mechanization service providers for improved mechanization service delivery. **Activity 1.1.6** community-based monitoring and evaluation (CBME) of implementation	1.1.3. training women in gardening and livestock production.	• % female ATTENDING each training to adopt introduced technologies		0%	30 %	50 %	PMU		
		• % female participating in decision over adoption of technologies that are labour-saving and suitable to their production		0%	10%	20%	PMU		
		• % female ADOPTING new technologies		0%	5%	10%	PMU		
		• % HHs ADOPTING one new non-rice crop (diversification)		0%	10%	20%	PMU		
		• Percentage of female-headed households participating in community-based action plans		0%	5%	10%	PMU		
		• Percentage of female-headed households participating in training on crop diversification		0%	10%	20%	PMU		
		• Percentage of women participating in project training who report reduced domestic workload and increased participation income generation activities.		0%	5%	10%	PMU		
		• Percentage of women who are trained under Training of Trainer to become a local change agent		0%	20%	40%	PMU		
		• Percentage of women who are interested in animal husbandry and who are trained chicken production		0%	10%	20%	PMU		
		• Percentage of women who are interested in Rice Farmers		0%	20%	40%	PMU		

Project activities	Responsive Activities	Indicators		Base-line	Mid-line	End-line	Responsible institutions	Timeline	Notes
		Field School and are trained on Rice FFS. ● Percentage of women who are interested in vegetable production and are trained vegetable production 1.1.3 ● At least 40% of people selected to attend project's TOT trainings will be female		0%	5%	10%	PMU		
Sub-Component 1.2 Climate adapted, value added, and market led agricultural investment									
Activity 1.2.1 Value chain study and planning **Activity 1.2.2** Establish District Multi-Stakeholder Platforms (MSPs) **Activity 1.2.3** Public Private Producer Partnership Facility (4PF)	1.2.1. Participatory gender and social inclusion analysis of value-chains is carried out to identify opportunities for women and youth in value-chain development and 4PF. 1.2.2. ● 4PF arrangements are selected and co-financed taking into account employment generation impact for women, men and vulnerable groups	1.2.1. ● Number of VC studies including attention to gender and social inclusion dynamics. ● N. of VCAPs reflecting detailed opportunities for women, youth and vulnerable groups. 1.2.2. ● Percentage of women/youth who establishes agri-business/rural enterprises ● Percentage of women/youth accessing grant scheme ● Percentage of female-headed households participating in value chains supported under the project ● Percentage of female-headed households participating in animal husbandry training under the project		1 1 0% 0% 0%	1 1 20 % 20% 20%	1 1 40% 40% 40%	PMU PMU PMU PMU PMU PMU		

Project activities	Responsive Activities	Indicators		Bas e-line	Mid-line	End-line	Responsible institutions	Timeline	Notes
	• Training women and youth in business skills development, processing etc.	• Percentage of women participating jointly with their husband in value chain supported under the project • Percentage of women/youth gaining employment in VC development and 4PF • Percentage of women-led enterprises (out of target 100 enterprise) to be supported under the project • Percentage of women receives grant based on their grant application • Percentage of youth receives grant based on their grant application		0% 0% 0% 0% 0% 0%	20% 20% 30% 30% 20% 15%	40% 40% 50% 50% 40% 30%	PMU PMU PMU PMU PMU		
Sub-Component 1.3 Improve enabling conditions, capacities and disaster risk management strategies									
Activity 1.3.1 Strengthening of agrometeorological stations in the project area **Activity 1.3.2** Establish Provincial-Level Coordination Platform for climate-informed agricultural advisories	1.3.1. 1.3.2. Extension workers are trained on the use of gender tools and approaches , also including attention to issues of joint decision-making and equitable workload distribution with a focus on	1.3.2. • Number of extension workers trained on the use of gender tools and approaches • Extension manuals integrate gender tools and approaches. 1.3.3. • Average % women/marginalized groups consulted		0% 0%	20 all	40 all	PMU PMU		

Project activities	Responsive Activities	Indicators		Base-line	Mid-line	End-line	Responsible institutions	Timeline	Notes
Activity 1.3.3 Building the capacities to deliver services **Activity 1.3.4 Establish the agrometeorology information and warning systems and the mechanisms including awareness raising**	climate smart agriculture. Activity 1.3.3. gender sensitive analysis and consultations are carried out to prepare a water, climate and agricultural early warning system that is also accessible to women and marginalized groups.	• At least 40% of individuals attending both formal and on-the-job training are women 1.3.4 • Percentage of farmers in the target command area (sub-scheme) receive agrometeorology information and warning systems • Percentage of farmers in the target command area (sub-scheme) report the agrometeorology information and warning systems help them take appropriate action to avoid/minimize adverse impact on their crop production • At least 40% of farmers receiving and using the information are women *Community based monitoring and evaluation* • A meeting will be held annually at each sub-scheme to agree with target community the		0% 0% 0% 0% 0%	40% 20% 20% 10%	40% 40% 40% 20%	PMU PMU PMU PMU PMU		

Project activities	Responsive Activities	Indicators		Base-line	Mid-line	End-line	Responsible institutions	Timeline	Notes
		issues they wish to monitor - as participatory monitoring			20%	40%			

Component 2: Upgrading and Climate-Proofing Water Infrastructure for Increased Resilience
OUTPUT:

- Flood proofing and functional drainage system operational
- Irrigation System modernized with climate resilient technologies
- Capacity of FWUC on water management increased

Sub-Component 2.1 Modernization of irrigation scheme and ponds

Project activities	Responsive Activities	Indicators		Base-line	Mid-line	End-line	Responsible institutions	Timeline	Notes
Activity 2.1.1 technical analysis, field survey and preparing plan for system upgrading **Activity 2.1.2** Implementation of infrastructure upgrading **Activity 2.1.3** Preparing canal O&M plans including application of ICT and SCADA for operation.	2.1.1. women's water needs are considered in technical analysis, field survey and preparation of plans for system upgrading 2.1.2. Infrastructure upgrading is informed by women's water needs; construction work generates employment for youth and vulnerable groups. 2.1.3. Preparation of canal O&M plans are carried out in consultation with women and youth.	2.1.1. ● Average % women consulted ● No. of plans for system upgrading incorporating specific features, which reflects women's water needs. ● 2.1.3.% women/youth consulted for canal O&M planning.		0% 0 0%	40% 5 (sub project level) 20%	40% 10 (sub project level) 40%	MOWRAW PMU PMU PMU		
Sub-Component 2.2 Flood-proofing and Drainage improvements									
Activity 2.2.1 Establish flood									

Project activities	Responsive Activities	Indicators		Base-line	Mid-line	End-line	Responsible institutions	Timeline	Notes
monitoring, information, and early warning systems **Activity 2.2.2** Strengthen and construction of flood control and drainage infrastructures. It will be implemented in an integrated manner with component 2.1 activities	Training on early warning system is delivered to women and marginalized groups	2.2.1. Average % women/marginalized groups/ethnic minorities participating in training on early warning system		0%	10%	20%	PMU		

Sub-Component 2.3 Establishments and training of Farmers Water User Communities

Project activities	Responsive Activities	Indicators		Base-line	Mid-line	End-line	Responsible institutions	Timeline	Notes
Activity 2.3.1 Formation of institutional strengthening of the FWUC **Activity 2.3.2** Build technical capacities of FWCU for canal O&M **Activity 2.3.2** Prepare long term financing plan for WUS and support its implementation.	2.3.1. FWCU are trained and develop gender and social inclusion plans, to promote the user rights of women and vulnerable groups and support women's participation and leadership in water governance 2.3.2. Training of women/youth/vulnerable groups trained in O&M	2.3.1. ● Average % FWUC with women in leadership positions. 2.3.2. ● Women/youth trained in O&M ● Youth/women/vulnerable groups gaining employment in O&M *Community based monitoring and evaluation* ● A meeting will be held annually at each sub-scheme to agree with target community the issues they wish to monitor - as participatory monitoring		0 0 0% 1	5 % 10% 20% 1	10% 20 % 40% 1	PMU PMU PMU PMU		
Output 2.4: Water information system (CIEWS and SCADA) established and operational in sub-projects									
Activity 2.4.1 Preparing all processes for establishment of the SCADA and Water information system									
Component 3: Institutional strengthening OUTPUT									

Project activities	Responsive Activities	Indicators		Base-line	Mid-line	End-line	Responsible institutions	Timeline	Notes
• Improved capacity of MoWRAM, NDA and stakeholder, and enhanced project sustainability • Enhanced project sustainability									
Sub-Component 3.1 Strenghtend MOE									
Activity 3.1.1 Activity 3.1.1 Strengthening River basin management and Water Accounting Systems **Activity 3.1.2** Communicating with stakeholders **Activity 3.1.3** Support to and guidance of Leaders of Commune, Village and FWUCs and Gender Awareness	3.1.1. Training of MOWRAM staff on gender and social inclusion	3.1.1. • Number of climate resilient design manuals for irrigation including attention to gender and social inclusion • Number of training on gender and social inclusion delivered to MOWRAM staff		0 0	All 4	All 8	MOWRAM		
Sub-Component 3.2 Strengthening of NDA and NCDD									
Sub-Component 3.2 Strengthened NCCDS at national and provincial level									
Activity 3.2 .1- Strengthening of NCDDS at the national level		3.2.1.							
Activity 3.2 .2 – Strengthening of the provincial teams									

Annex 3 – Summary of stakeholders consulted for gender assessment during project preparation

1. *Summary of type of stakeholder consultation sessions conducted for gender assessment*

1.1 Round 1 of 2 (June-December 2023)

No.	Time	Number of Community meeting	Number of participants		
			Male	Female	Total
1	26-39 June 2023	7	270 (82%)	58 (18%)	328 (100%)

Province	Irrigation Schemes	Between from Canal Up to 100 M from a Canal (Close to Canal)	Between 100 M to 2.5 KM	Beyond 2.5 KM	Total
		N	N	N	n
Pursat	O Ta Paong Tanay Scheme	74	64	14	152
	Sub-total 1	**74**	**64**	**14**	**152**
Kampong Chhnang	Lum Hach Scheme	65	43	23	131
	Krang Ponley Kandal Scheme	6	3	5	14
	Anlong Chrey Scheme	37	14	2	53
	Sub-total 2	**108**	**60**	**30**	**198**
Kampong Speu	Brambei Mom Scheme	25	42	20	87
	Kra Peu Troum Scheme	22	17	19	58
	Krang Ponley Kandal Scheme	11	7	16	34
	Sub-total 3	**58**	**66**	**55**	**179**
Kandal	Krang Ponley Kandal Scheme	34	51	16	101
	Sub-total 4	**34**	**51**	**16**	**101**
	Grand Total	**274**	**241**	**115**	**630**

1.2 Round 2 of 2 (July – September 2024)

Participants consulted for the ESCIA, including additional gender assessment

Levels	Household survey (Beneficiaries and Potentially Affected)	Focus Group Discussions	Key Informant Interviews	Total
Village level	272 (134 Females)	309 (180 Females)	32 (3 Females)	613 (317 females)
Commune level			35 (11 Females)	35 (11 Females)

District level			28 (10 Females)	28 (10 Females)
Provincial level			51 (12 Females)	51 (12 Females)
			Total	**727 (350 Females)**

Types of stakeholders, and topics of the consultations is shown below:

No.	Levels	Key topics	Methods	Timeframe	Participants	Concerns
1	Village	Beneficiary Households	Household Survey	10 - 18 August 2024	A household survey was carried out to gather information from local people living in the 31 villages neighbouring the ESCIA project area.	Refer to concerns and suggestions from consulted people in section 9.2.2.
		Potentially Affected Households				
		Rice farmers	FGDs	10 - 18 August 2024	309 persons (129 male and 180 female)	
		Vegetable producers				
		Chicken raisers				
		Duck raisers				
2	Commune	Commune Committee for Women and Children	Key Informant Interviews	26 - 28 July 2024 03 - 09 August 2024	31 persons (20 male and 11 female)	
		Agricultural Cooperatives				
		Water Farmer User Community				
		Fishermen				
		Community Forestry				
		Community Fishery				
3	District	Offices of Development and Planning, Agriculture, Environment, Women Affairs,	Key Informant Interviews	21 - 23 August 2024	28 persons (18 male and 10 female)	
4	Province	Provincial Hall, Departments of Land, Water, Agriculture, Environment, Women Affairs, Labour, and Culture	Key Informant Interviews	26 - 28 July 2024 21 - 23 August 2024	51 persons (39 male and 12 female)	

2. *Summary of feedback from consulted participants for gender assessment*

Concerns

- Participants are concerned that the presence of workers which could increase the risk of sexual abuse of women and children living near the project site,

- Drug trafficking and consuming among workers may cause unsafe environment for both workers and surrounded residents,
- The construction activities may generate dust and slippery road conditions which may lead to traffic incidents,
- Waste generated from worker camps and construction activities during the project could degrade water quality, negatively impacting human health, livestock, and crops.
- Unequal water distribution among farmers leads to partisanship and disputes, often because farmers do not adhere to water distribution instructions or announcements from authorities, and
- Manure for fertiliser is declining due to reduced animal raising and lower cattle prices are prompting farmers to reduce livestock raising.

Suggestions from the consulted people

- Relevant government departments should provide training on gender issues, violence prevention, labor laws, and social protections to their staff and workers,
- The project should conduct gender education at the district to prevent and address gender-based violence.
- Contractors should promote gender awareness, enforce stricter punishments for offenders, and prevent drug trafficking in the workplace,
- Women should be engaged and be promoted as the members of Water User Community to promote water distribution equality regarding water allocation from the system,
- Construction companies should prioritize hiring local workers, comply strictly with laws, and take steps to mitigate concerns about gender-based and sexual violence.
- Facilitate the development of additional agricultural markets.
- The construction companies must collaborate with local authorities in all activities to make a proper management, control and monitoring during construction,
- The construction companies should regularly place warning signs and water the roads, as well as schedule construction times to minimize noise pollution that could disrupt the community, and
- Separate accommodations for men and women should be provide.

See more gender neutral feedback from stakeholder consultation provided at Annex 3 of stand-alone Stakeholder Engagement Plan.